



2.0 Tension and Speed Control

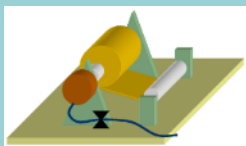
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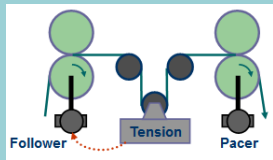
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WH Fundamentals

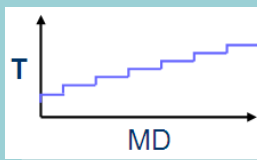
2: Tensioning



**Tensioning
Elements**



**Tension
Control
Systems**



**MD Tension
Profile**

+ Tension Control

- ✓ What is the right tension for your web or process?
- ✓ What controls tension in your process?
- ✓ What determines the optimum number of tension zones?

2.0 Tensioning

2.1 Tension Variations

2.2 Tensioning Systems

2.3 Tension Zones

2.4 Speed, Length, Diameter

2.5 Special Topics

What is Tension? Tension Control?

Tension is:

- ▶ The act or process of stretching something tight.
- ▶ The condition of so being stretched.
- ▶ A force tending to stretch or elongate something.

Tensioning:

- ▶ The process of creating and tension (hopefully, more than zero and less than the breaking point).

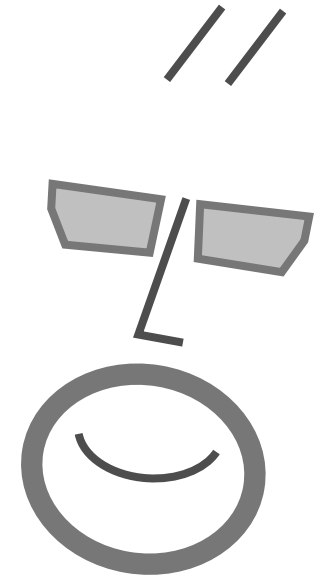
Tension Control is:

- ▶ The process of creating tension or elongation within a desired range about a set point.

The Right Tension Qs

Why tension a web?

What problems are caused by poor tension control?



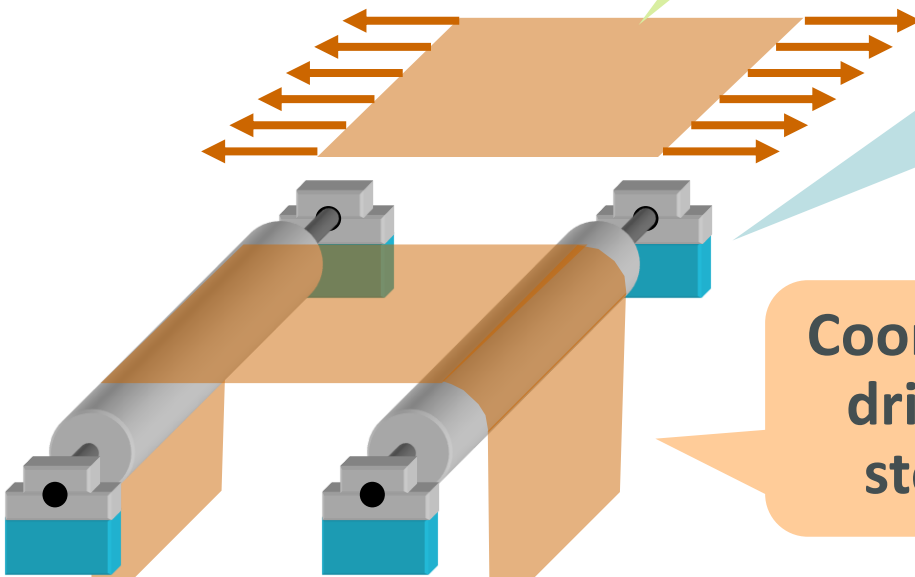
A: Benefits of Good Tension and Speed Control

Controls web **stretch (strain)** to pull out bagginess without damaging the web.

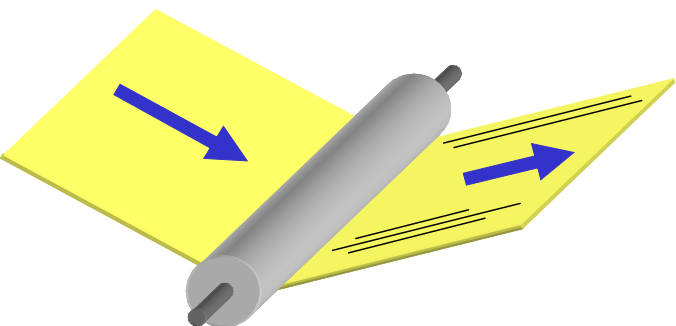
Stiffens the web to reducing sag and wrinkles.

Creates **traction** on rollers and in winding rolls to control slip, scratching, and tracking.

Coordinates web **speed** at different drive points during acceleration, steady-state, and deceleration.



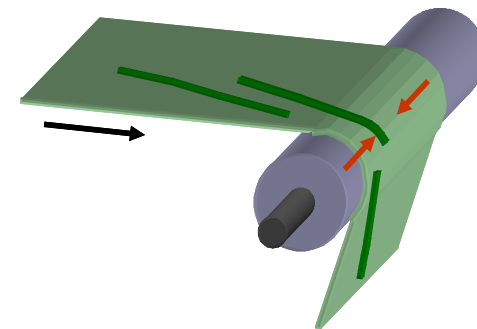
Problems from Poor Tension Control



Slip, Scratching



Web Breaks



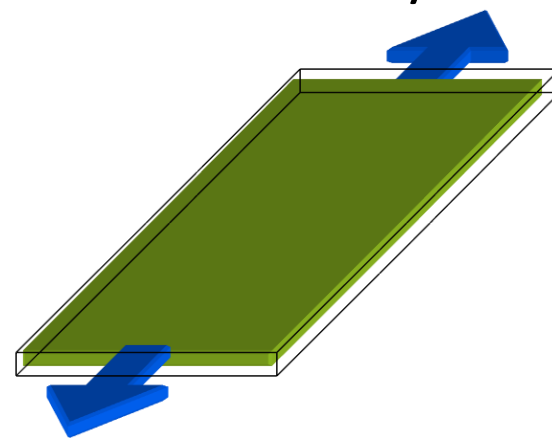
Increased Wrinkle Sensitivity



Loose or Tight Roll Defects



Web Curl



Width Variation

Problems of Low Tension

- ▶ Web fails to peel and wraps rollers, breaking the web or damaging the equipment.
- ▶ Web slips on idlers, causing scratches, lateral shifting, and guiding system failure.
- ▶ Web slips on driven rollers, especially drive rollers without nips or vacuum assistance, causing:
 - ▶ Loss of tension control or draw ratio
 - ▶ Tension varying with web-roller traction
 - ▶ Loss of speed control, if the slipping drive roller is the process pacer.

Problems of Low Tension

- ▶ Web sags in long, non-vertical spans.
- ▶ Web sails laterally in air floatation ovens and turns.
- ▶ Bagginess is revealed, wrinkling at nip points, sailing or shifting with even slightest air drag, fluttering in lanes or edges at zero or low tension.
- ▶ Web flutters, especially in long spans.
- ▶ Wound rolls are too loose, telescoping in handling, winding, or unwinding.

Problems of High Tension

- ▶ **Width Variations or Loss:** Web MD elongation creates CD width contraction from Poisson's ratio effect, causing width-related problems, such as:
 - ▶ Loss of edge trim
 - ▶ Mismatched widths at laminating
 - ▶ Loss of edge margin at coating
 - ▶ Roll sidewall misalignment at winding

Problems of High Tension

- ▶ **Elastic Width Change:** For most web handling tensions, width contraction is elastic and will be recovered if tension is lowered or removed. Width contraction will be more significant for low modulus webs and webs with high Poisson's ratios (some, but not all tissues, textiles, nonwovens, and film).
- ▶ **Inelastic Width Change:** Necking is the extreme case of width loss, causing a permanent width changes that accompanies permanent length change of yielding.

Problems of High Tension

- ▶ Web or splice breaks.
- ▶ Wrinkle sensitivity increases with added web-roller friction.
- ▶ Winding or unwinding rolls cinching and telescope.
- ▶ Wound rolls are too tight, leading to tight roll defects, such as blocking, core crushing, starring, spoking, and yielding (that creates bagginess).
- ▶ Web tension over radius increases pressure beyond cushion pressure of air floatation systems (air turns, floatation nozzles), making undesired contact, scratching, contamination, or web breaks.

Problems of CD Tension Variations

- ▶ Web breaks or yields when edge stress is too high.
- ▶ Wrinkles on rollers (especially equipment-induced one-sided tension variations or loose center baggy web in long spans).
- ▶ Wrinkles at nips (if portions of the web are at or near zero tension).
- ▶ Wide rolls or adjacent slit rolls have quality variations as function of CD position.

Always Under Tension?

Manufacturing Process Advice

Automated Assembly:

- ▶ Grab a part and don't let it go.
 - ▶ The transition from a free part to a properly fixed and positioned part is crucial to automated assembly processes. Letting go of a part gives up control that is difficult to get back.

Web Handling:

- ▶ Tension a web and don't let go.
 - ▶ The transition from a loose to tensioned and laterally aligned web is crucial to web handling processes. Letting go of tension, which also releases lateral position, gives up control that is difficult to get back.



2.2 Tension Settings and Variations

What is the right tension?

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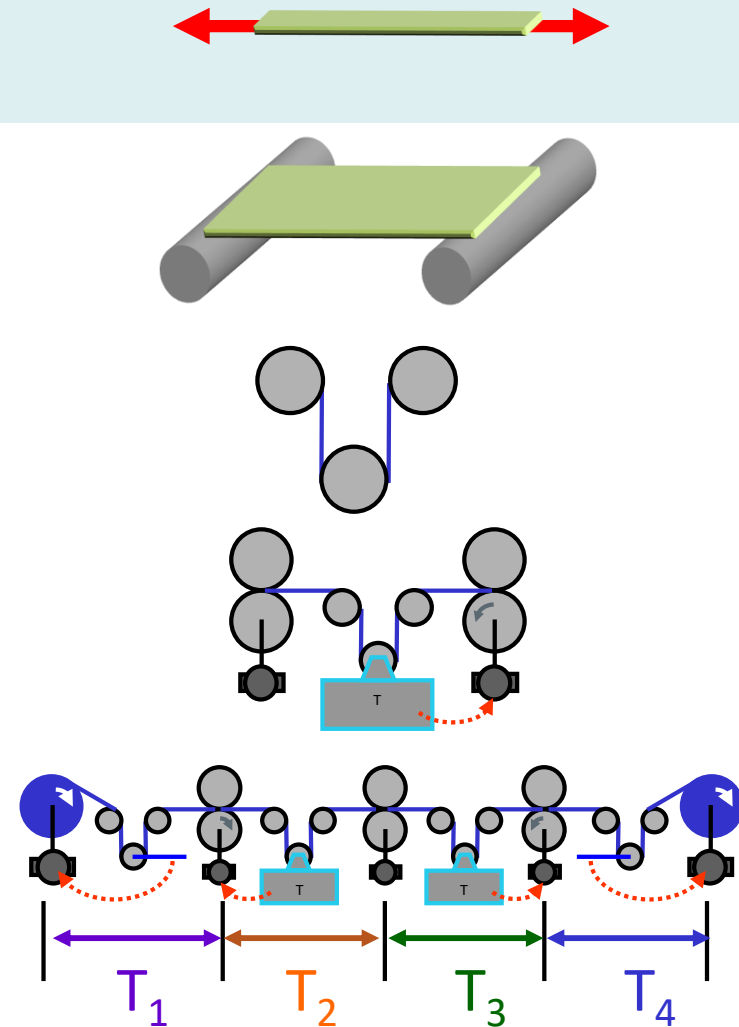
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2.2 Learning Objectives

- ▶ What is tension? What is compression?
- ▶ Why is tension important to web handling?
- ▶ What problems occur if tension is too high?
- ▶ What problems occur if tension is too low?
- ▶ What tests are used to determine the right tension for a given product or material?
- ▶ What is strain? Why is strain important?

Tensioning Hierarchy

- ▶ Tension Fundamentals
- ▶ CD Tension Variations
- ▶ MD Tension Variations
- ▶ Tension Zone
- ▶ Tension in Multiple Zones



The Right Tension?

- ▶ Strain? When you tension it, how much does it stretch?
- ▶ Force and Tension (Force per Width) are fine, but Stress is what predicts web behavior.
- ▶ Stress is Tension per Thickness or Force over Cross-Sectional Area
- ▶ What stretch will damage it (yield or break)?
- ▶ Convert errors into percent and compare to damaging strain.
- ▶ Set average tension to 10-20% of trouble, maybe higher if variations are low. Understand tension variations (setting, CD, MD, time)

Typical Tensions

The 'magic' tension is 1 lb/in or 1 PLI.

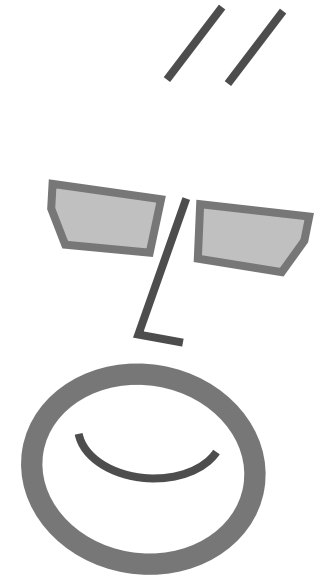
175 N/m or 18 kgf/m

80-90% of processes run
between 0.3 and 3.0 PLI

50 N/m or 500 N/m

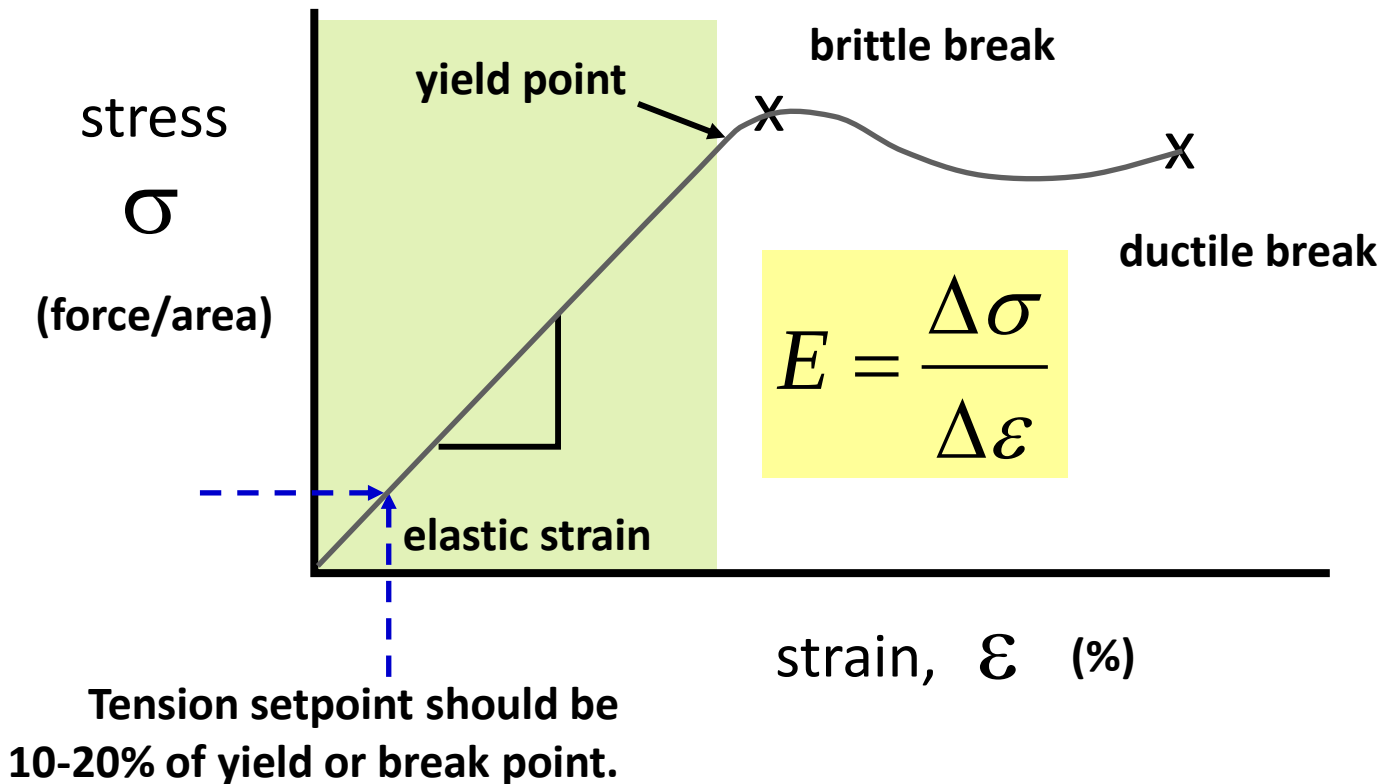
95-98% of processes run
between 0.1 and 10.0 PLI

17 N/m or 1750 N/m



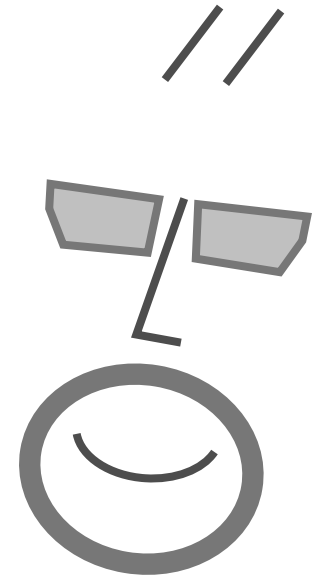
Web Handling Stress

Typical web handling tensions are 10 to 20% of a web's yield or break stress.



The Right Tension Qs

What are typical tension
by product?



Typical Tensions – Paper, Paperboard

$$1\text{pli} = 175 \text{ N/m}$$

Paper Basis Wt	Tension	Paperboard	Tension
15 lbs/ream	0.5 pli	8 point	3.0 pli
20 lbs/ream	0.75 pli	12 point	4.0 pli
30 lbs/ream	1.0 pli	15 point	5.0 pli
40 lbs/ream	1.5 pli	20 point	7.0 pli
60 lbs/ream	2.0 pli	25 point	9.0 pli
80 lbs/ream	2.5 pli	30 point	11.0 pli
		40 point	14.0 pli
		50 point	16.0 pli
		60 point	18.0 pli

$$T_{\text{UNWIND}} (\text{in pli}) = 0.035 \text{ basis wt}$$

$$T_{\text{WIND}} (\text{in pli}) = 0.055 \text{ basis wt}$$

$$\text{ream} = 3000 \text{ ft}^2$$

$$\text{lbs/ream} = 0.615 \text{ g/m}^2$$

$$\text{pli} = 1.75 \text{ N/cm}$$

$$\text{point} = 0.001'' = 25.4 \mu\text{m}$$

Courtesy of TAPPI

Example Tension – Films, Foils

$$1\text{ pli/mil} = 7 \text{ N/m}/\mu\text{m}$$

Material

Tension

Polyester (PET)

0.5 - 1.5 pli/mil

Aluminum (Al) Foil

Cellophane

0.5 - 1.0 pli/mil

Polystyrene

1.0 pli/mil

Polypropylene

0.25 - 0.50 pli/mil

Polyethylene

Nylon

0.10 - 0.25 pli/mil

Vinyl

0.05 - 0.2 pli/mil

pli = lbs/in (pounds per lineal inch of width)

(Sometimes referred to as PIW)

mil = 0.001"

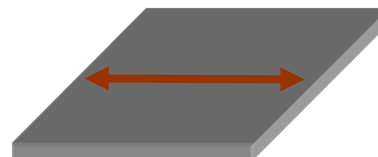


PET

t = 0.001"

w = 2"

F = 2 lbs



Al Foil

t = 0.003"

w = 5"

F = 15 lbs

Vinyl

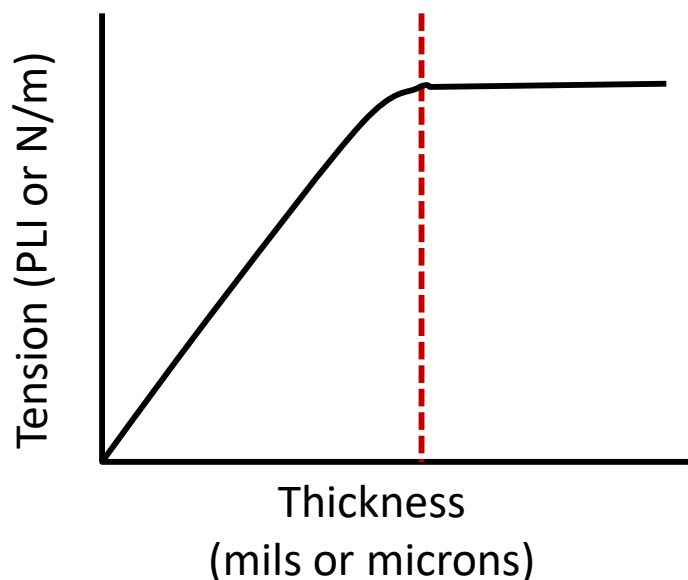
t = 0.005"

w = 24"

F = 12 lbs



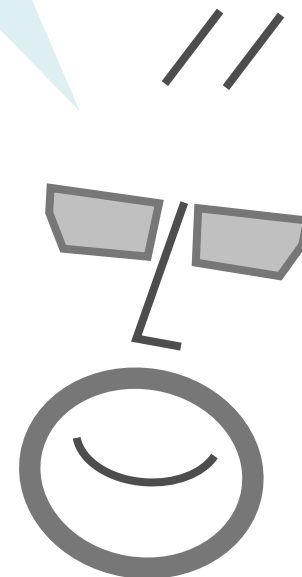
Tension, Width, and Thickness



If width doubles, tension force should nearly always double.
How about if thickness doubles?

Tension (i.e. force/width) is usually proportional to thickness, especially for matching laminating strains. However, at greater thicknesses, tension may not be increased due to:

- Equipment tensioning limits
- Less need for tension stiffening in thick webs
- Limited ability to pull out bagginess in thick webs

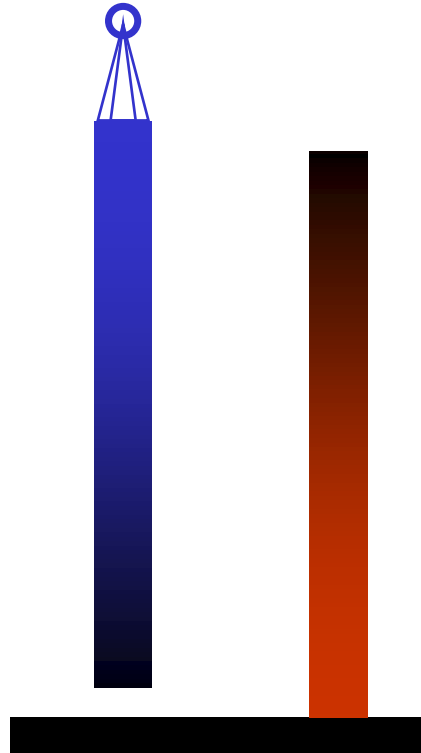


Tension vs. Compression

Tension: A force tending to stretch or elongate a solid material.

Here, the hanging beam develops tension due to gravity.

The gravity-induced tension will go from zero at the bottom to the full weight of the beam at the top.

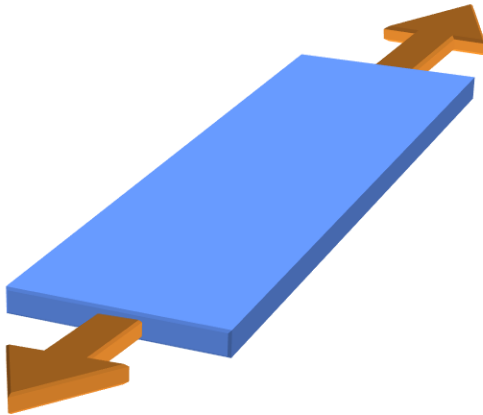
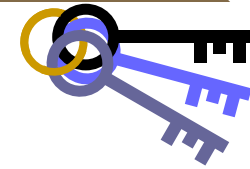


Compression: A force tending to shrink or shorten a solid material.

Here, the standing beam develops compression due to gravity.

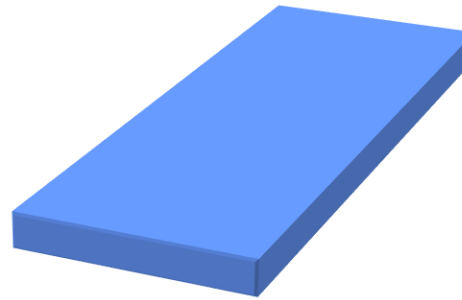
The gravity-induced compression will go from zero at the top to the full weight of the beam at the bottom.

Tensile and Compressive Stress

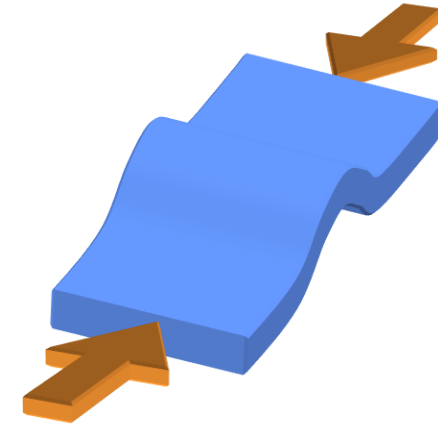


Tensile Stress

Many webs can handle high tension stress without damage.



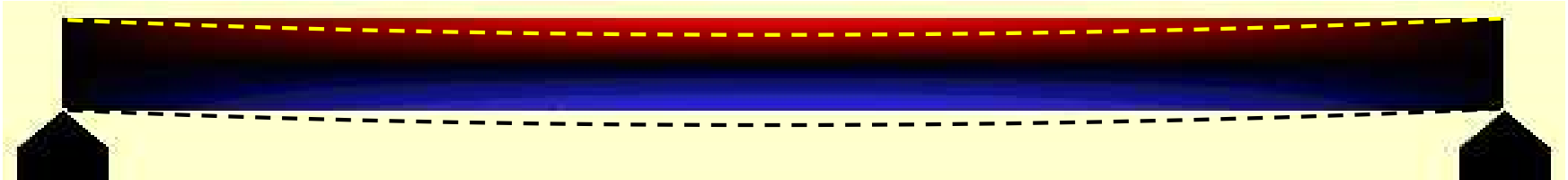
No Stress



Compressive Stress

Most webs will buckle or wrinkle under even low compressive stress.

Tension vs. Compression



Tension and **compression** can occur at the same time.

This beam is simply-supported at both ends.

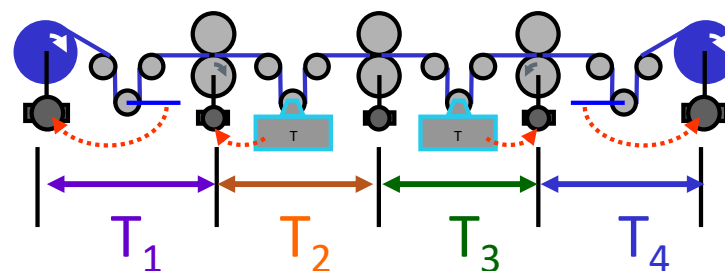
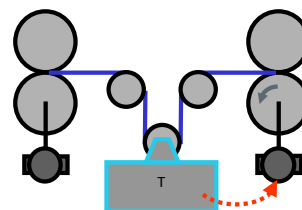
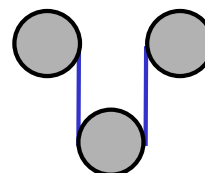
Gravity pulls down on the beam,
causing it to subtly sag in the middle.

The sagging curvature elongates the bottom of the beam, creating
tension or tensile stress.

Also, the top of the beam is forced to a shorter length,
causing **compression**..

Tensioning Hierarchy

- ▶ Tension Fundamentals
- ▶ CD Tension Variations
- ▶ MD Tension Variations
- ▶ Tension Zone
- ▶ Tension in Multiple Zones

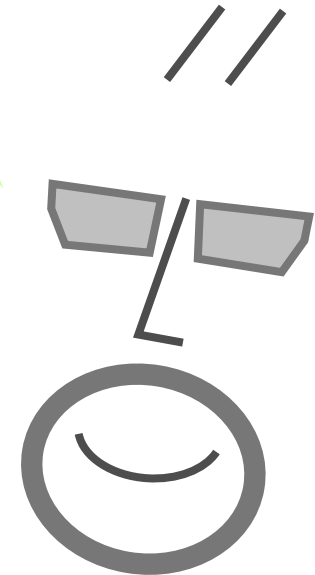


Tension, Strain,... Qs

Why is a 5:1 or 10:1 tension safety factor needed?

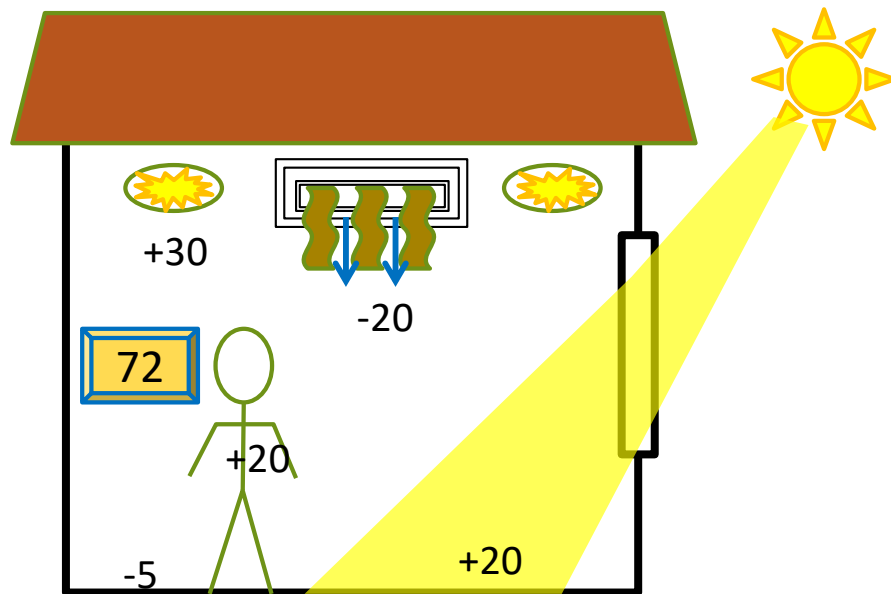
Tensioning systems control average tension.

Web and equipment imperfections can create larger MD and CD variations from average tension.

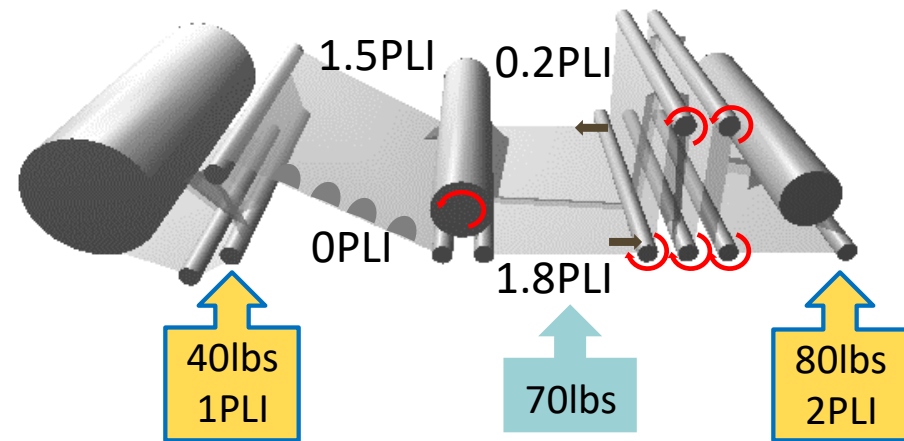


Visualizing Tension Variations

A thermostat will indicate the temperature at one point, but the temperature will vary from point-to-point within the house.

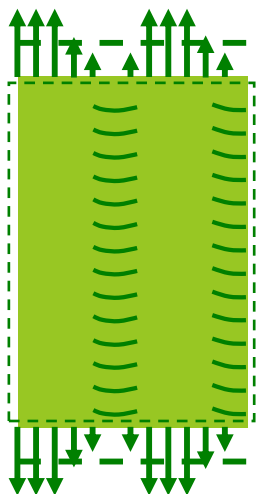
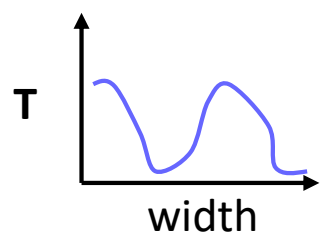


A load cell or dancer roller will control or indicate the tension at one point in a process, but the tension will vary through the process, both MD and CD.

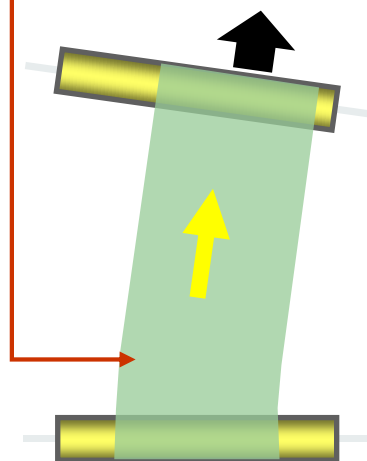
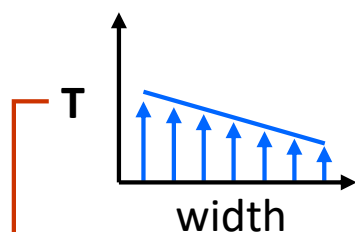


Sources of MD and CD Tension Variations

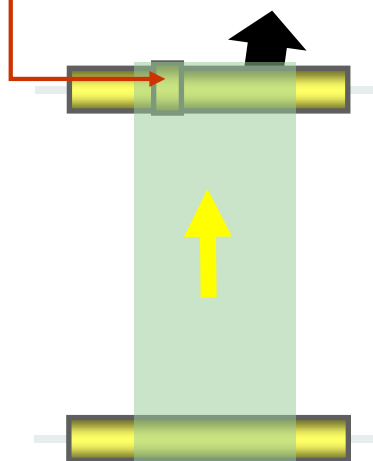
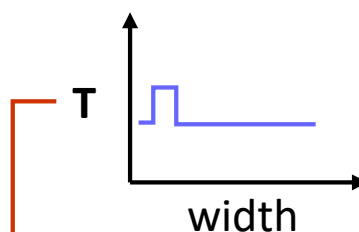
Baggy Web



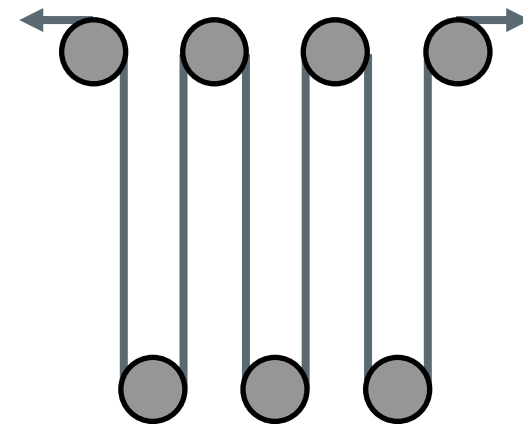
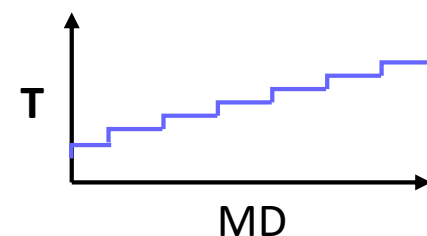
Misalignment



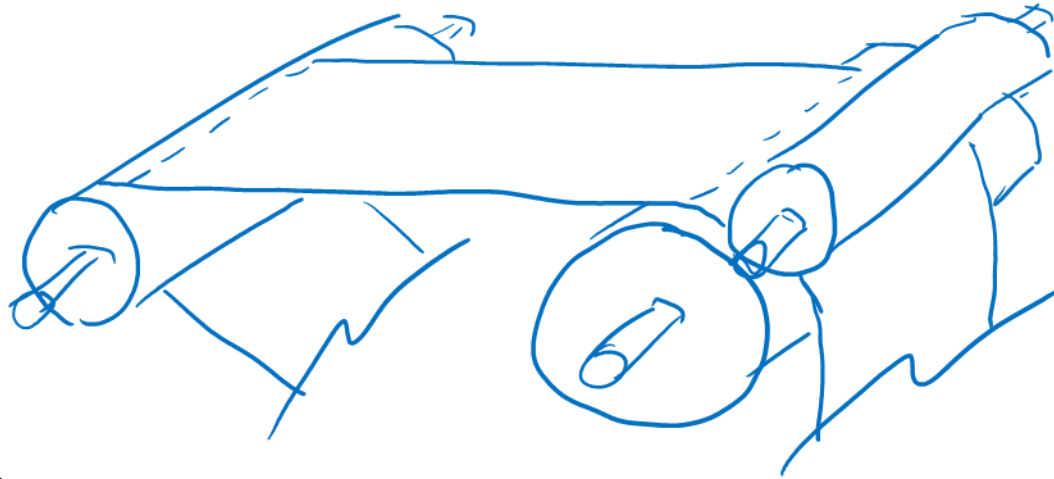
Diameter Variations



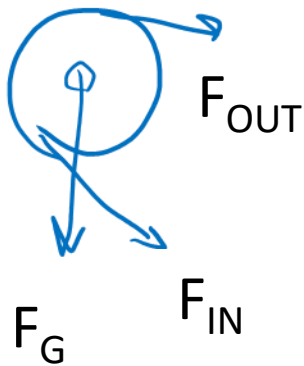
Drag and Inertia



Free Body Diagrams



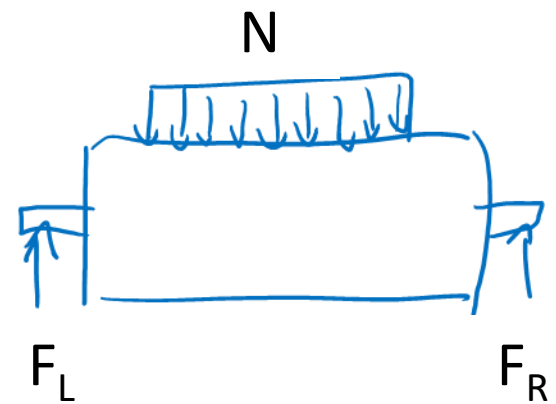
Rollers



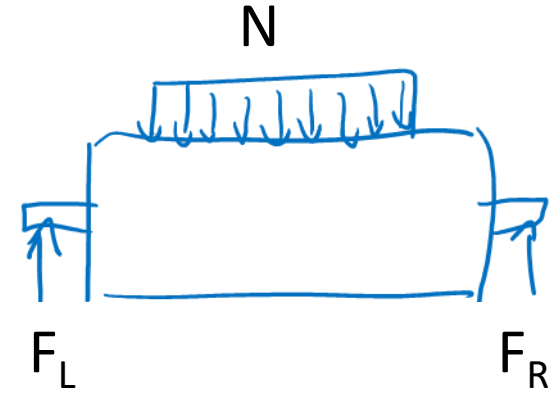
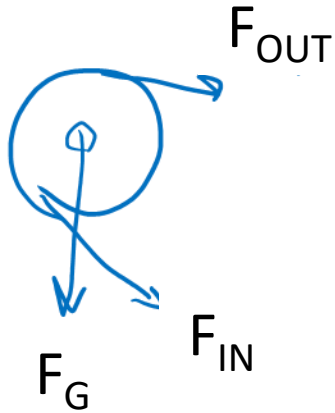
Spans



Nips



Free Body Diagrams



Rollers

- Calculate load on bearing
- Size tension load cells.

Spans

Find causes of:

- CD shifting
- Wrinkling, spreading

Nips

- Calculate roller and core bending
- Find nip pressures and variations.

Spans: Tensioning from Span Ends

Tensioning from Rollers:

- Straightness
- Concentricity
- Eccentricity
- Level
- Squareness

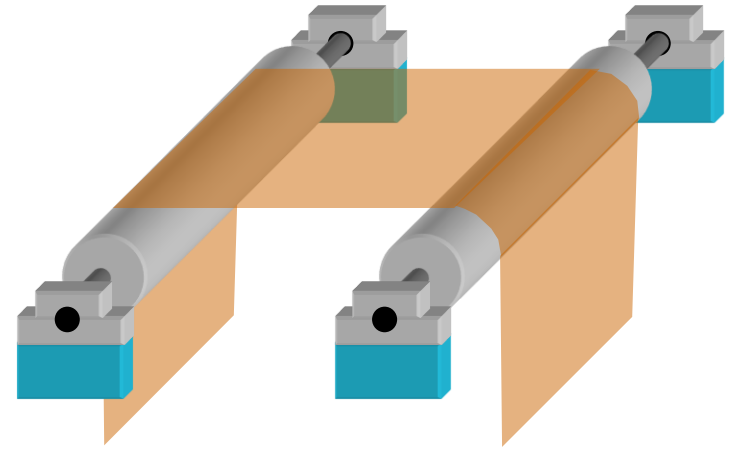
Web path length variations



Stretch (strain) variations

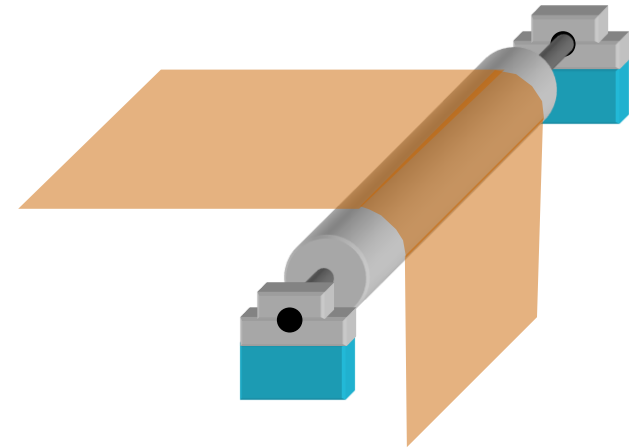


Tension (stress) variations



Rollers: Straining to Conform

The web stretches, attempting to conform to the roller's shape.



Diameter variations



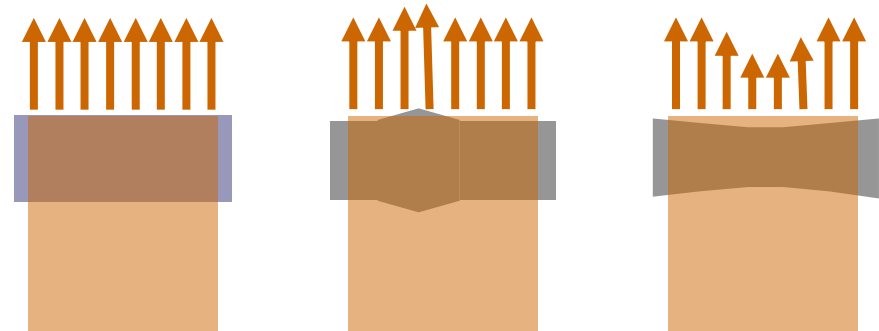
Speed variations



Stretch (strain) variations



Tension (stress) variations

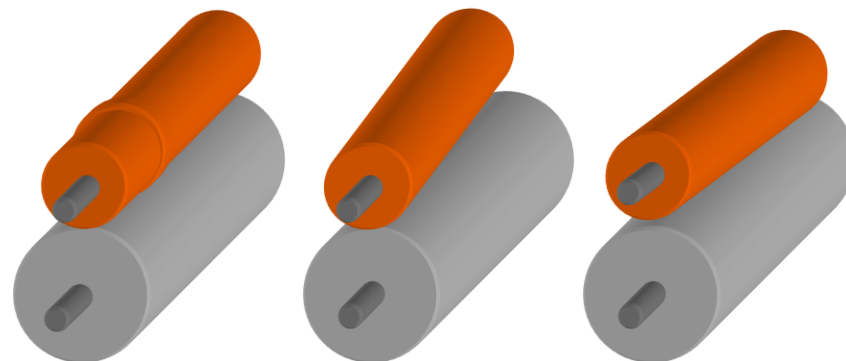
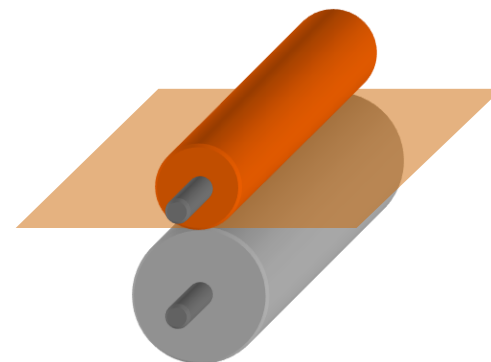


Nips: Under Pressure

Nips create the highest pressures
on the web.

Nips variations can create BIG
pressure variations.

- Diameter variations
- Alignment variations
- Loading variations
- Hardness variations
- Deflection variations
- Product thickness variations



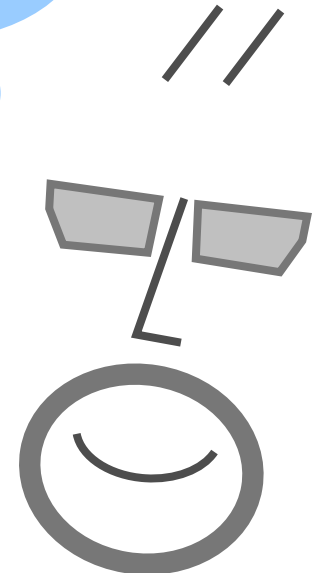
Strain: The Secret of WH!

How do you know how much tension change will be caused by variation X?

What is the secret to web handling?

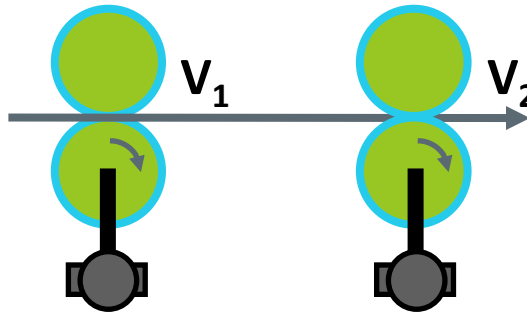
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STRAIN!

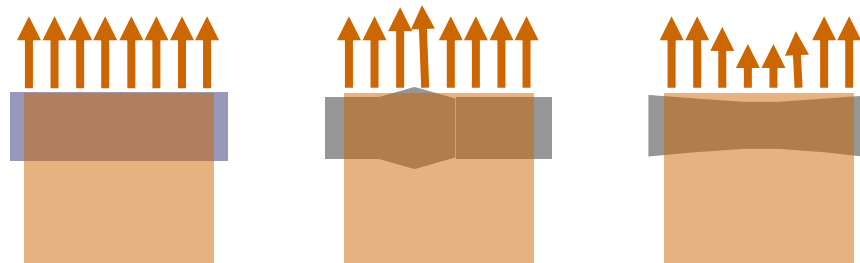


Why is Strain Important?

1. **In tensioning** elastic materials (most webs), strain creates tension...and strain or speed variations create tension variations.

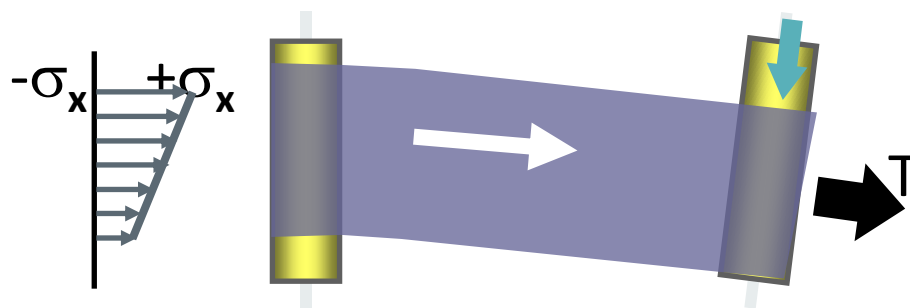


2. **In handling**, web strain determines sensitivity to roller diameter variation, deflection, and misalignment.

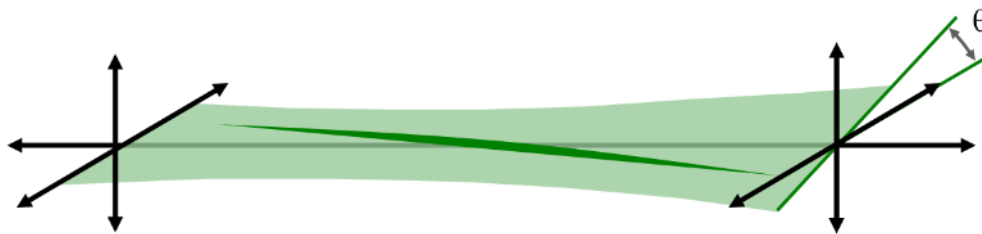


Why is Strain Important?

3. In **bending**, large tension changes occur in sub-visible misalignment.

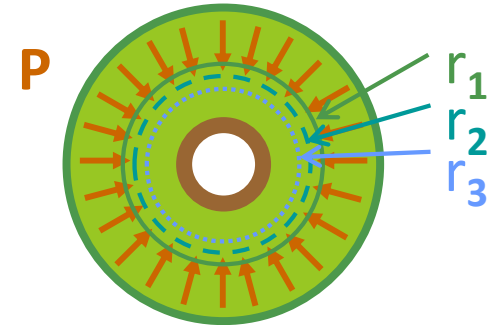


4. In **twisting**, large angles are needed to cause significant tension changes.



Why is Strain Important?

5. In **winding**, web strain determines how much a roll loosens from radial compression.

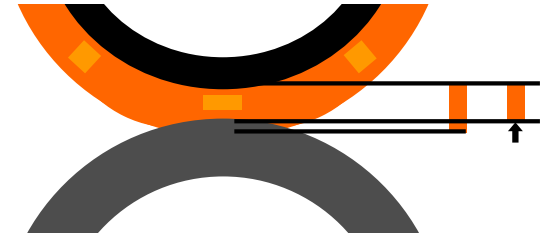


6. In **laminating**, matching strains eliminates curl.



Why is Strain Important?

7. In **nipped rollers**, pressure is proportional to rubber strain and pressure variations are caused by strain variations.

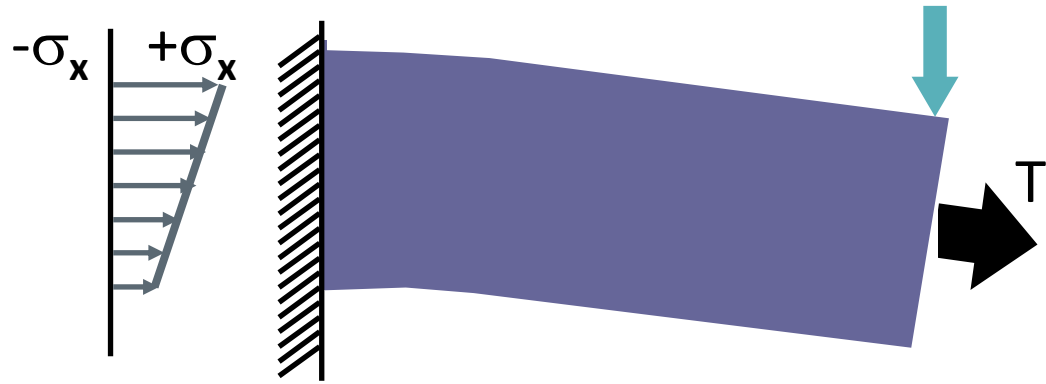


8. In **slitting**, web strain will determine how much width is recovered when tension is removed, increasing slit width.

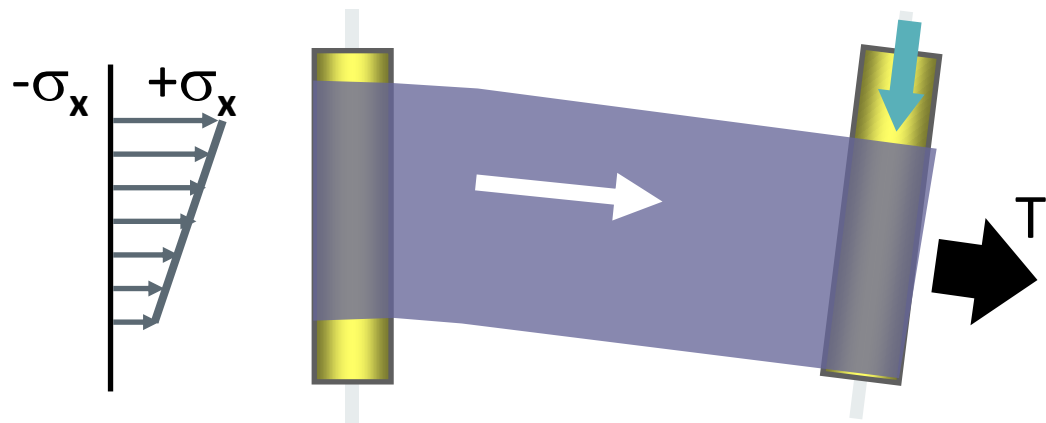


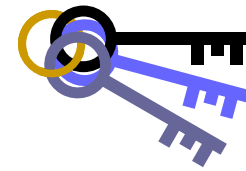
Web Bending Stress Analysis

Lateral bending requires lateral force and creates uneven stresses at the pivot end.



Web bending to reach parallel entry requires web/roller traction and creates similar upstream stress variations.

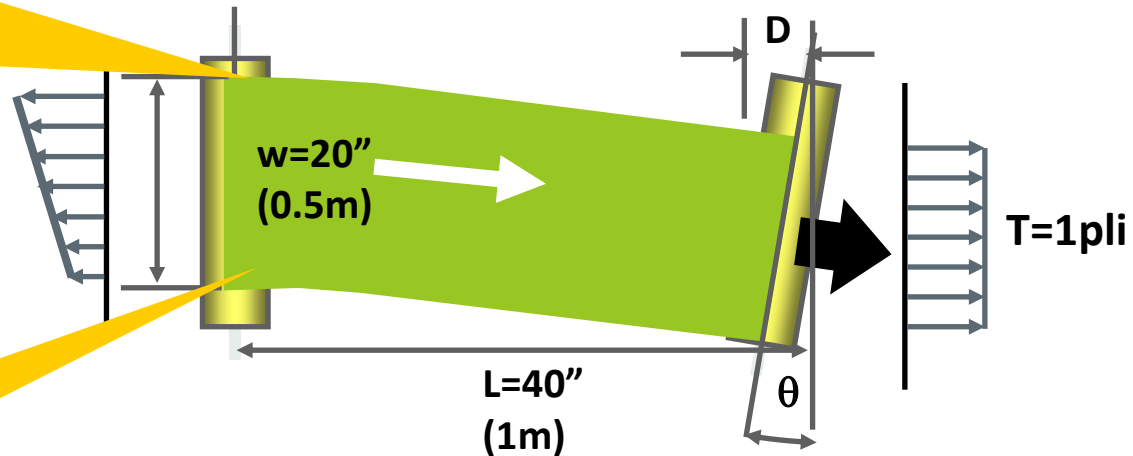




Web Bending Stress Analysis

At the slack misalignment angle, tension will be 2X on one edge and...

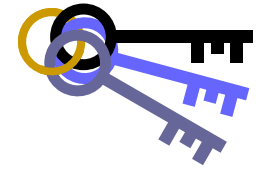
...slack (zero tension) on the other edge.



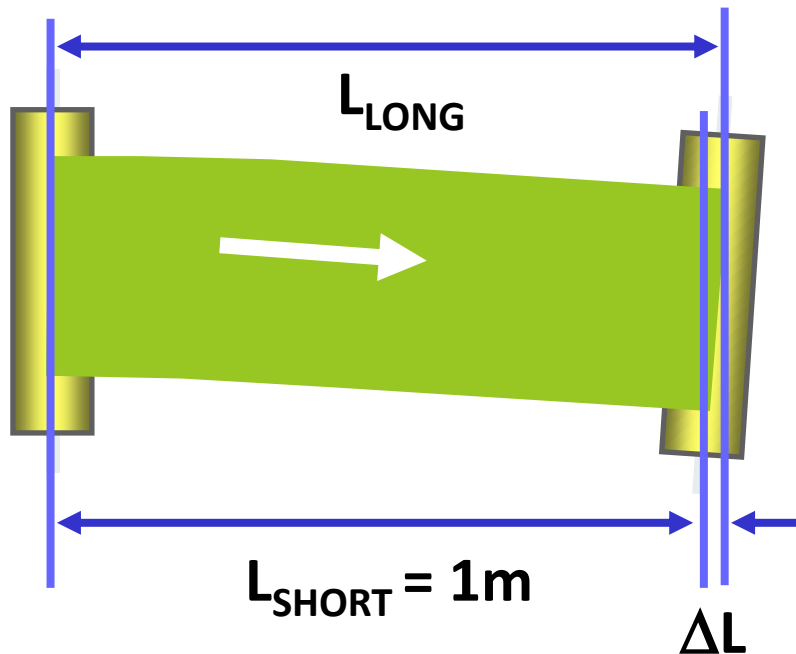
$$\theta_{\text{slackedge}} = \bar{\varepsilon} \frac{L}{w}$$

$$D_{\text{slackedge}} = w \theta_{\text{slackedge}} = \bar{\varepsilon} L = \frac{TL}{tE}$$

t, thickness	= 0.001" (25microns)
E, modulus	= 500000 psi (3.4 GPa)
ε , strain	= (1)/(.001)(500000) = 0.2%
$\theta_{\text{slackedge}}$	= (0.002)(40)/(20)
	= 0.004 radians
	= 0.22 degrees
$D_{\text{slackedge}}$	= (20)(.004) = 0.080" (2mm)



Strain Variation and Tension



$$\Delta L = 100 \mu\text{m} = 0.0001 \text{ m}$$

$$\Delta \varepsilon = \Delta L / L = 0.0001 \text{ or } 0.01\%$$

If Polyethylene, ΔL creates 1% crossweb tension variation

If Polyester, ΔL creates 5% crossweb tension variation

If Aluminum, ΔL creates 100% crossweb tension variation

Strain Change from Bending ☹️

$$\Delta \varepsilon = \frac{\delta}{L_B} = \frac{(L_A - L_B)}{L_B}$$

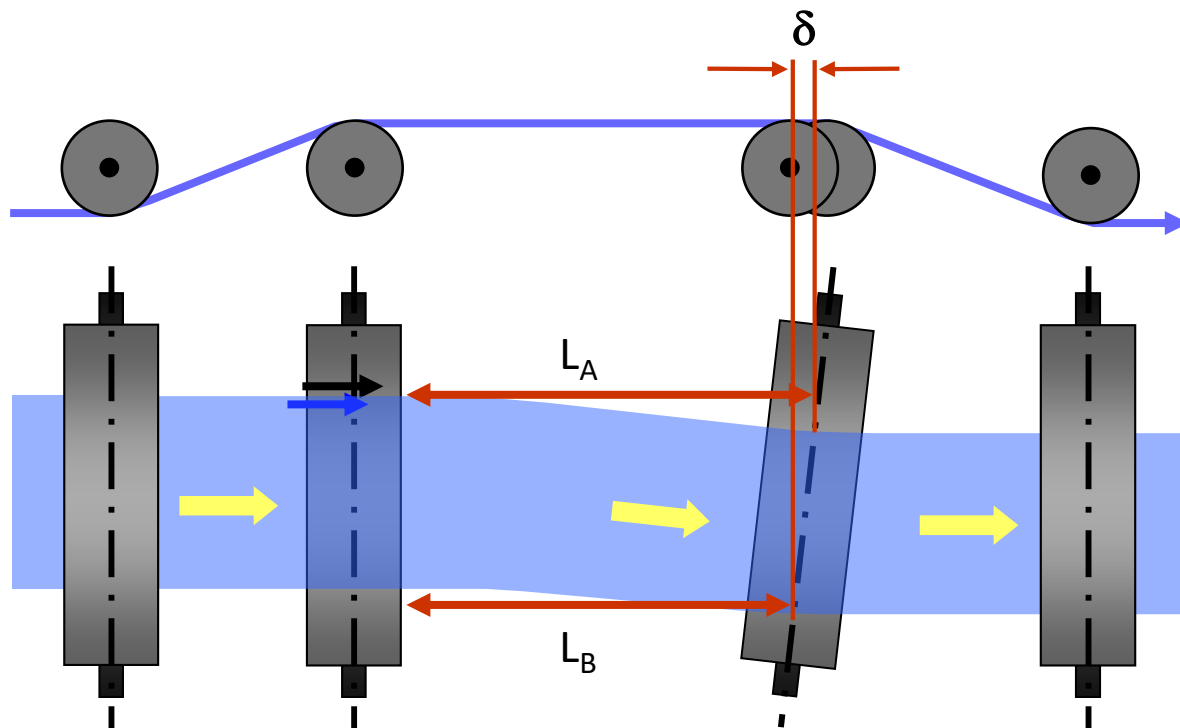
For misalignment that induces bending, the strain change will equal the ratio of the error to the span length.

Example:

$$\delta = 0.4 \text{ in (10mm)}$$

$$L_B = 40 \text{ in. (1000mm)}$$

$$\Delta \varepsilon = 0.4/40 = (10)(1000) = 1\%$$



Strain Change from Twisting 😊

$$L_A^2 = L_B^2 + \delta^2$$

$$\Delta \varepsilon = \frac{(L_A - L_B)}{L_B} = \frac{\sqrt{L_B^2 + \delta^2} - L_B}{L_B}$$

For misalignment that induces twisting, the strain change will be nearly insignificant.



Example:

$$\delta = 0.4 \text{ in. (10mm)}$$

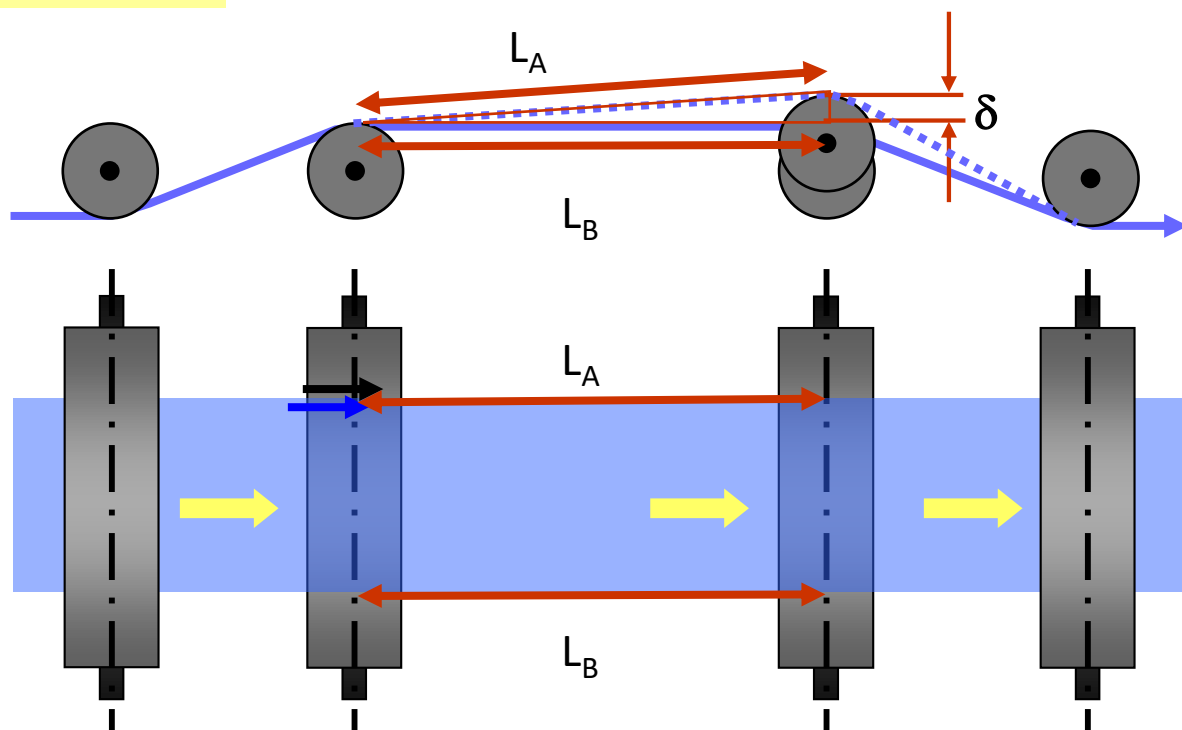
$$L_B = 40 \text{ in. (1000mm)}$$

$$L_A = 40.002 \text{ in. (1000.05)}$$

$$\Delta L = 0.002 \text{ (50 microns)}$$

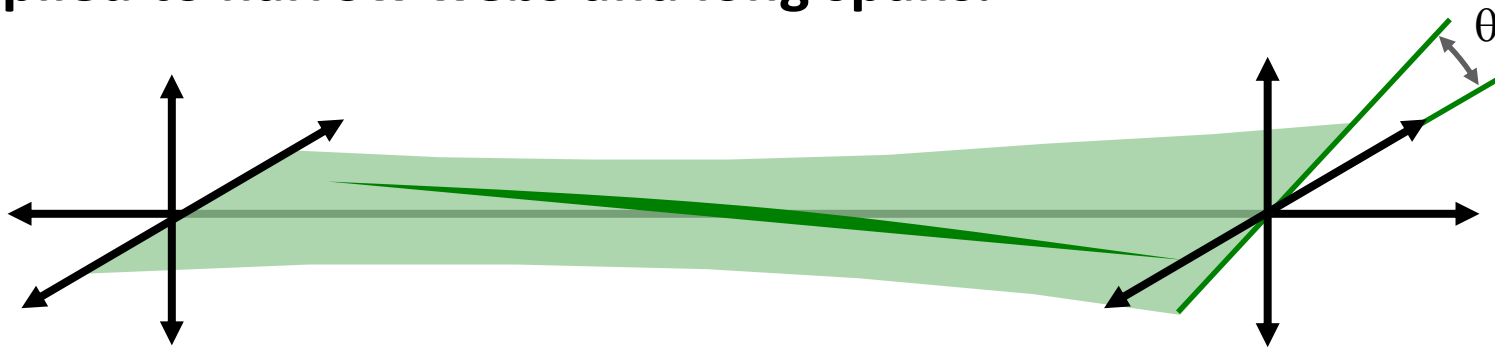
$$\Delta \varepsilon = 0.002/40$$

$$= 0.005\%$$



Stresses from Twisting

Web twisting is stable and non-deforming when applied to narrow webs and long spans.



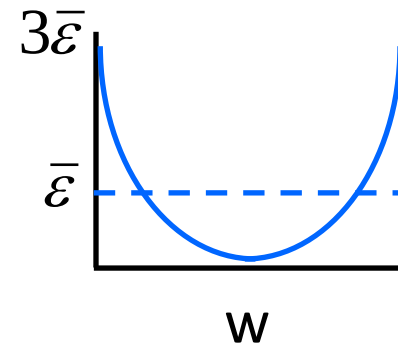
Assuming a parabolic tension distribution, the center web tension is zero at the minimum twisting span, $L_{\min}$, or maximum twisting angle, θ_{crit} :

$$L_{\text{slack_edge}} = w\theta \sqrt{\frac{twE}{24F_T}} = \frac{0.2w\theta}{\varepsilon^{0.5}}$$

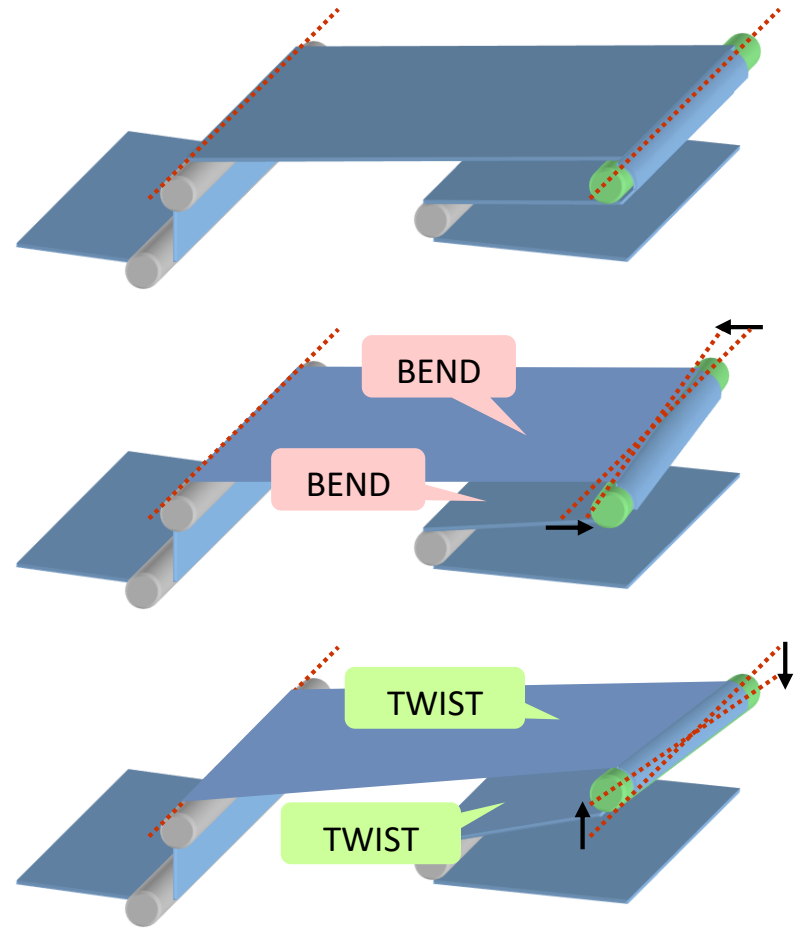
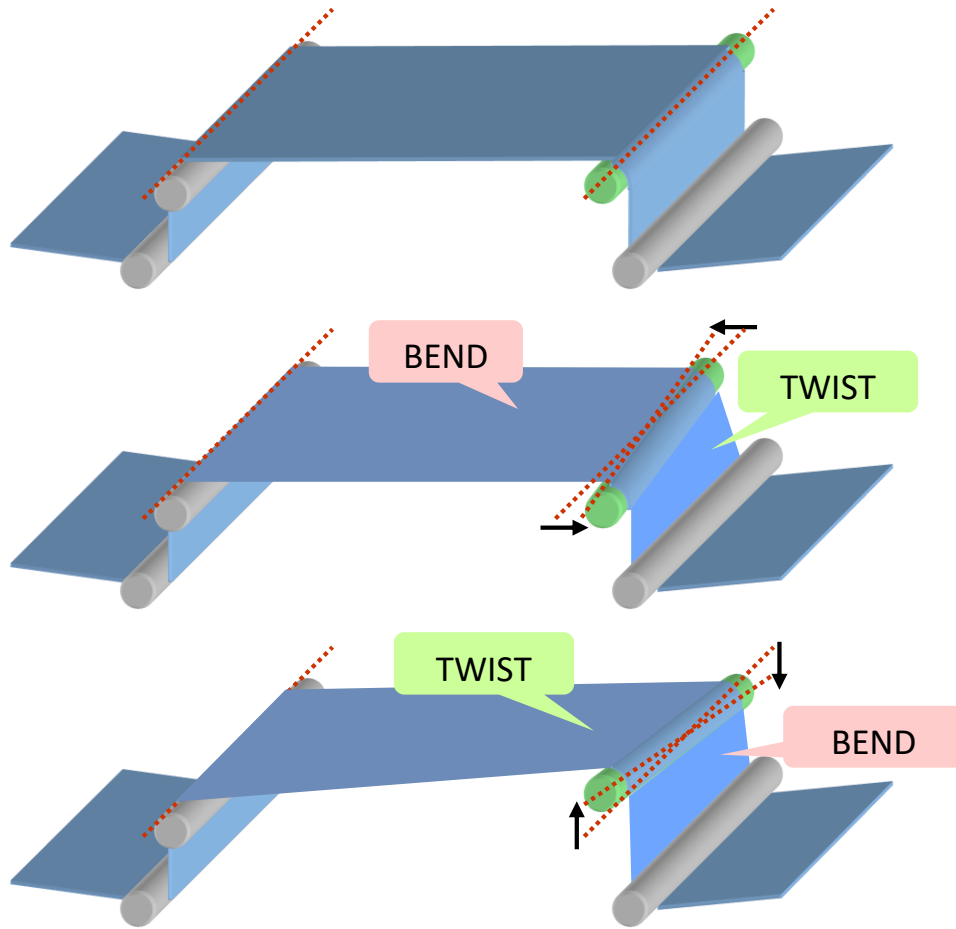
$$\theta_{\text{slack_edge}} = \frac{L\varepsilon^{0.5}}{0.2w} = \frac{(40)\sqrt{0.002}}{(0.2)(20)}$$

= 0.44 radians

= 25 degrees



Bending & Twisting Geometries



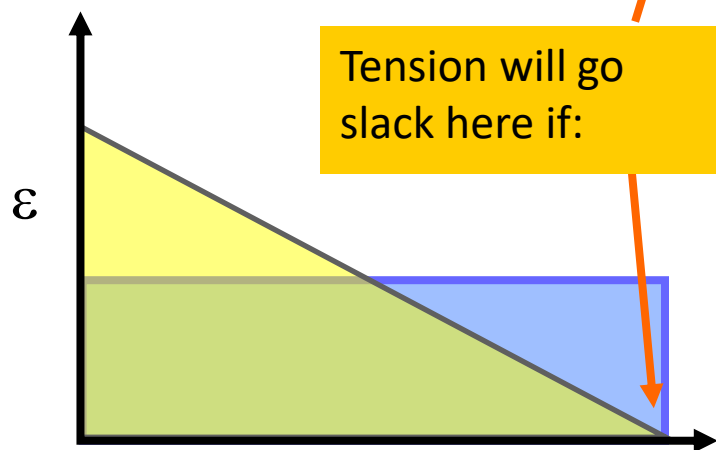
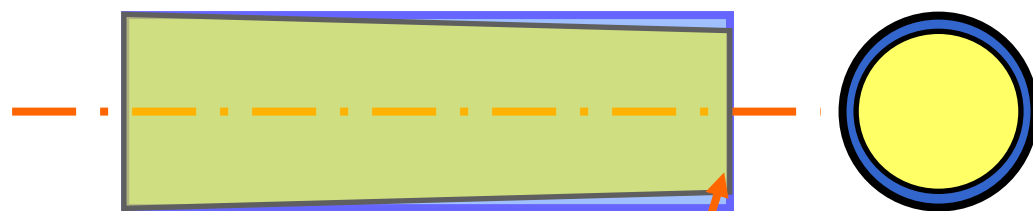
Tapered Rollers

Tapered rollers shift the web's strain and tension from the small to large radius side. Δr_{slack} is the change in radius where the small diameter edge tension is zero.

Radial variations should be viewed in % difference and compared to damaging strain.

$$\Delta \varepsilon_{\text{left/right}} = \frac{\Delta r}{r}$$

$$\Delta r_{\text{slack_edge}} = 2r\bar{\varepsilon} = \frac{2rT}{tE}$$



Tension will go slack here if:

$$\begin{aligned} t &= 0.001'' \\ T &= 1 \text{ PLI} \\ E &= 500000 \text{ psi (PET, Paper)} \\ r &= 2'' \end{aligned}$$

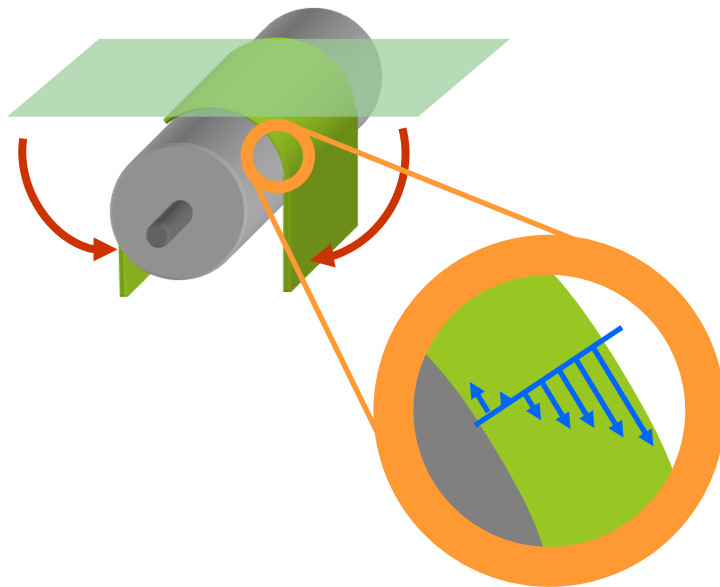
$$\Delta r_{\text{slack}} = (2)(2)(1)/(.001)(500000) = 0.008''$$

Stresses from Z-Direction Curvature

A web wraps a roller creates a stress differential through the thickness of the web.

$$\Delta \varepsilon_{in/out} = \frac{t}{r}$$

$$\Delta \sigma_{in/out} = \frac{tE}{r}$$



ε = strain, -

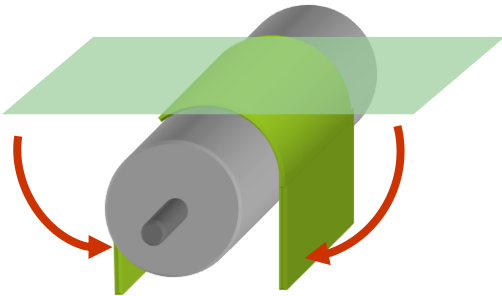
σ = stress, psi or N/mm² (force/area)

E = modulus, psi or N/mm² (force/area)

t = web thickness, in or mm (length)

r = radius of curvature, in or mm (length)

Example: Wrapping a Roller



For zero tension case:

$$r_{YIELD} = \frac{tE}{\sigma_{YIELD}}$$

$$t = 0.003''$$

$$E = 500000 \text{ psi (PET, Paper)}$$

$$\text{For } \Delta\sigma < \sigma_{YIELD}$$

$$\text{Assume } \sigma_{YIELD} = 10000 \text{ psi}$$

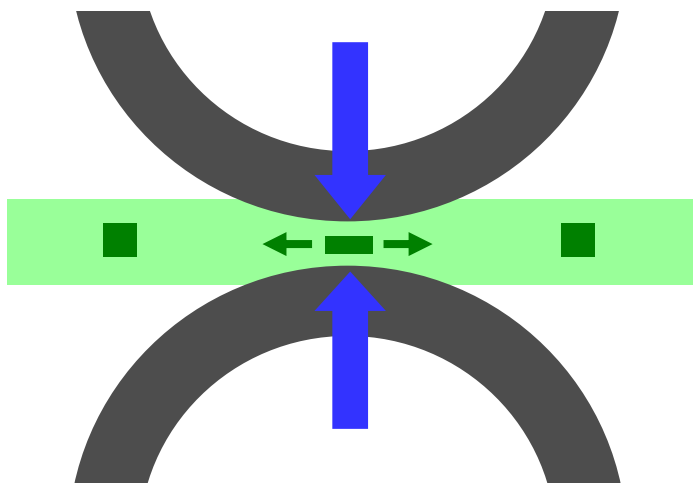
$$r_{YIELD} = tE/\sigma_{YIELD}$$

$$\begin{aligned} r_0 &= (0.003)(500000)/(10000) \\ &= 0.15'' \end{aligned}$$

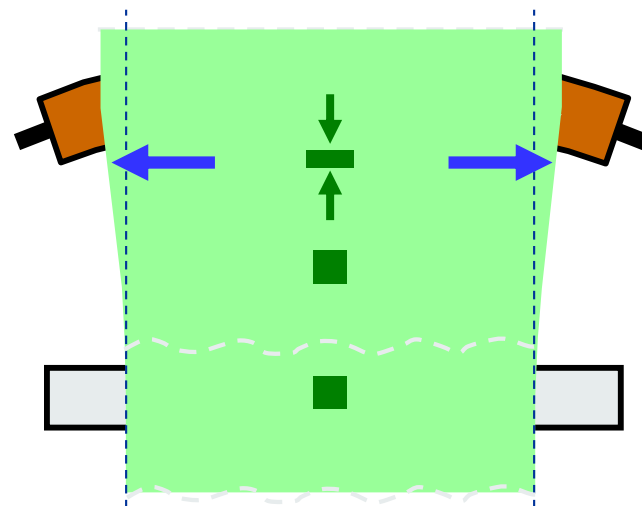
Strains from Nipping and Spreading

Tensioning in the machine direction (X) changes the strain in the crossweb (Y) and thickness (Z) directions.

Similarly, nipping and spreading may change the machine direction tension.



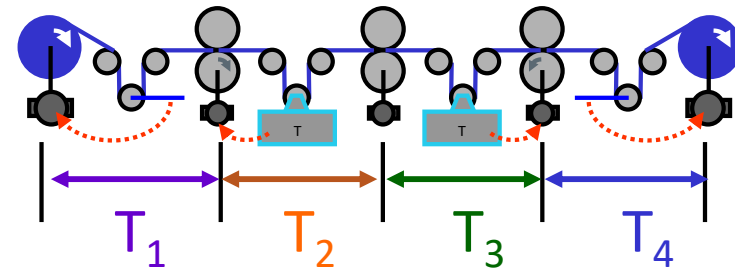
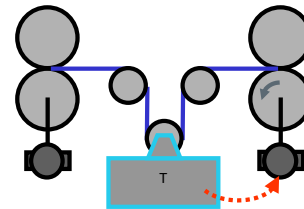
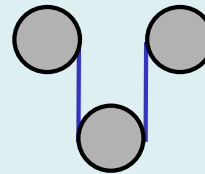
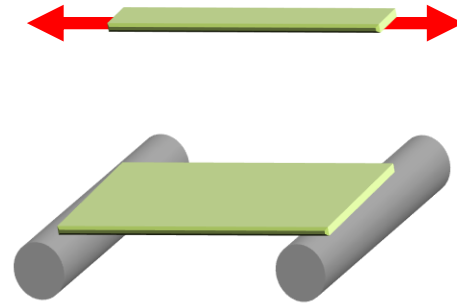
Nipping creates MD web strain.



Spreading may tighten baggy lanes

Tensioning Hierarchy

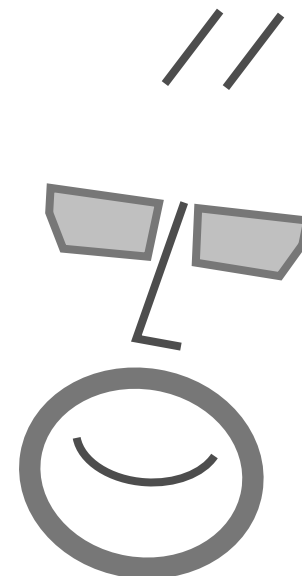
- ▶ Tension Fundamentals
- ▶ CD Tension Variations
- ▶ MD Tension Variations
- ▶ Tension Zone
- ▶ Tension in Multiple Zones



The Right Tension Qs

What causes MD tension variations?

- Roller drag
- Process drag
- Roller inertia (during speed changes)
- Web-roller bond (adhesion)
- Gravity
- Bending (more important in thick webs)
- Air drag (only significant at high speeds)



Web Tension Changes at Idler Rollers

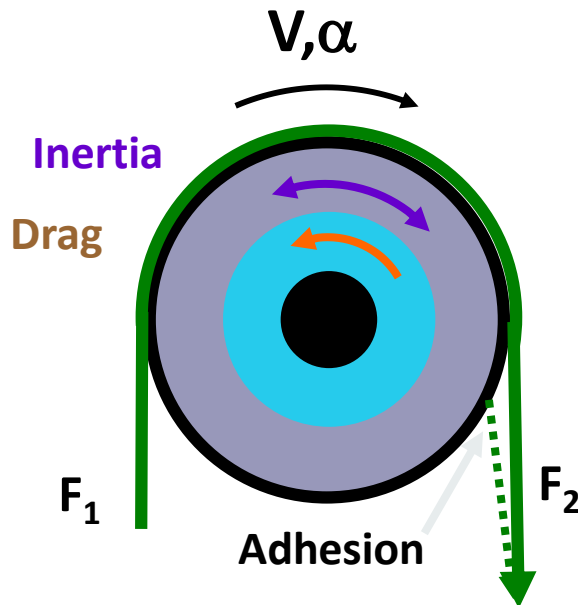
The tension change across any roller will be a function of applied torque, inertia (during speed changes), and adhesion.

More on this topic in the
Rollers and Traction section.

Applied Torque = M_{app} .

Inertial Torque = $I\alpha$

Adhesive Losses = $f_{peel}r$



$$\Sigma M = 0$$

$$F_2 r = F_1 r + M_{App} \pm I\alpha + f_{peel} r$$

$$\Delta F = \frac{M_{App} \pm I\alpha + f_{peel} r}{r}$$

Applied torque includes effective torque of motor, clutches, of brakes (including bearing drag).

Tension in Vertical Spans

Tension will increase in vertical spans due to web span weight.

$$F_{T_{\max}} = F_0 + W_{SPAN}$$

$$W_{SPAN} = \rho t w L$$

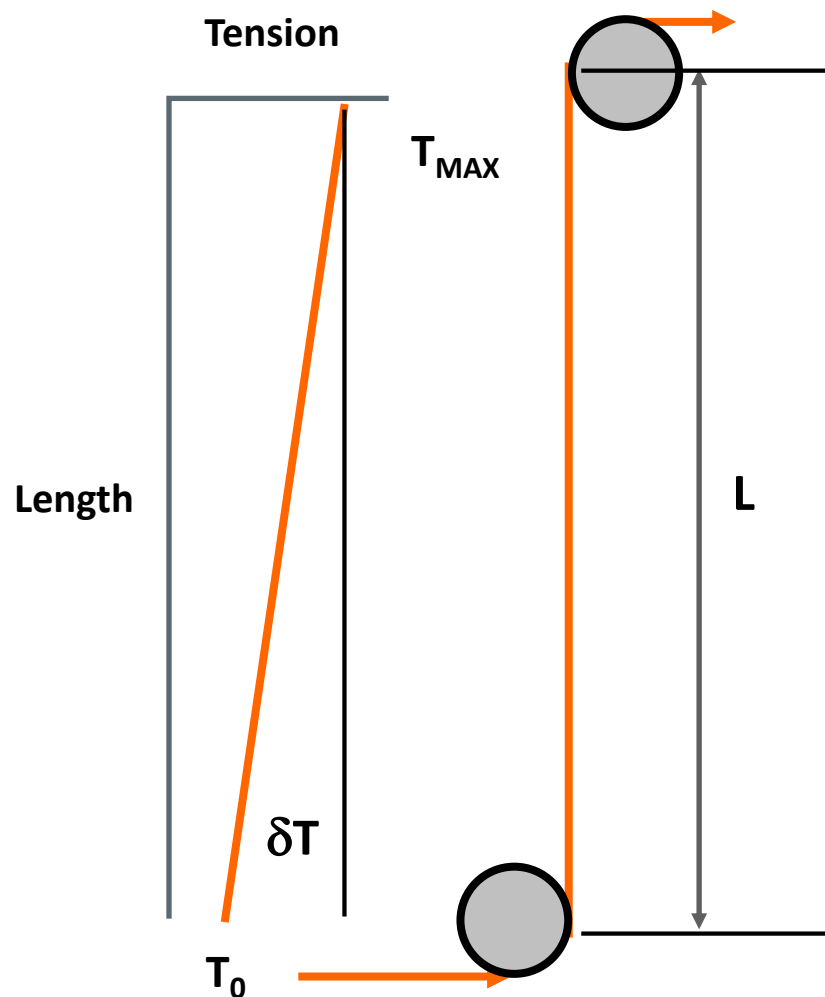
T = Tension, pli

ρ = Density, lbs/in³

t = Thickness, in.

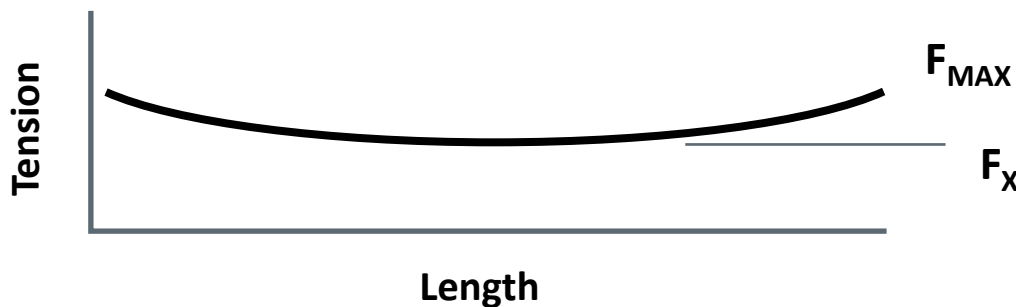
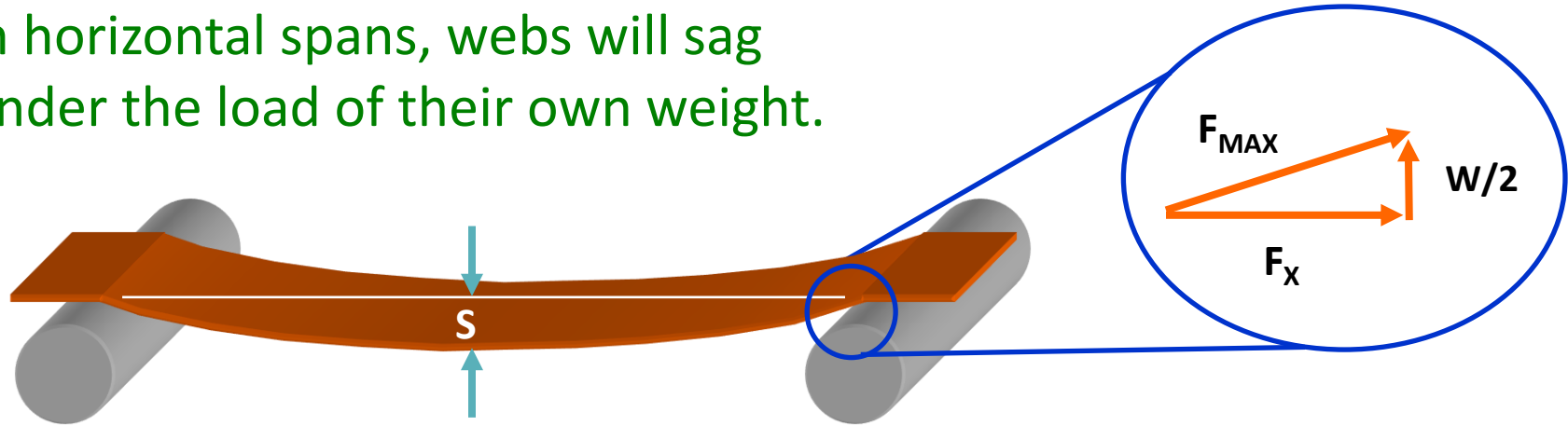
w = Width, in.

L = Span Length, in.



Catenary Sag and Tension

In horizontal spans, webs will sag under the load of their own weight.



$$F_X = \frac{\rho t w L^2}{8S}$$

$$S = \frac{\rho t w L^2}{8F_X}$$

$$S_x = S \left(1 - \frac{4x^2}{L^2} \right)$$

- S = Catenary Sag, inches
- ρ = Web Density, lbf/in³
- t = Web Thickness, inches
- w = Web Width, inches
- L = Length, inches
- T = Tension, lbf



2.4 Tensioning Methods

Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgcumbe Road, Saint Paul, MN 55116

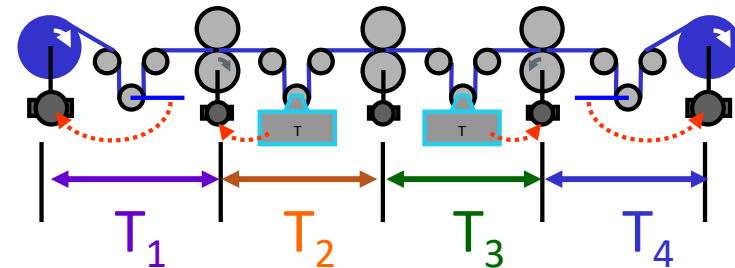
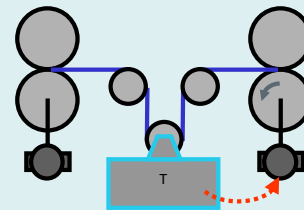
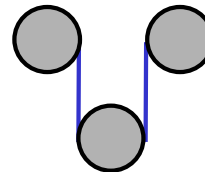
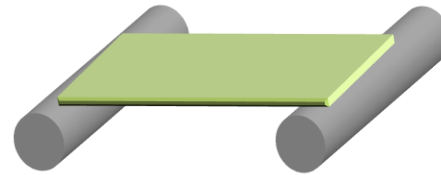
tjwalker@tjwa.com www.webhandling.com 651.686.5400

2.4 Learning Objectives

- ▶ What are the three ways to create tension?
- ▶ How do torque devices create tension?
- ▶ How does speed ratio create tension?
- ▶ What devices are used for closed-loop tension control?
- ▶ What are advantages and disadvantages of load cell roller feedback?
- ▶ What are advantages and disadvantages of dancer roller feedback?

Tensioning Hierarchy

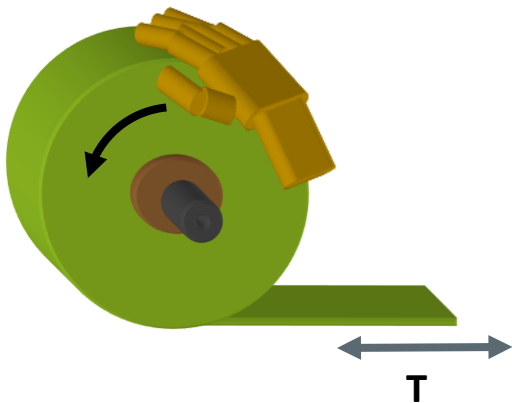
- ▶ Tension Fundamentals
- ▶ CD Tension Variations
- ▶ MD Tension Variations
- ▶ Tension Zone
- ▶ Tension in Multiple Zones



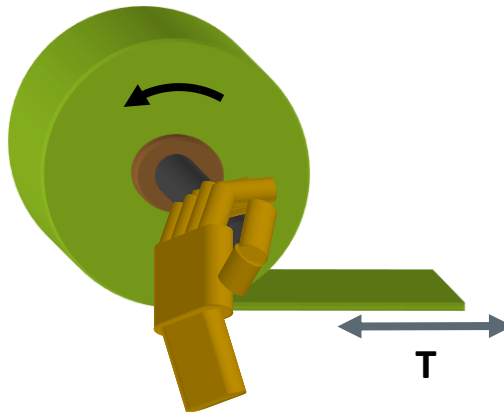
Creating Tension

Tension is created via force, torque, or stretch.

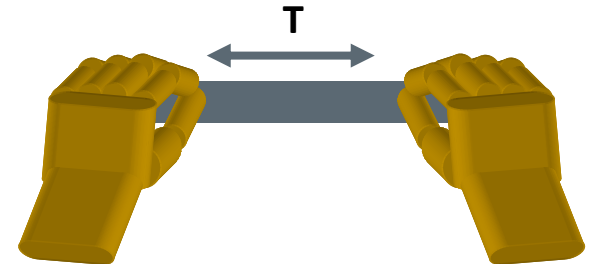
Force



Torque



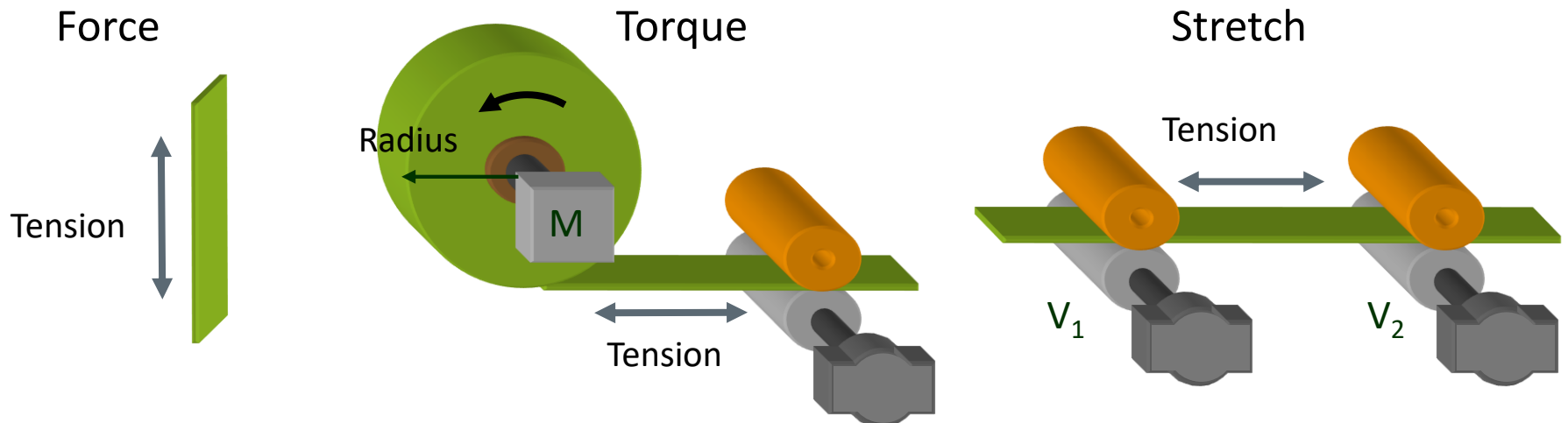
Stretch



Creating Tension

Tension is created by:

1. Direct force lbs, N
2. Torque over radius in-lbs/in, N-m/m
3. Stretch (strain) via speed ratios %



Creating Tension

► Force

Tension can be created from directly applied forces of gravity (in vertical spans), pneumatically loaded rollers, and surface friction.

► Torque

A rotating roller or winding shaft can have forward or back torque applied to create increasing or decreasing tension.)

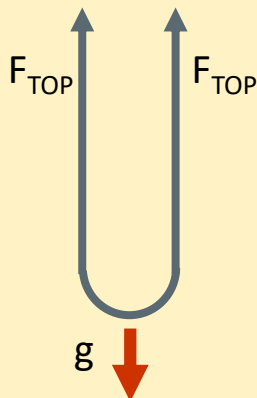
► Speed Ratio

Controlling a differential speed of two non-slipping drive points can increase or decrease web strain and associated tension

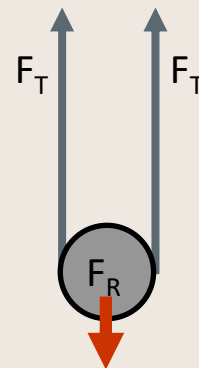
Creating Web Tension: Force

Force of Gravity:

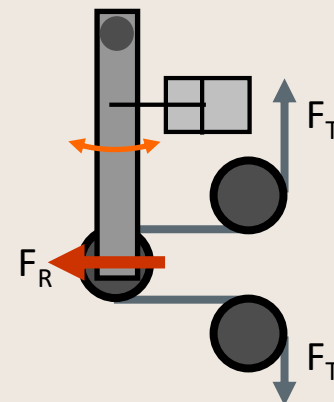
In a vertical span, the weight of the web will create tension proportional to length.



$$F_{TOP} = \rho L t w$$



$$F_T = F_R / 2$$



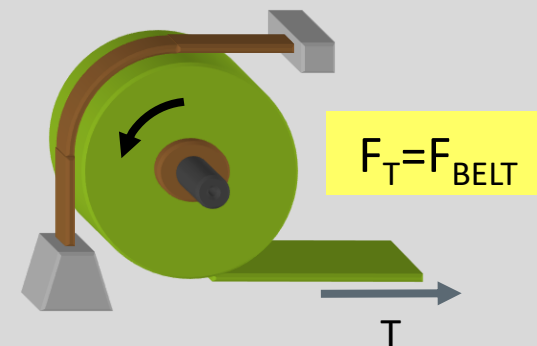
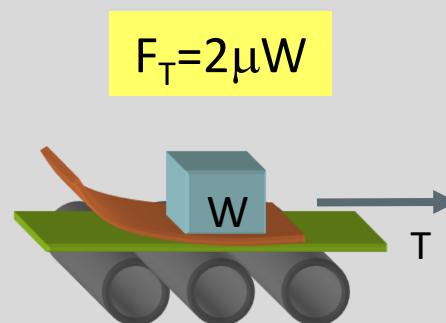
$$F_T = F_R / 2$$

Force of Roller:

The combination of gravity and pneumatic cylinder forces can load a roller into the web.

Force of Friction:

Create tension by loading the surface of the web, unwinding or winding roll with a frictional force.

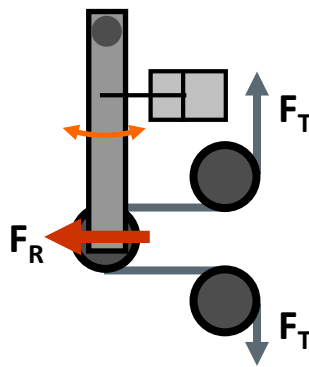
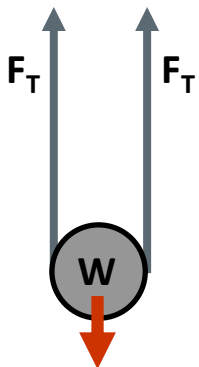


Creating Web Tension: Force

With a 180° wrap, the loading force is opposed by two web spans, so the web tension is ½ the loading force.

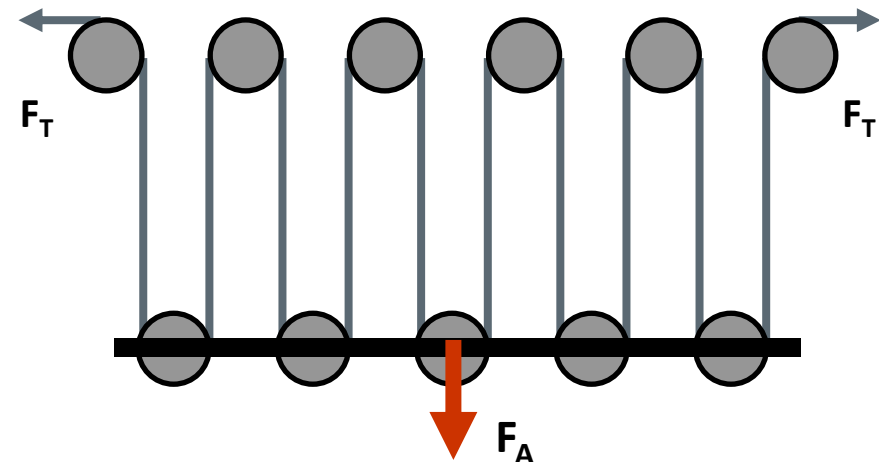
$$F_T = \frac{W}{2}$$

$$F_T = \frac{F_R}{2}$$



If more than two webs carry the load, the tension is the loading force (F_A) divided by the number of webs (n).

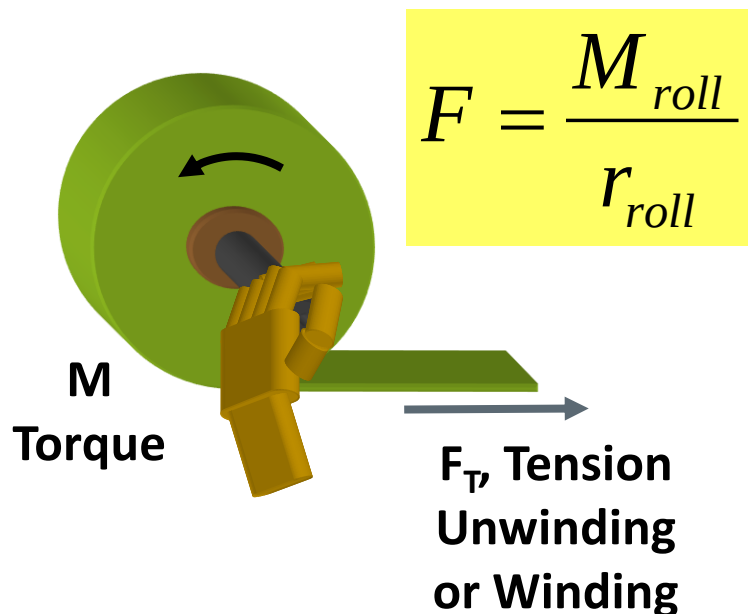
$$F_T = \frac{F_A}{n}$$



Creating Web Tension: Torque

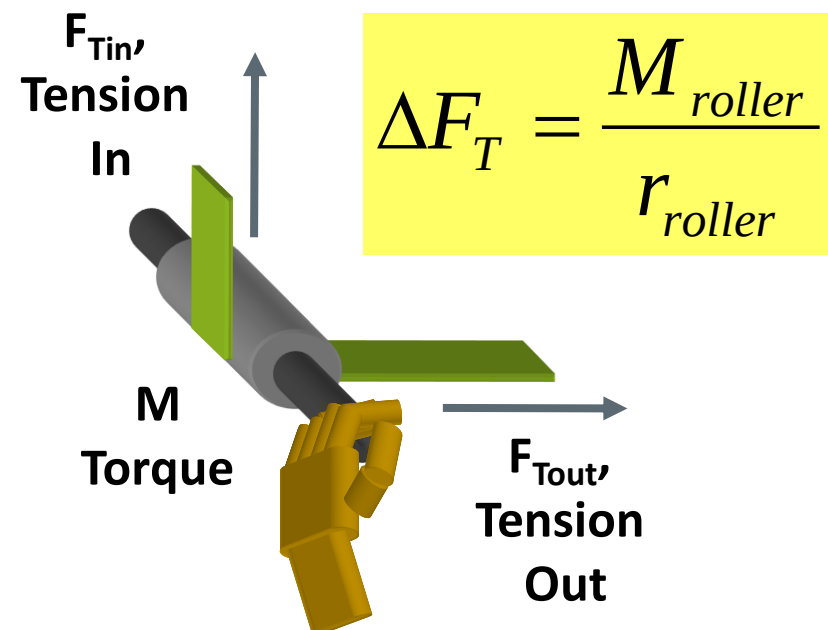
Torque & Rolls:

Create tension by applying torque to the shaft or core of a roll .



Torque & Rollers:

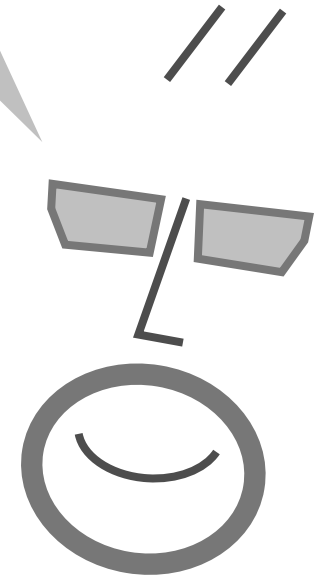
Create a change in tension by applying torque to the shaft a roll roller.



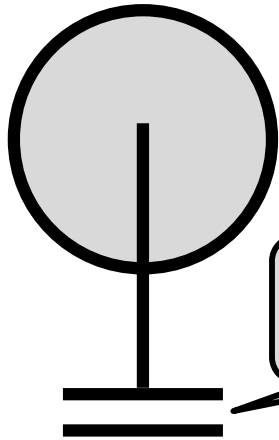
Torque and Speed Control Qs

How is tension created in converting equipment?

Converting processes create tension using motors, clutches, and brakes connected to the web through a roller or roll.

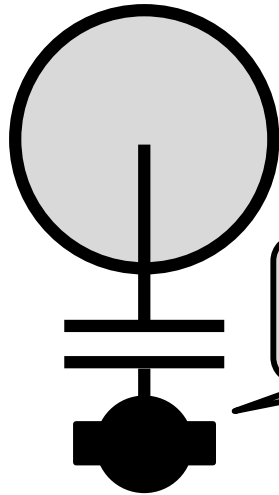


Apply Torque to Rollers and Rolls



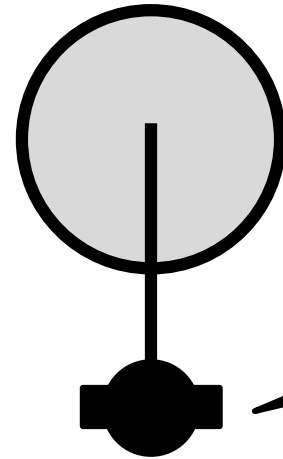
Symbol for
Brake

Braked Rollers/Rolls



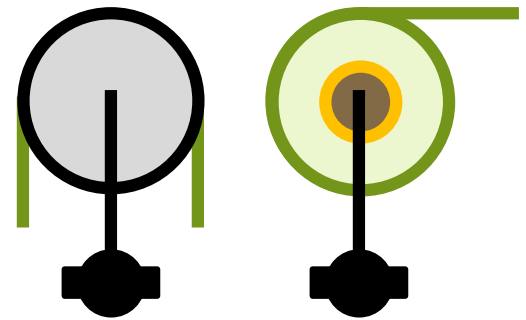
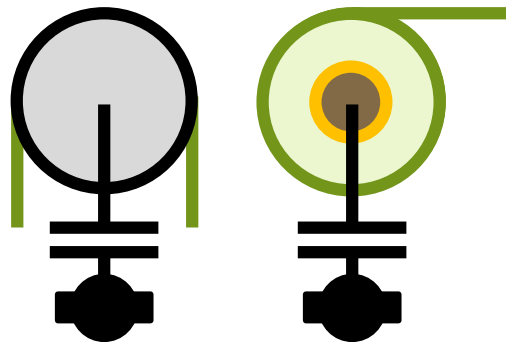
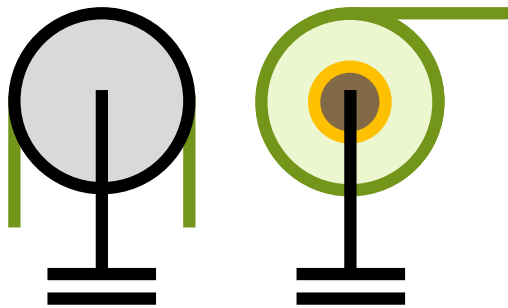
Clutch and
Motor

Clutched Rollers/Rolls

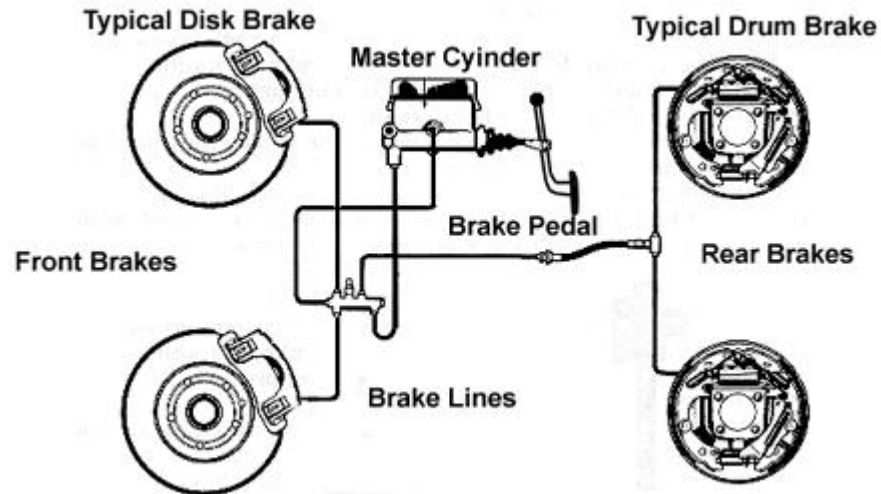
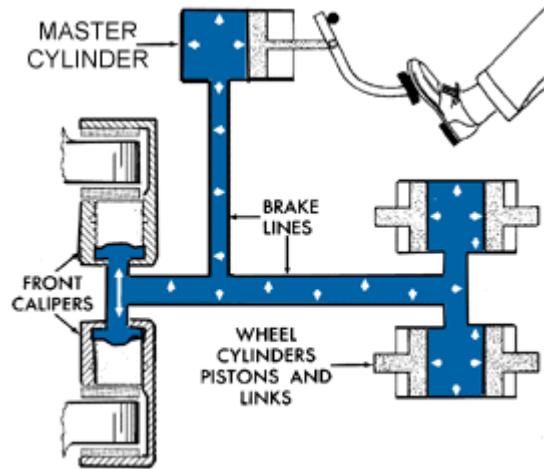


Motor

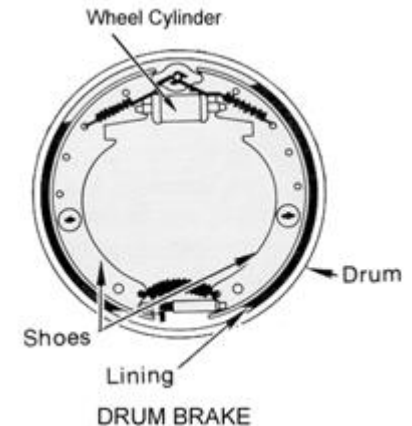
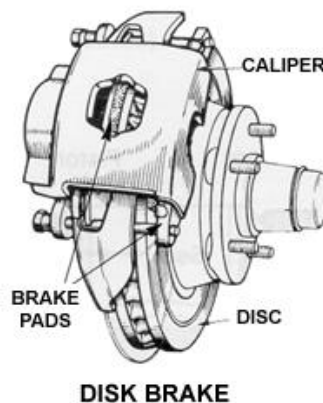
Driven Rollers/Rolls



Automobile Braking Systems



Most automobiles have brakes on all four tires, typically using disc brakes on the front tires and drum brakes on the rear tires.



Torque and Speed Control Qs

What devices create tension?

Motors



Image courtesy of Siemens



Image courtesy of Allen Bradley

Clutches

(including differential shafts)



Image courtesy of Converttech



Images courtesy of Montalvo



Image courtesy of Dusenbury



Images courtesy of Magpowr

Brakes



Torque and Speed Control Qs

What devices create tension?

Control Options	Motors	Clutches	Brakes
Controls Torque	X	X	X
Controls Speed	X		
Braking	X	X	X
Driving	X	X*	

* Clutch plus a motor

Tensioning Decisions

Speed Devices



or



Torque Devices

Motors, Belted / Geared Rollers

Commonly controlled to 0.1%

Typical torque range < 20:1

Positive control independent of
torque losses / variations.

Clutches, Brakes

Typical torque range < 40:1

Multi-disc brakes, 240:1

Usually simpler and less expensive
than speed control.

Tension Options

The answer to these two questions leads you to one of three tension options.

Open-Loop Control
No Tension Feedback

Will it have tension
feedback?

Closed-Loop Control
Tension Feedback by
Load Cell or Dancer Roller

What Variable is Adjusted?

Torque

Relative Speed

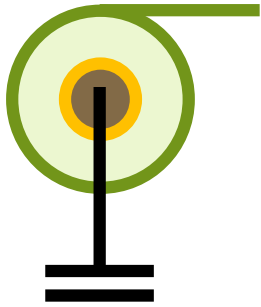
Torque
Control

Draw
Control

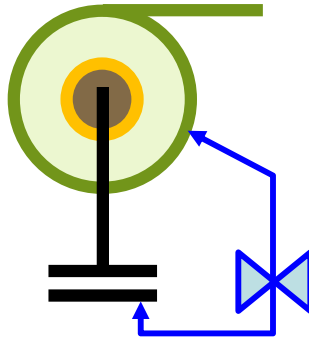
a.k.a. Speed Ratio Control

Tension Control

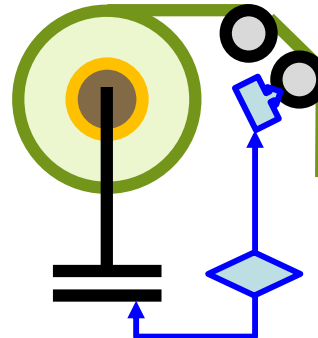
Unwinding Options



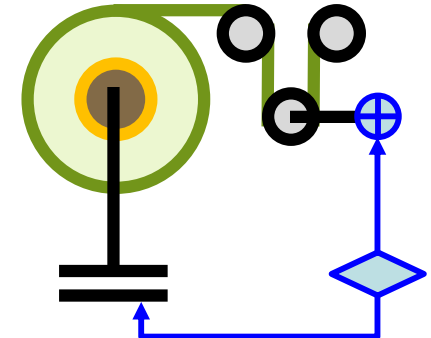
Open-Loop
Brake



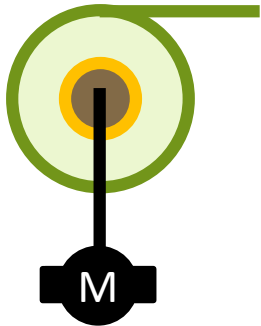
Brake with
Diameter Feedback



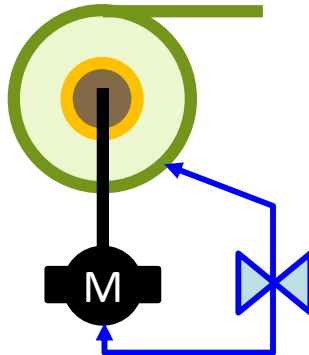
Brake with
Load Cell Feedback



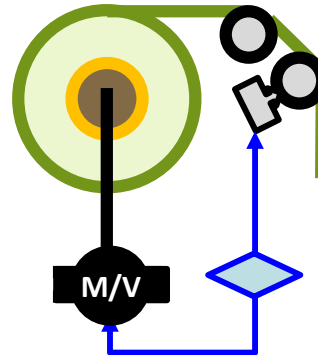
Brake with
Dancer Feedback



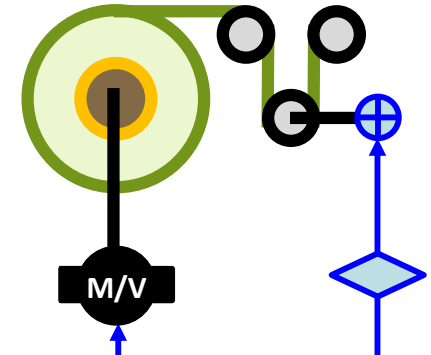
Open-Loop
Torque Motor



Motor with
Diameter Feedback



Motor with
Load Cell Feedback



Motor with
Dancer Feedback

Torque Control

What Variable is Adjusted?

Torque

Relative Speed

Open-Loop Control
No Tension Feedback

Will it have tension
feedback?

***Torque
Control***

**Draw
Control**

a.k.a. Speed Ratio Control

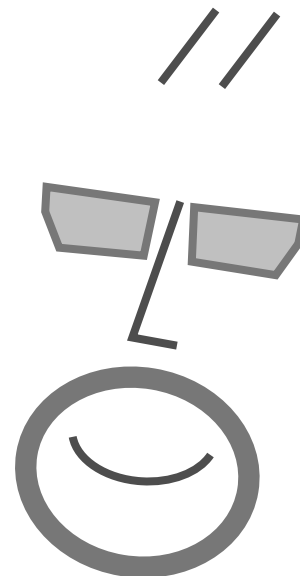
Closed-Loop Control
Tension Feedback by
Load Cell or Dancer Roller

Tension Control

Torque and Speed Control Qs

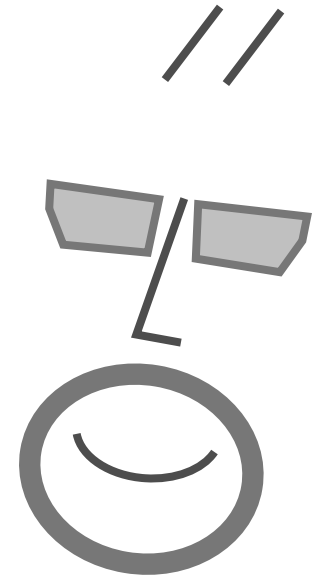
What is open-loop tensioning?

Setting a variable that directly changes tension (surface force, torque or speed ratio), but without tension feedback.



Tensioning Qs

How does torque create tension?



Tension vs. Torque

Tension acts linearly.

Force of tension units are lbs.

Web Tension traditionally defined in force per width, lb/in or PLI.



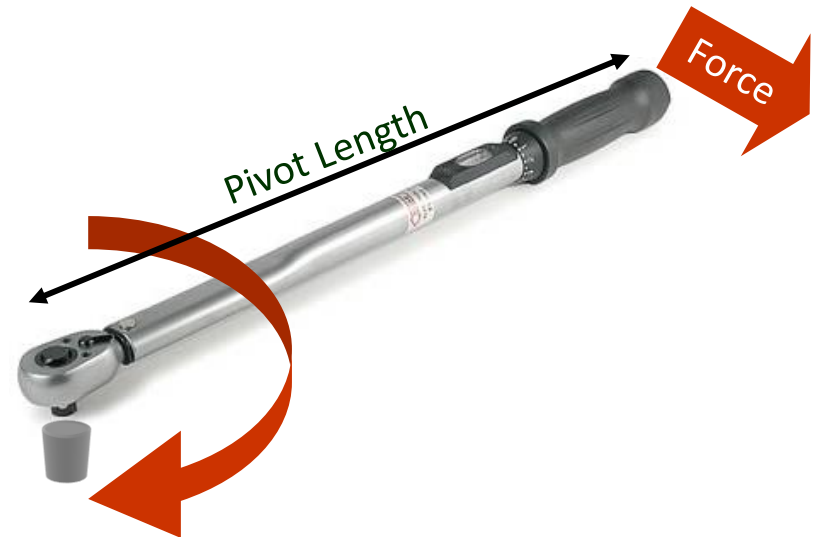
Torque acts rotationally.

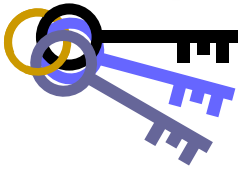
Torque units are in-lbs

A force at a length from a pivot point creates torque.

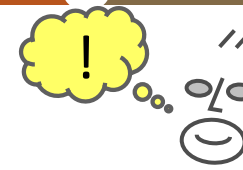
-or-

Torque can create a force at a length from a pivot point.





KEY CONCEPT

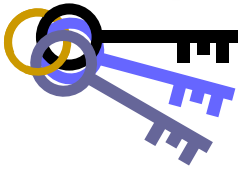


Tension

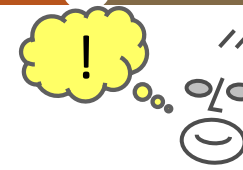
Tension, T , is a force per width tending to stretch or elongate something.

Use: Calculate the tension.

Example: An unwinding web has a 80 lbs force applied across a 40-in wide web. The tension is $80 \text{ lbs} / 40\text{-in} = 2 \text{ lbs per inch}$ or 2 PLI.



KEY CONCEPT



Torque

Torque, M, is a turning or twisting force (units are force x length), a force at a radius that tends to produce rotation.

$\text{Torque(in-lbs)} = \text{Tension(lb/in)} \times \text{Width(in)} \times \text{Radius(in)}$

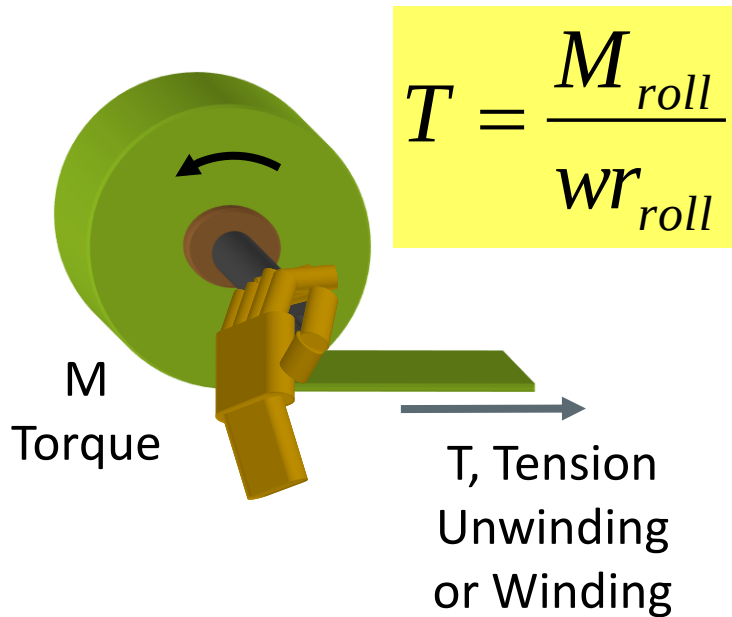
Use: Calculate the tension created from torque.

Example: An center unwind brake that applies 800 N-cm will create 80 N tension a 10 cm radius or 8N at 100 cm radius.

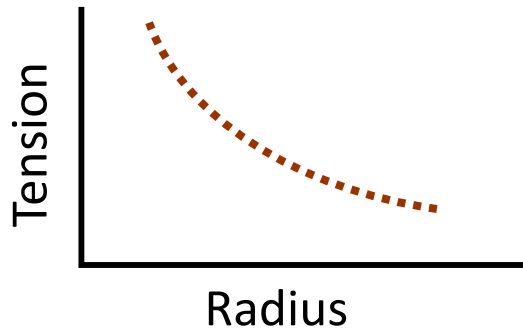
Use: Calculate torque required for winder or driven roller.

Example: To apply a tension differential of 100N, the 20 cm radius drive roller will need 2000 N-cm of torque.

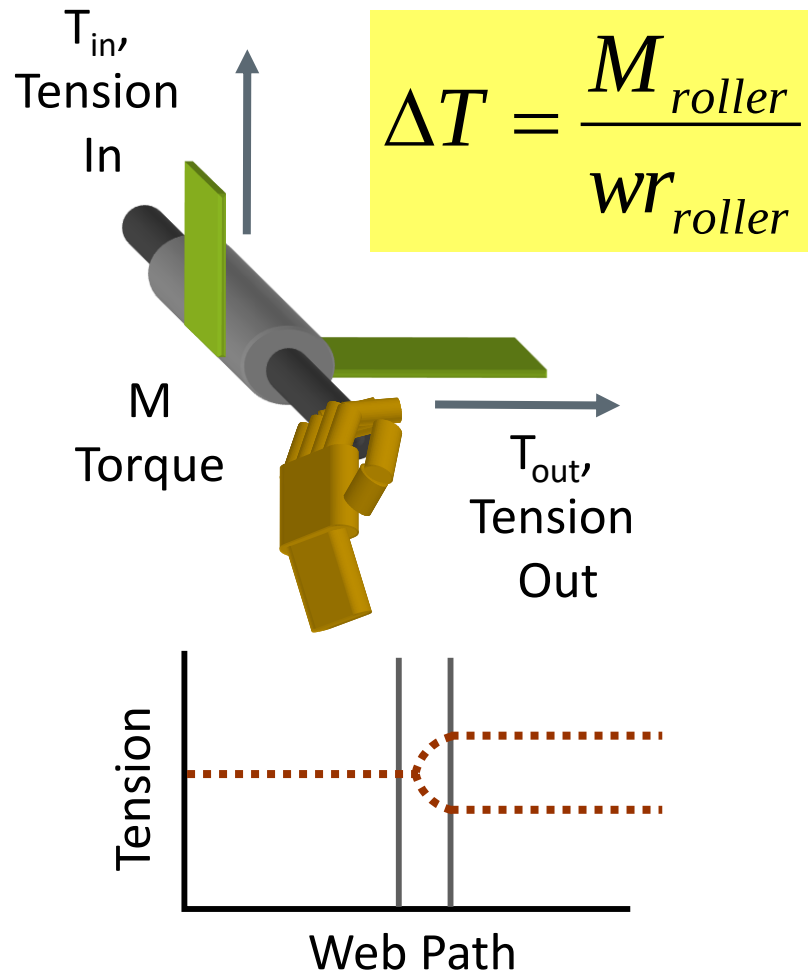
Creating Web Tension: Torque



At unwinding or winding, applied torque will directly create tension



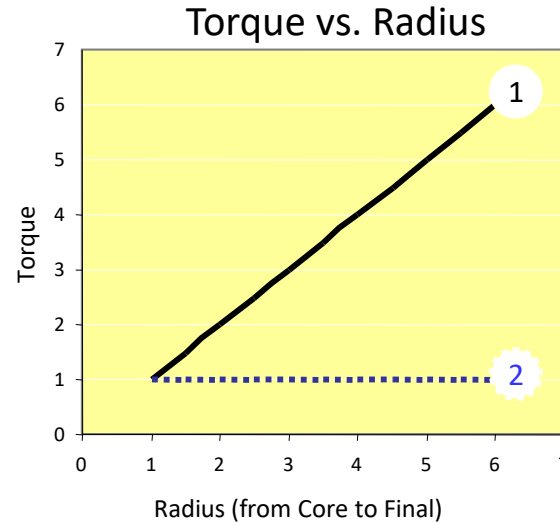
Creating Web Tension: Torque



At a roller, applied torque creates a tension differential.

Torque and Tension vs. Radius

Constant tension
requires torque to
increase proportional to
radius.



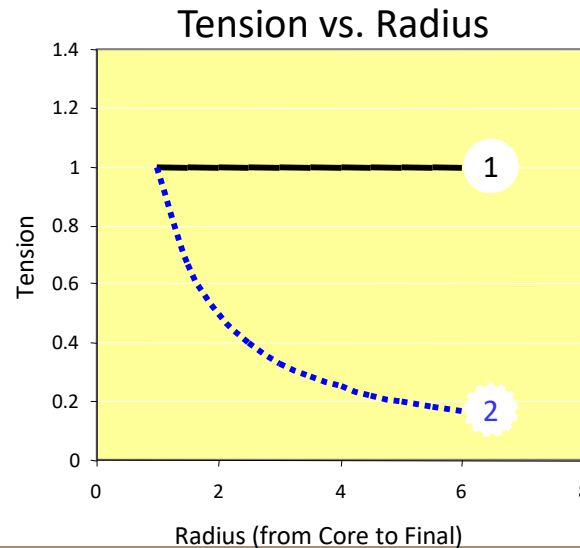
$$M = Twr$$

Legend

1 Constant Tension

2 Constant Torque

Constant torque creates
tension to change
inversely with radius.

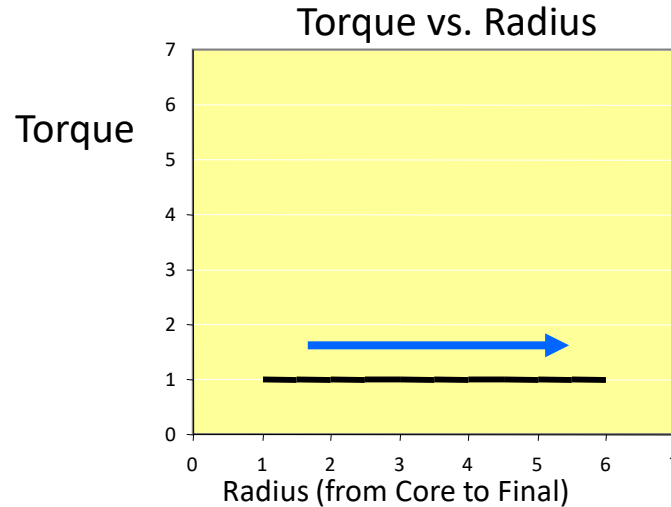


$$T = \frac{M}{rw}$$

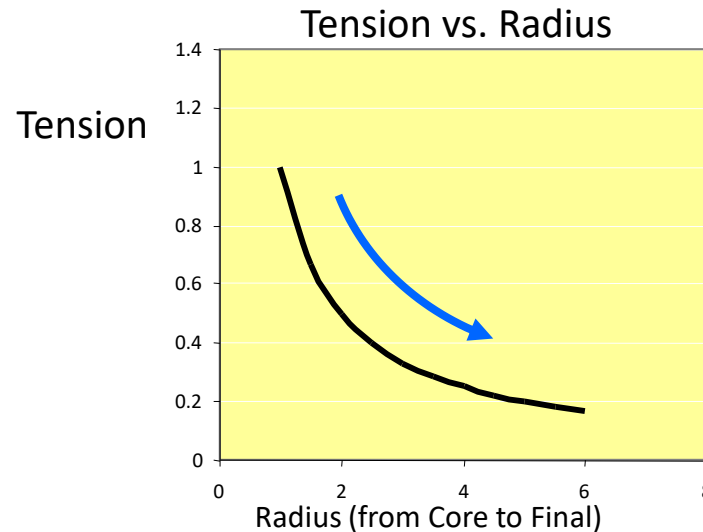
Torque and Tension vs Radius

If center torque is held constant, tension will change inversely with roll radius.

Constant center torque creates decreasing tension at winding.



$$M = Twr$$

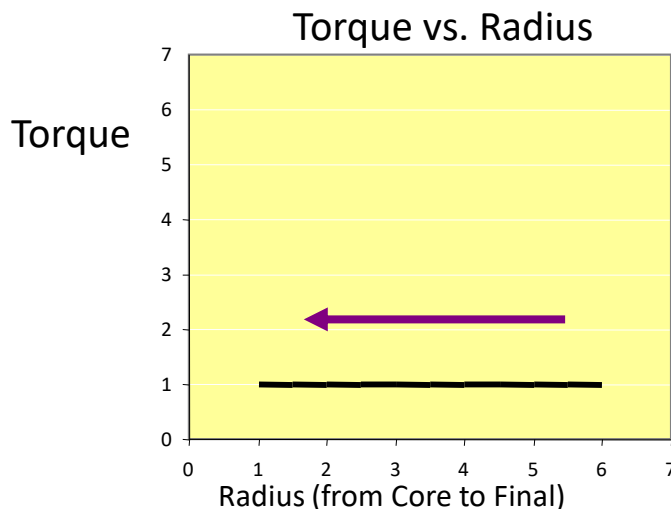


$$T = \frac{M}{rw}$$

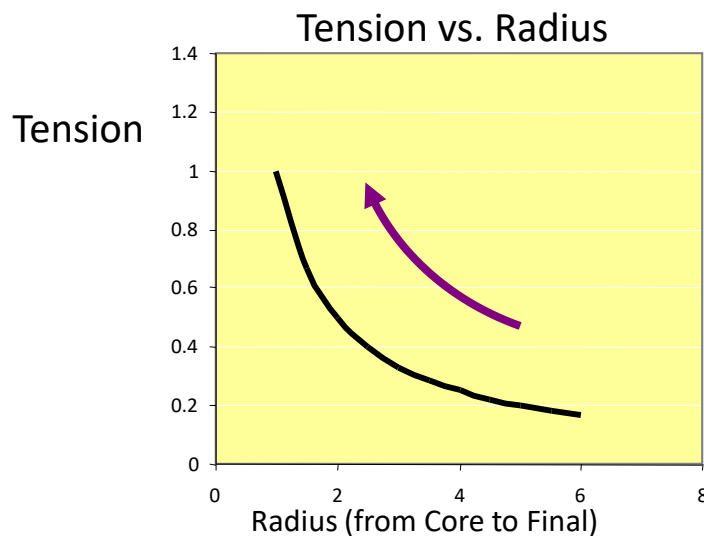
Torque and Tension vs Radius

If center torque is held constant, tension will change inversely with roll radius.

Constant center torque creates increasing tension at unwinding.



$$M = Twr$$



$$T = \frac{M}{rw}$$

Torque Control

What's Needed?

Where is it used?

Torque Control

A torque device:
 Friction brake or clutch
 Magnetic hysteresis clutch
 Magnetic particle clutch/brake
 DC motor
 AC motor
 Leather strap and weight
 (Optional diameter sensor)

Unwinds
 Winders
 Differential shafts
 Center-surface winding
 Dual driven nips
 (steel direct drive with torque controlled rubber nip)

Torque Control

Advantages

Disadvantages

Torque Control

Simple, inexpensive.

Creates constant unwind tension with diameter feedback (assumes inertia effects are small)

May be the most compact option.

Wind with natural taper tension.

No web feedback.

Tension will vary with system torque losses from inertia and drag.

Maintenance of wearing elements.

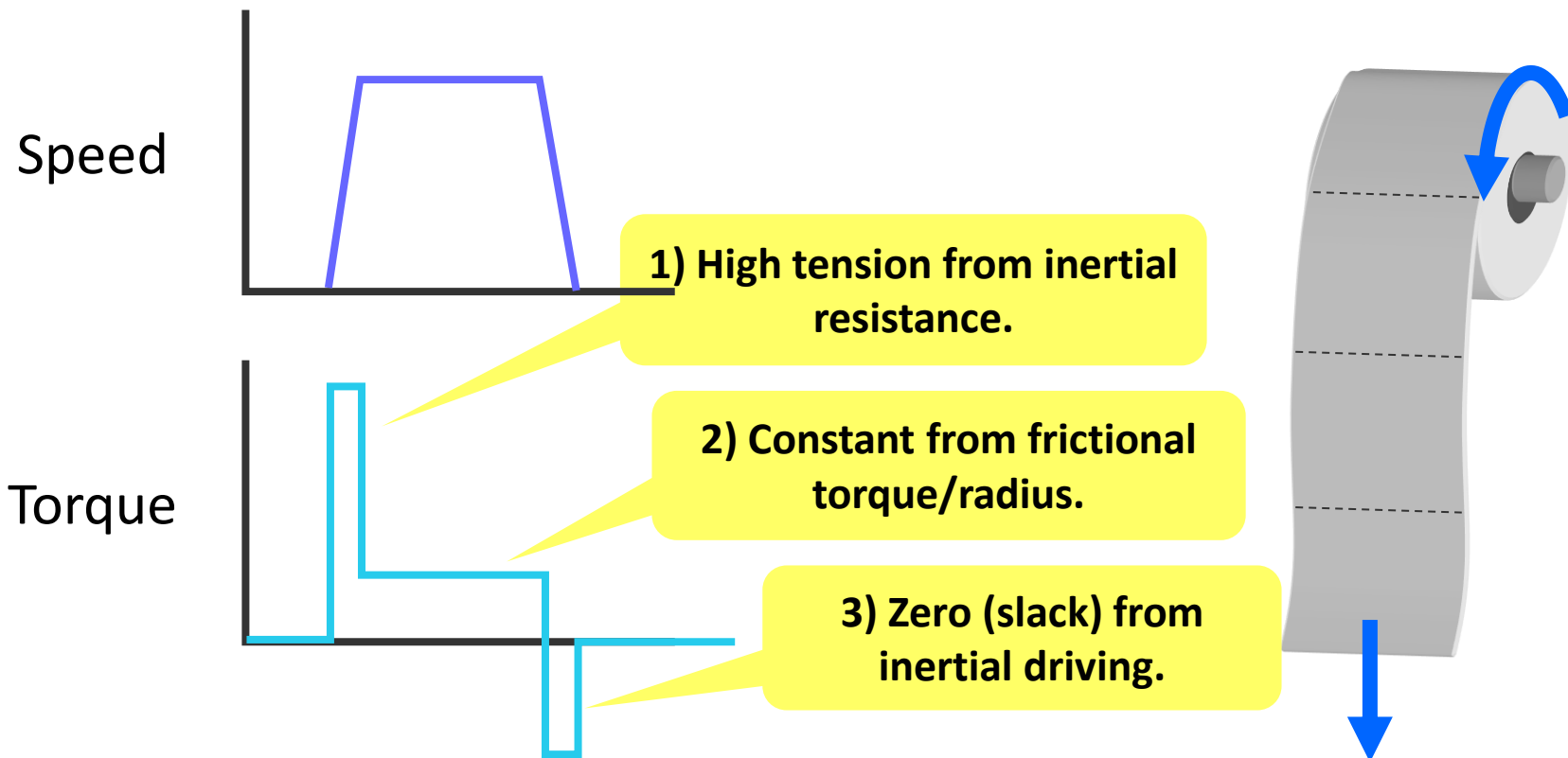
Heat generation.

Noise?

Contamination

Tensioning and Inertial Torques

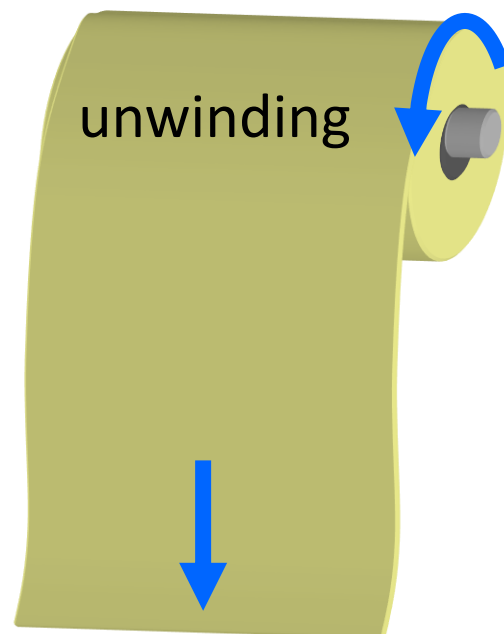
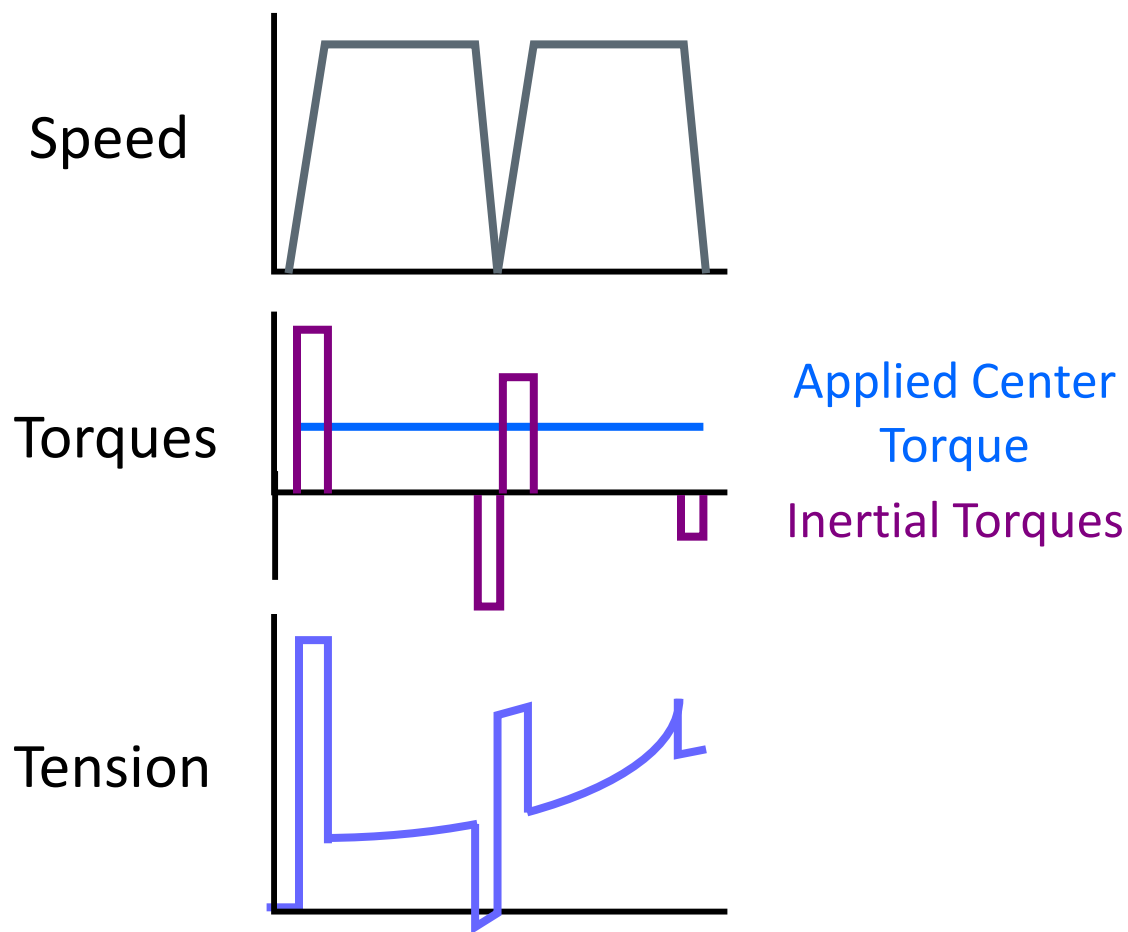
What is the tension of an unwinding toilet paper roll?



Conclusion: Achieving constant tension takes effort.

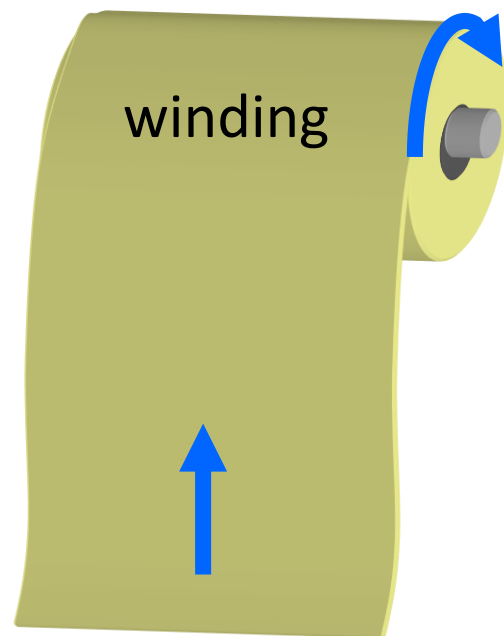
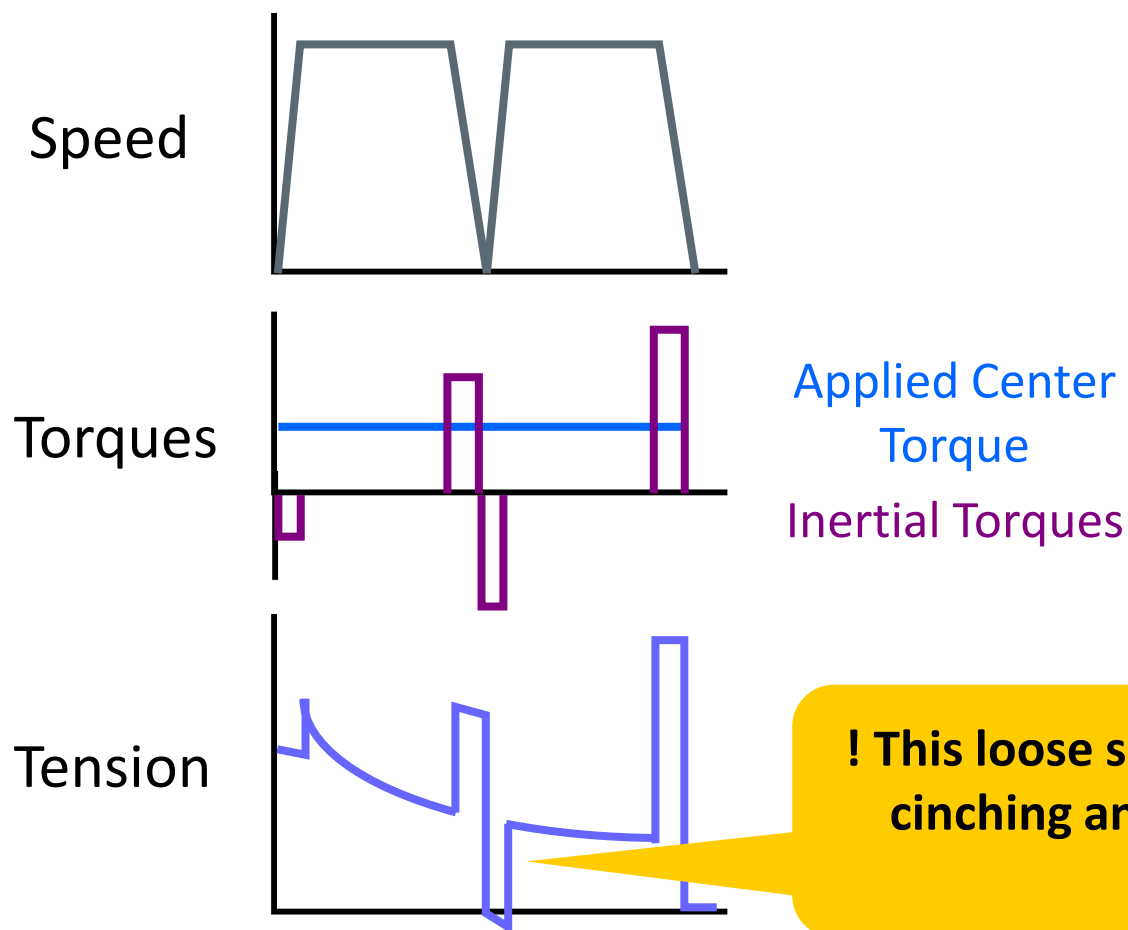
Center Torque, Inertia, and Tension

What if the unwinding roll is stopped mid-roll?



Center Torque, Inertia, and Tension

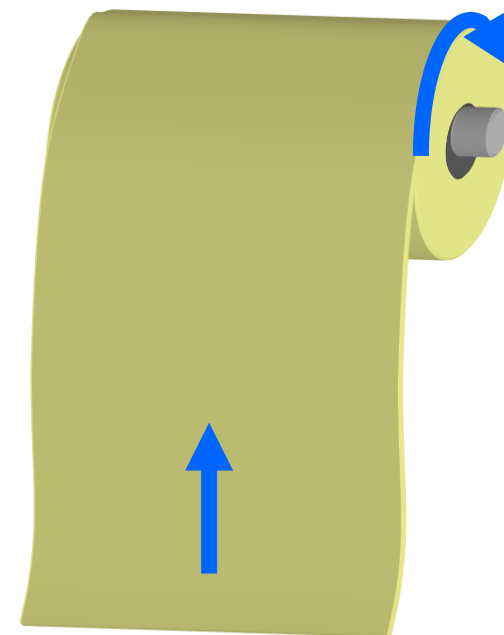
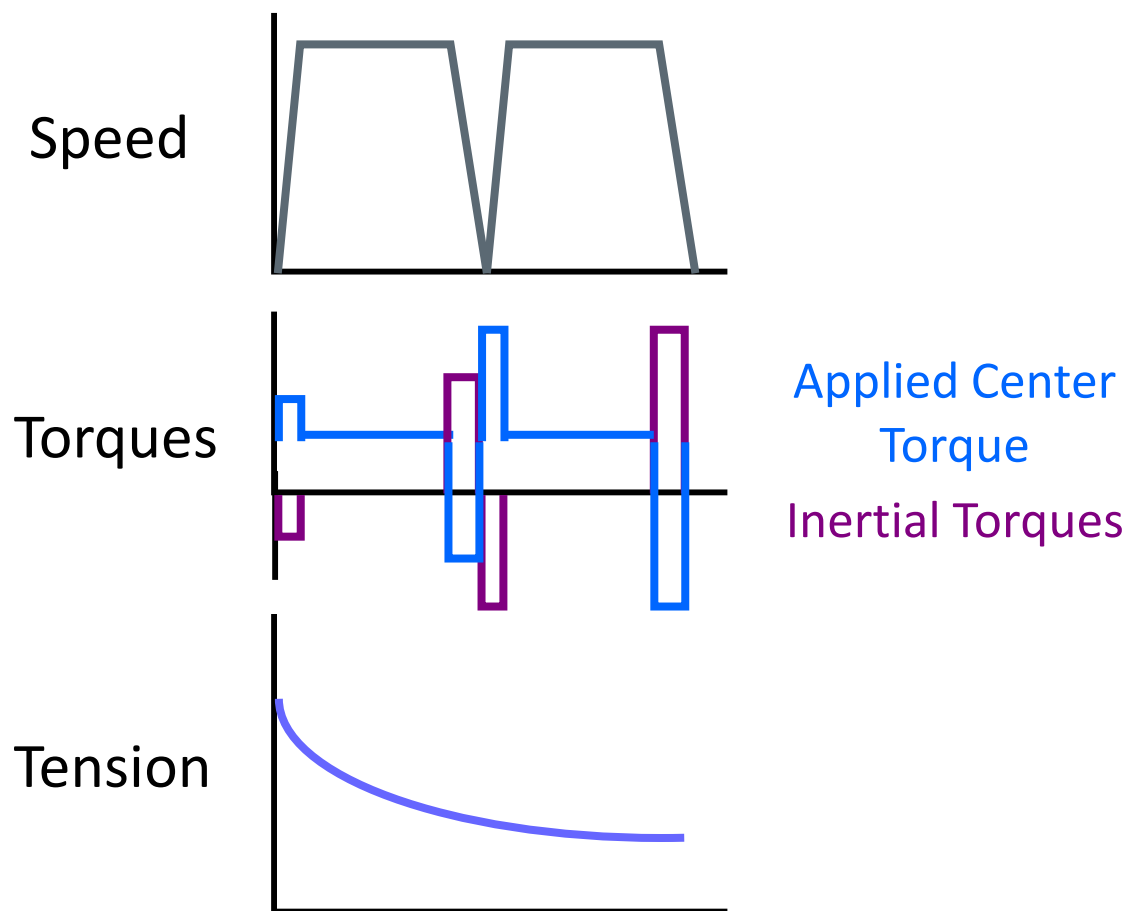
How about winding roll?



! This loose spot in the roll will likely cause cinching and telescoping at winding or unwinding!

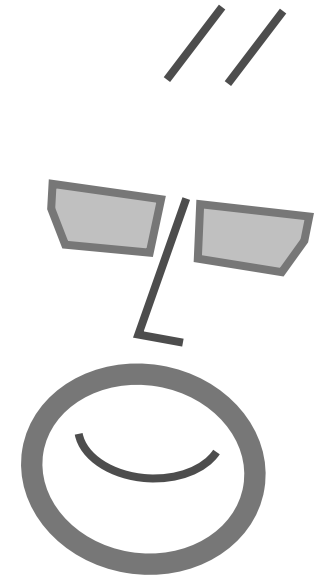
Center Torque, Inertia, and Tension

Inertia compensation reduces tension variations.



Torque Control Qs

How much torque is needed?



Winding Torque and Tension

Torque ratio equals the maximum torque divided by the minimum torque.

Center winder specifications can exceed practical torque ranges.

Example:

Tension (@core) = 50 to 300 N/m (6:1)

Tension (@final) = 50 to 300 N/m

Width = 0.3 to 1.5m (5:1)

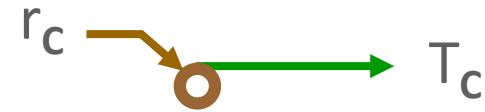
Radius = 0.05 to 0.5m (10:1)

Torque(min) = $(50)(0.3)(0.05) = 0.75 \text{ N-m}$

Torque(max) = $(300)(1.5)(0.5) = 225 \text{ N-m}$

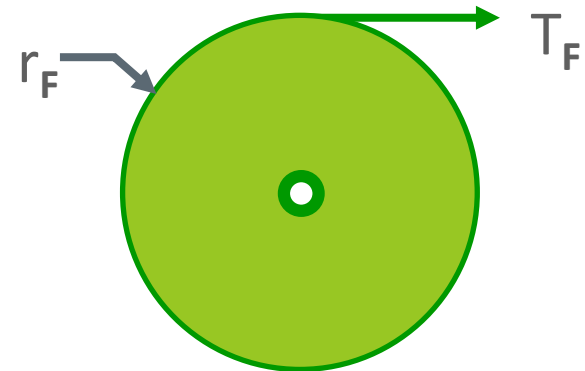
Requested Torque Range = $(225)/(0.75) = 300:1$

$$\text{Torque_Range} = \frac{M_{MAX}}{M_{MIN}}$$



$$M_{MIN} = T_{CORE,MIN} w_{MIN} r_{MIN}$$

$$M_{MAX} = T_{FINAL,MAX} w_{MAX} r_{MAX}$$



Winding Torque and Tension

In specifying a winder, the combination of tension, width, and radius ranges must fit within a reasonable torque range (usually 20 or 30:1).

The tension range can be reduced using “taper tension”, where the maximum final tension is lower than the maximum core tension.

Example:

Tension (@core) = 50 to 300 N/m

Tension (@final) = 50 to 150 N/m (3:1)

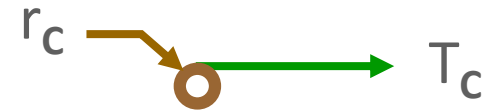
Width = 0.75 to 1.5m (2:1)

Radius = 0.1 to 0.5m (5:1)

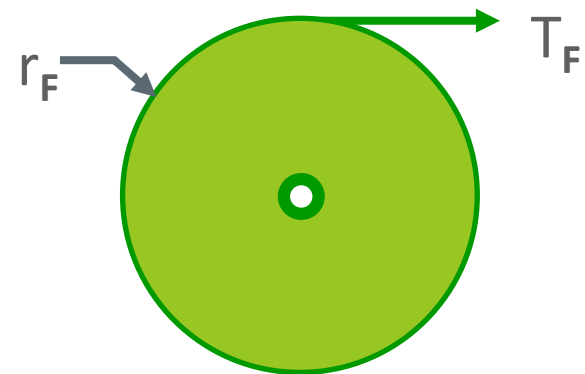
Torque(min) = (50)(0.75)(0.1) = 3.75 N-m

Torque(max) = (150)(1.5)(0.5) = 112.5 N-m

Requested Torque Range = (112.5)/(3.75) = 30:1



$$Torque_Range = \frac{M_{MAX}}{M_{MIN}}$$



Winding Torque and Tension

Torque ratio equals the maximum torque divided by the minimum torque.

Center winder specifications can exceed practical torque ranges.

Example:

Tension (@core) = 0.5 to 3 PLI

Tension (@final) = 0.5 to 3 PLI (6:1)

Width = 12 to 60" (5:1)

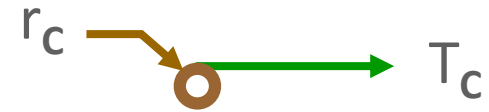
Radius = 2 to 20" (10:1)

Torque(min) = (0.5)(12)(2) = 12 in-lbs

Torque(max) = (3)(60)(20) = 3600 in-lbs

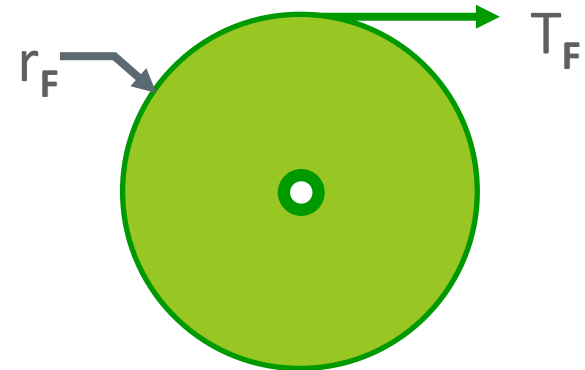
Requested Torque Range = (3600)/(12) = 300:1

$$\text{Torque_Range} = \frac{M_{MAX}}{M_{MIN}}$$



$$M_{MIN} = T_{CORE,MIN} w_{MIN} r_{MIN}$$

$$M_{MAX} = T_{FINAL,MAX} w_{MAX} r_{MAX}$$



Winding Torque and Tension

In specifying a winder, the combination of tension, width, and radius ranges must fit within a reasonable torque range (usually 20 or 30:1).

The tension range can be reduced using “taper tension”, where the maximum final tension is lower than the maximum core tension.

Example:

Tension (@core) = 0.5 to 3 PLI

Tension (@final) = 0.5 to 1.5 PLI (3:1)

Width = 30 to 60” (2:1)

Radius = 4 to 20” (5:1)

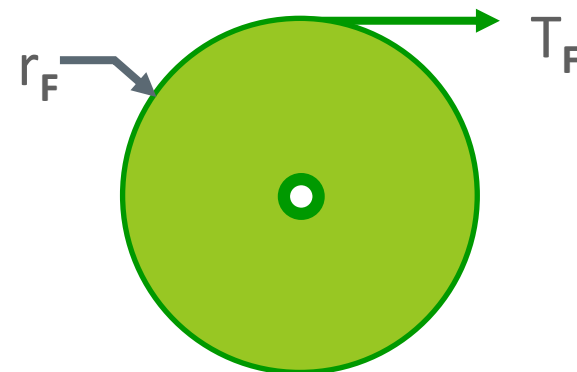
Torque(min) = (0.5)(30)(4) = 60 in-lbs

Torque(max) = (1.5)(60)(20) = 1800 in-lbs

Requested Torque Range = (1800)/(60) = 30:1



$$Torque_Range = \frac{M_{MAX}}{M_{MIN}}$$



Equipment Supplier's Challenge

Asking for a 300:1 torque range is unreasonable.

Ask for it anyway and rate the potential suppliers response.

- ▶ Some suppliers will turn the project down.
- ▶ Some suppliers will bid the project, basing the winder design on the maximum torque requirement (and add 20%). Result: Big motors and terrible low end control.
- ▶ The smart suppliers will do these calculations and call you to work out a reasonable torque range.

Constant tension winding and radial buildup do not increase motor HP or kW. As diameter increases, torque of constant tension may go up, but rpms go down proportionally.

Calculating Power

Kilowatt Equations

P Power, W

T Tension, N/m

w width, m

V Speed in m/s

M Torque in N-m

I Inertia, kg-m²

α Acceleration, rad/s²

M_{lost} Torque losses via
drive train, nip load.

$$P = TwV = M\omega$$

$$P(kW) = (M)(radians / s)$$

$$M = \Delta Twr + I\alpha + M_{lost}$$

Calculating Power

Horsepower Equations

T	Tension, pli
w	width, in.
V	Speed in ft/min
M	Torque in ft*lbs
I	Inertia, ft-lbs-s ²
α	Acceleration, rad/s ²
M_{lost}	Torque losses via drive train, nip load.

$$HP = \frac{(Tw)(V)}{33000}$$

$$HP = \frac{(M)(rpm)}{5250}$$

$$HP \cong 0.75kW$$

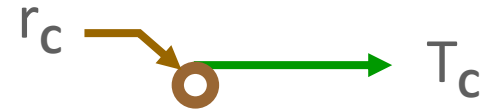
$$M = \Delta Twr + I\alpha + M_{lost}$$

Web Speed vs Rotational Speed

Motor size is based on motor rotational speed.

What is the winding roll rotational speed?

$$n = \frac{V}{2\pi r_{core}}$$



Example:

$$V = 500 \text{ fpm} = 100 \text{ in/s}$$

$$\text{Radius} = 2\text{-in} = 0.17\text{-ft}$$

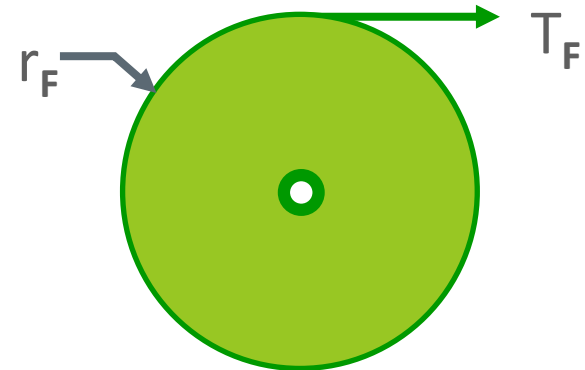
$$n_{core} = (100 \text{ in/s}) / (2)(3.14)(2\text{-in}) = 8 \text{ rps}$$

$$n_{core} = (500 \text{ fpm}) / (2)(3.14)(0.17\text{-ft}) = 480 \text{ rpm}$$

$$n_{max} = (100 \text{ in/s}) / (2)(3.14)(20\text{-in}) = 0.8 \text{ rps}$$

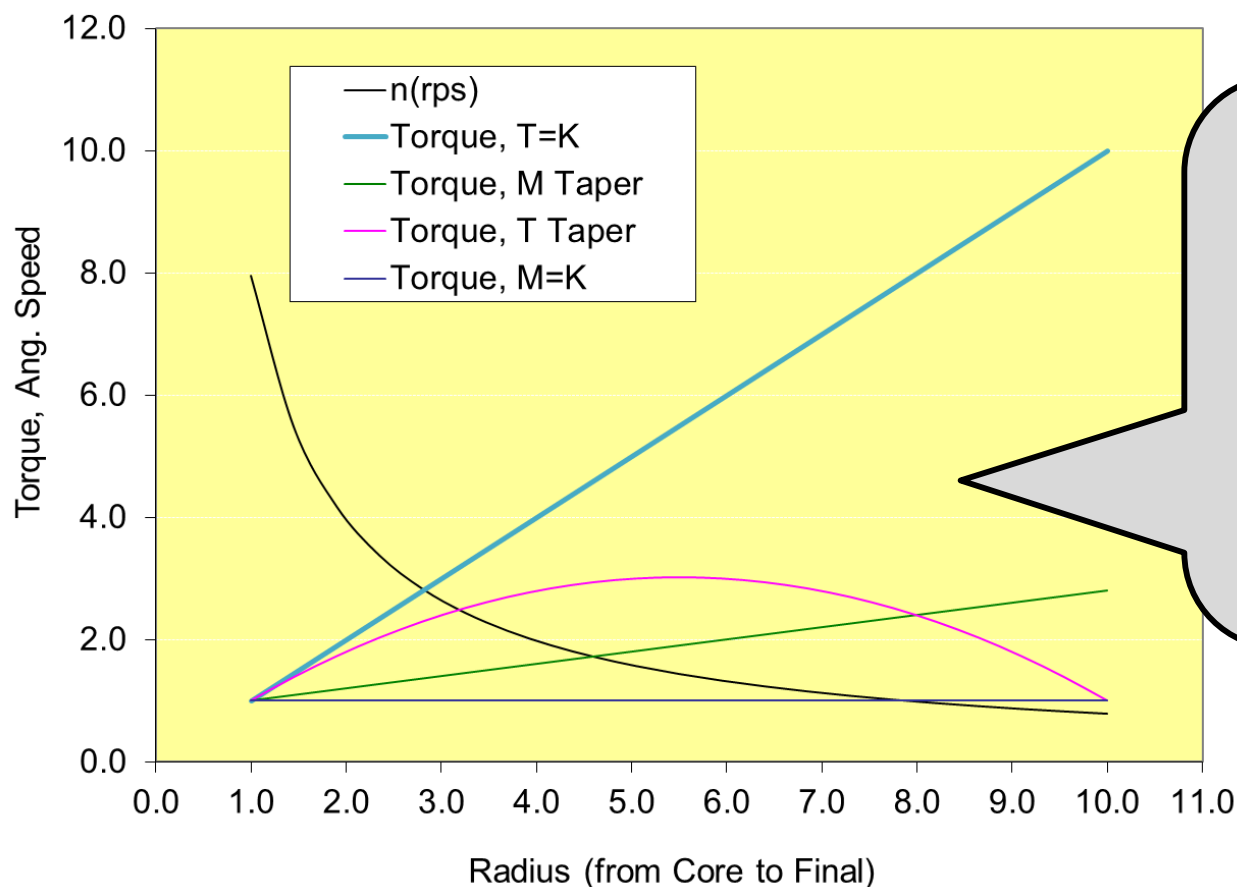
$$n_{max} = (500 \text{ fpm}) / (2)(3.14)(1.67\text{-ft}) = 48 \text{ rpm}$$

$$n = \frac{V}{2\pi r_{max}}$$



Web Speed vs Rotational Speed

Torque vs. Radius

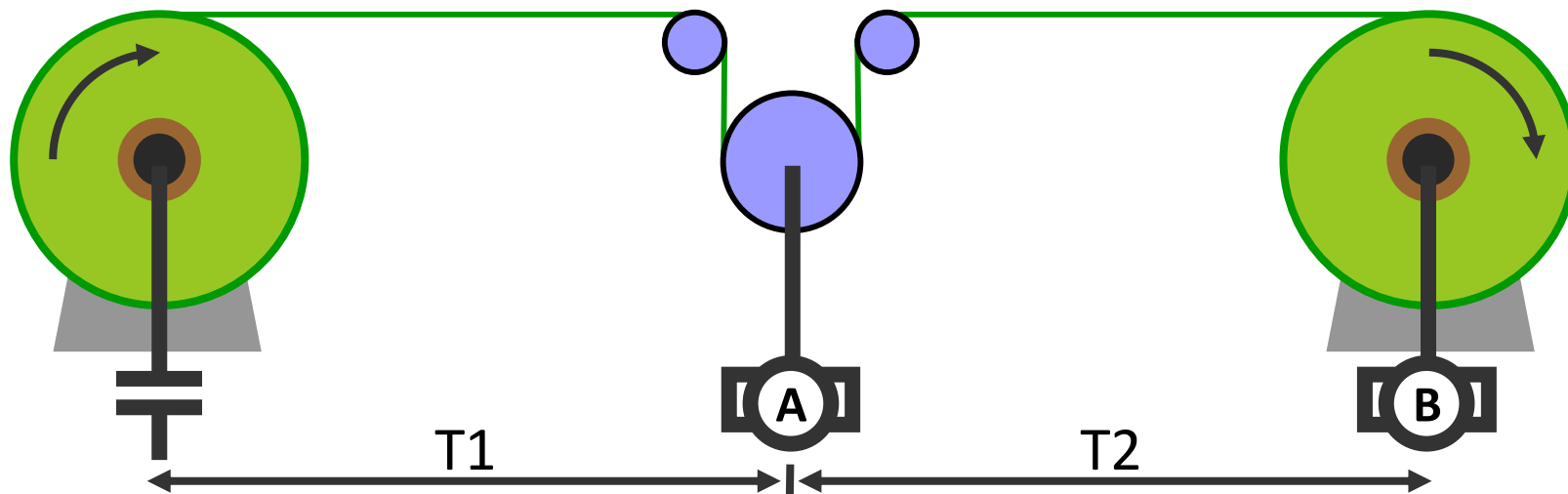


With constant tension winding, motor HP or kW do not change from core to final roll. With increasing radius, the increasing torque demand is offset by decreasing rpms.

Torque Demands

Unwinding

Winding



$$M_A = (T_2 - T_1)wr + I\alpha + M_{lost}$$

$$M_B = T_2wr + I\alpha + M_{lost}$$

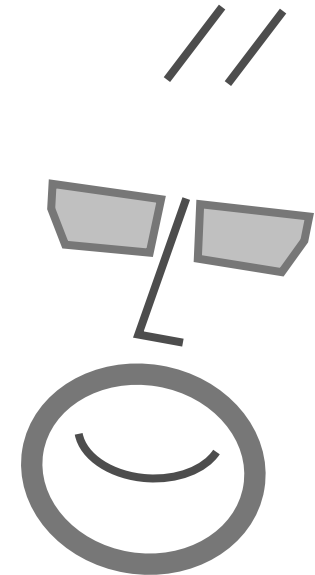
Torque demand for unwinds and winders will be higher than intermediate driven rollers.

$$M_B > M_A$$

Speed Control Qs

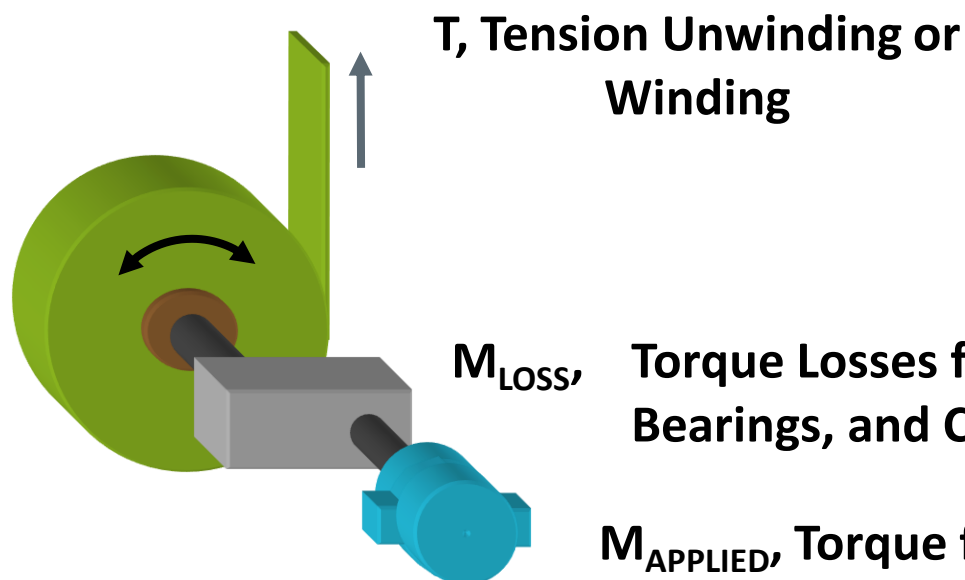
When is open loop torque control not a good idea?

When torque losses between the torque device and the web are variable or unknown.



Creating Web Tension: Torque

Torque devices (motors, brakes, clutches) usually have good accuracy, but a complex mechanical coupling system, such as a gearbox, may create torque losses and tension variations.



T, Tension Unwinding or Winding

$$F_T = \frac{(M_{APPLIED} - M_{LOSS})}{r_{roll}}$$

$$T = \frac{(M_{APPLIED} - M_{LOSS})}{wr_{roll}}$$

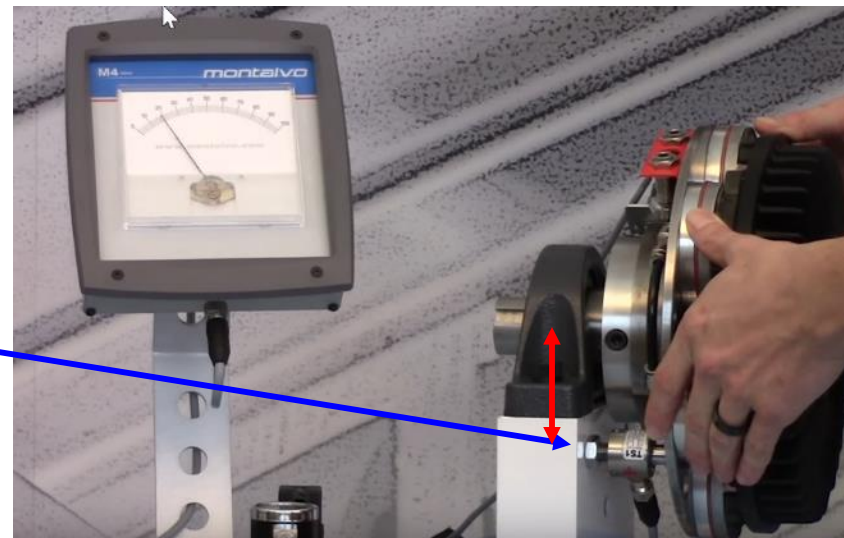
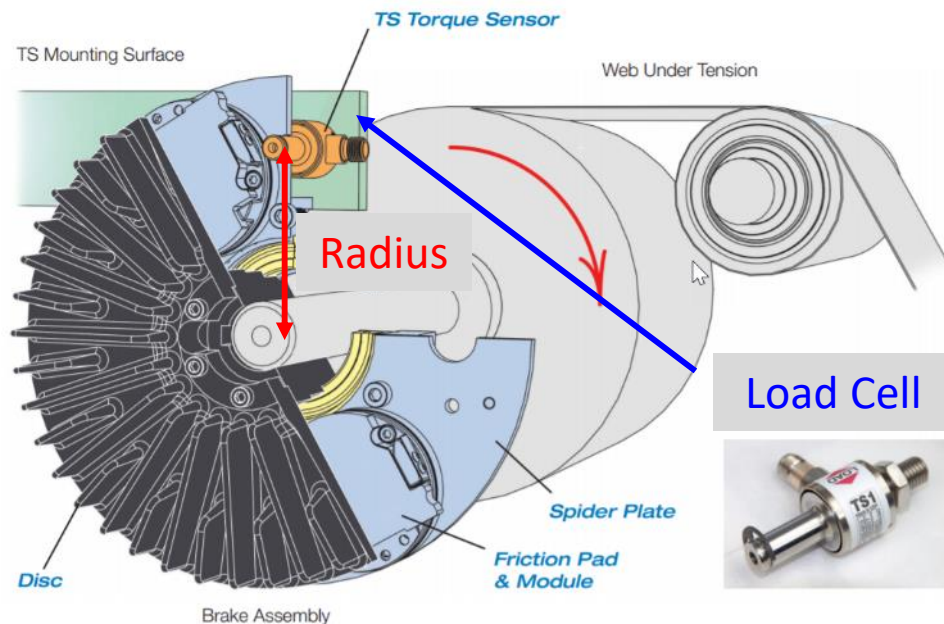
M_{LOSS}, Torque Losses from Gearbox, Bearings, and Couplings

M_{APPLIED}, Torque from Motor, Clutch, or Brake

Torque Sensing (Force Sensing x Radius)

A brake or clutch is mounted such its rotation is restricted by a load cell. Applied torque is calculated as the force to restrain the brake or clutch multiplied by the radial distance from the load cell to the shaft's center.

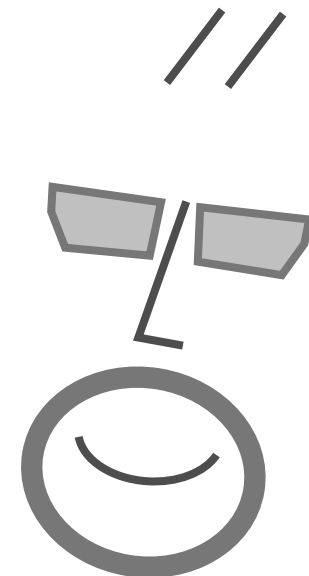
$$\text{Torque} = F(\text{Load Cell}) \times (\text{Radius})$$



Courtesy of Montalvo

Torque Control Qs

What type of brakes are used in web handling?



Disc Brake in Web Handling

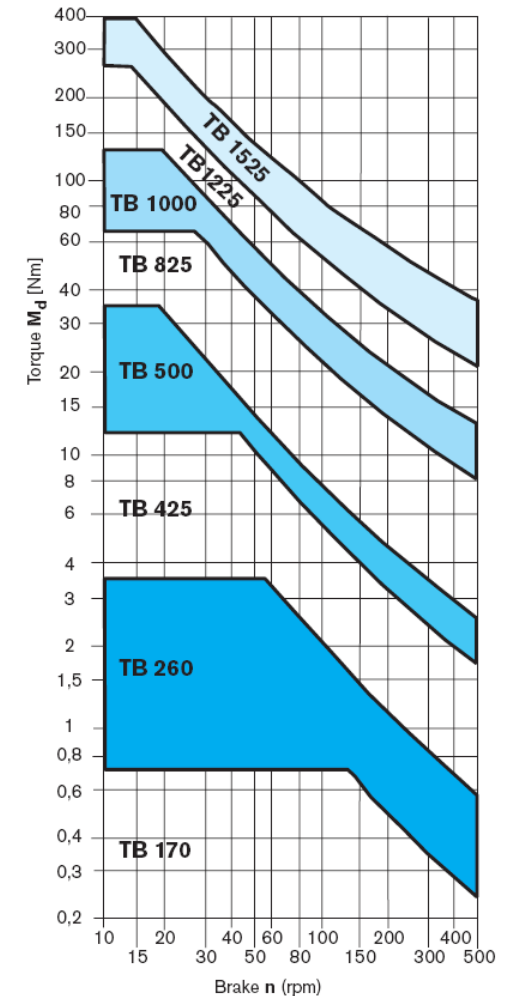
Disc brakes are commonly used for controlling unwinding roll tension.

Due to heat generation, all brakes have a 'thermal horsepower' or 'slip watts' limit that is a function of torque and rpms.

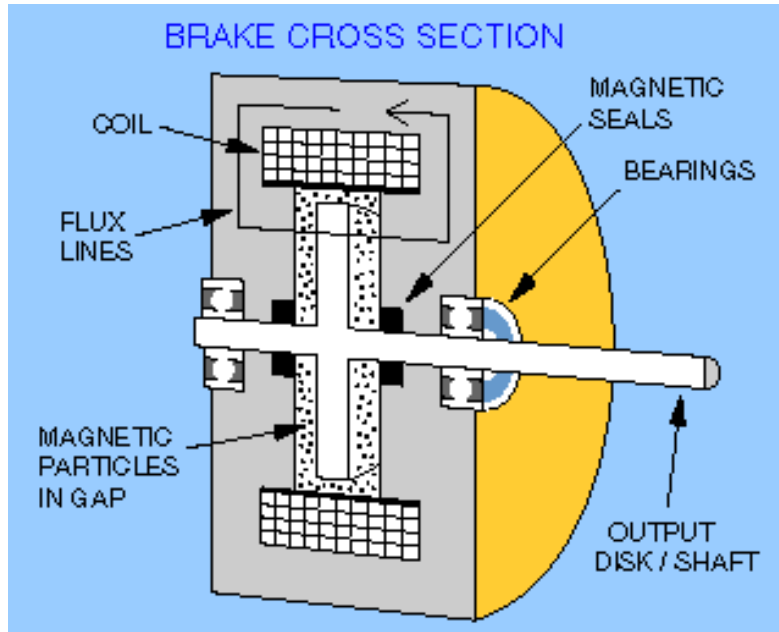


Images courtesy of Warner Electric

Dynamic braking torque
TB170 - TB1525



Magnetic Particle Brake



Courtesy of Placid Industries

“The output disk/shaft does not touch the body. The gap in between is filled with a fine, dry stainless steel powder.

“The powder is free flowing, until a magnetic field is applied from the coil. The powder particles form chains along the magnetic field lines, linking the disk to the body.

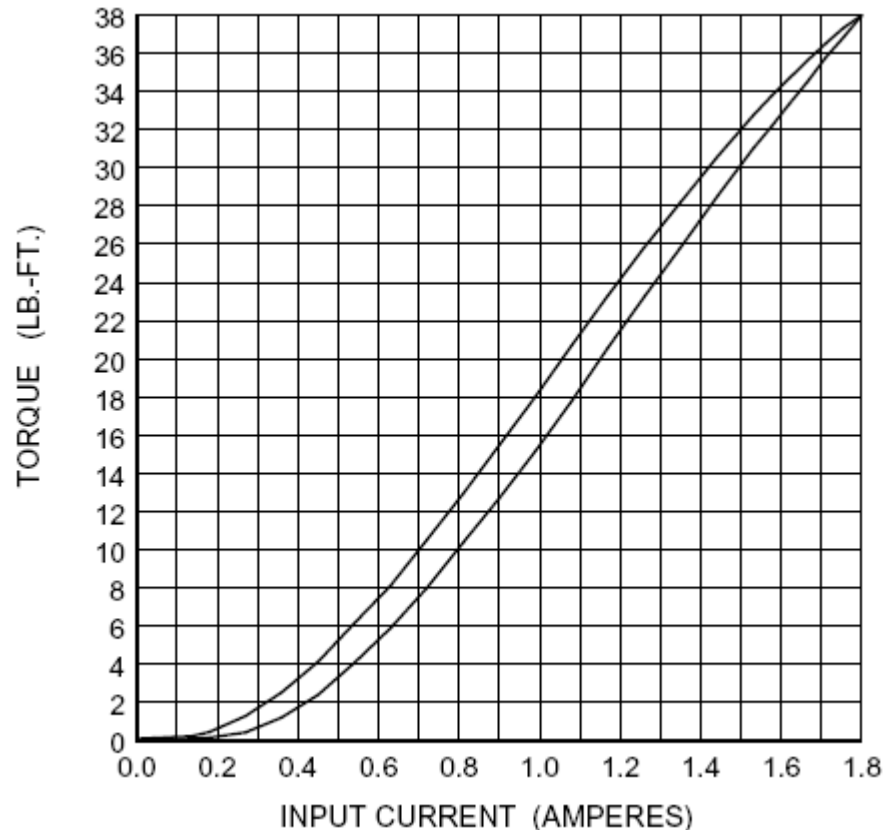
“The torque is proportional to the magnetic field strength, and therefore to the applied D.C. input current.”

Magnetic Particle Brake - Example



Courtesy of Placid Industries

Torque range	1 to 38	lb.-ft.
Maximum RPM	1800	RPM
Heat dissipation, @ 100 RPM	190	watts
Heat dissipation, @ 1000 RPM	275	watts
Heat dissipation, w/ piped air . .	425	watts



$$\text{HEAT (watts)} = \text{RPM} \times \text{TORQUE (lb.-ft.)} \times 0.14$$

Torque Control Qs

Why use motors vs. brakes or clutches?

Answer 1:

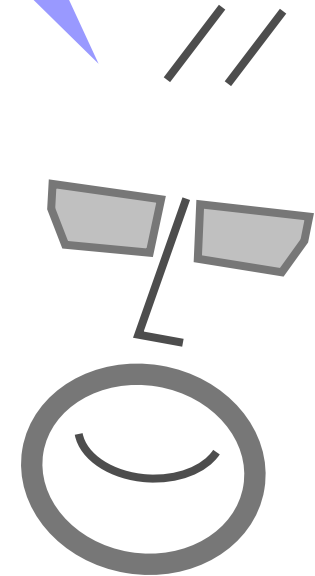
Motors can function in either direction, applying torque either with or against the direction of rotation.

Answer 2:

Motors can function in speed or position control, which is vital to MD registration or length control.

Answer 3:

Braking motors can share a common bus and recover the unwinding generated energy.



Draw Control

What Variable is Adjusted?

Torque

Relative Speed

Open-Loop Control
No Tension Feedback

Will it have tension
feedback?

Torque
Control

***Draw
Control***

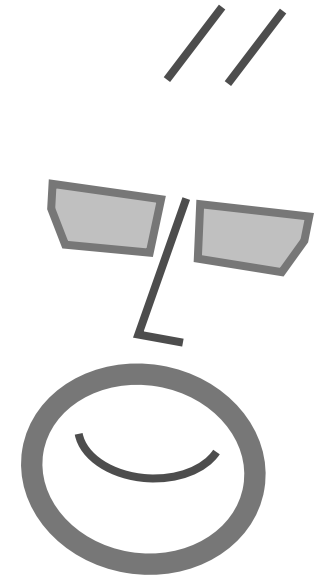
a.k.a. Speed Ratio Control

Closed-Loop Control
Tension Feedback by
Load Cell or Dancer Roller

Tension Control

Tensioning Qs

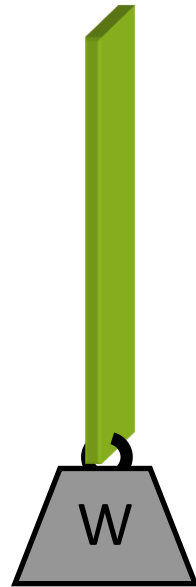
How does a draw or speed ratio create tension?



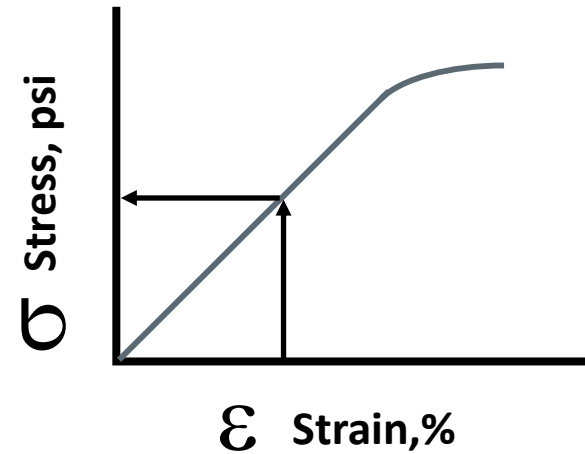
Creating Tension

Force Control

$$\sigma = \frac{F}{tw}$$



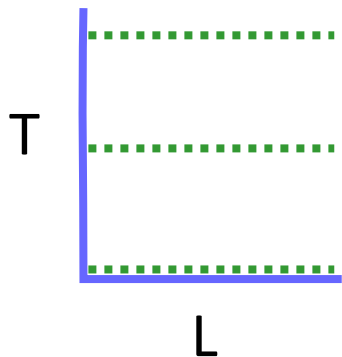
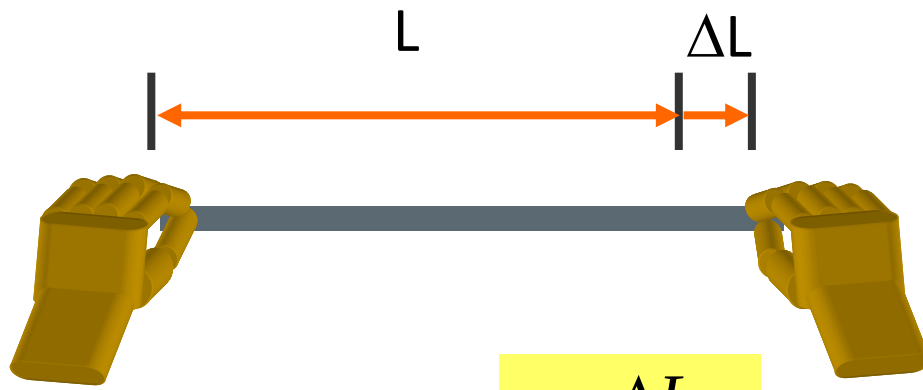
Stretch Control



$$\sigma = \epsilon E$$

Creating Web Tension: Strain

Stretching

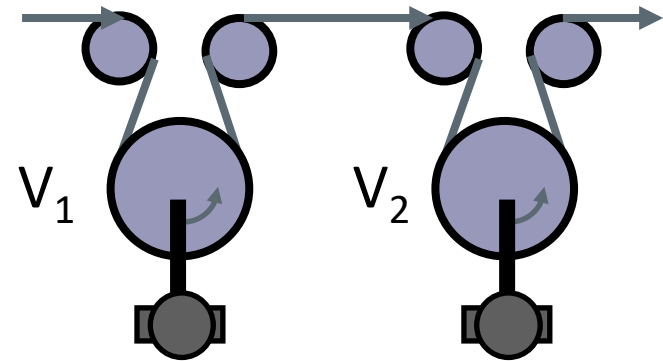


$$\varepsilon = \frac{\Delta L}{L_0}$$

$$\Delta T \propto \varepsilon$$

$$\Delta T = \varepsilon Et$$

Speed Increase



$$Draw = \frac{\Delta V}{V_1}$$

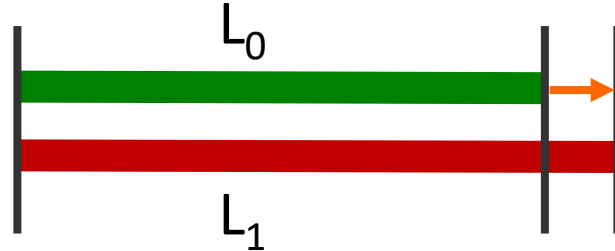
$$\Delta \varepsilon \propto Draw$$

$$\Delta T \propto (Draw)Et$$

Tensioning the Moving Web

Tensioning a static web

$$\varepsilon = \frac{L_1 - L_0}{L_0} = \frac{\Delta L}{L_0}$$

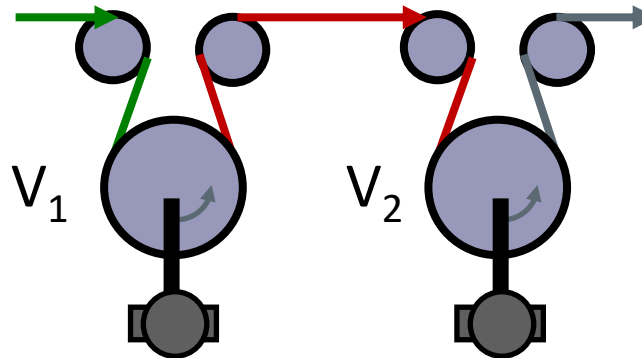


$$L_1 = L_0(1 + \varepsilon_1)$$

$$k = \frac{L_0}{(1 + \varepsilon_0)} = \frac{L_1}{(1 + \varepsilon_1)}$$

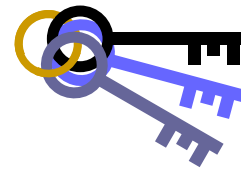
$$T_x = \varepsilon_x Et$$

Tensioning a moving web



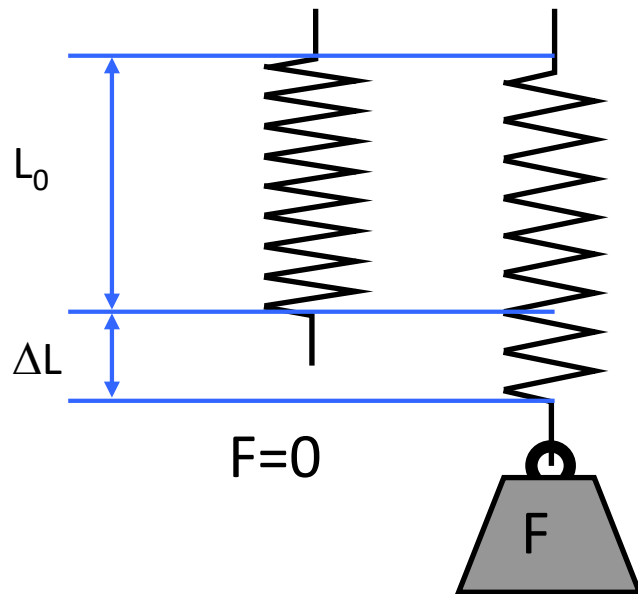
$$k = \frac{V_1}{(1 + \varepsilon_1)} = \frac{V_2}{(1 + \varepsilon_2)}$$

$$T_x = \varepsilon_x Et$$



Hooke's Law

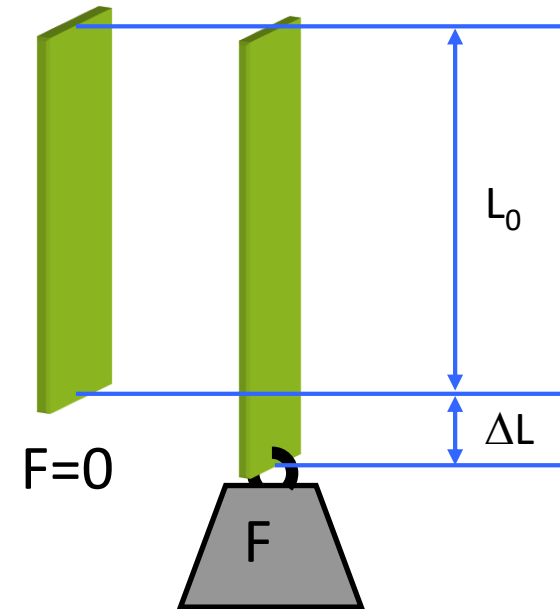
Spring



$$F = k \frac{\Delta L}{L_0} = k\varepsilon$$

The web's spring constant is the product of its thickness, width and modulus.

Web

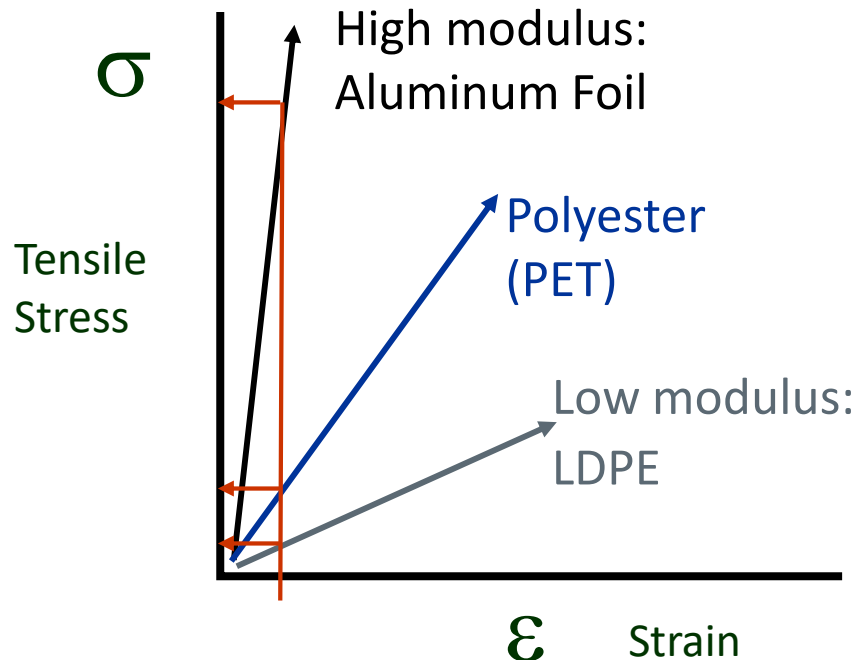


$$F = k \frac{\Delta L}{L_0} = (twE)\varepsilon$$

Tension from Strain

For a given strain (% stretch):

1. A higher modulus web will have higher tension than a lower modulus web.
2. A thicker web will have more tension than a thinner web.



For strain, $\epsilon = 0.001 = 0.1\%$

$E(\text{Al}) = 10 \text{ Mpsi (69 GPa)}$

Tension = 10PLI/mil

$E(\text{PET}) = 500 \text{ kpsi (3.4 GPa)}$

Tension = 0.5 PLI/mil

$E(\text{LDPE}) = 100 \text{ kpsi (6.9 MPa)}$

Tension = 0.1 PLI/mil

Draw Control

Draw is an equipment property determined by the rpm and diameter of sequential driven rollers.

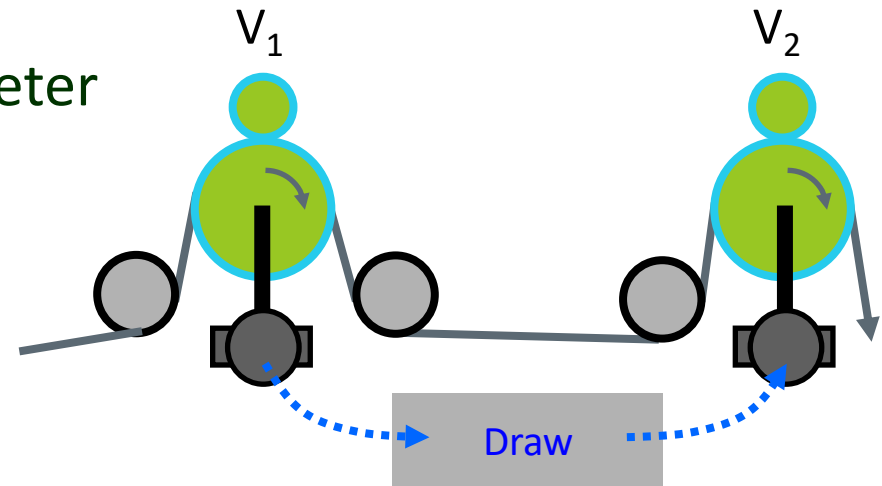
Draw
Ratio

$$N_{DR} = \frac{V_2}{V_1}$$

$$Draw = \frac{(V_2 - V_1)}{V_1}$$

$$Draw = N_{DR} - 1$$

$$\%Draw = 100 \frac{(V_2 - V_1)}{V_1}$$



However, Draw $\neq$ Strain.

Web Strain from Draw

Strain is a web property.

Draw increases web strain.

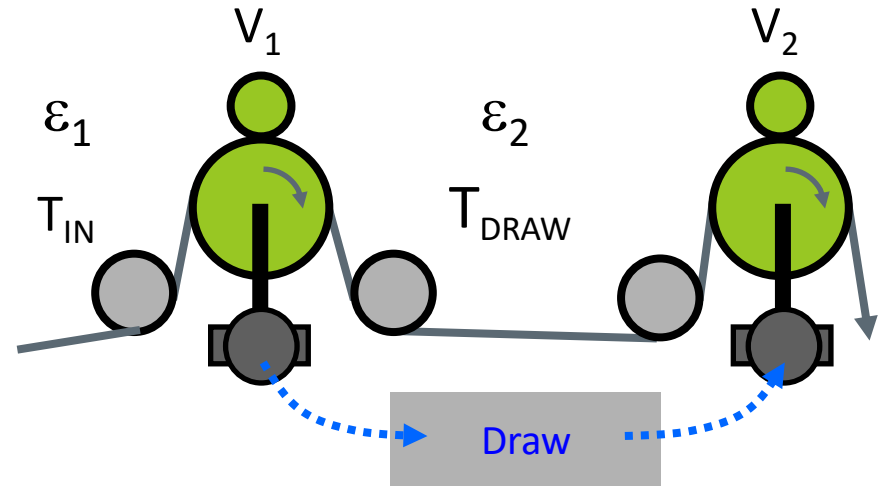
For elastic webs...

$$\frac{V_1}{(1 + \varepsilon_1)} = \frac{V_2}{(1 + \varepsilon_2)}$$

$$\varepsilon_2 = Draw + \frac{V_2 \varepsilon_1}{V_1}$$

Simplified:

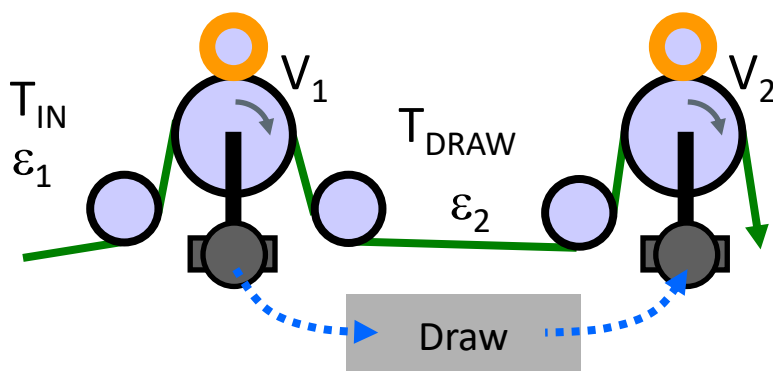
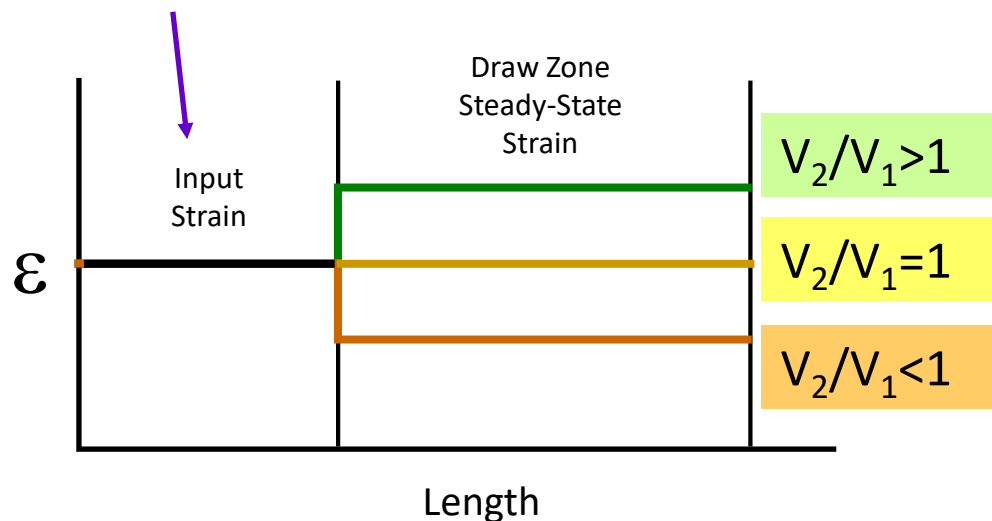
$$\varepsilon_2 = Draw + \varepsilon_1$$



Strain and tension in the draw zone is a function of V_1 , V_2 , and the upstream zone's strain.

Draw Strain and Velocities

The input web strain is the baseline for the draw zone strain.

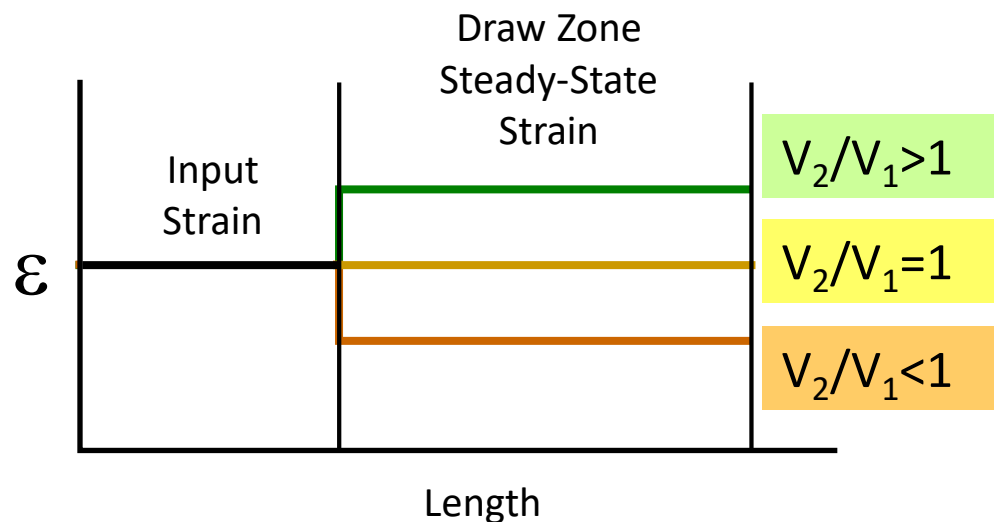


A draw ratio greater than one will increase draw zone strain.

If the draw ratio is 1:1, the draw zone strain will equal the input strain.

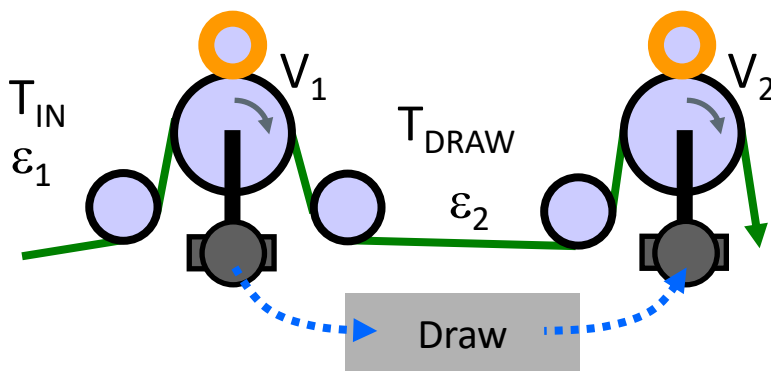
A draw ratio less than one will decrease draw zone strain.

Draw Strain and Velocities



Draw zone strain is always relative to the upstream strain.

Changes in upstream tension or strain will feed into and change the draw zone tension and strain.



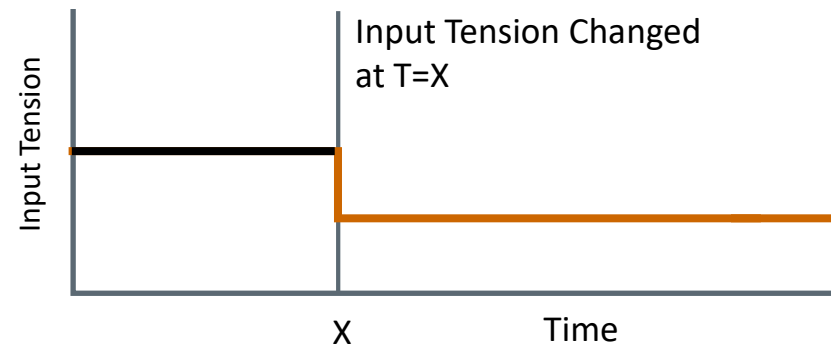
Draw Zone Time Constant

A draw zone does not react as quickly as tension controlled zone.

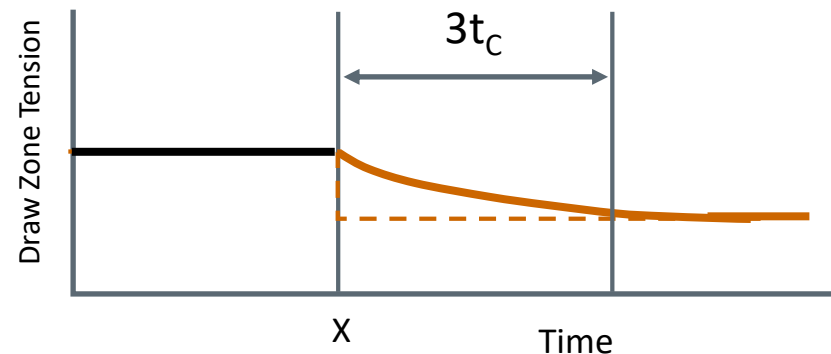
A draw zone must wait for the strain transport mechanism to flush out the old strain level and move to new steady-state strain level.

The time required to reach the new steady-state is dependent on the draw zone length and web speed.

$$t_c = L/V$$

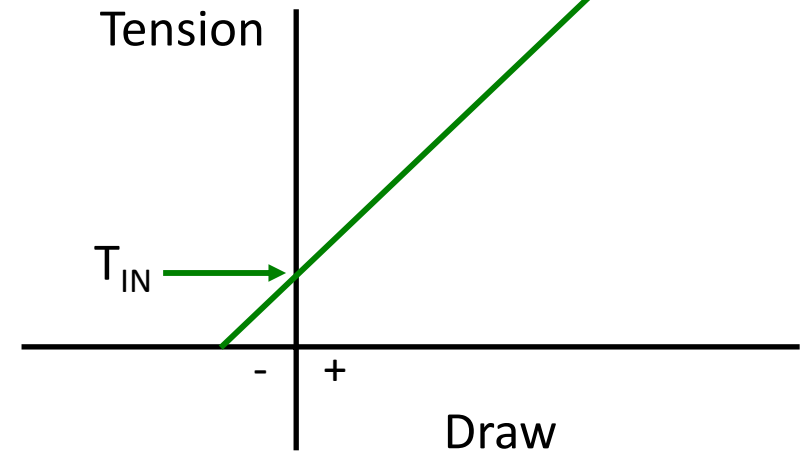
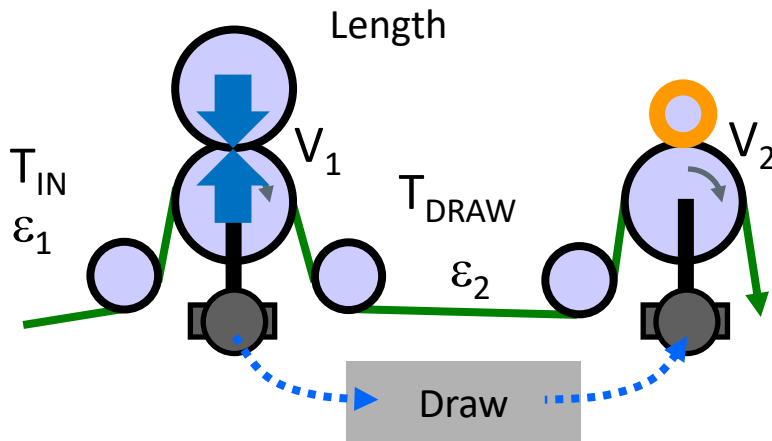


Assume $V_2=V_1$

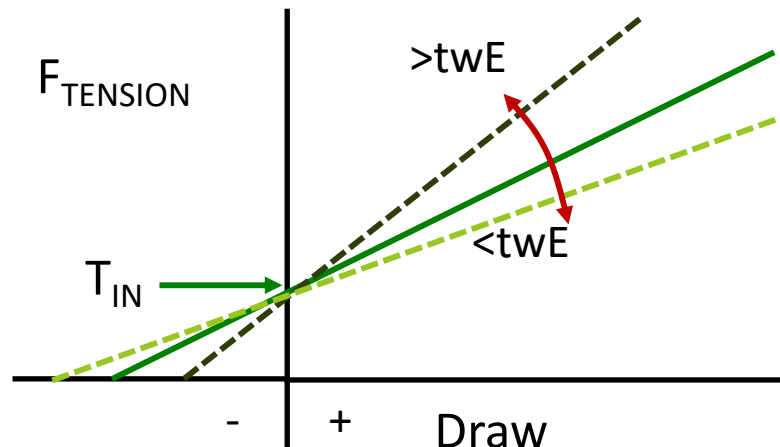
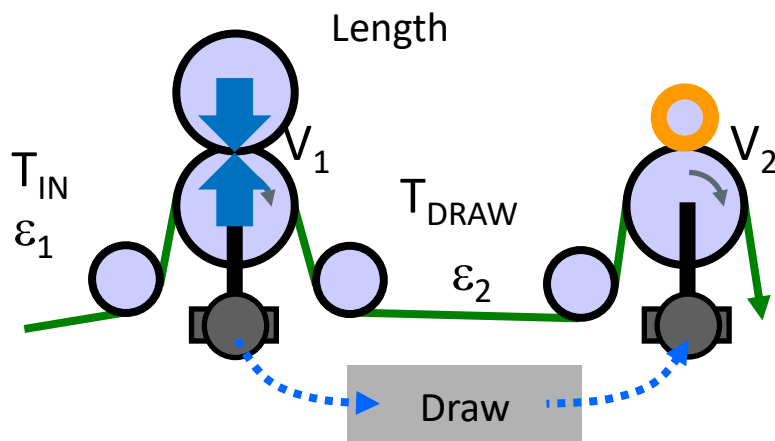


Draw Control and Yielding

If upstream tension creates 1% strain, draws greater than -1% will avoid slackness.



Draw Control and Spring Constant



$$F_{TENSION} = (twE)\varepsilon$$

F_{TEN} = tension force, lbs or N

t = thickness, in or mm

w = width, in or mm

E = Young's Modulus, psi or Pa

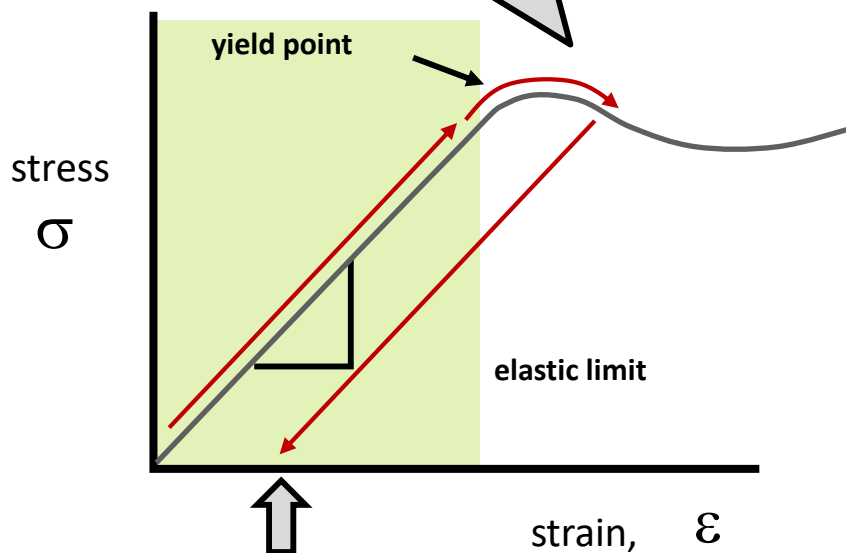
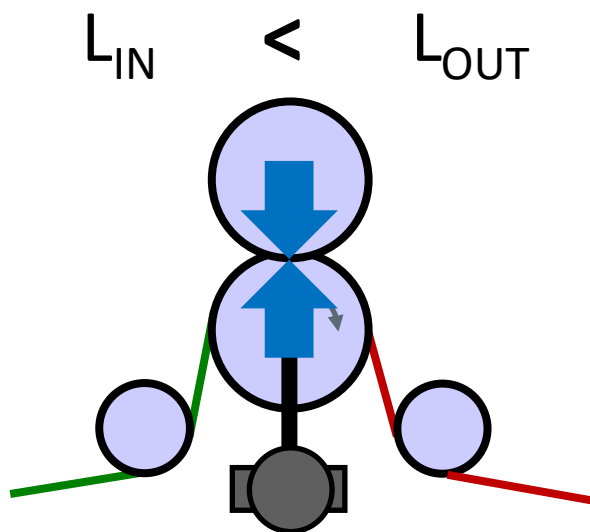
ε = strain, --

The tension from strain or draw is a function of the web's spring constant (thickness x width x modulus). For a given draw, the tension will increase with thicker, wider, or higher modulus webs.

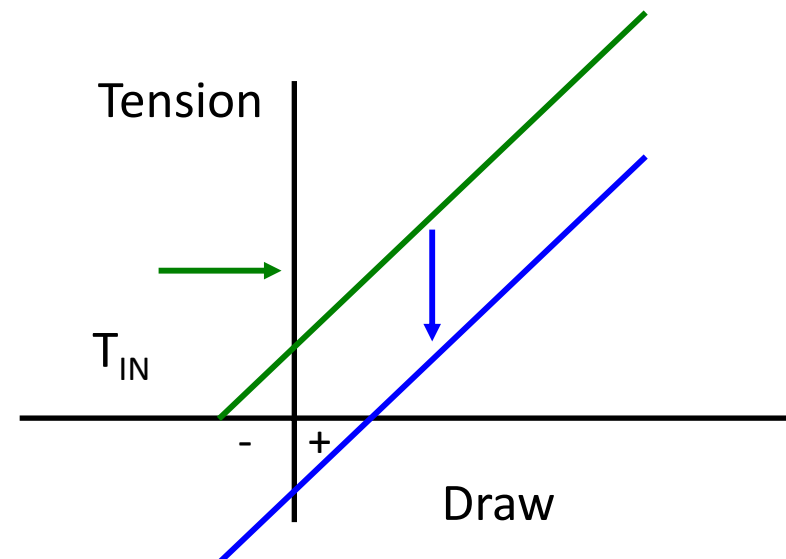
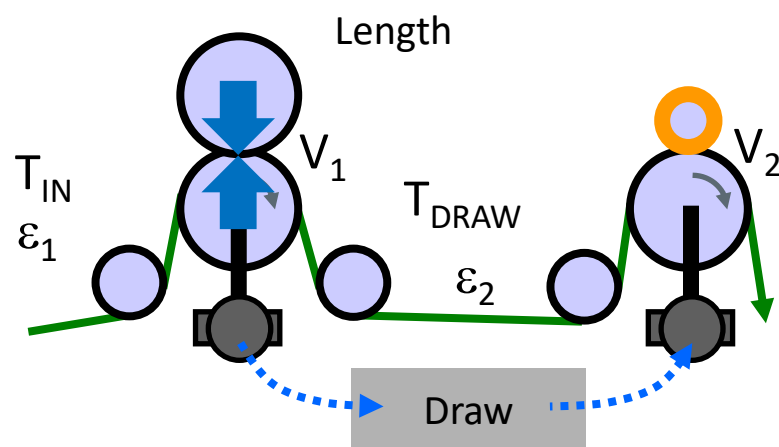
For polymer films, tension in a draw zone may change with temperature as the modulus changes.

Draw Control and Length Changes

Some processes may permanently change the web length (calendering, embossing, heating, stretching).

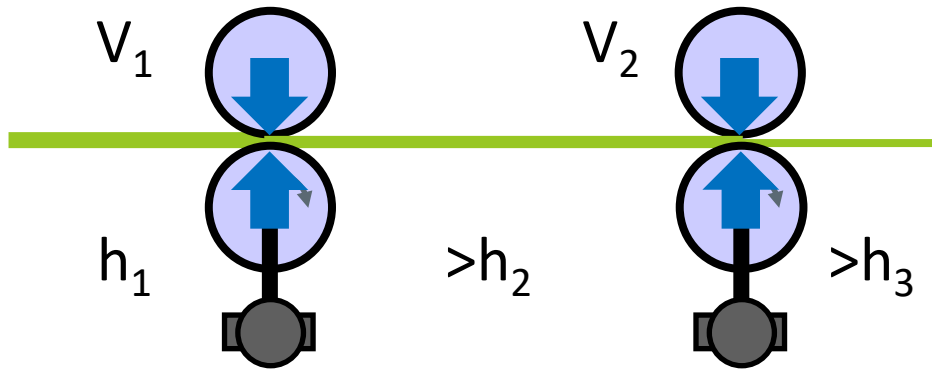


Draw Control and Yielding



If upstream tension creates 1% strain and the process elongates the web 2%, +1% draw will be required to avoid slackness and +2% draw will be needed to maintain upstream tension.

Draw Control and Yielding



$$L_0 = V_0 t = \left(\frac{V_1}{1 + \varepsilon_1} \right) t$$

If web thickness decreases, but density and width are constant, length will increase.

Increasing length in 1:1 draw will reduce tension. If the length change exceeds the entering strain, the web will become slack.

Constant
Density
Volume

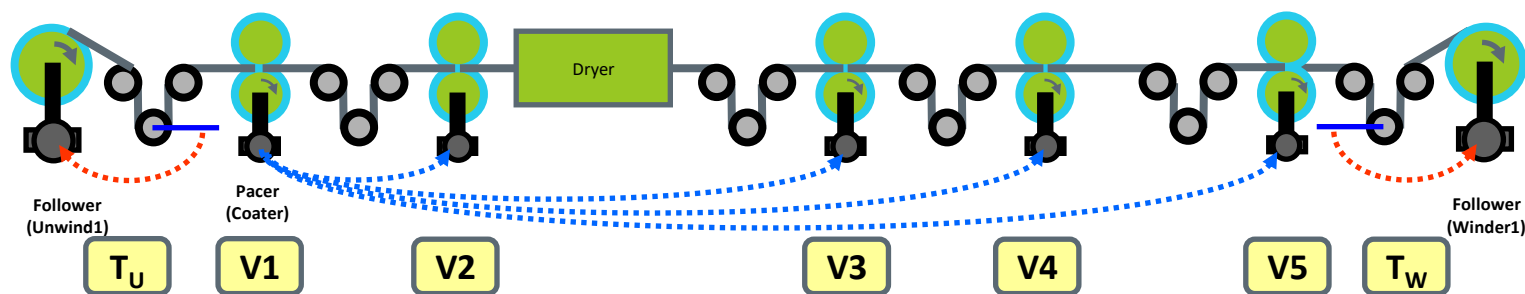
$$= L_1 w_1 h_1 = L_2 w_2 h_2$$

Constant
Mass Flow

$$= V_1 w_1 h_1 = V_2 w_2 h_2$$

Master Reference Draw Control

Multiple Draw Zone Weblines: Case 1



Baseline: All draws relative to reference roller
(usually the pacer or first draw pull roller)

$$Draw = \frac{V_X - V_{REF}}{V_{REF}}$$

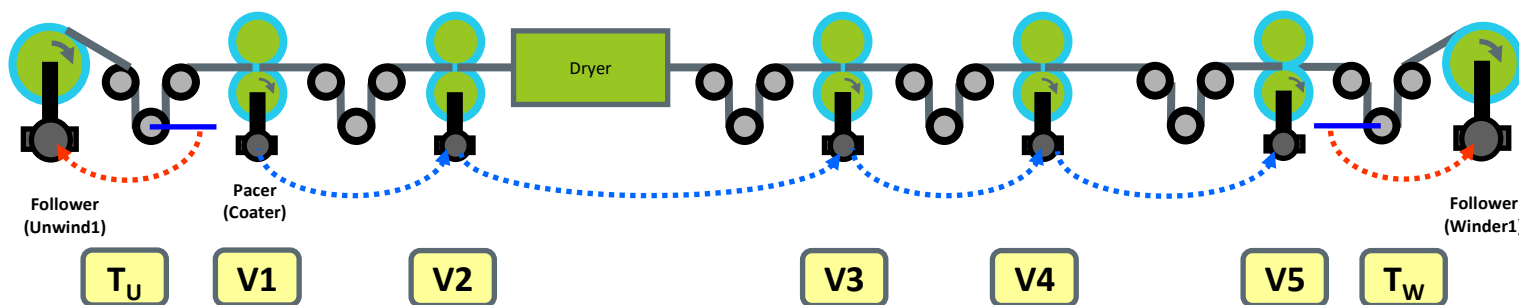
Advantage:

Total draw is easy to understand.

Best reflects what the web responds to.

Cascading Draw Control

Multiple Draw Zone Weblines: Case 2



Cascade: Each draw is set relative to previous driven roller

$$Draw = \frac{V_X - V_{X-1}}{V_{X-1}}$$

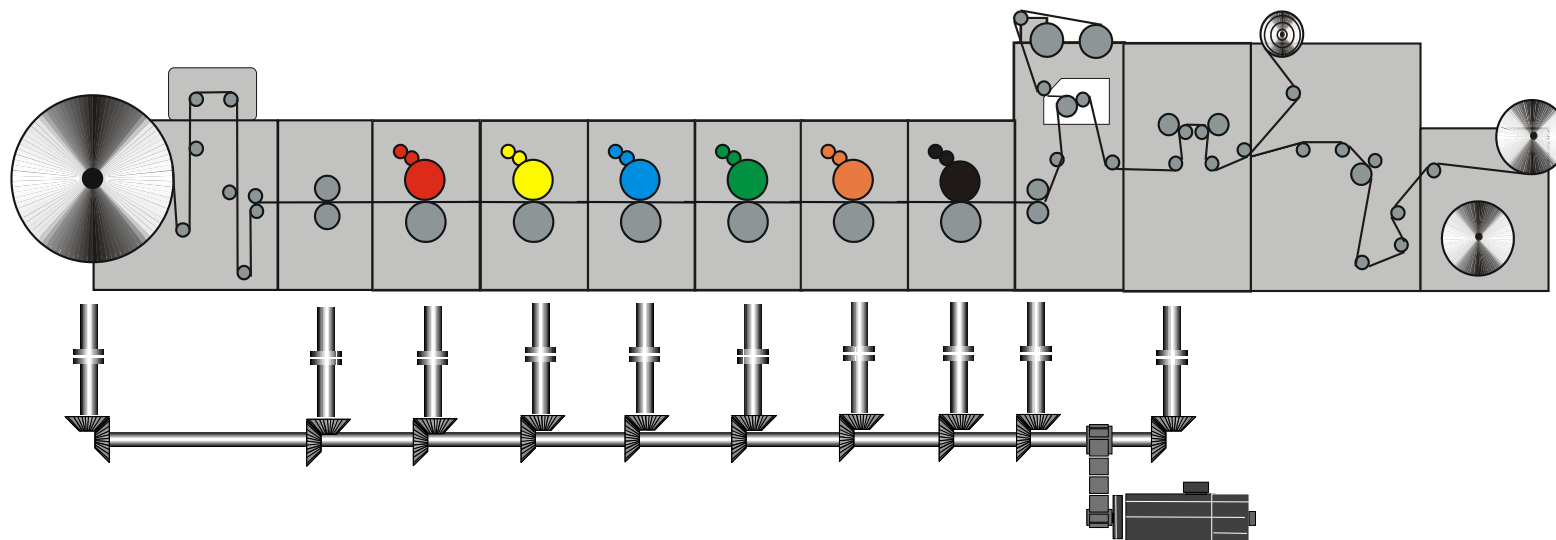
Advantage:

Easier to understand local draw.

Disadvantage:

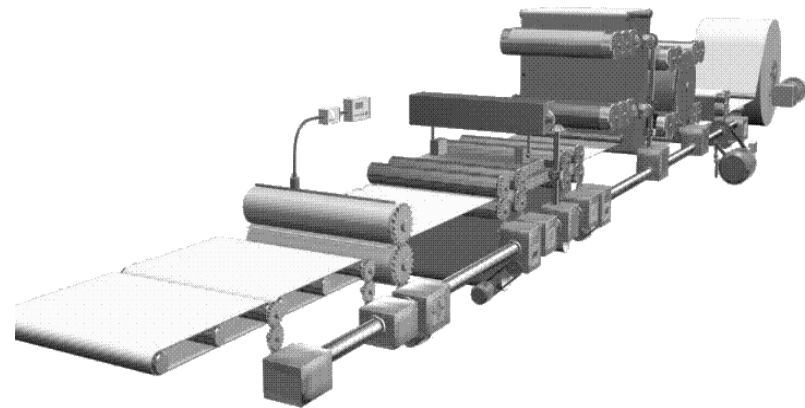
Tends to lead to excessive draw stack up.

Printing Press with Mechanical Line Shaft



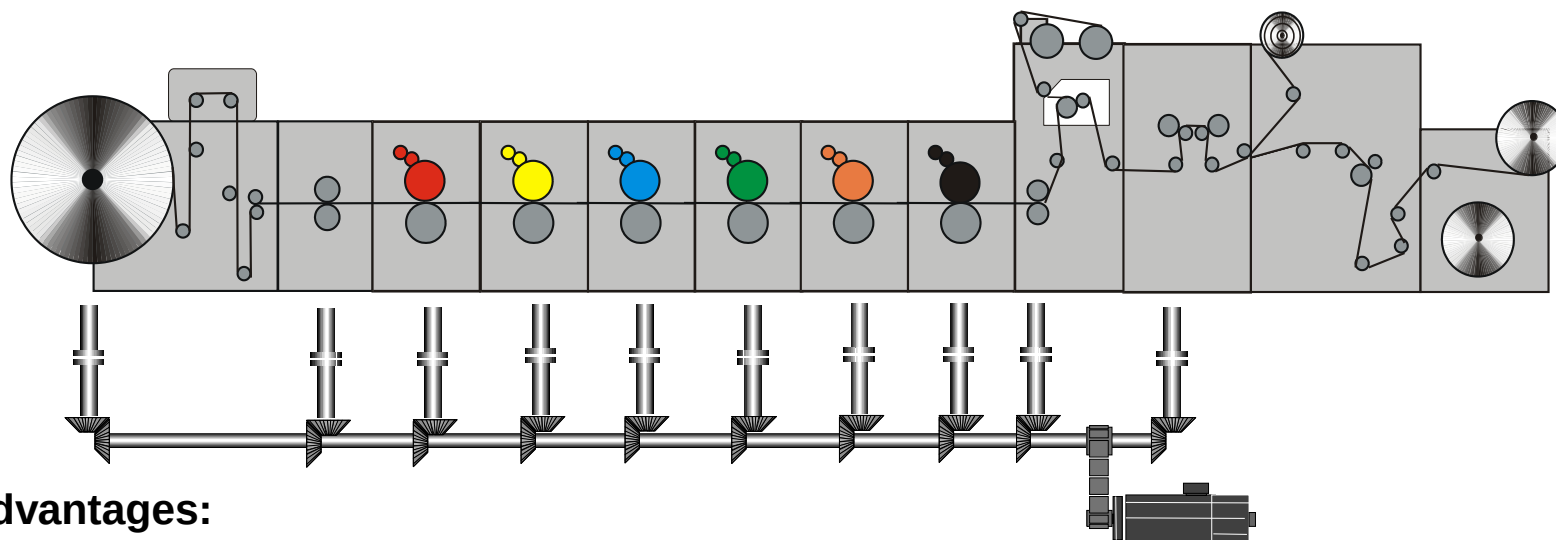
Mechanical press drive systems for:
Label & Package printing, Commercial printing, Newspaper printing and Converting machines...

- One motor powers the entire machine
- Horizontal and vertical line shaft, gears and toothed belts, distribute the drive torque to all machine units.
- The mechanical linkage ensures synchronization of all cylinders and rollers.



Benefits of Electronic Line Shafting - Bosch Rexroth

Mechanical Line Shaft: Advantages and Disadvantages

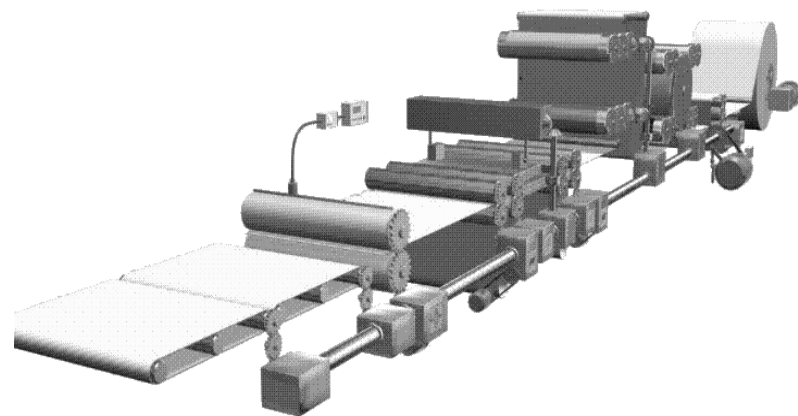


Advantages:

- Mechanical machines are sturdy and reliable.
- Press operators and service staff are familiar with the machine components for routine maintenance and breakdown.

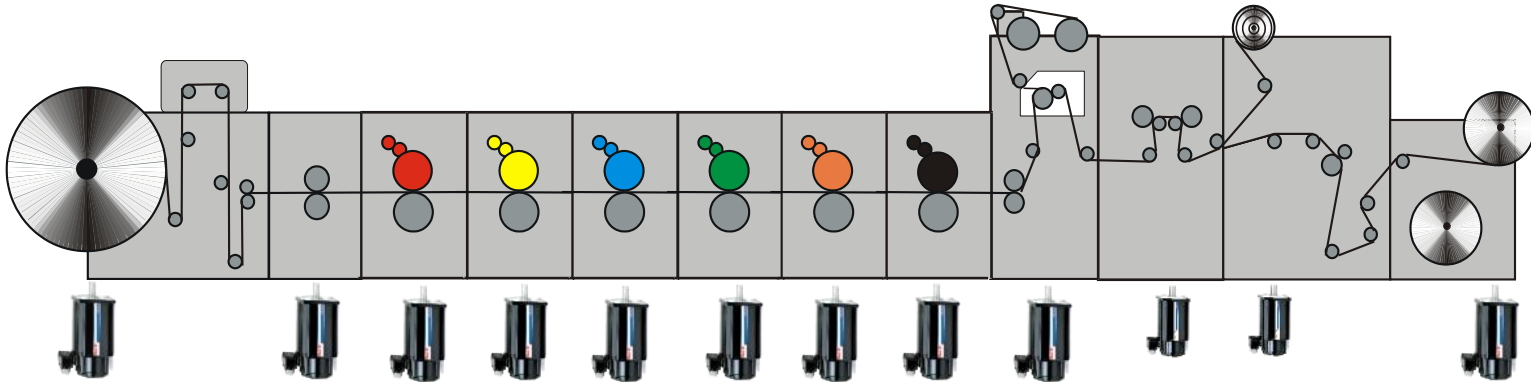
Disadvantages:

- Mechanical machines are not flexible for today's demands of short run, quick changeover and make ready.
- Print quality and accuracy are sometimes not sufficient due to mechanical wear and torque disturbances between units.



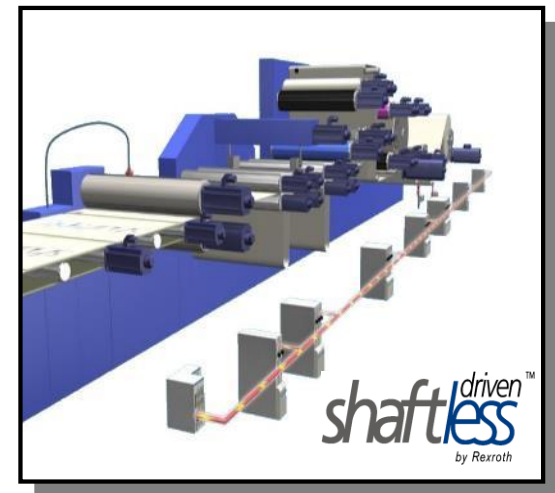
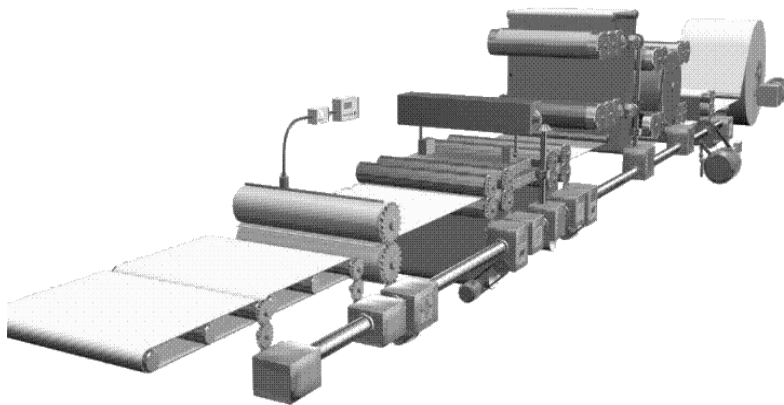
Benefits of Electronic Line Shafting - Bosch Rexroth

Printing Press with Electronic Line Shaft



ELS - Distributed Drive System

The Electronic Line Shaft does not transmit force or torque disturbances between units



Draw Control

What's Needed?

Two speed controlled rollers with either:

Independently controlled motors
—or—

One motor coupled to two rollers via line shaft, gears, chains, or timing belts.

Draw
Control

Where is it used?

MD registration,

S-wrap or multi-roller drive sections,

Shear slitting,

Stretchy, easily yielding, or visco-elastic webs.

Draw Control

Advantages

Disadvantages

Draw Control

Simple.

Draw is proportional to tension increase for elastic webs.

Limits yielding in non-elastic webs.

Limits overspeed if slip occurs.

Small speed errors create can create large tension changes.

Slow response depending on time constant (L/V).

Tension Control

What Variable is Adjusted?

Torque

Relative Speed

Torque
Control

Draw
Control

a.k.a. Speed Ratio Control

Tension Control

Open-Loop Control
No Tension Feedback

Will it have tension
feedback?

Closed-Loop Control
Tension Feedback by
Load Cell or Dancer Roller

Why Tension Control is Challenging (1of2)

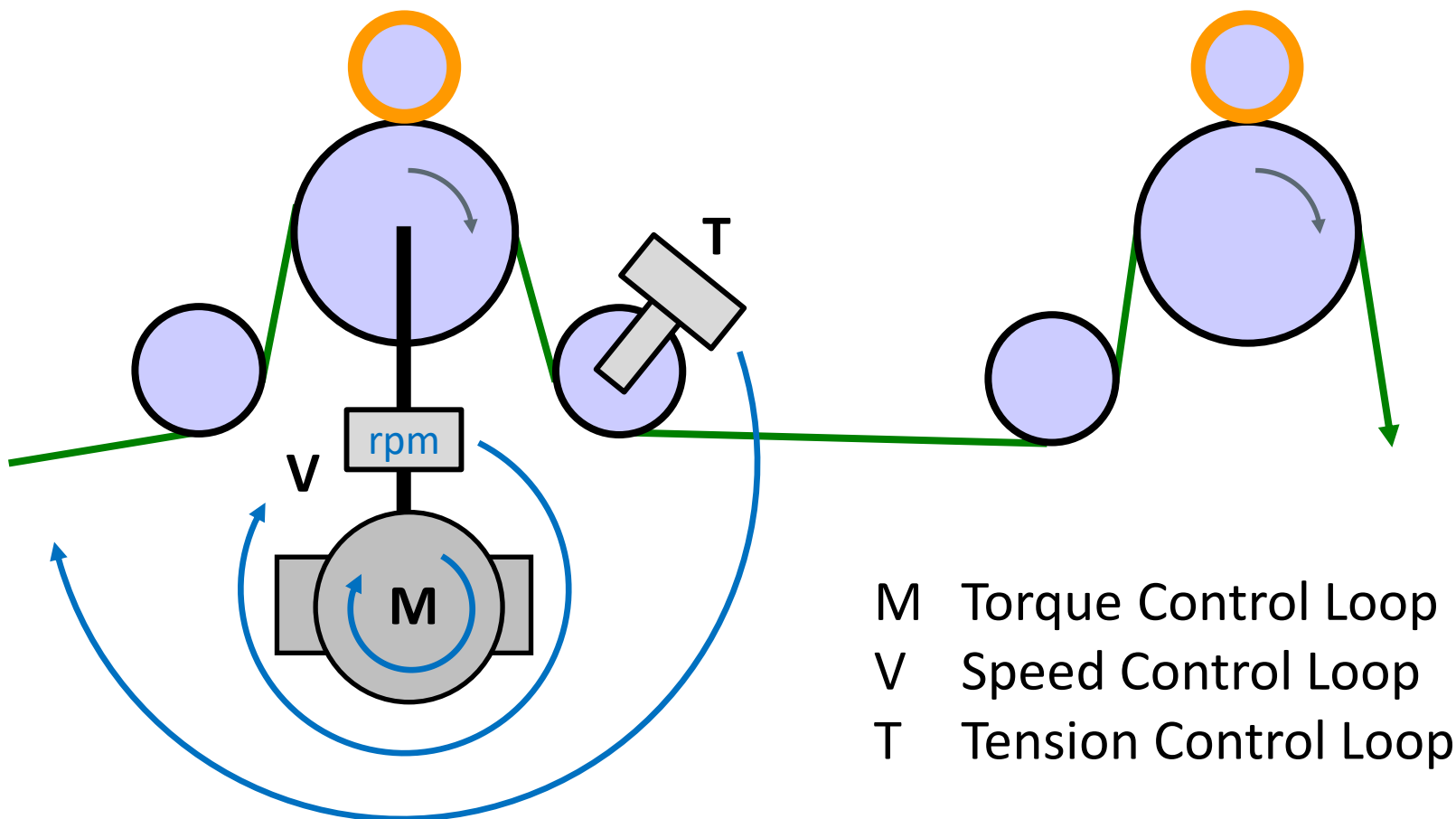
- ▶ Torque to tension relationship is complicated by other torque demands (inertia, process, nipping, transmission losses).
- ▶ Speed control is difficult with changing torque demands.
- ▶ Inertia of winding/unwinding rolls changes with diameter, width, and density.
- ▶ Multi-motor system need to be performance matched.

Why Tension Control is Challenging (2of2)

- ▶ Speed to tension is high gain system (small changes in speed are BIG changes in tension).
- ▶ Speed to tension gain varies with material properties and geometry (thickness, width, and modulus of elasticity).
- ▶ Tension control loops are interdependent with upstream tension changing downstream steady-state.

Loops of Closed Loop Tension Control

Closed-loop tension control required three control loops for torque, speed, and tension with each outer loop slower than the inner loop.



Motor Torque and Speed Control

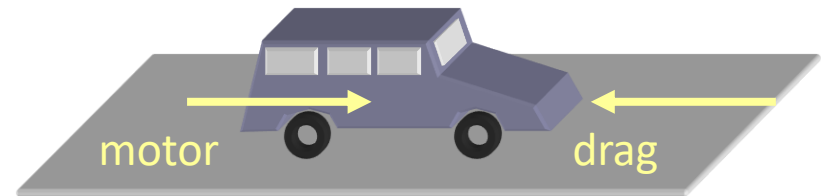
A motor's output is torque (in-lbs). Applied through the rotation of the tires, the torque is converted to force.

(Force=Torque/Radius)

Force = mass x acceleration -or- Force / mass = accel.

Therefore, the result of motor torque is not speed, but acceleration.

If the car has no resistance, the applied torque will continue to accelerate the car to infinite speed.



A car will reach a steady speed only if the motor-created forces are opposed by an equal opposite force, such as air drag.

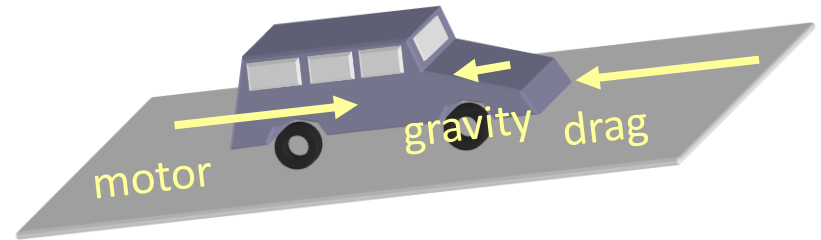
Motor Torque and Speed Control

If the balance of motor vs. drag is upset, for instance by driving up or down a hill, the force imbalance will lead again to acceleration or deceleration.

Therefore, to keep at constant speed, the motor must increase torque output driving up hills and decrease motor torque driving down hills.

In driving your car, either:

- 1) You control this by watching the speedometer and pressing more or less on the gas pedal, or
- 2) The cruise control monitors speed and adjusts the fuel to the motor, as needed.

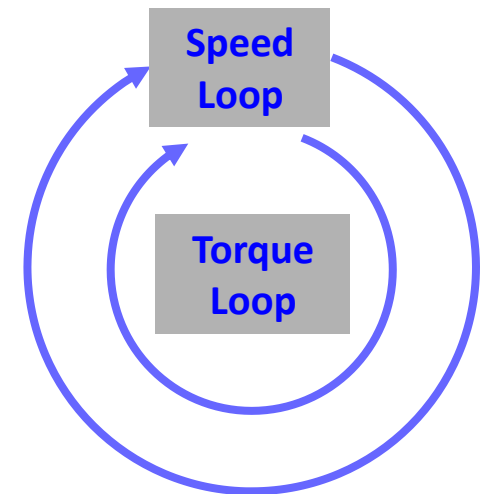
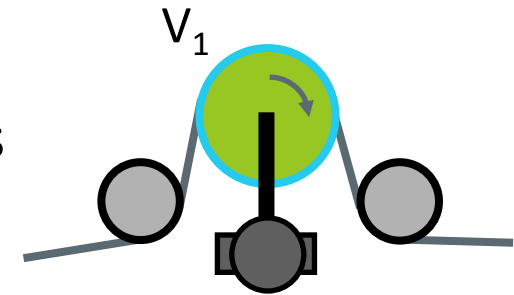


Speed Control

Motor speed control is like a car's cruise control.

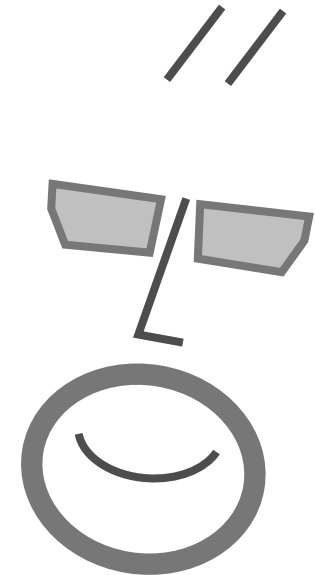
An encoder or tachometer continuously monitors the actual motor speed. If there is an error in the actual speed from the set point, the drive will request more or less torque to reach the desired speed.

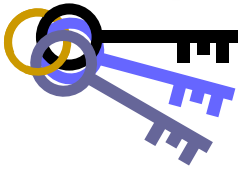
If there is an upset in the force balance opposing the motor torque, such as a change in tension or web break, the motor would accelerate or decelerate unless a speed control loop adjusts motor torque accordingly.



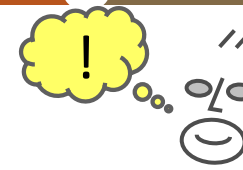
Closed-Loop Tension Control Qs

How does closed-loop tension control work?





KEY CONCEPT



Closed-Loop Tension Control

Closed-loop system designed to keep web tension a desired set point by adjusting the speed or torque of a roller (or roll) in response to the reaction of a tension-sensing roller.

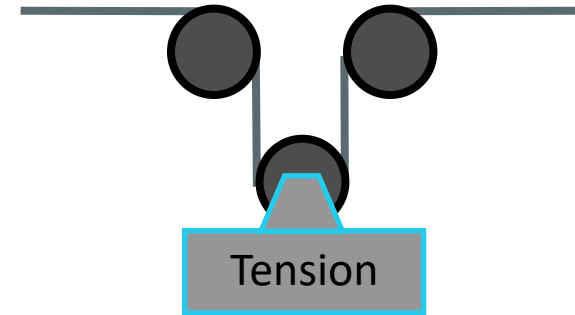
Use: For advance control of a tension zone, a transducer or dancer roller provides a proportional signal of tension variations.

Example: Transducer rollers and speed-controlled driven rollers are the most common form of closed-loop tension control.

Feedback Options

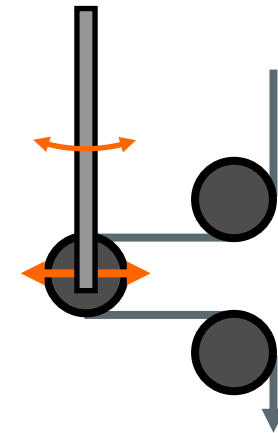
Load Cell Rollers

A calibrated force measuring system.



Dancer Rollers

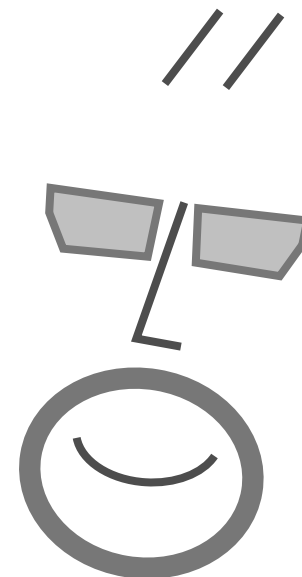
A force-loaded position feedback device used to respond to speed errors of consecutive pull rollers.



Closed-Loop Tension Control Qs

How does a load cell roller work?

How is tension measured?

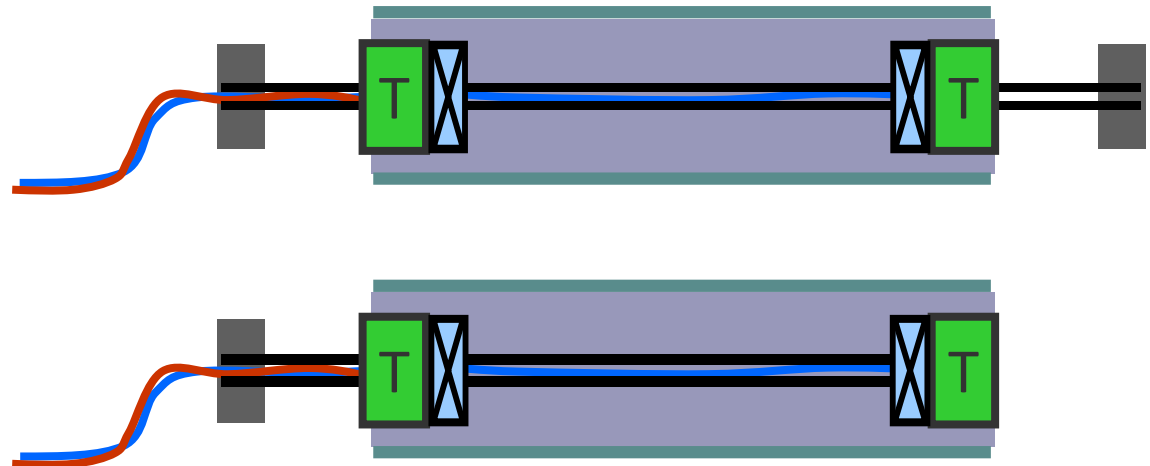


Load Cell / Tension / Transducer Roller Designs

Mount Style
Transducers



Internal
Transducers

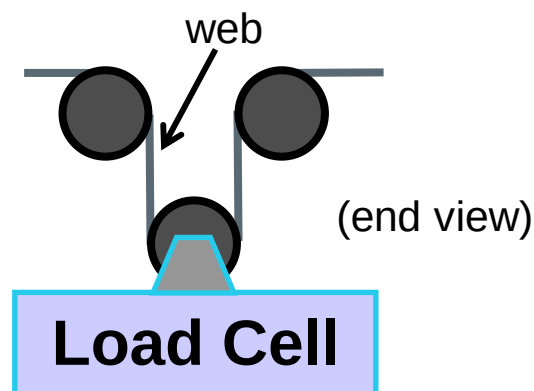


Measuring Tension

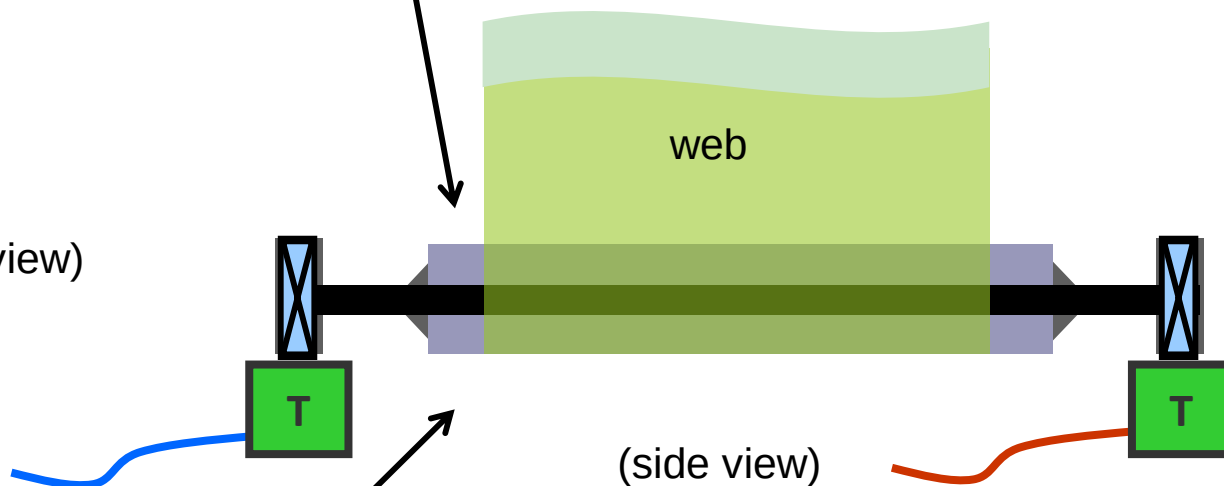
Tension force is measured using load cell rollers.

A load cell roller has two load cells, one at each end, which are summed.

A load cell measures force



Load Cell Roller



Tension Transducer Rollers

LVDTs have too much deflection for use in tension rollers.

Strain Gauge: Foil



Strain gauge-based tension rollers are the most common, having less deflection and better accuracy.

LVDT

(Linear Variable Differential Transducers)



Image courtesy of Comptrol

Strain Gauge: Semiconductor



Images courtesy of Cleveland Machinery



Images courtesy of Dover Flexo

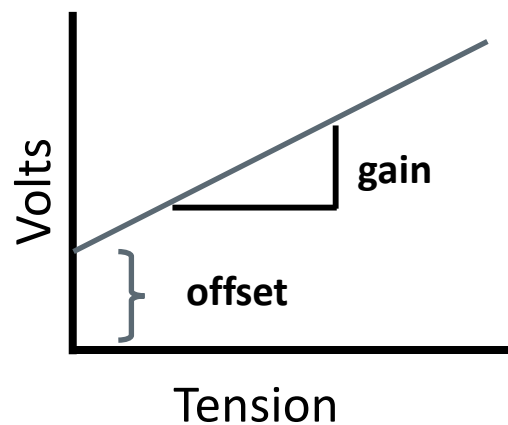
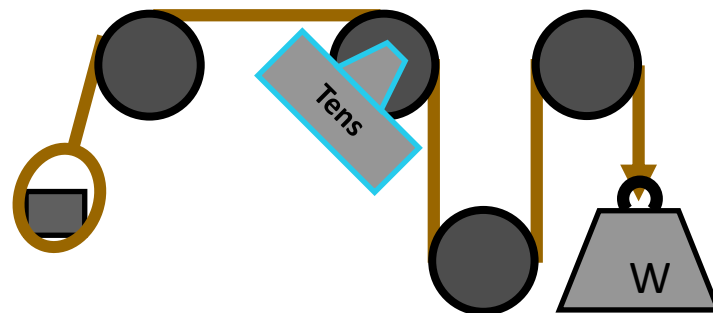
Pressductor®



Images courtesy of ABB

Calibrate a Load Cell Roller 1

1. Adjust the zero offset with no load.
2. Fix a rope at one end.
3. Feed the rope through the web path that defines the load cell roller's wrap angle.
4. Hang a weight (or other method to exert a known load) on the rope.
5. Adjust the system gain.
6. Re-check the zero offset, repeat 2-5 if needed.



Roller Load Geometry

Total load on the roller bearings and tension measuring load cells is the sum of weight and tension vectors.

SLIM CELL SIZING CALCULATION

$$MWF^* = T \times K \times \sin (A/2) \pm (W/2) \times \sin (B)^{**}$$

MWF = Maximum Working Force (lbs.)

T = Maximum Total Tension (lbs.)

K = Transient Tension Overload Factor
(normally 1.5 for most applications)

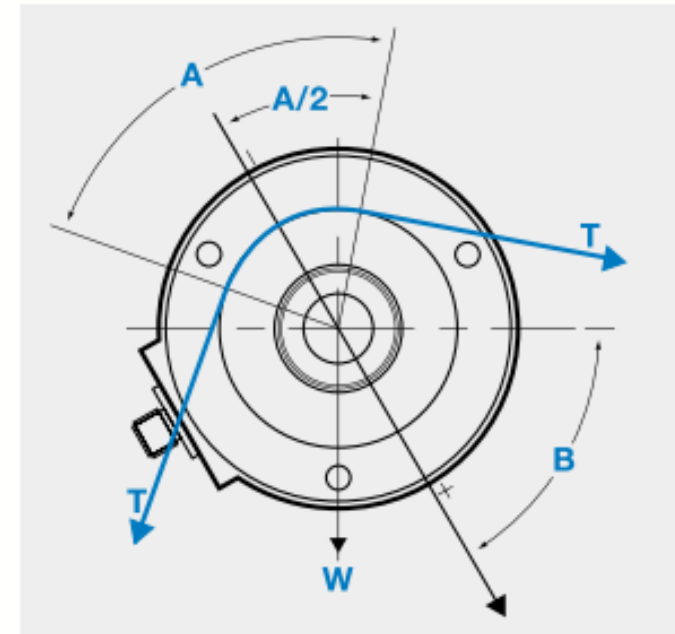
A = Wrap Angle (degrees)

B = Angle of Tension Force (degrees)

W = Weight of Roller

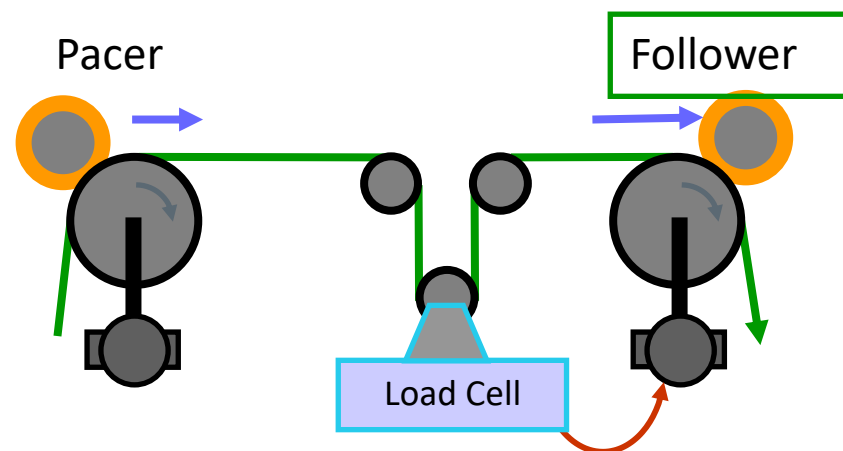
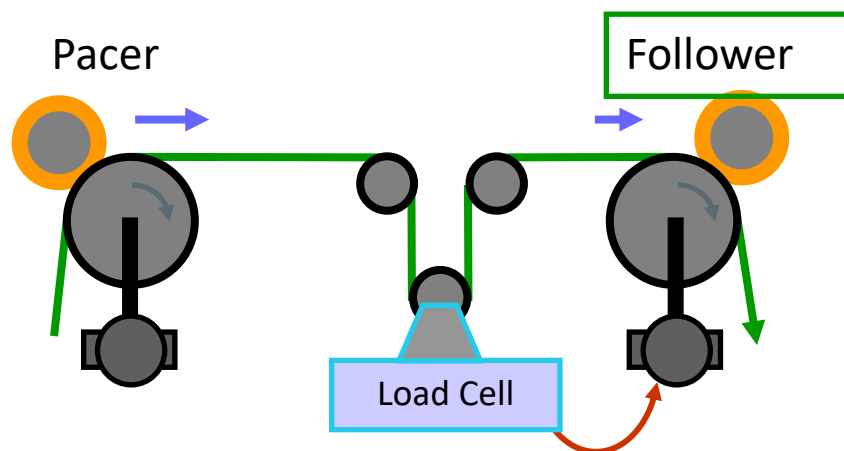
* The MWF calculation defines the force on each individual load cell.

** If Angle B is below horizontal use + in calculation. If Angle B is above horizontal use - in calculation.



This equation calculate the force on each of two load cells on either end of a roller.
Total force is $MWF \times 2$.

Downstream Follower & Load Cells



If the load cell measures

$$T_{\text{ACTUAL}} > T_{\text{SET POINT}}$$

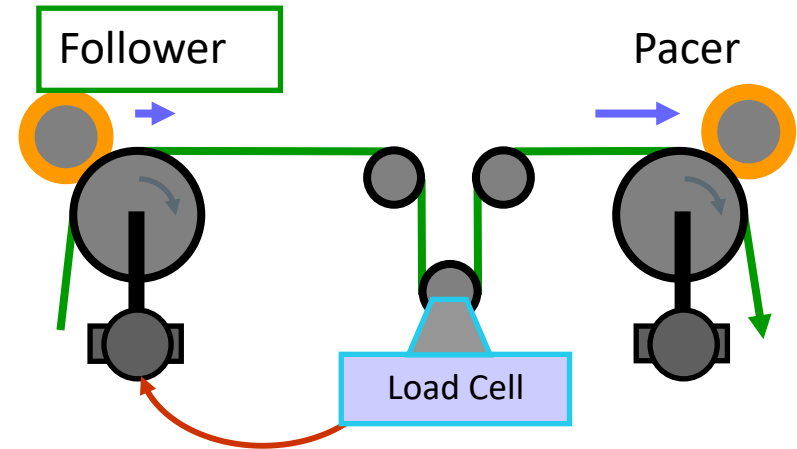
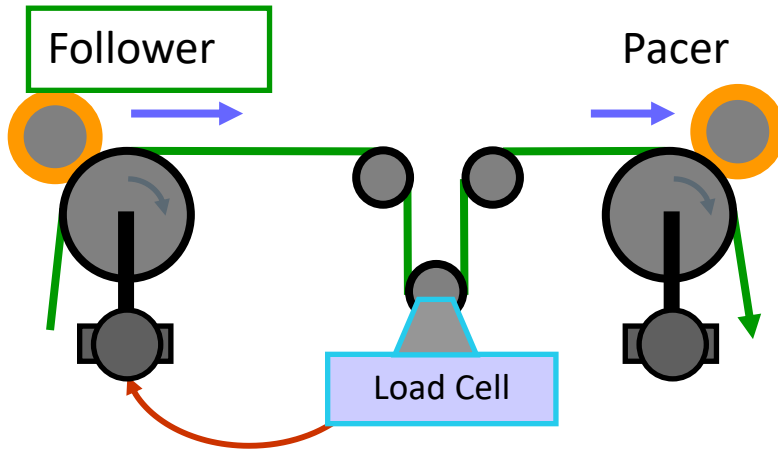
the tension control loop will reduce V_F or M_F to bring the tension back to the set point.

If the load cell measures

$$T_{\text{ACTUAL}} < T_{\text{SET POINT}}$$

the tension control loop will increase V_F or M_F to bring the tension back to the set point.

Upstream Follower & Load Cells



If the load cell measures

$$T_{\text{ACTUAL}} > T_{\text{SET POINT}}$$

the tension control loop will increase V_F or reduce M_F to get tension back to the set point.

If the load cell measures

$$T_{\text{ACTUAL}} < T_{\text{SET POINT}}$$

the tension control loop will reduce V_F or increase M_F to get tension back to the set point.

Load Cells Filtering

Very important:

For tension feedback and control, use the unfiltered output.

Many load cell amplifiers have two outputs:

- a filtered output suitable for display purposes,
- an unfiltered output to be used for control purposes.

If a load cell roller output is used for control, it is vital that the filtering on that output is turned off. Using the filtered output will probably cause control system instability!

Load Cell Deflection

Another important aspect of load cells used for tension measurement is their stiffness. All load cells deflect under a force load.

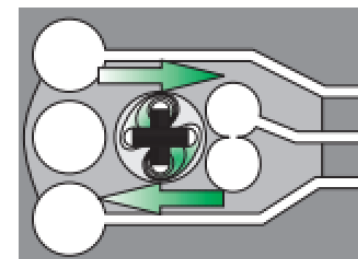
When a cambered web is run over a load cell with insufficient stiffness, the roller becomes misaligned.

Recommendations for alignment sensitive products and processes:

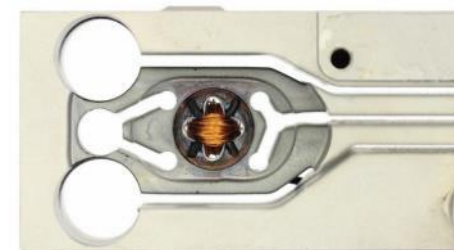
- ABB Pressductor (low deflection load cells)
- Specify load cells 2x larger than your need.



Pressductor™ Details



Pressductor® Technology:
Mechanical force alters magnetic field.



Mini Series PillowBlock Load Cells rely on electromagnetic changes in the transducer, not on physical movement, to sense fluctuations in web tension. The Pressductor Technology operating principle provides exceptional improvements in load cell performance characteristics, including reliability (notably absence of drift), durability, repeatability, and wider measurement range.

Machined from a solid block of steel, Mini Series load cells are rugged and stiff, affording high overload protection as well as an extended measurement range above the nominal load. And they won't contribute to machine vibration, even at high speeds.

Since the transducer action – the magnetic flux – takes place inside a steel core, environmental factors like dirt or fluids can't degrade performance and reliability. These stainless steel load cells don't require any physical seals.

Images and text courtesy of ABB

Load Cell Deflection - Stiff

ABB Pressductor:
2-5 mils (50-125 μm)

Properties

Nominal load (rated capacity)					
	kN	0.1	0.2	0.5	1.0
	Lbs.	22	45	112	225
Extended load ¹⁾					
	kN	0.15	0.3	0.75	1.5
	Lbs.	33	67	169	337
Permitted load					
Transverse direction					
(vertical) h=135 mm					
	kN	0.3	0.6	1.5	3.0
	Lbs.	67	135	337	674
Overload capacity ²⁾					
Measurement direction					
(horizontal)					
	kN	0.3	0.6	1.5	3.0
	Lbs.	67	135	337	674
Transverse direction					
(vertical) h=135 mm					
	kN	0.5	1.0	2.5	5.0
	Lbs.	112	225	562	1125
Deflection ³⁾					
	mm	0.05	0.05	0.07	0.13
	1/1000 in.	2.0	2.0	2.8	5.0
Spring constant					
	kN/mm	2.0	4.0	7.0	8.0
	1000 Lbs./inch	11.3	22.4	39.7	44.6

Magpowr TSU:
9 mils (225 microns)

Specifications

Gage Resistance	350 Ω
Excitation Voltage	10 VDC nominal
Output Signal	21 mVDC nominal per load cell at full load rating
Operating Temperature	-30° to 95° C
Temperature Effect on Zero	.02% of rating per °C
Combined Non-linearity and Hysteresis	.5% of full scale maximum
Repeatability	.2% of full scale maximum
Overload Stops	Internal at 105% to 150% of full load rating
Deflection at Full Load	.009 inch (23 mm)

Load Cell Deflection – Less Stiff

Magpowr TSU:
9 mils (225 microns)

Specifications

Gage Resistance	350 Ω
Excitation Voltage	10 VDC nominal
Output Signal	21 mVDC nominal per load cell at full load rating
Operating Temperature	-30° to 95° C
Temperature Effect on Zero	.02% of rating per °C
Combined Non-linearity and Hysteresis	.5% of full scale maximum
Repeatability	.2% of full scale maximum
Overload Stops	Internal at 105% to 150% of full load rating
Deflection at Full Load	.009 inch (23 mm) ←

Vishay:
3-20 mils (75-500 μm)

SPECIFICATIONS

Performance (% Rated Output)

Rated Capacity	20, 50, 100, 200, 500 lb (9.1, 22.7, 45.4, 90.1, 227kg)
Rated Output (RO), nom.	2.000 mV/V $\pm$ 0.25%
Nonlinearity	0.25%
Hysteresis	0.10%
Repeatability	0.02% RO
Creep (20 Minutes)	0.05% RO
Zero Balance	5.0% RO

Electrical

Input Resistance	350 ohms $\pm$ 3 ohms
Output Resistance	350 ohms $\pm$ 3 ohms
Insulation Resistance	< 5000 megohms
Recommended Excitation	10Vac/dc

Temperature

Operating Range	-40 to 220°F (-40 to 105°C)
Compensated Range	+30 to 130°F (-1 to 54°C)
Effect on Zero Balance	0.0050% RO/°F
Effect on Rated Output	0.0050% of load/°F

Overload Rating

Safe Load	200% Rated Capacity
Ultimate Load	500% Rated Capacity
Safe Side Load	100% Rated Capacity

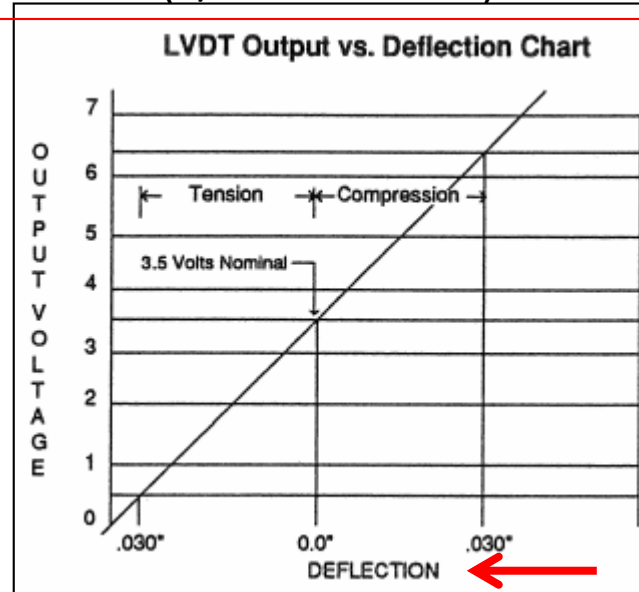
Materials

All Capacities	All Stainless Steel
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Deflection at Rated Capacity

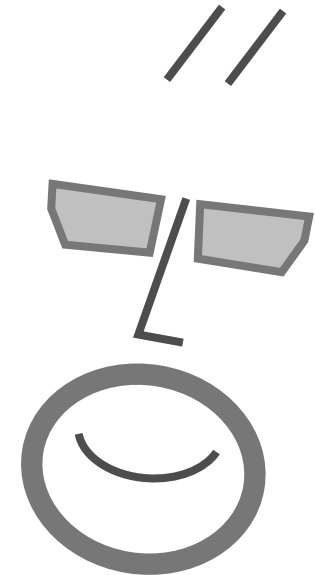
All Capacities	0.003 - 0.020 inches ←
----------------	------------------------

Comptrol:
+/- 30 mils (+/- 750 microns) !



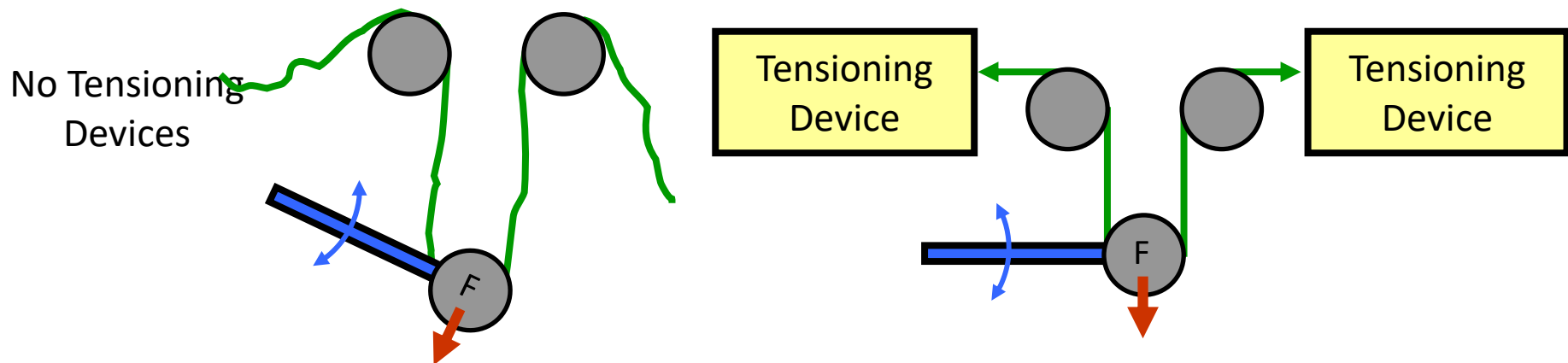
Closed-Loop Tension Control Qs

What is a dancer roller? How does it work?



Tensioning: It Takes Two

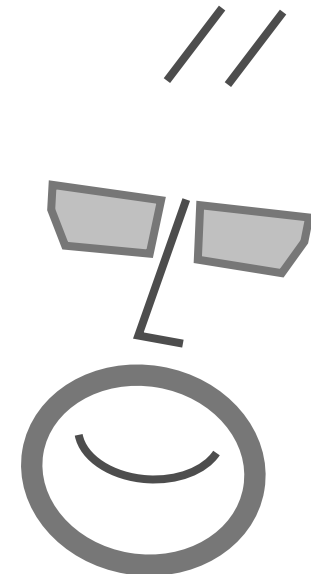
A single force can't create tension.
It needs another force to oppose it.
The gravity or pneumatic load of a dancer
roller, does not create tension by itself.



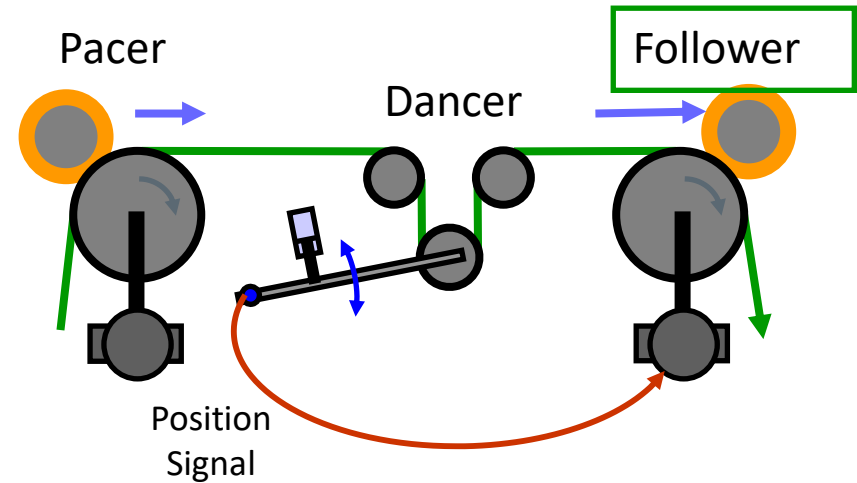
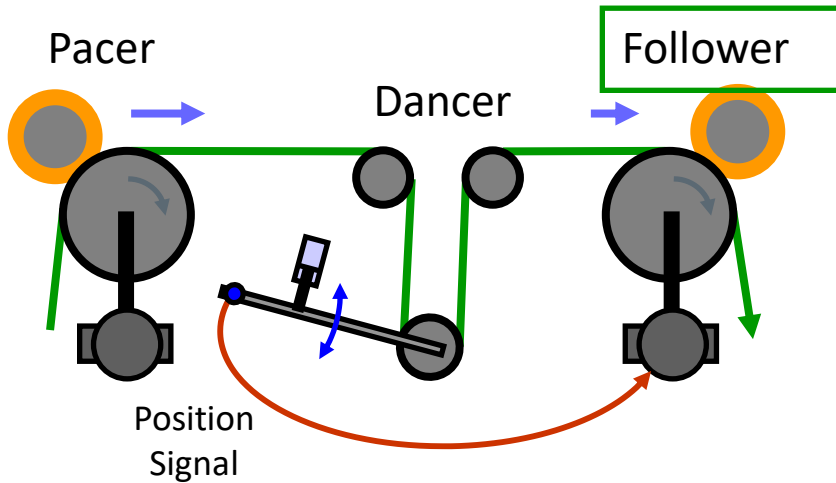
Closed-Loop Tension Control Qs

What is a dancer roller? How does it work?

A dancer loads the web to a desired tension set point. The dancer moves in response to differences in upstream to downstream web speed. The dancer accumulates web if the V_{in} is too fast relative to V_{out} and dispenses web if the V_{in} is too slow relative to V_{out} .



Downstream Follower & Dancer



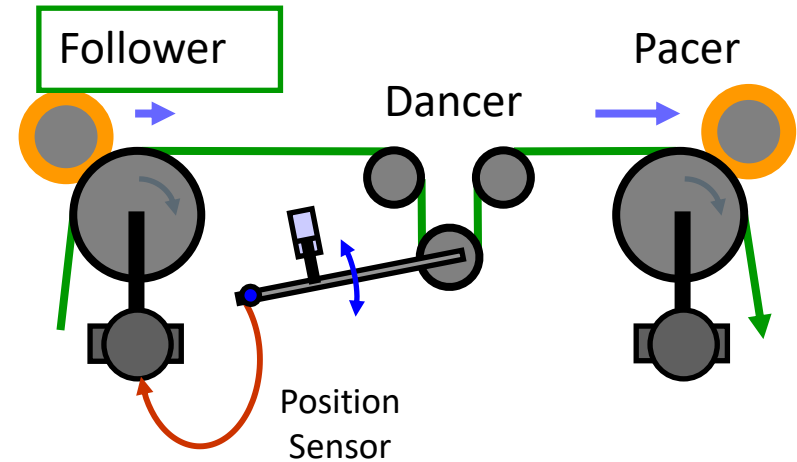
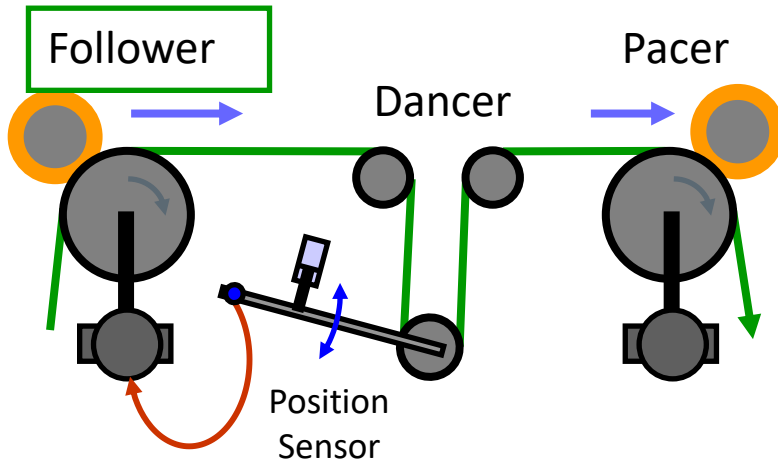
If V_F is less than V_P ,
the dancer accumulates web.

The tension control loop will
increase V_F to raise the dancer back
to the null position.

If V_F is greater than V_P ,
the dancer dispenses web.

The tension control loop will
decrease V_F to lower the dancer
back to the null position.

Upstream Follower & Dancer



If V_F is greater than V_P ,
the dancer accumulates web.
The tension control loop will
decrease V_F to raise the dancer
back to the null position.

If V_F is less than V_P ,
the dancer dispenses web.
The tension control loop will
increase V_F to lower the dancer
back to the null position.

Calculate Dancer Tensioning

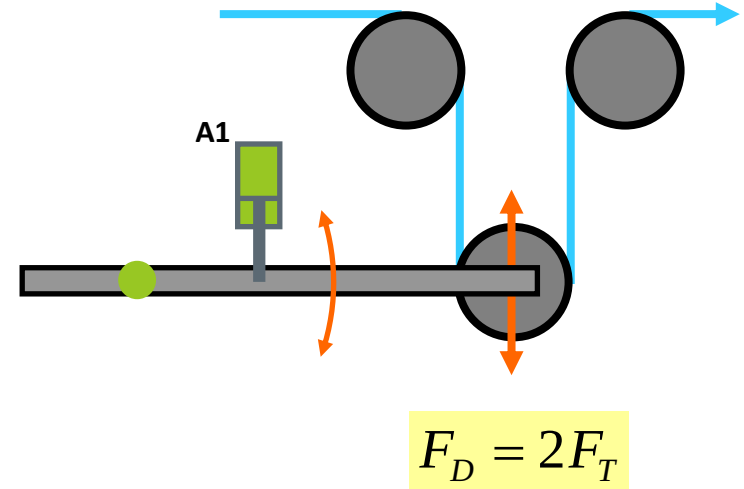
Dancer controlled tension can be estimated from the system elements.

1. Measure all pivot lengths.
2. Check air cylinder bore diameters.
3. Estimate the dancer roller weight.

$$\Sigma M = 0$$

$$\Sigma M = W_D L_D + P_{CYL} A_{CYL} L_{CYL} - 2F_T L_D$$

$$F_T = \frac{1}{2} \left[W_D + P_{CYL} A_{CYL} \left(\frac{L_{CYL}}{L_D} \right) \right]$$

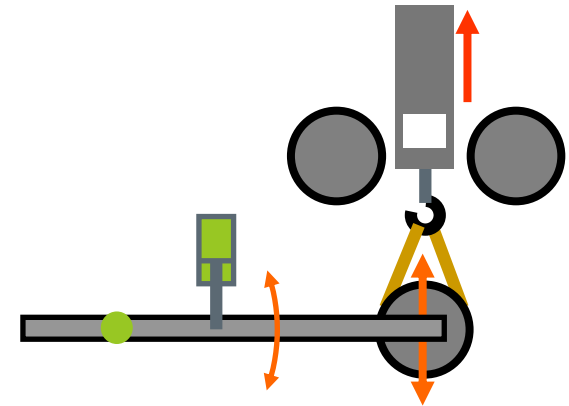


Measure Dancer Force

Measure the dancer system forces with a rope and a force gauge.

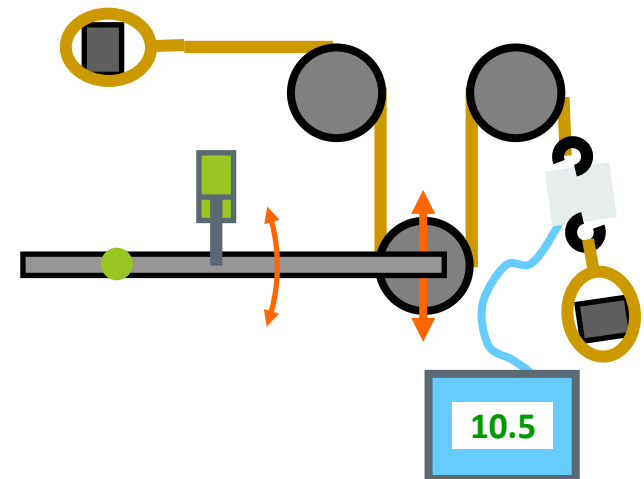
Method 1:

1. Wrap a rope around the dancer.
2. Attach the rope to a handheld force gauge.
3. Measure the force at various input pressures.
4. Convert to web tension by dividing by two.



Method 2:

1. Fix a rope through the web thread path, around the dancer, to another fixed point, with a load cell in-line.
2. Measure the rope (web) tension with no air pressure and at various pressures in the range of normal web tension.



Dancer Angle Measurement

- ▶ Most older systems use an encoder or rotational potentiometer to measure dancer position or angle.
- ▶ A new options is a precision inclinometer.



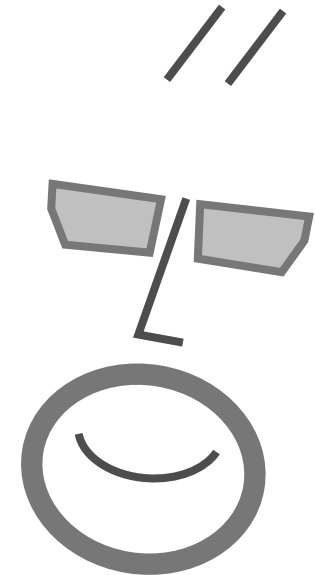
DPS Dancer Position Sensor /
Inclinometer



Image courtesy of Montalvo

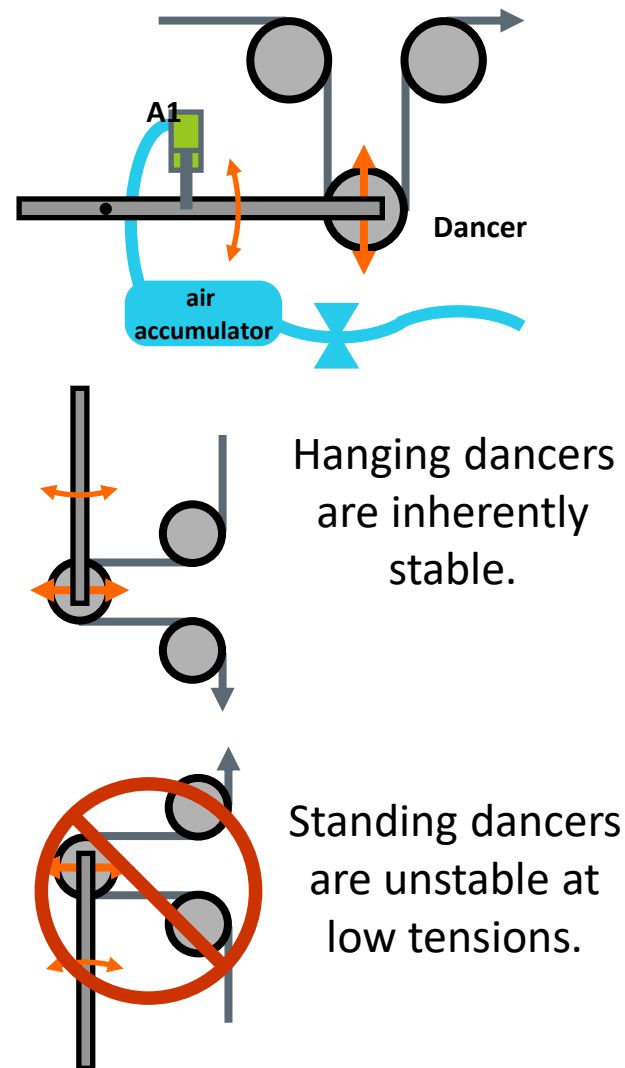
Closed-Loop Tension Control Qs

What makes a good dancer
roller system?



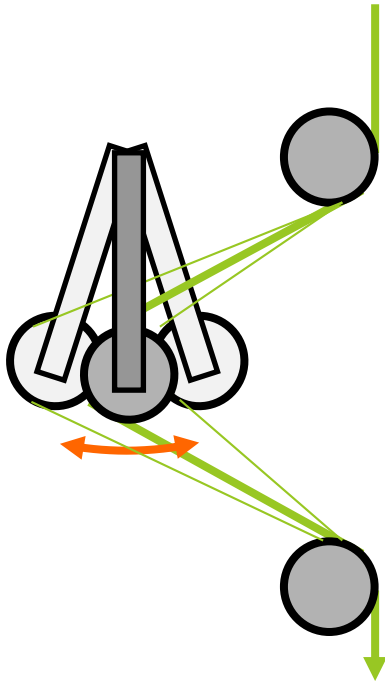
Dancer Design

- ✓ Minimize friction and inertia.
- ✓ Don't counterbalance roller with dead weight.
- ✓ Reduce pneumatic system hysteresis with air accumulators, large piping, and high volume flow controllers.
- ✓ Default wrap angle should be 180°.
- ✓ Magnify dancer motion with geared up coupling (or use precision inclinometer).
- ✓ Pivoting dancers are preferred to linearly sliding dancers.



Dancer Wrap Angle

The default design of dancer roller is 180-degree wrap angle controlled by the placement of the entry and exit rollers. The dancer shown below could be criticized for a wrap angle that changes as it accumulates and dispenses, creating tension variations. However, a well-controlled dancer-motor system will have only small dancer motion, limiting the wrap angle and tension variations. Inertia and friction are typically larger source of tension variation than wrap angle.

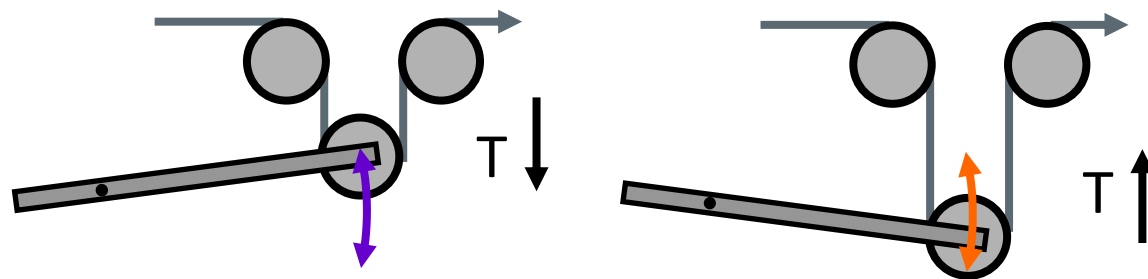


Courtesy of Comexi Group.

Dancer Hysteresis (from Friction and Inertia)

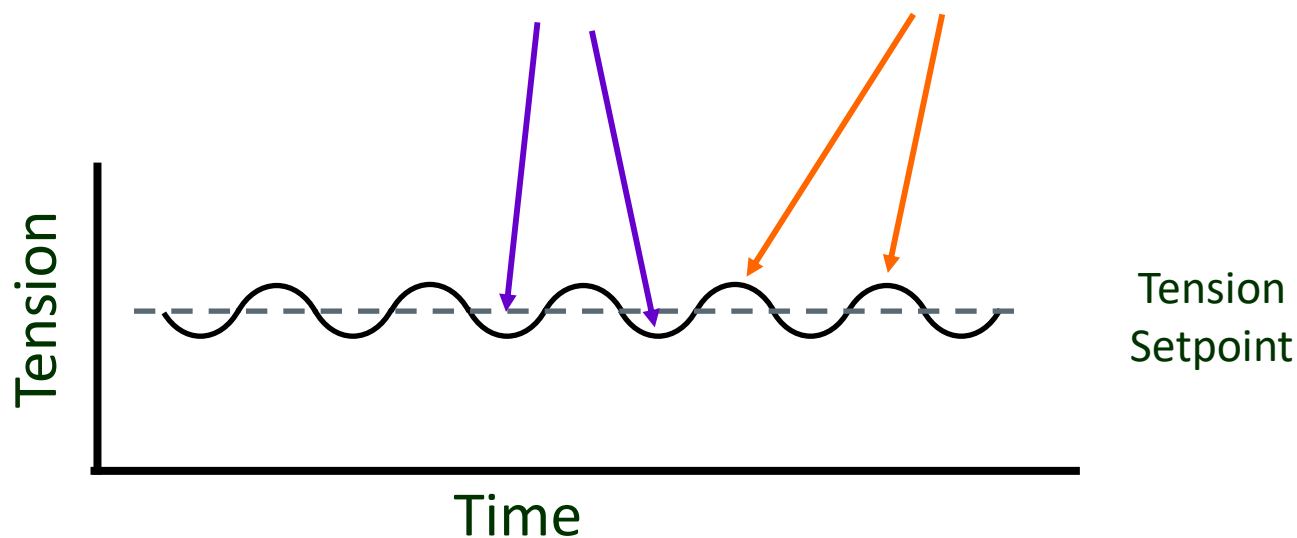
Dancers cause tension variations.

A dancer's friction, inertia, and pneumatics create hysteresis in dancer load.



Tension decreases while accumulating.

Tension increases while dispensing.



Dancer Roller Low Drag Air Cylinders

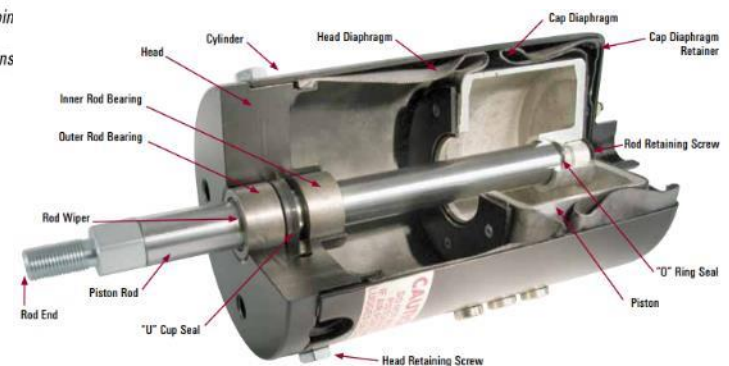
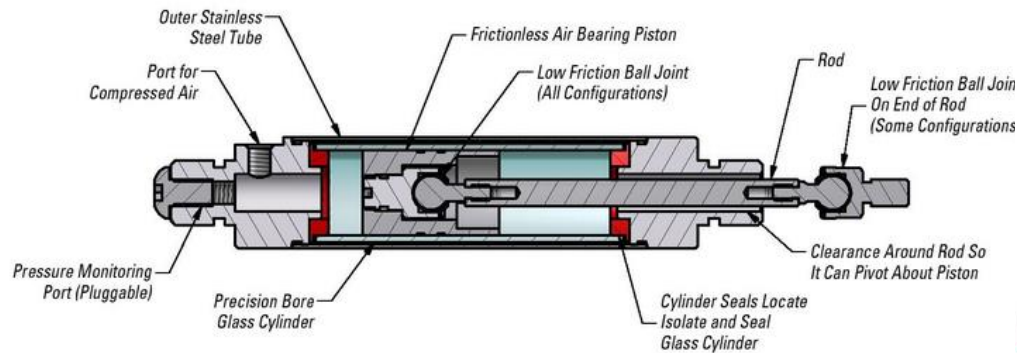
Reduce dancer friction with low drag air cylinders.



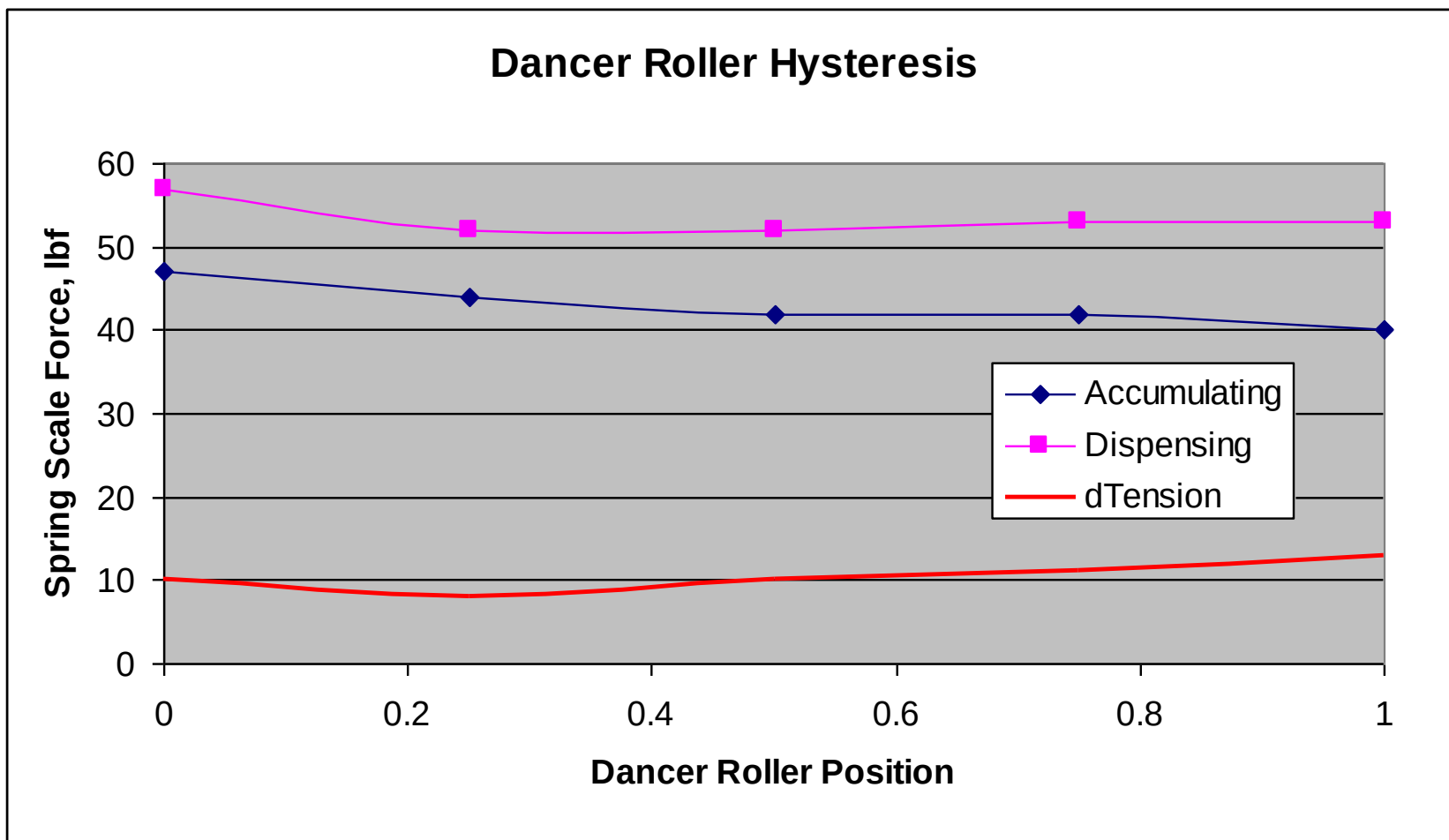
www.airpot.com



www.marshbellofram.com



Dancer Hysteresis



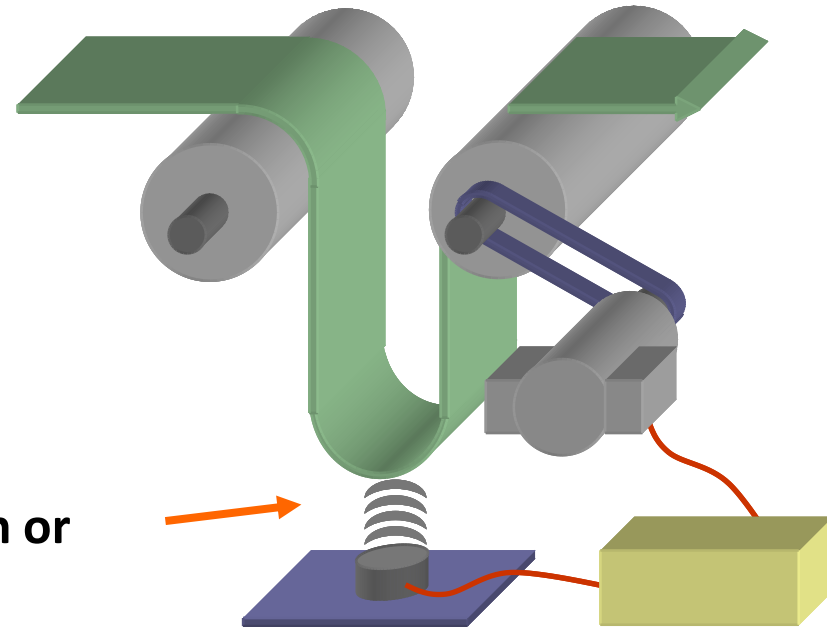
Droop Dancer

Droop control works similar to a vacuum dancer.

Tension is proportional to web weight, only.

$$T = \frac{\rho t w L}{2}$$

- Web position is sensed with a light curtain or ultrasonic sensor.
- **Hysteresis is from web and air inertia only.**
- Potential problem: wrinkles and lateral web shifting required high web stiffness



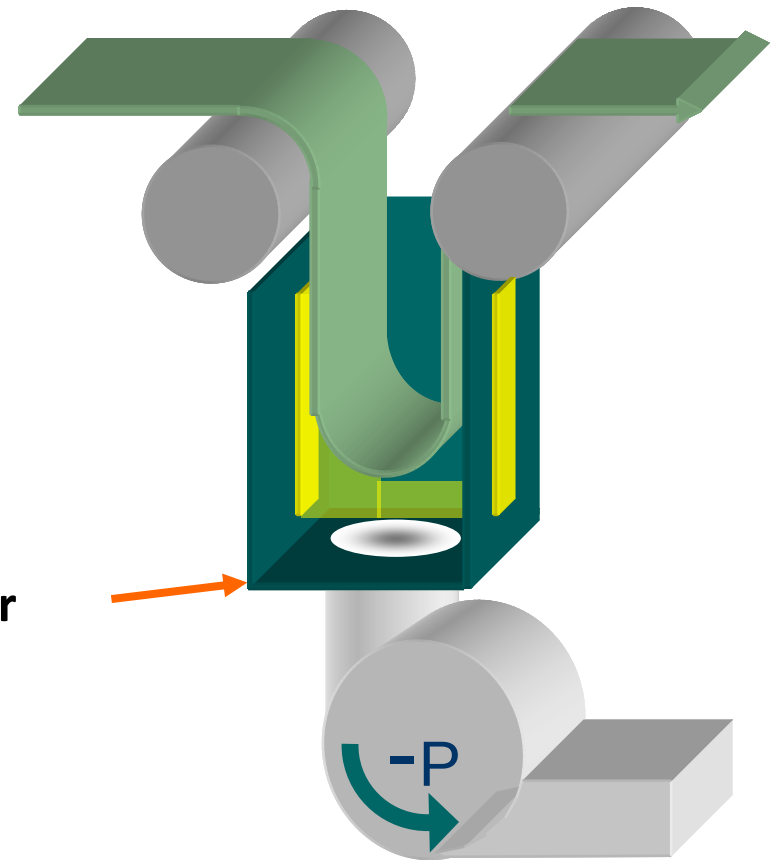
Vacuum Dancer

A pressure drop is pulled across a loop of web.

Tension is proportional to web weight + the pressure drop and area.

$$T = \frac{\Delta PA + \rho twL}{2}$$

- Web position is sensed with a light curtain or ultrasonic sensor.
- **Hysteresis is from web and air inertia only.**
- Potential problem: wrinkles and lateral web shifting required high web stiffness

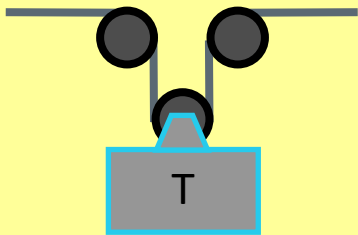


Advantages / Disadvantages

Advantages

Disadvantages

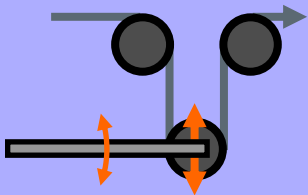
Transducers



Actually measures tension
Often small wrap angles and one-sided contact
Fast and accurate
Low maintenance

Small accumulation, web path changes create large tension changes
Poor response to high frequency upsets

Dancers



Accumulate & dispense web
Minimal tension change with web path changes.
Compensates for poor motor control.

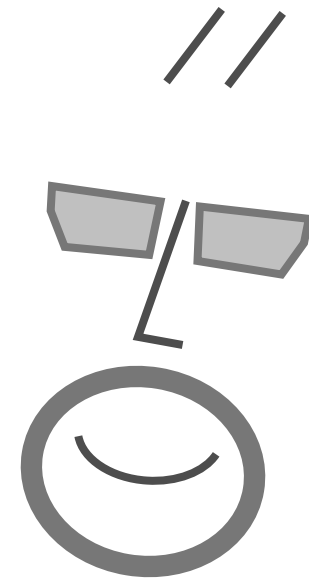
Doesn't measure tension
Large wrap angle + difficult to align = wrinkle sensitive
Creates tension variation from friction and inertia
Poor response at high or resonant frequencies
Higher maintenance

Tension Feedback Advice

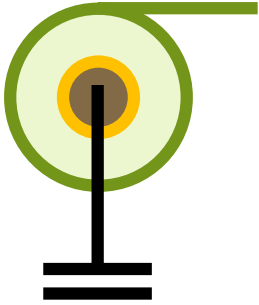
Dancers are a mechanical system devised to compensate for poor controls.

Dancer rollers are overused.

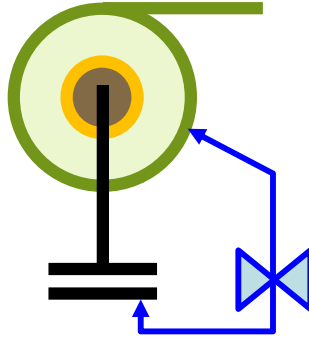
In most cases, today's motors and control systems don't need dancers, but many suppliers and converters are slow to change.



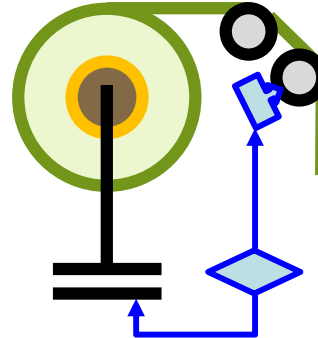
Unwinding Options



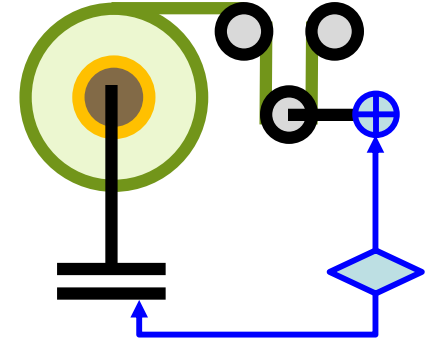
Open-Loop
Brake



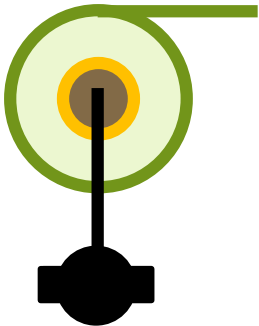
Brake with
Diameter Feedback



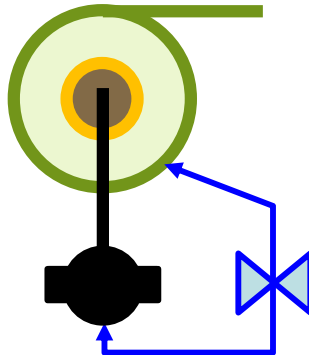
Brake with
Load Cell Feedback



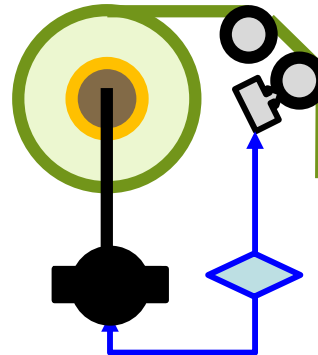
Brake with
Dancer Feedback



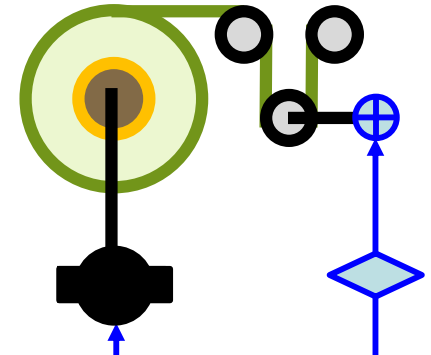
Open-Loop
Torque Motor



Motor with
Diameter Feedback

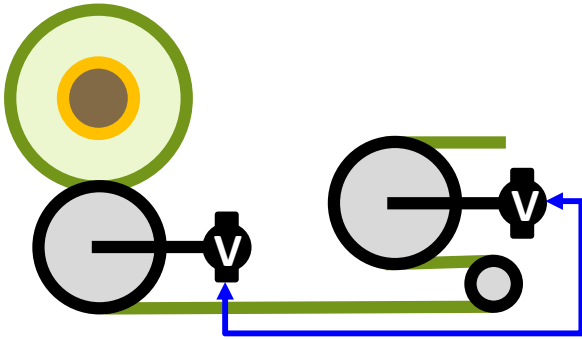


Motor with
Load Cell Feedback



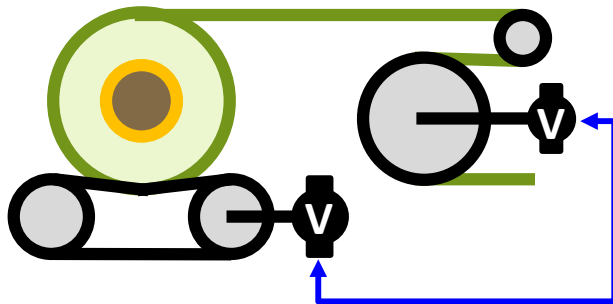
Motor with
Dancer Feedback

Unwinding Options

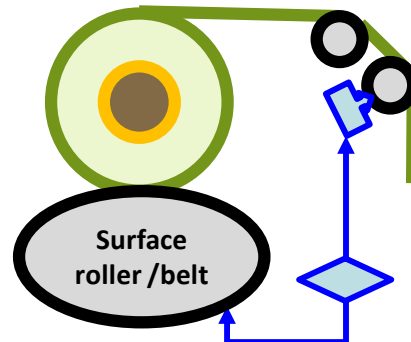


Surface Roller
Driven Unwind

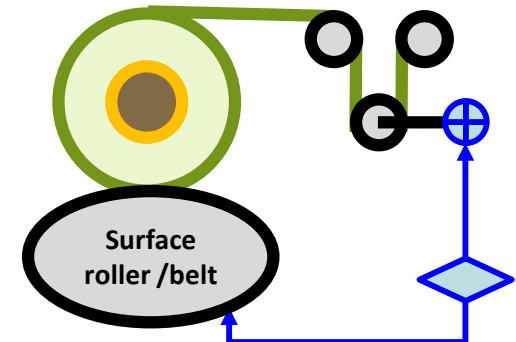
Surface driven unwinds are typically speed ratio controlled, but could be closed-loop tension control with load cell or dancer feedback.



Surface Belt
Driven Unwind

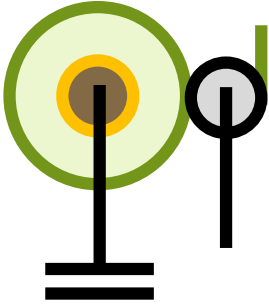


Brake with
Load Cell Feedback



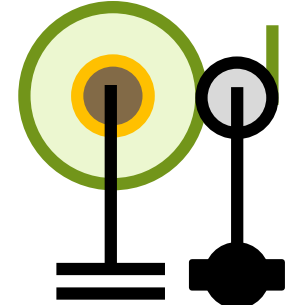
Brake with
Dancer Feedback

Unwinding Peel Roller Options

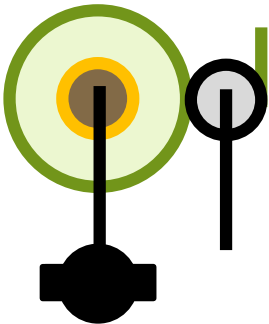


Braked Unwind with
Idling Peel Roller

Peel roller are used to reduce web path variations from peel force variations (for unlinered adhesive coated webs or other products with side A to B bonding forces).

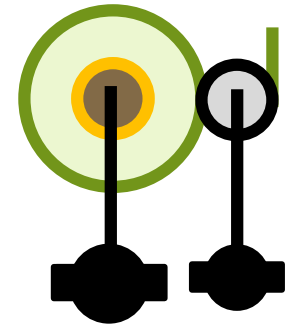


Braked Unwind with
Driven Peel Roller



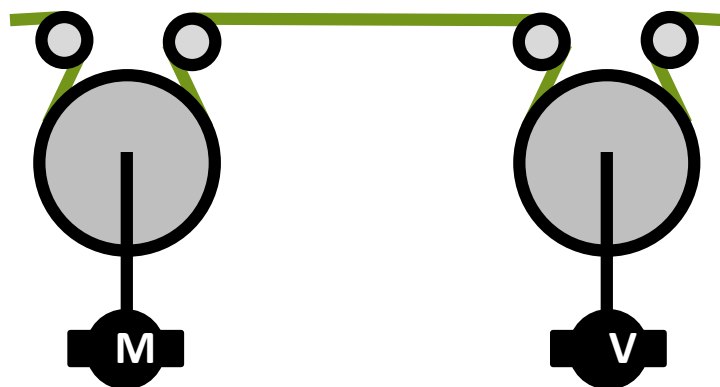
Driven Unwind with
Idling Peel Roller

Peel rollers are lightly loaded against the unwinding roll and may be driven to reduce the tension increase due to peel force.



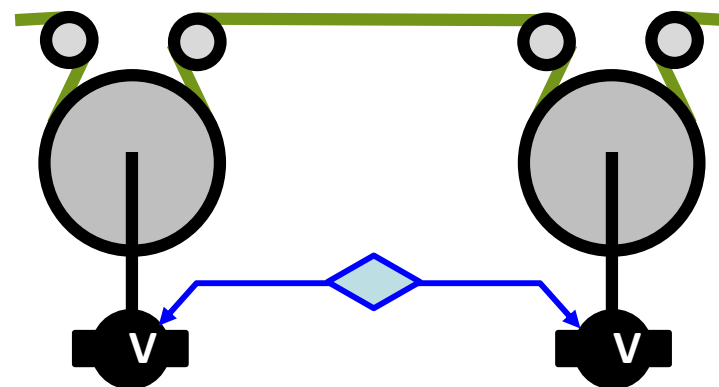
Driven Unwind with
Driven Peel Roller

Intermediate Tension Zone Options – Downstream Pacer



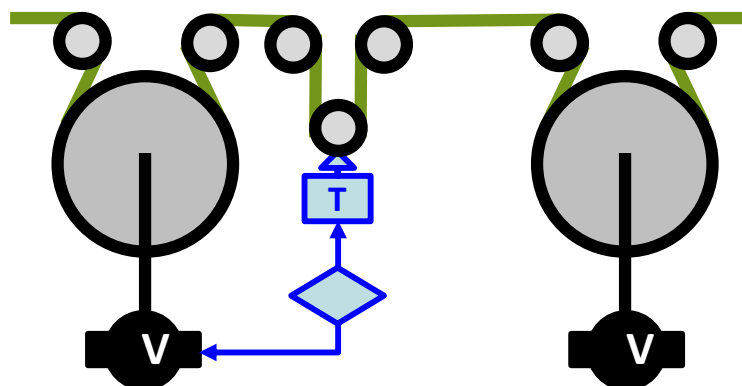
Open-Loop
Torque Control

Pacer



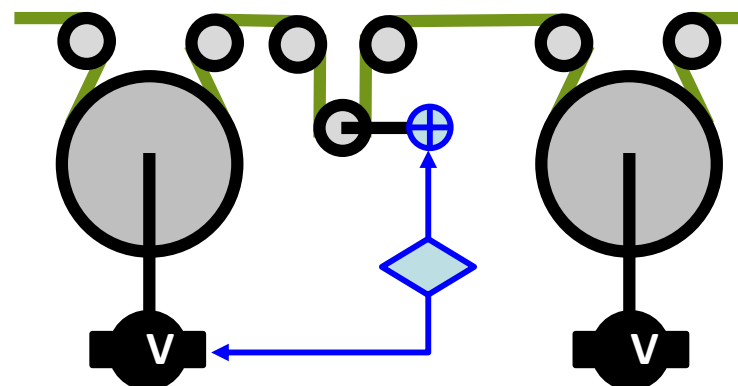
Open-Loop
Draw Control

Pacer



Closed-Loop Tension Control
with Load Cell Feedback

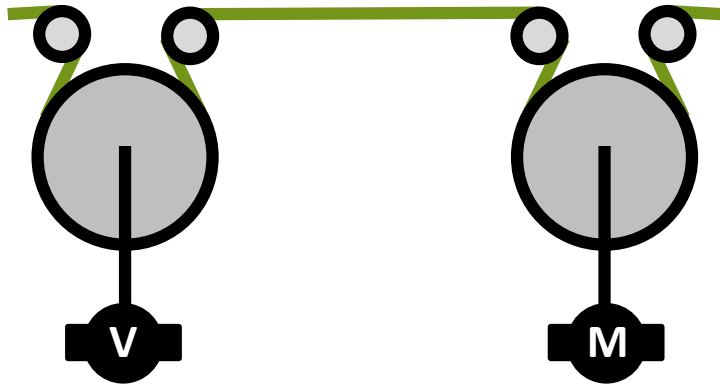
Pacer



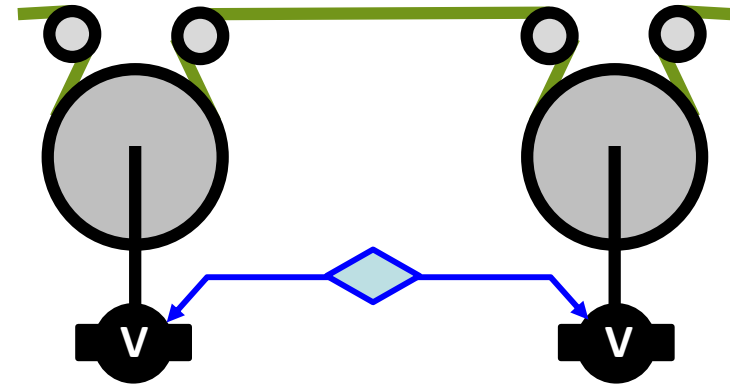
Closed-Loop Tension Control
with Dancer Feedback

Pacer

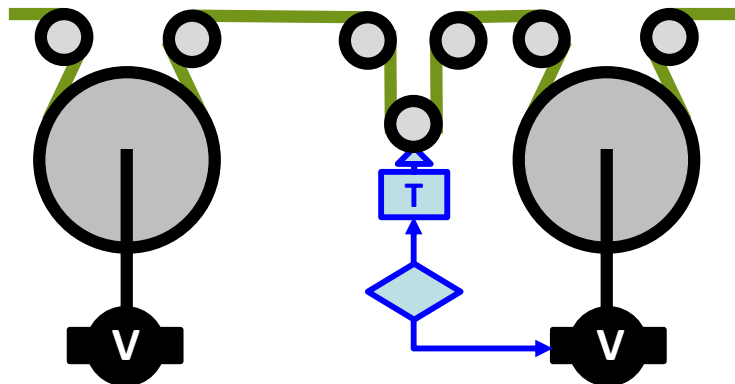
Intermediate Tension Zone Options – Upstream Pacer



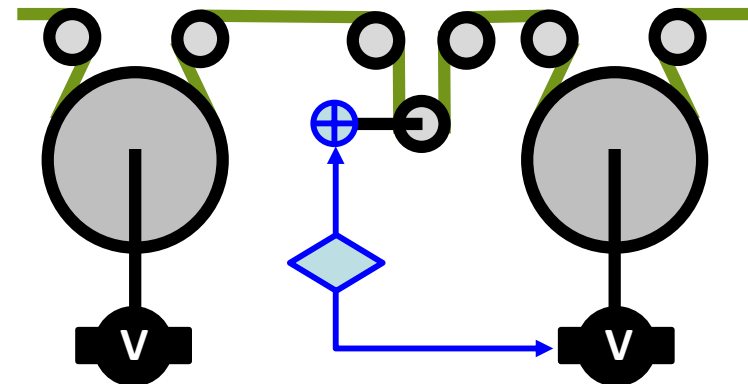
Pacer

Open-Loop
Torque Control

Pacer

Open-Loop
Draw Control

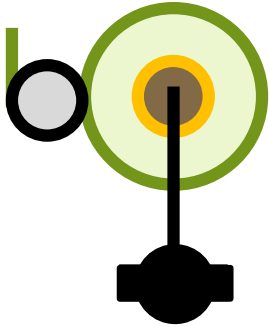
Pacer

Closed-Loop Tension Control
with Load Cell Feedback

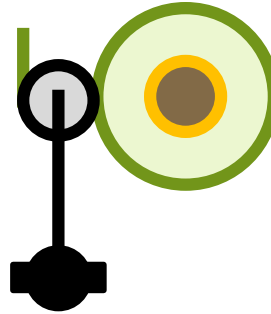
Pacer

Closed-Loop Tension Control
with Dancer Feedback

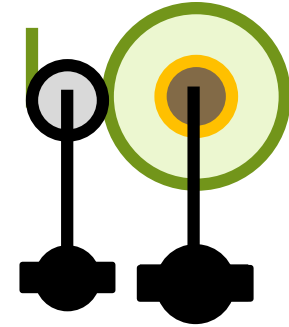
Winding Options: Center Driven



Center Driven
Winder

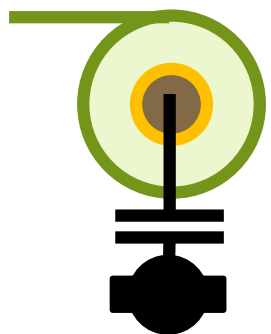


Surface Driven
Winder

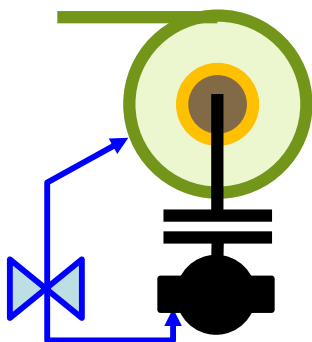


Center+Surface
Driven Winder

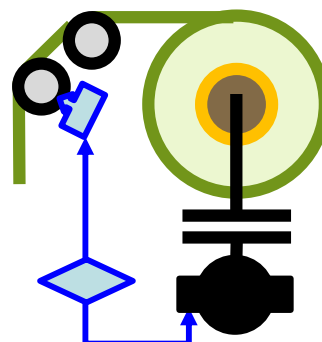
Winding Options: Center Driven



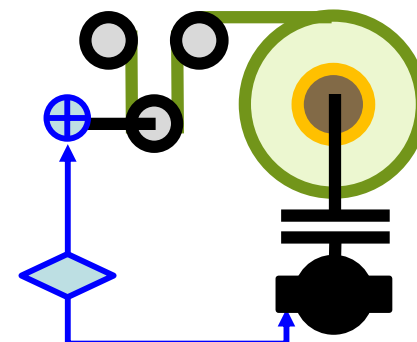
Open-Loop
Clutch



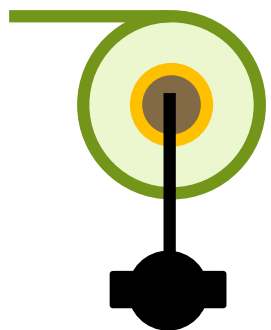
Clutch with
Diameter Feedback



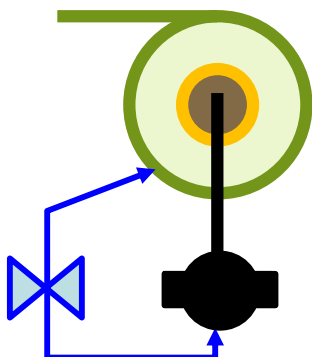
Clutch with
Load Cell Feedback



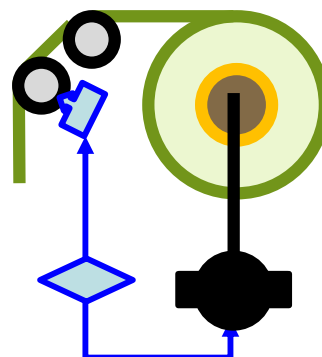
Clutch with
Dancer Feedback



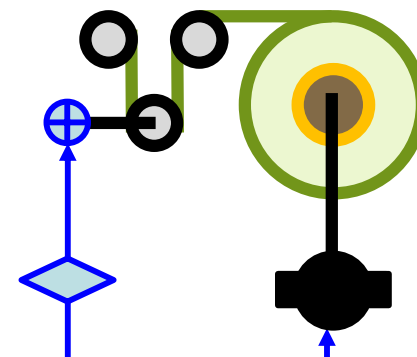
Open-Loop
Torque Motor



Motor with
Diameter Feedback

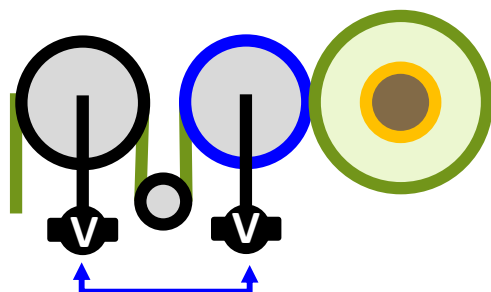


Motor with
Load Cell Feedback

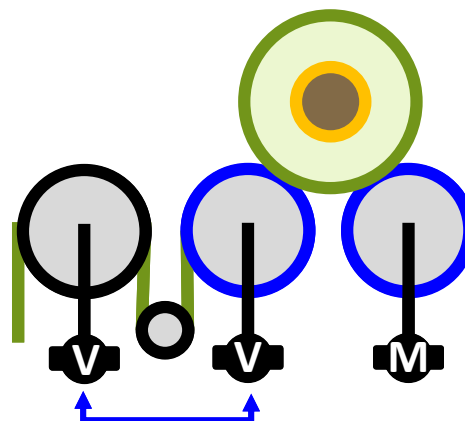


Motor with
Dancer Feedback

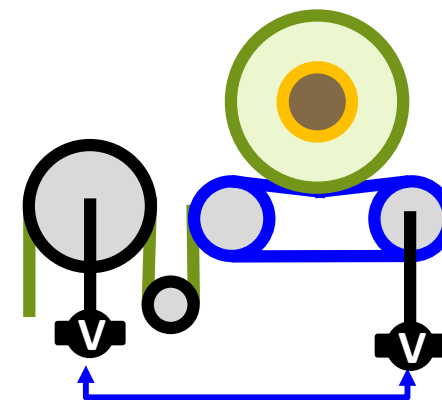
Winding Options: Surface Driven



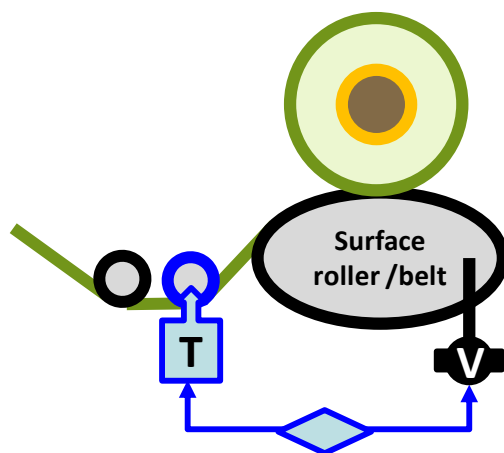
Surface Roller
Driven Winder



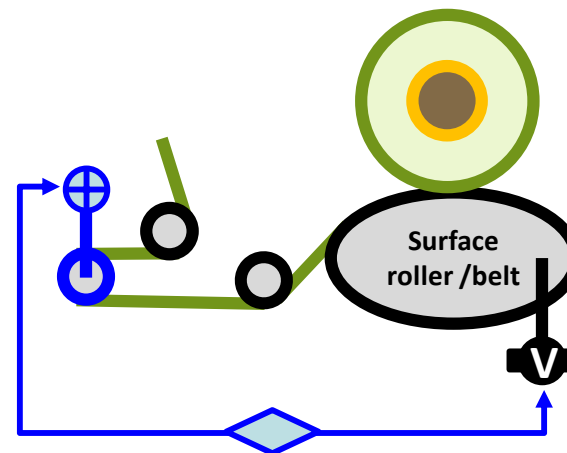
2-Drum Surface Roller
Driven Winder



Surface Belt Driven Winder

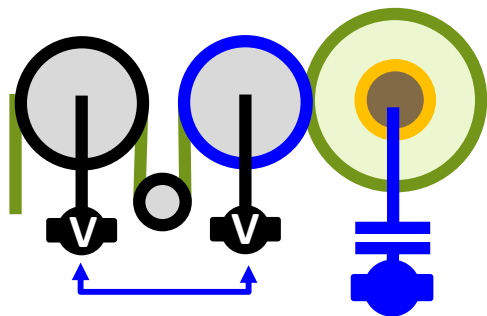


Motor with Load Cell Feedback



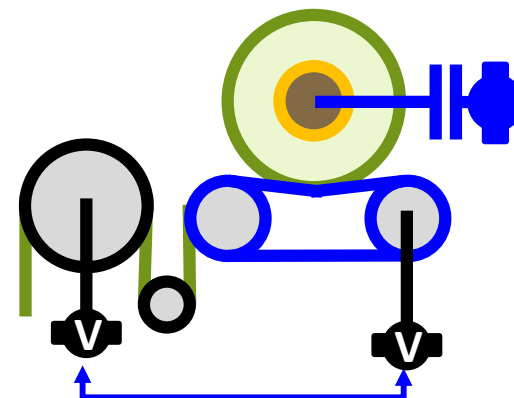
Motor with Dancer Feedback

Winding Options: Surface-Center Driven

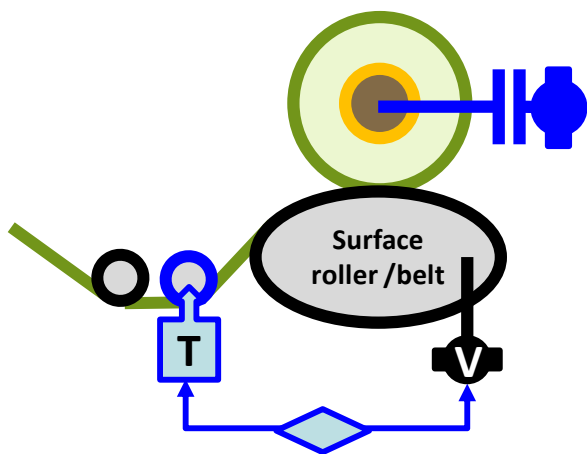


Surface Roller Driven Winder

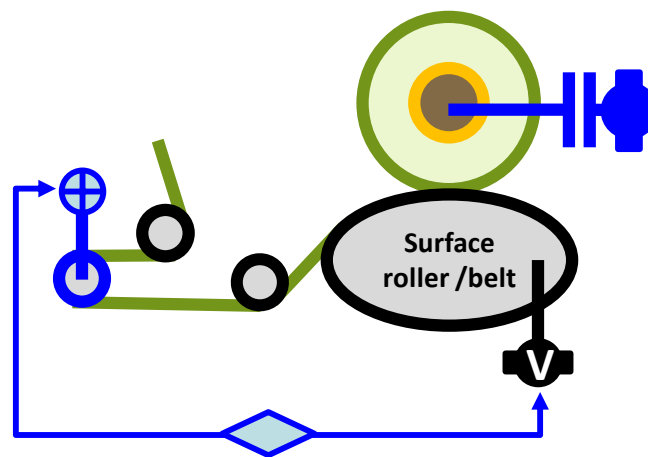
Add torque assist
to center of any
surface winder.



Surface Belt Driven Winder



Motor with Load Cell Feedback



Motor with Dancer Feedback

Where is Closed-Loop Tension Control Used?

- Just about anywhere.

It's easier to say where it isn't used:

- If space, web path, or environmental condition prohibit load cell or dancer rollers.
- If tension is so low it is difficult or impossible to:
 - ✓ Measure with a load cell
 - ✓ Design a dancer roller where friction and inertia is small relative to tension.

Where is Torque Control Used?

Tensioning:

- Simple unwinders and rewinders
- Provides a natural taper tension at winding
- Slip shaft winding after slitting

Non-Tensioning, Torque-Assist:

- ✓ Tendency drive roller
- ✓ Rubber rollers in nipped systems
- ✓ Winding core in center-surface winding

Where is Draw Control Used?

When the process requires it:

- Registration processes (printing, die cutting)
- Length change processing (film orientation, stretching)

When Strain is easier to control than tension:

- Low spring constant webs (low thickness, width, modulus) where speed differential creates small tension changes
- Control non-elastic or easily yielding webs (non-wovens before bonding, viscoelastic films, crepe paper)

Draw is the simplest intermediate tension control option:

- Draw control rollers minimize percent slippage (vs. torque or closed-loop control).
- Multi-roller driven sections (S-wrap, bridle rollers)
- Slitter-rewinder transport rollers (avoid losses from inertia during frequent acceleration and deceleration)

Slippage?

What controls the process tension and roller speed if a roller slips in...

- ▶ A torque controlled system?
- ▶ A draw control system?
- ▶ A follow driven roller in closed-loop tension control system?
- ▶ A pacer driven roller in closed-loop tension control system?



2.5 Tension Zones

How many tension zones are needed in a process?

Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgcumbe Road, Saint Paul, MN 55116

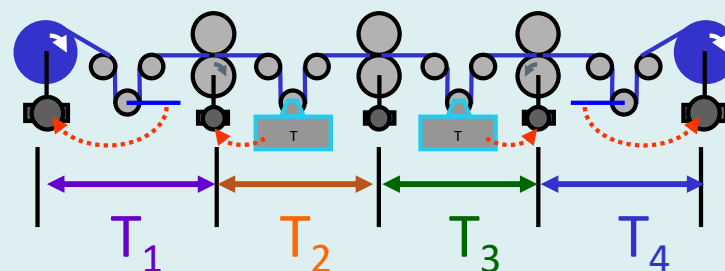
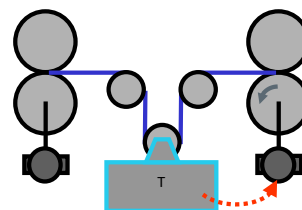
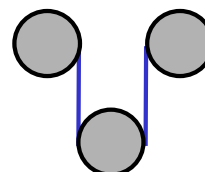
tjwalker@tjwa.com www.webhandling.com 651.686.5400

2.3 Learning Objectives

- ▶ What is a tension zone?
- ▶ How many are needed in a process?
- ▶ What is the master, pacer, or lead drive or section?
- ▶ What is a follower drive or section?
- ▶ How do you identify the drive plan of existing converting equipment?

Tensioning Hierarchy

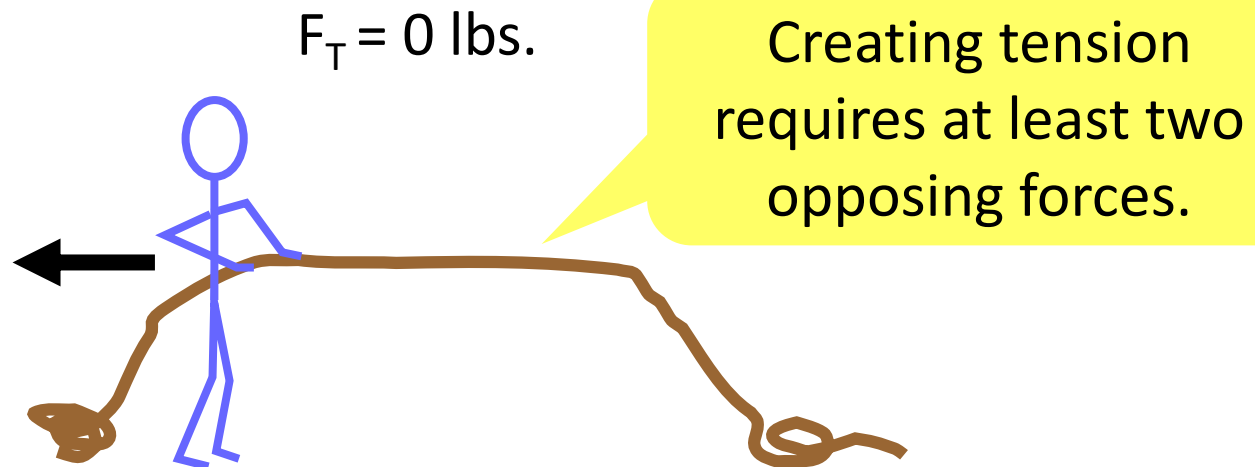
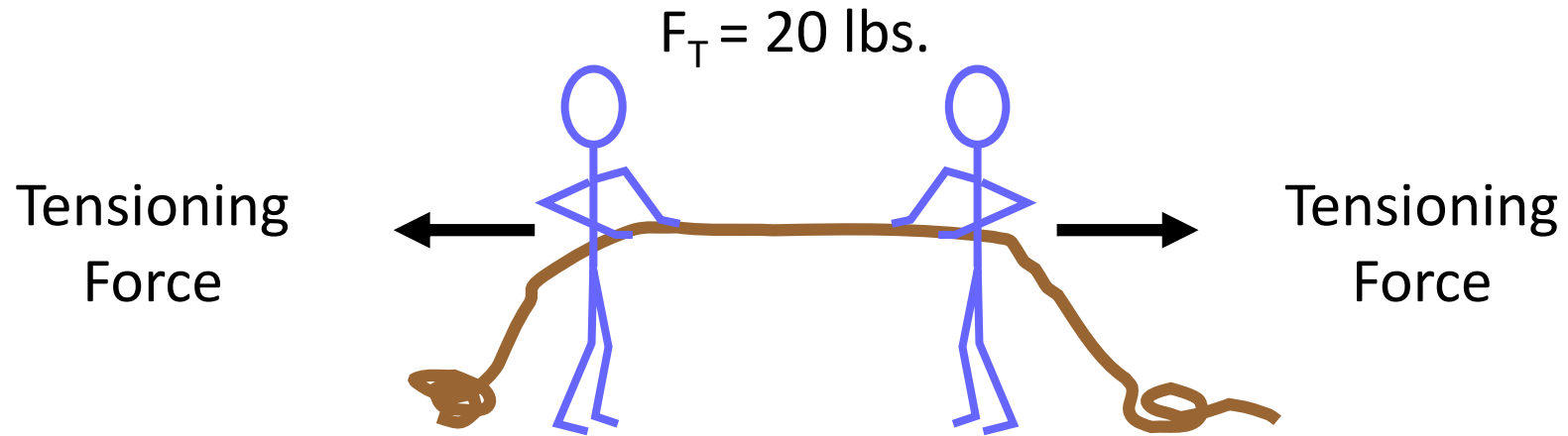
- ▶ Tension Fundamentals
- ▶ CD Tension Variations
- ▶ MD Tension Variations
- ▶ Tension Zone
- ▶ Tension in Multiple Zones



Your Tensioning System

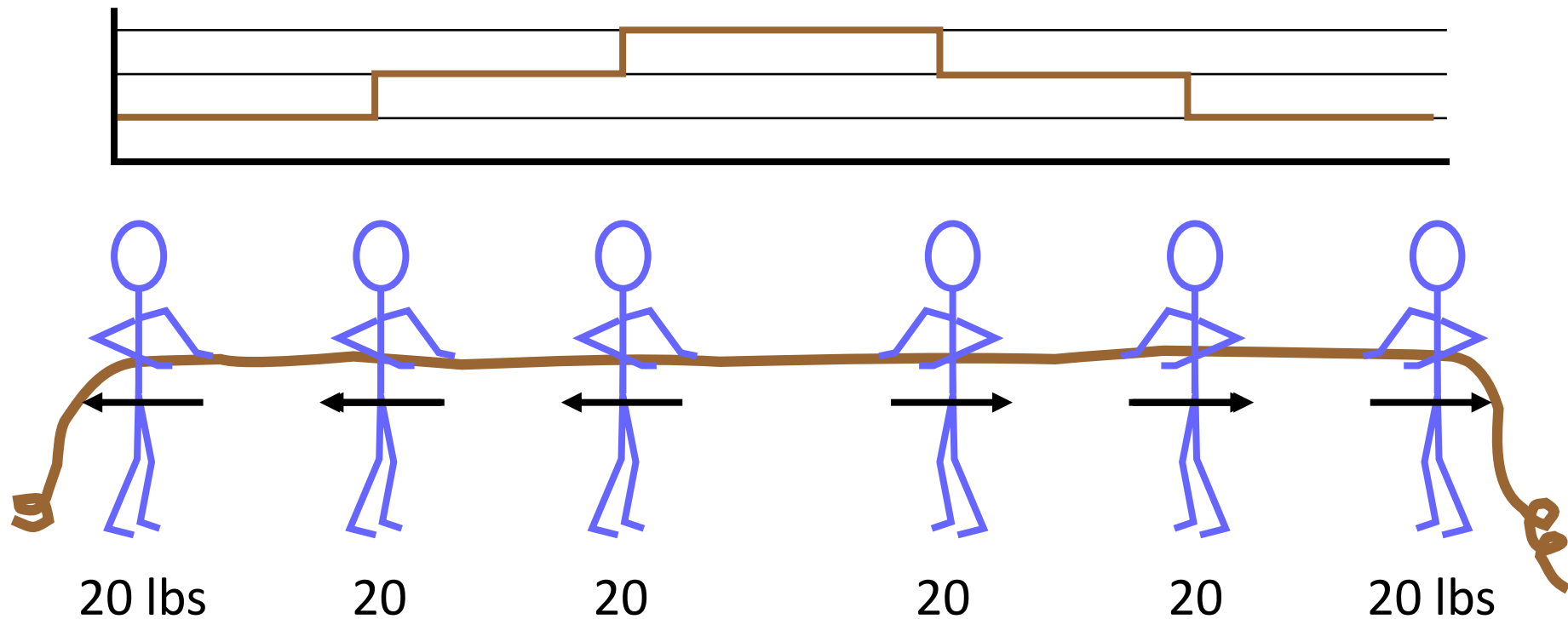
- ▶ Be able to map out the tension control system of any process.
- ▶ Understand how tension is created by torque or draw and open vs. closed-loop systems.
- ▶ Know when each makes sense. Know how to estimate tension when it is not known.
- ▶ Verify tension feedback or readouts.
- ▶ Divide processes into logical tension zones.
- ▶ Ensure drive rollers between zones have sufficient traction to avoid slip.
- ▶ Understand lead sections and followers.

Tension Control = Tug-of-War?

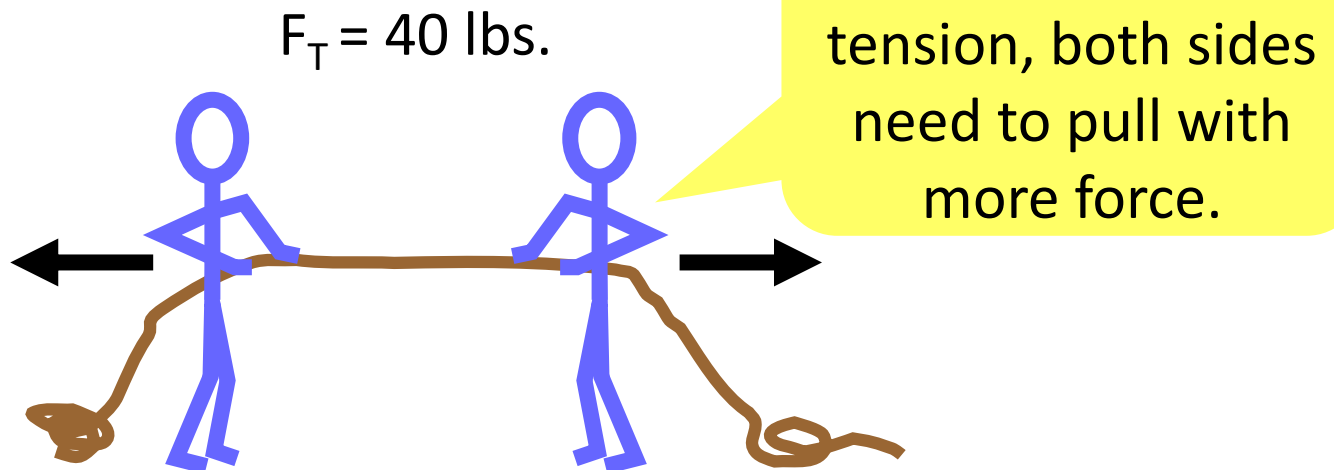
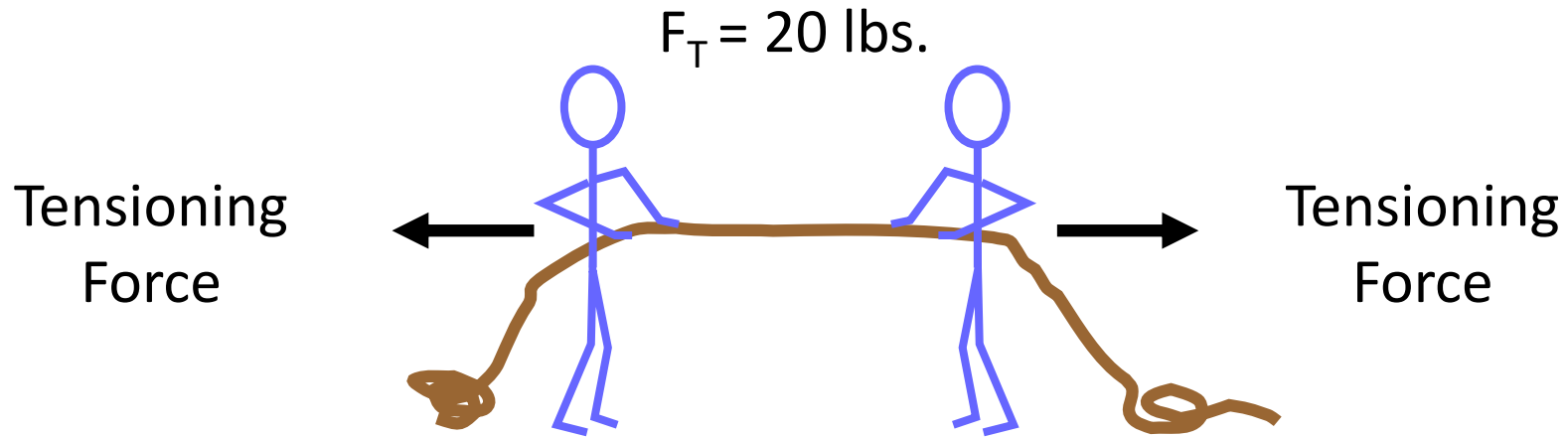


Tension Control = Tug-of-War?

T_1 T_2 T_3 T_4 T_5
20 lbs 40 60 lbs 40 20 lbs

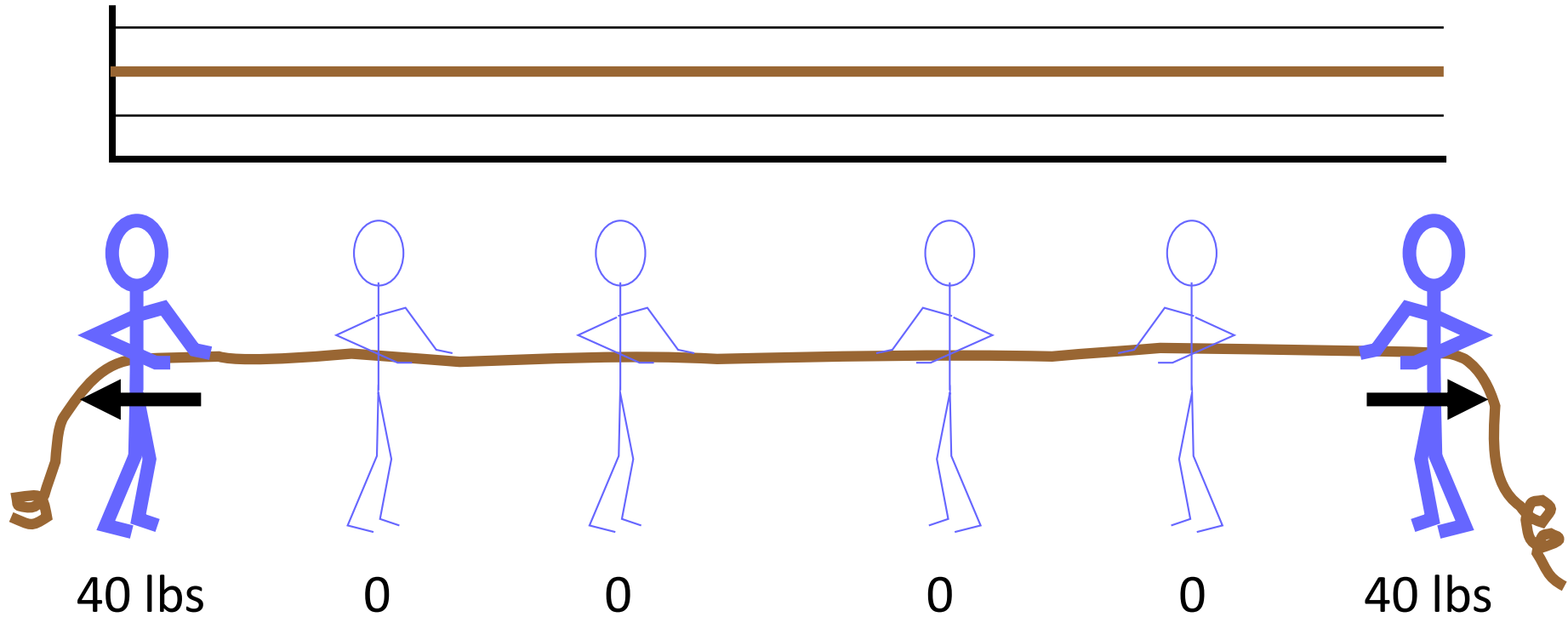


Tension Control = Tug-of-War?

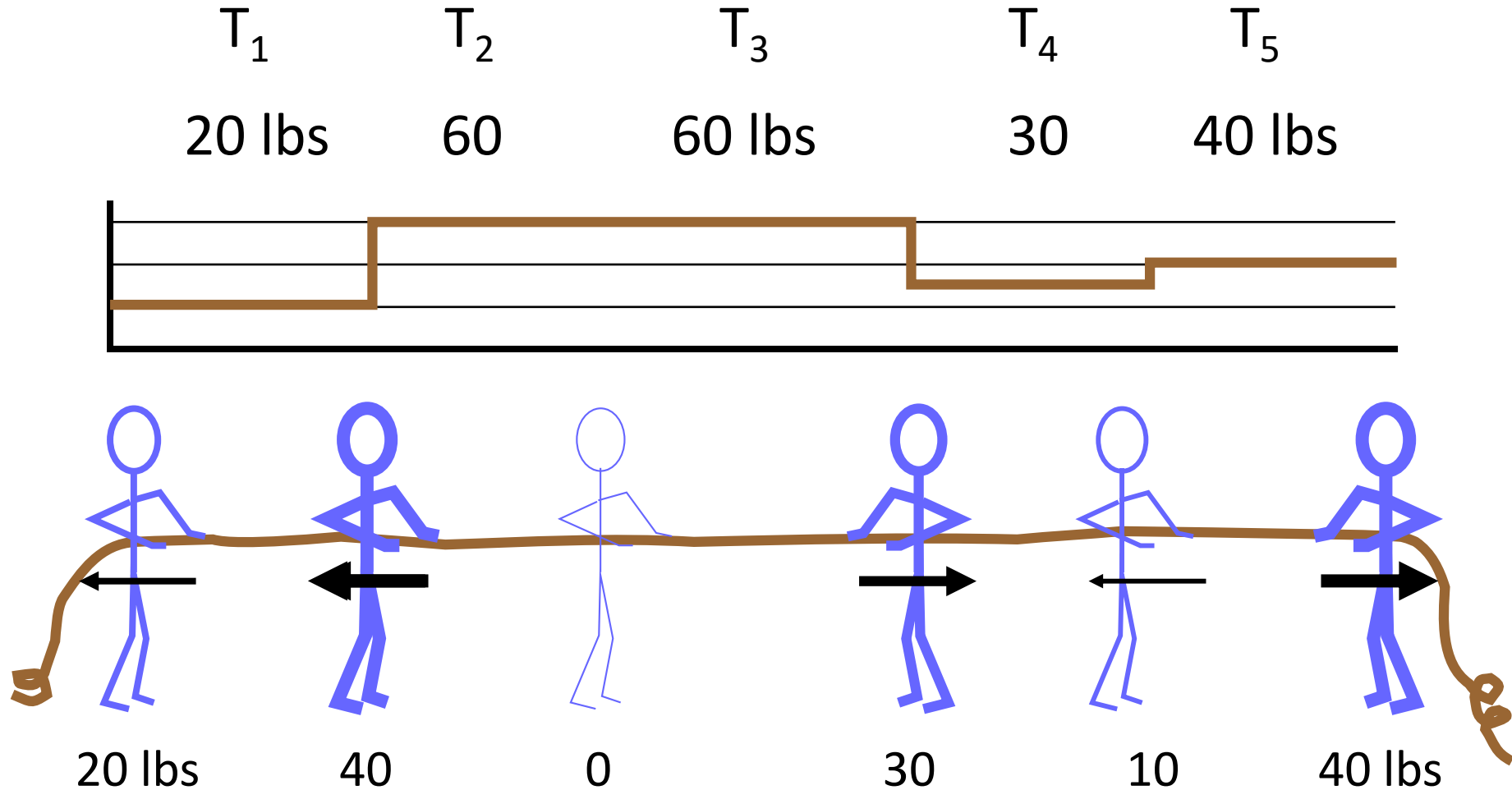


Tension Control = Tug-of-War?

T_1	T_2	T_3	T_4	T_5
40 lbs	40	40 lbs	40	40 lbs

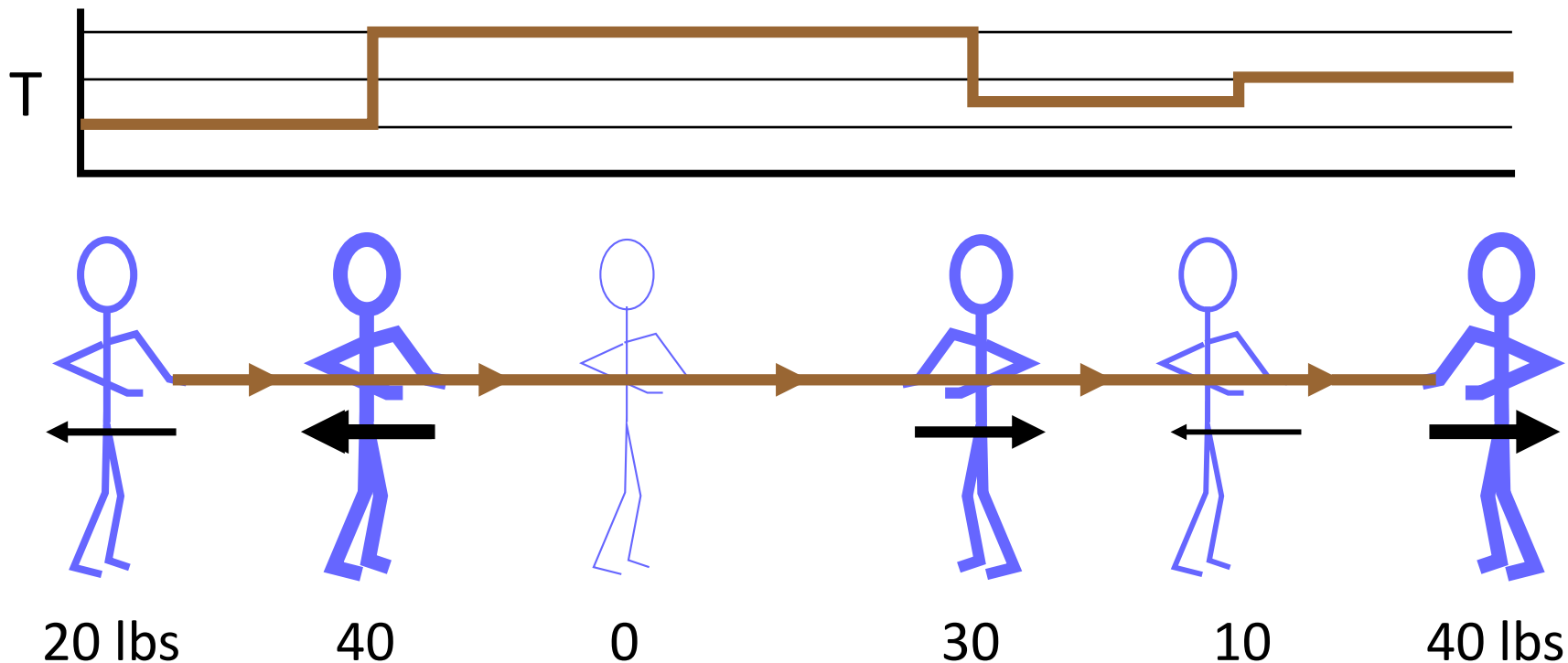


Tension Control = Tug-of-War?



Tension Control = Tug-of-War?

The challenge of web handling is to play Tug-of-War while the rope is moving.

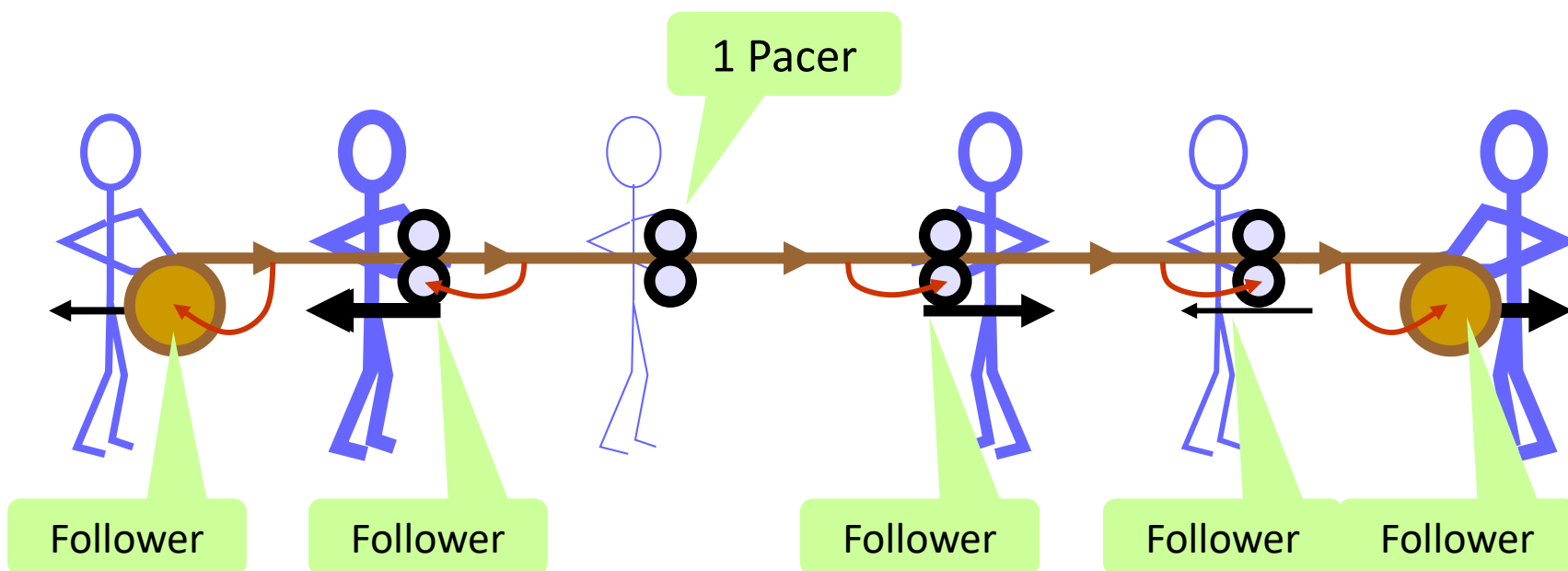


Tension Control

Every two drive points make one tension zone.

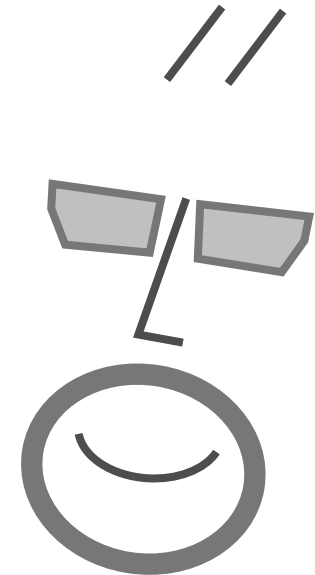
Here we have six drive points.

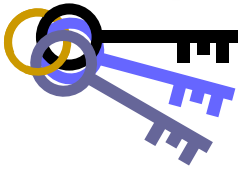
One point is the pacer, the other five are in charge of one zone.



Tensioning Zone Qs

What is a tension zone?





KEY CONCEPT



Tension Zone

A section of the web line, between two torque or speed controlling devices, where the tension is set by design (non-adjustable) or by control (adjustable).

Use: Design integrated web lines to vary web tension by process step.

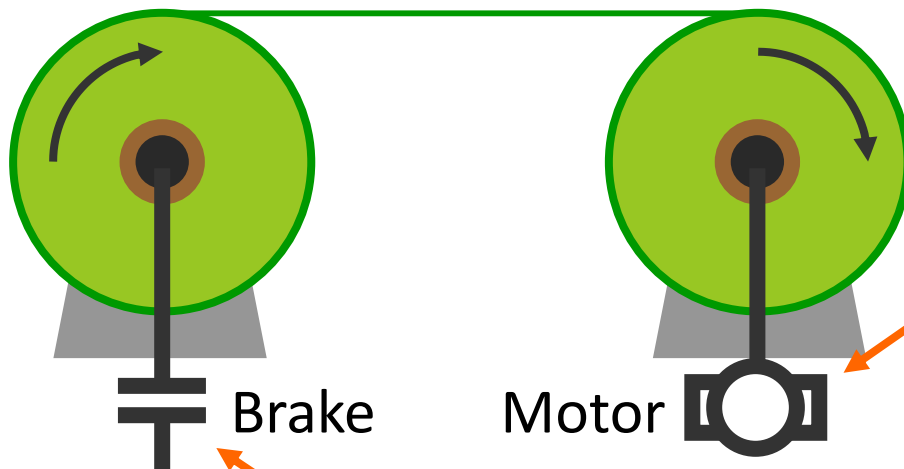
Example: For film processing, coating and drying are usually divided into two tension zones to allow a tight, wrinkle-free web at coating and a low tension when the film is heated.

One Tension Zone Web Line

The simplest web line has one tension zone.

Unwinding

Winding

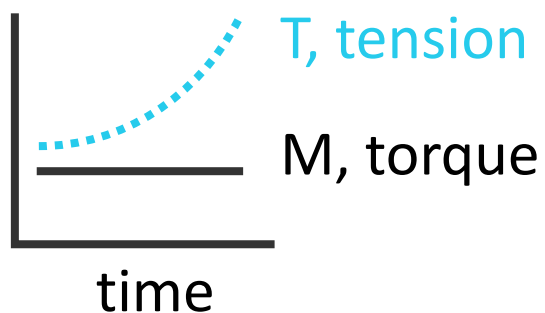
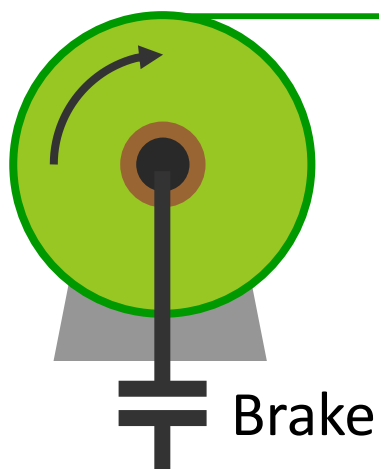


A motor drives the winding core, setting the process speed.

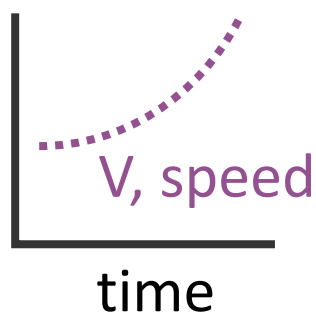
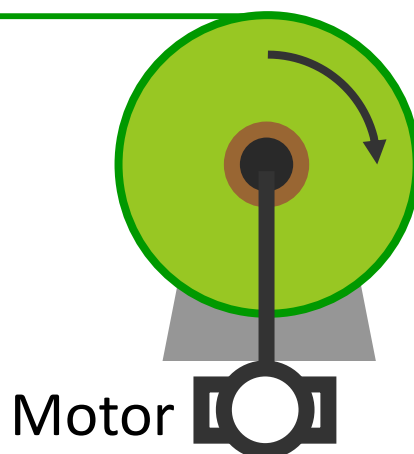
Tension is increased or decreased by adjusting the unwinding roll's brake torque.

Single Zone Web Line Problems

Unwinding



Winding



Problem #1:

The changing speed is a problem for many process.

Problem #2:

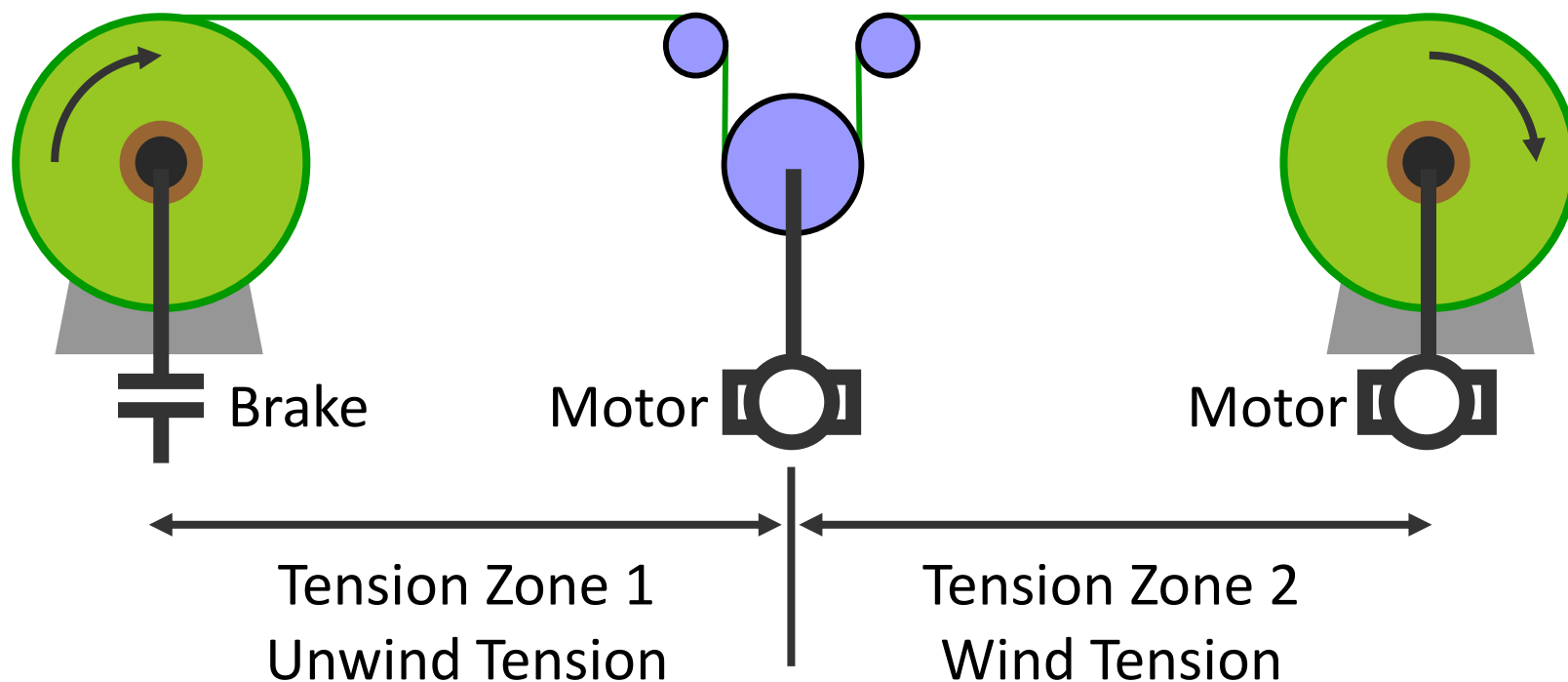
The increasing tension will create winding defects in many products.

Two Tension Zone Web Line

To avoid single zone problems, most web lines add an intermediate drive point (a.k.a. a pacer).

Unwinding

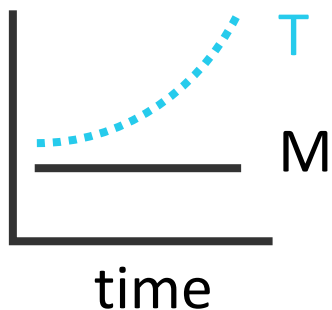
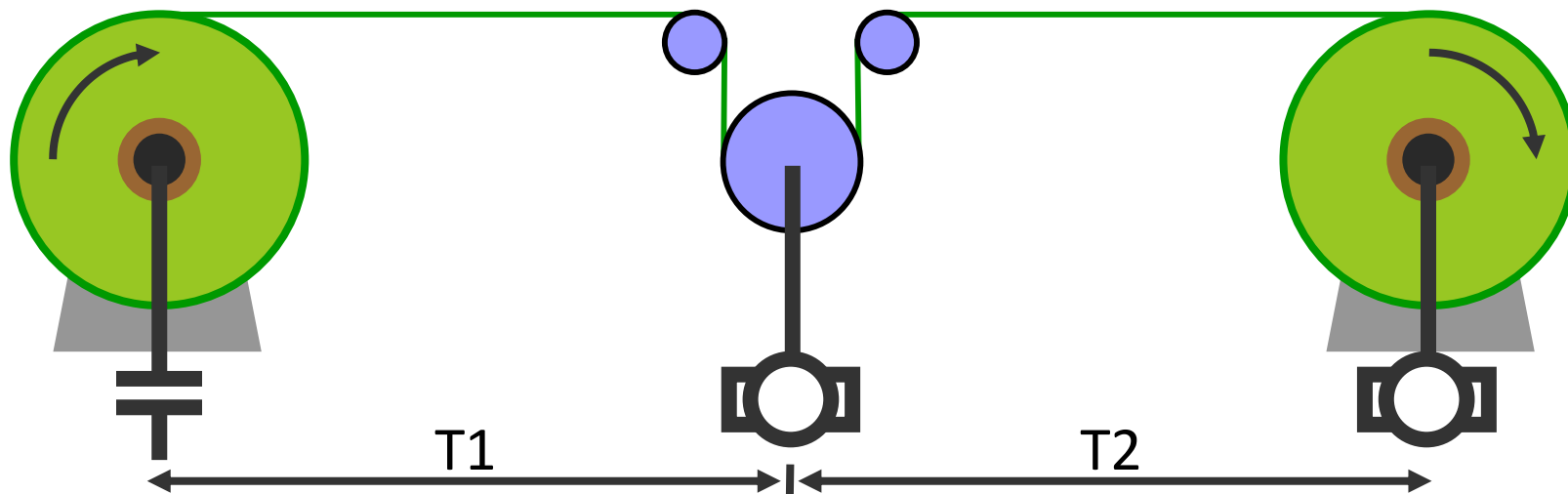
Winding



Two Tension Zone Web Line

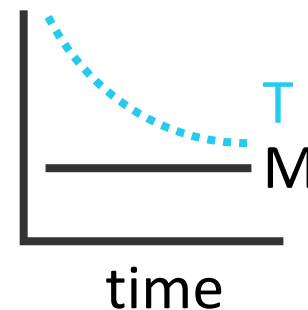
Unwinding

Winding

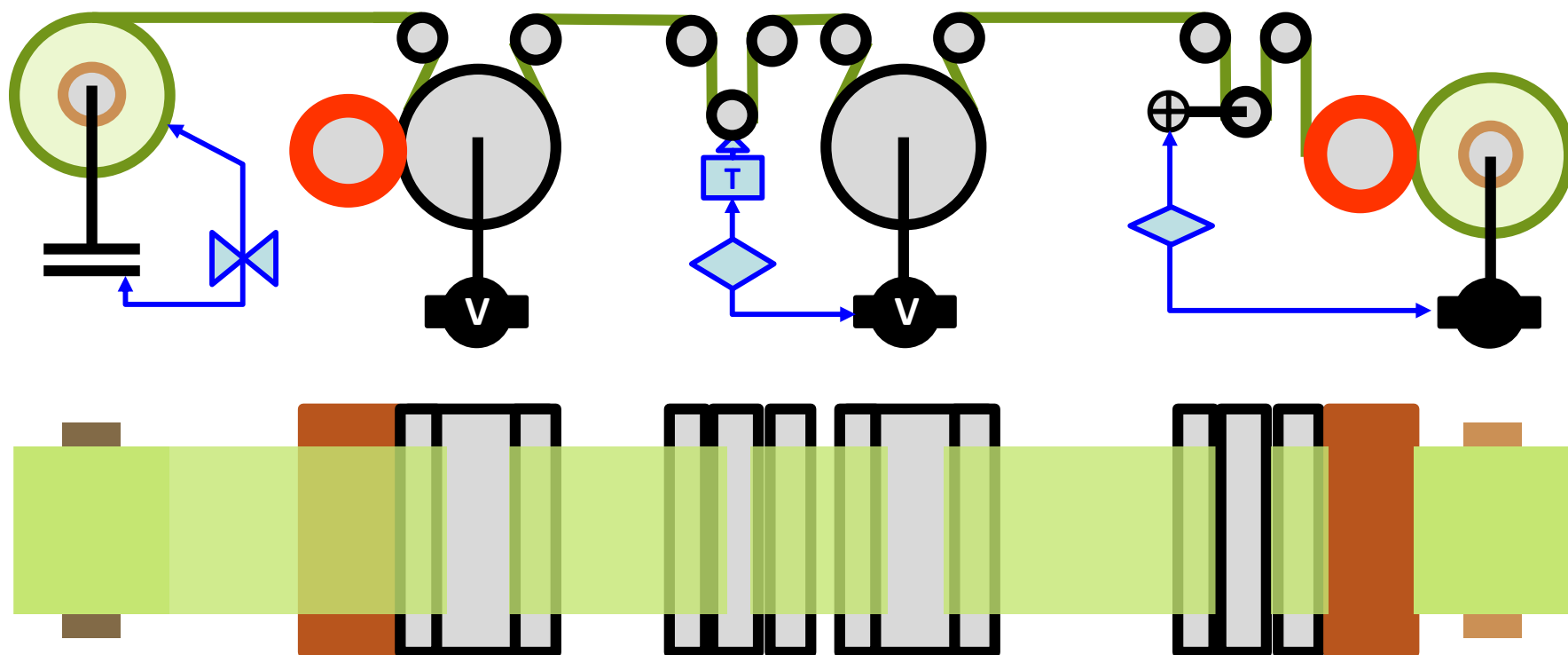


Speed is independent of roll radius.

Constant torque winding creates desirable hard-to-soft wind tension.



Three Tension Zones



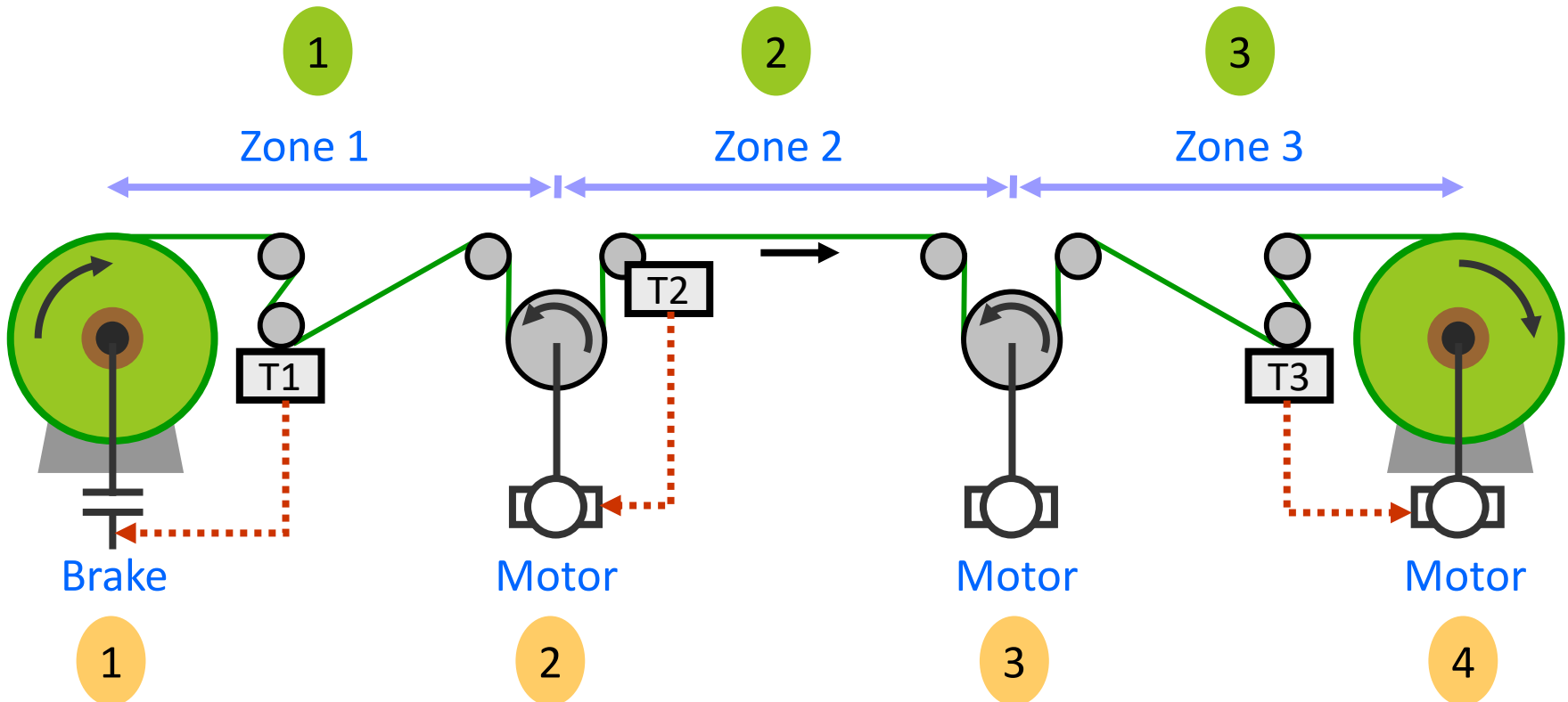
Tension Zones

of Tension
Zones

=

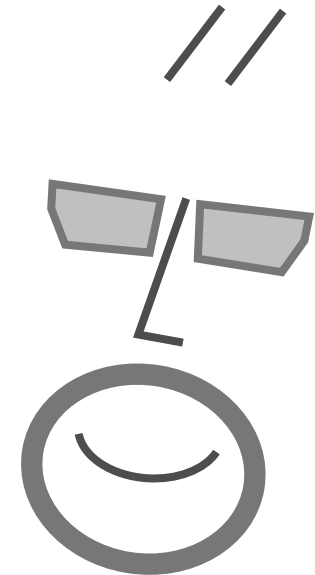
of Brakes,
Clutches, Motor

- 1



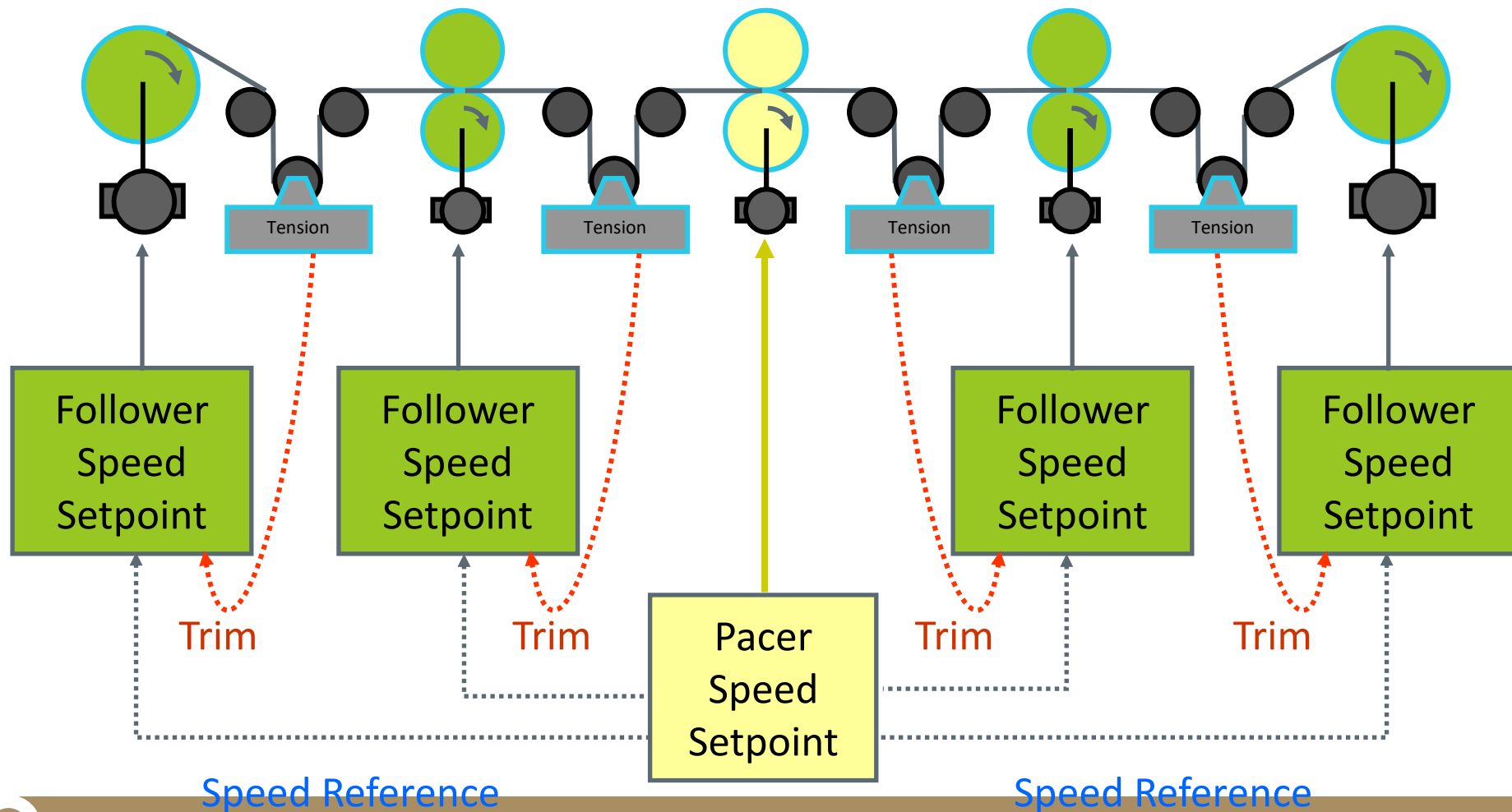
Tensioning Zone Qs

What are pacers and followers?



Line Speed Reference

Line speed reference provides baseline information to all drives.

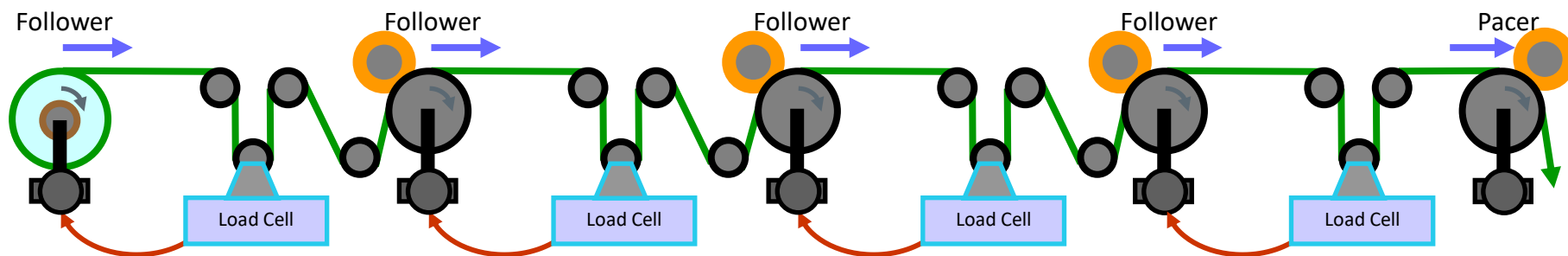
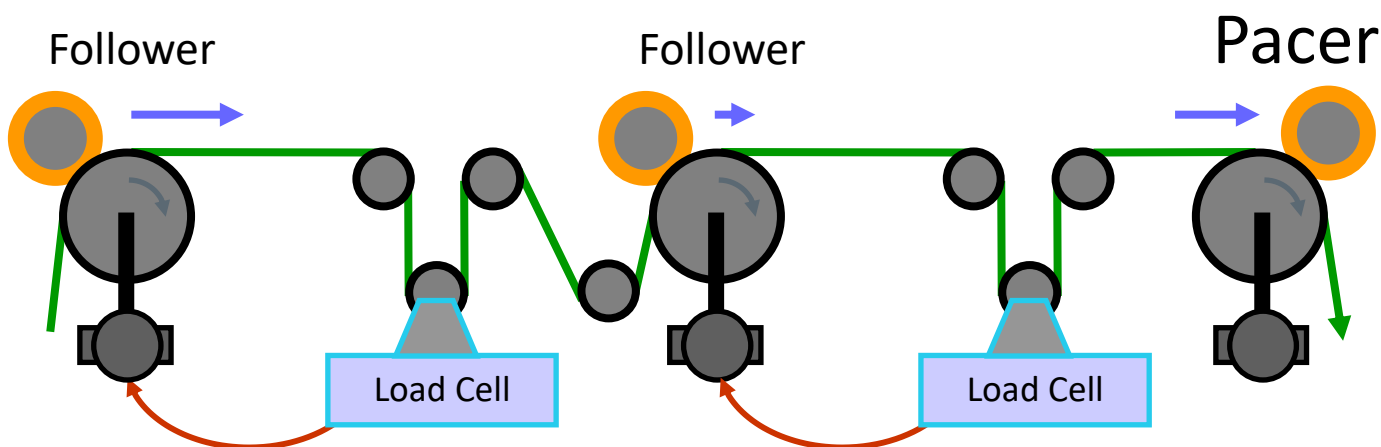


Speed Reference

Speed Reference

Series of Follower & Load Cells

Additional followers and tension zones can be added as needed.



Pacer Fundamentals

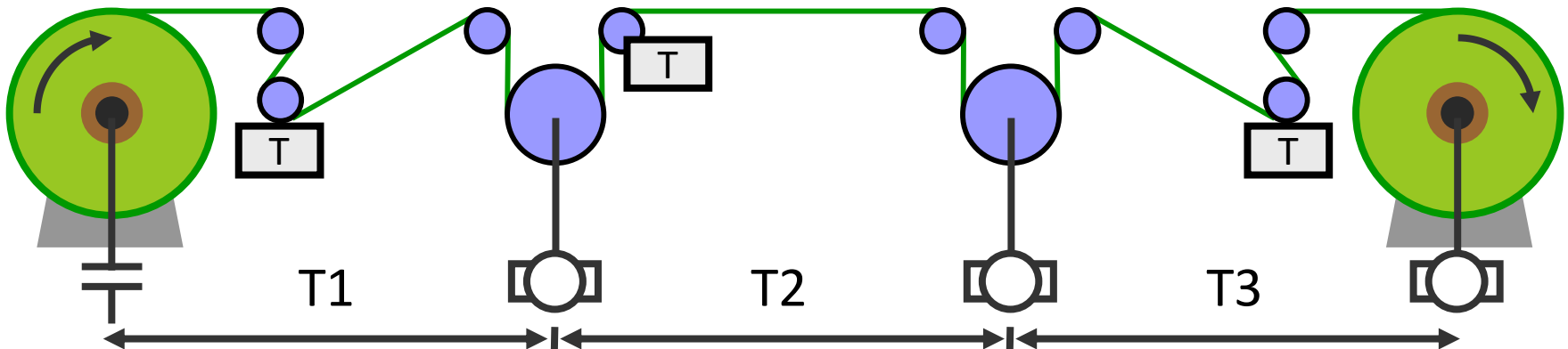
Pacer Selection Criteria

- The web must contact with the pacer.
- The web must not slip on the pacer.
- Locate the pacer near your speed-sensitive process.
- A large diameter pacer will create less web speed variations for the same absolute diameter or web thickness variation.
- Make your highest inertia component your pacer.

More Than Two Tension Zones?

When would more than two zones be beneficial?

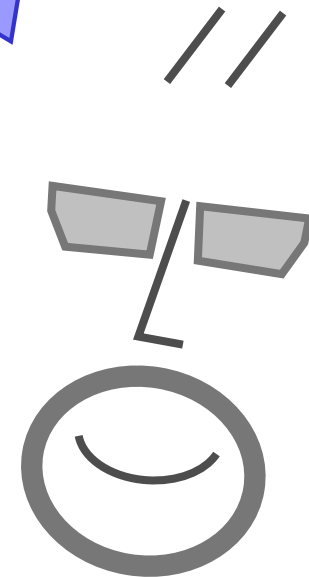
1. When total roller drag and inertia losses require too much tension.
2. To isolate a critical process (such as coating) from unwind and rewind tension variations.



Tension Zone Advice

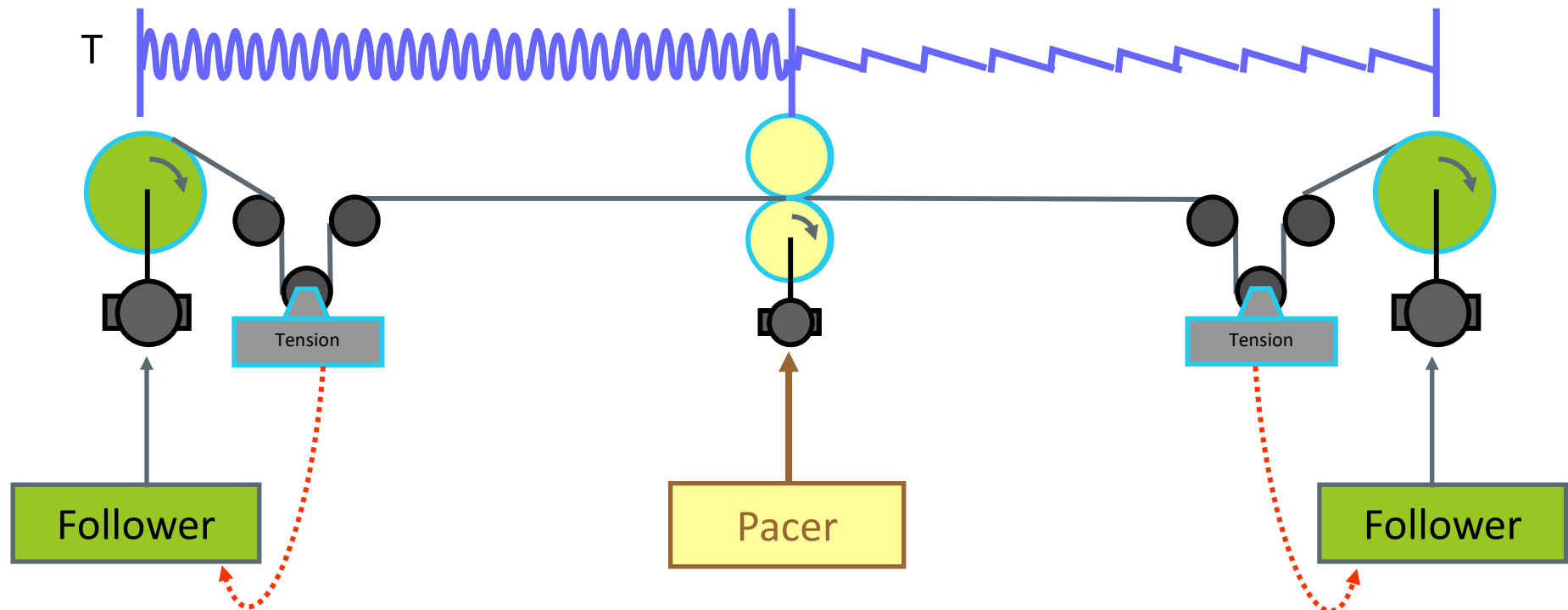
Many web lines would benefit from having unwind and winder pull roller units.

Unwind and winder pull rollers help to optimize handling and winding tensions independent of process tensions.



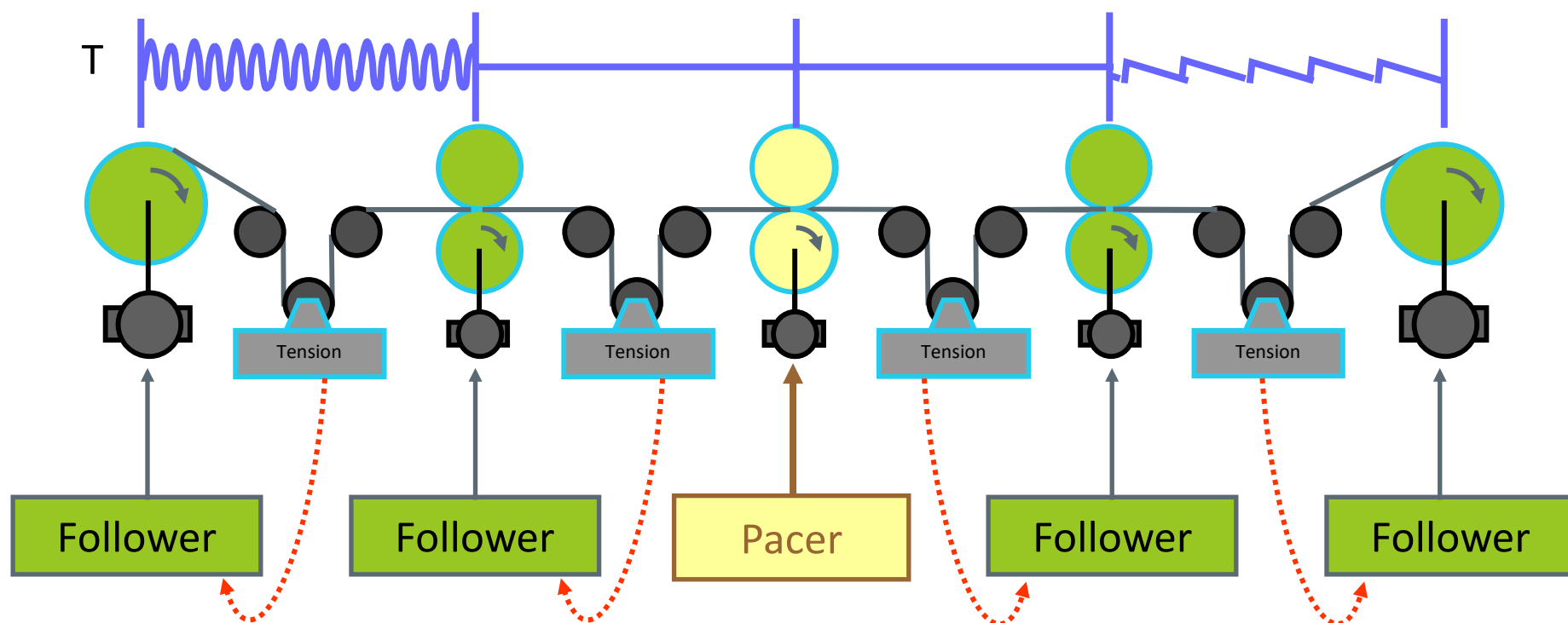
Benefit of Winder Pull Rollers

Simple: Both unwind and winder tension zones have tension variations from either unwind runout or winder tension taper.



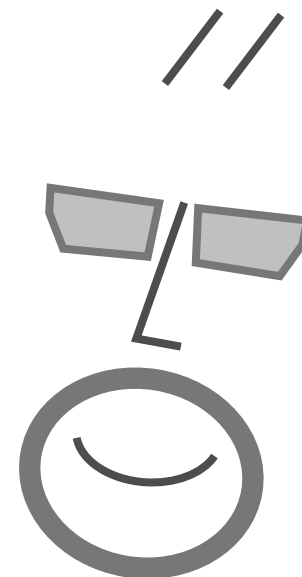
Benefit of Winder Pull Rollers

Improved: The two additional tension zones isolate the critical process from unwind and winder tension variations, allowing handling and winding process each to have optimized tension.



Tensioning Zone Qs

How many tension zones are needed?



How Many Tension Zones?

1. Identify Process Steps
(include unwinding and winding as process steps)
2. Specify Tension Requirement of Each Step
(nominal tension and variation sensitivity)
3. Group Compatible Consecutive Steps.
4. Tension Zones = Number of Compatible,
Consecutive Groups.

How Many Tension Zones?

Undriven Coater with Drying - Heavy Weight Paper

Step 1

Processes =

Unwind

Coat

Dry

Wind

Step 2

Nominal T =

Low

High

Med-High

High → Med

ΔT Allowed =

High

Low

Med-Low

Med-High

Step 3

Compatible?

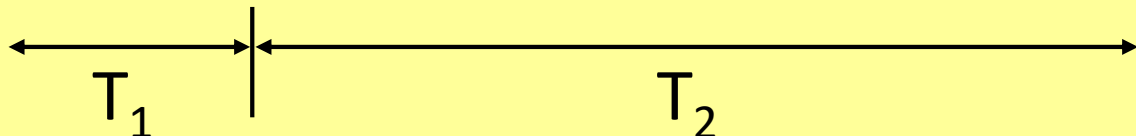
No

Yes

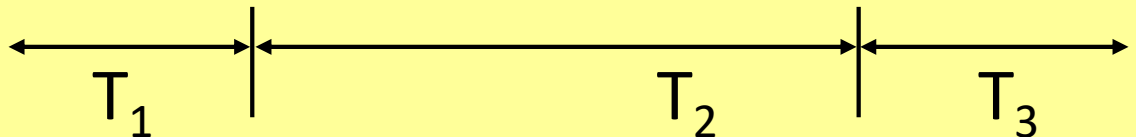
Yes

Step 4

Tension Plan



Alternate Plan



How Many Tension Zones?

Driven Coater with Drying - Heavy Weight Paper

Step 1

Processes =

Unwind

Coat

Dry

Wind

Step 2

Nominal T =

Low

High

Med-High

High → Med

ΔT Allowed =

High

Low

Med-Low

Med-High

Step 3

Compatible?

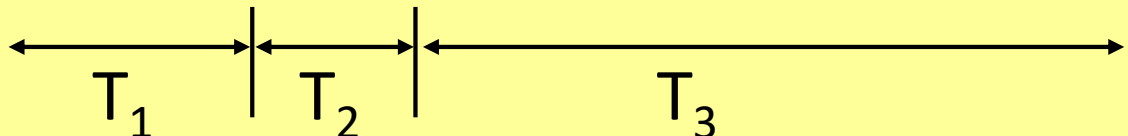
No

Yes

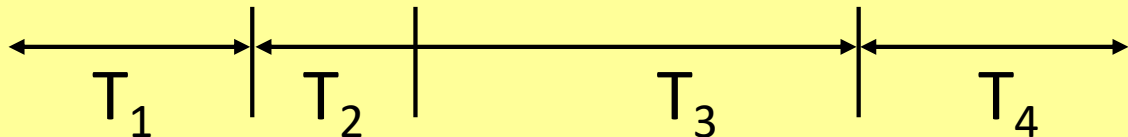
Maybe?

Step 4

Tension Plan

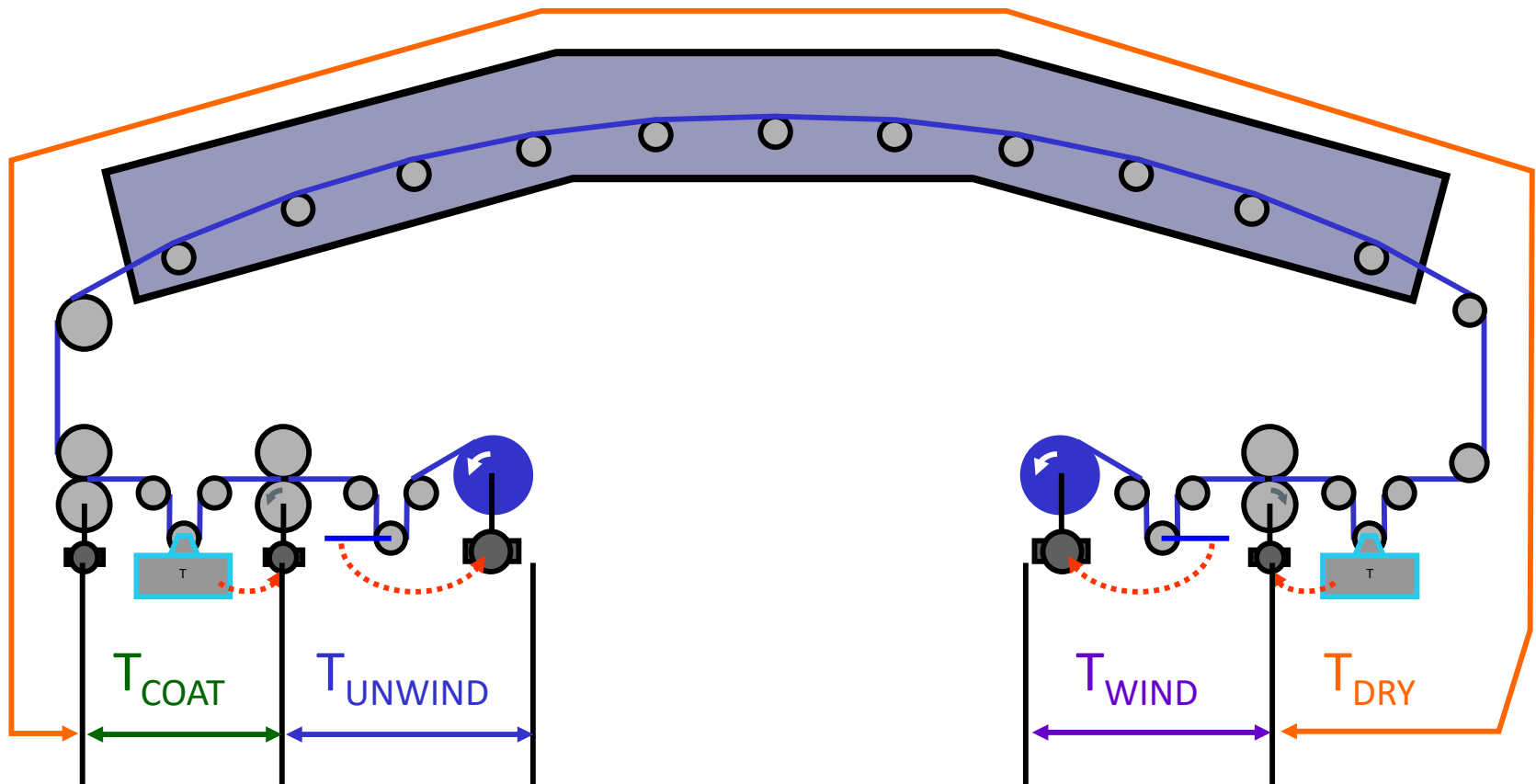


Alternate Plan



Typical Coater Tension Control

Coaters have tension zones to meet the needs of the integrated process steps.



How Many Tension Zones?

Driven Coater with Drying – Thin Films

Step 1

Processes =

Unwind

Coat

Dry

Wind

Step 2

Nominal T =

Low

High

Low

High → Med

ΔT Allowed =

High

Low

Med-Low

Med-High

Step 3

Compatible?

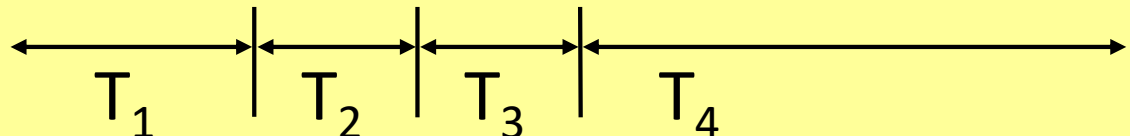
No

No

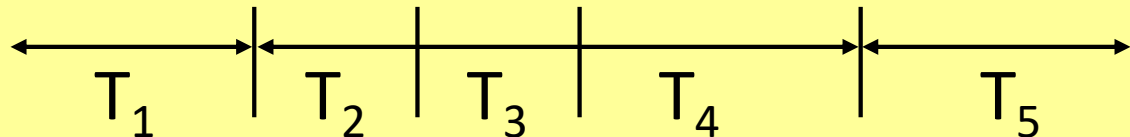
Maybe?

Step 4

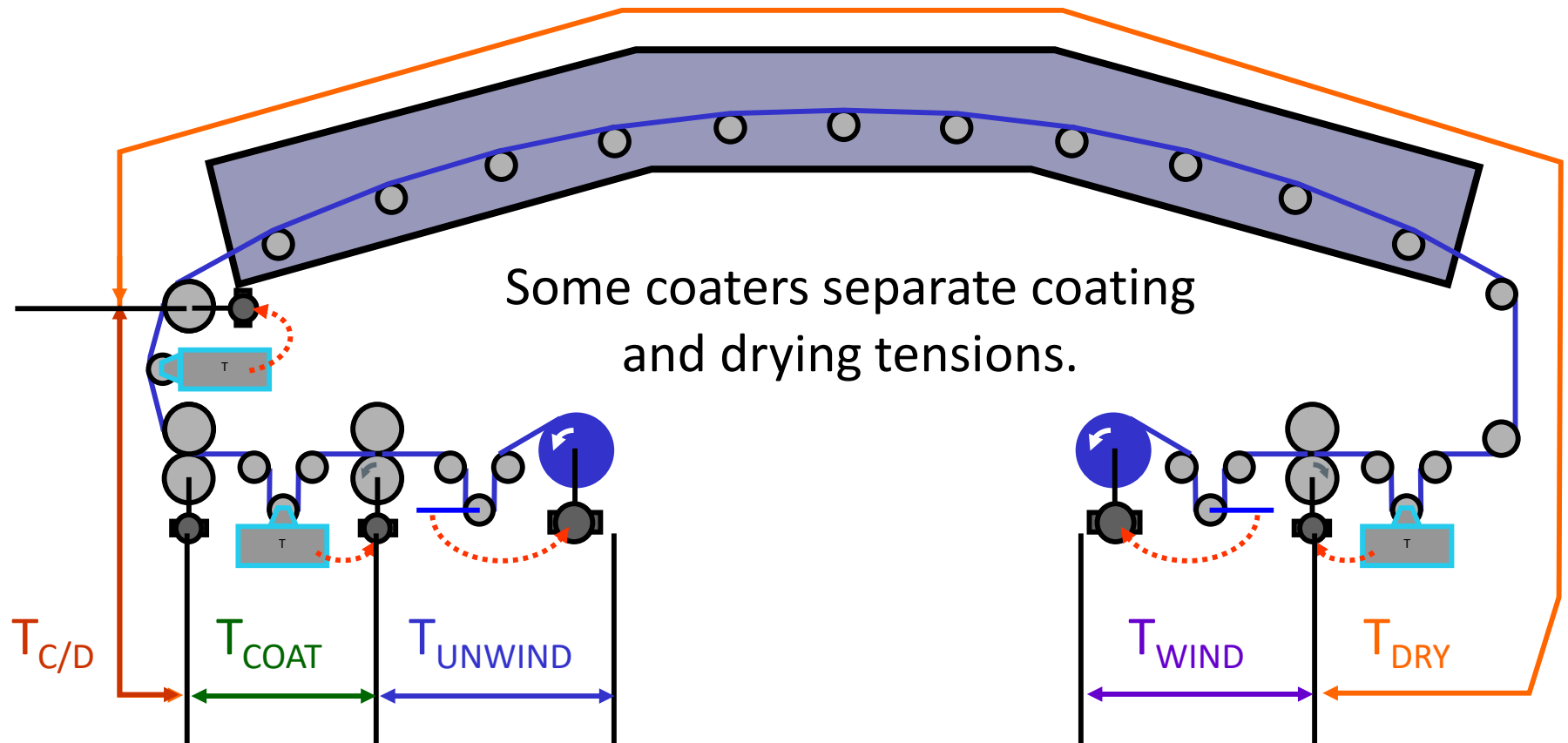
Tension Plan



Alternate Plan



Typical Coater Tension Control



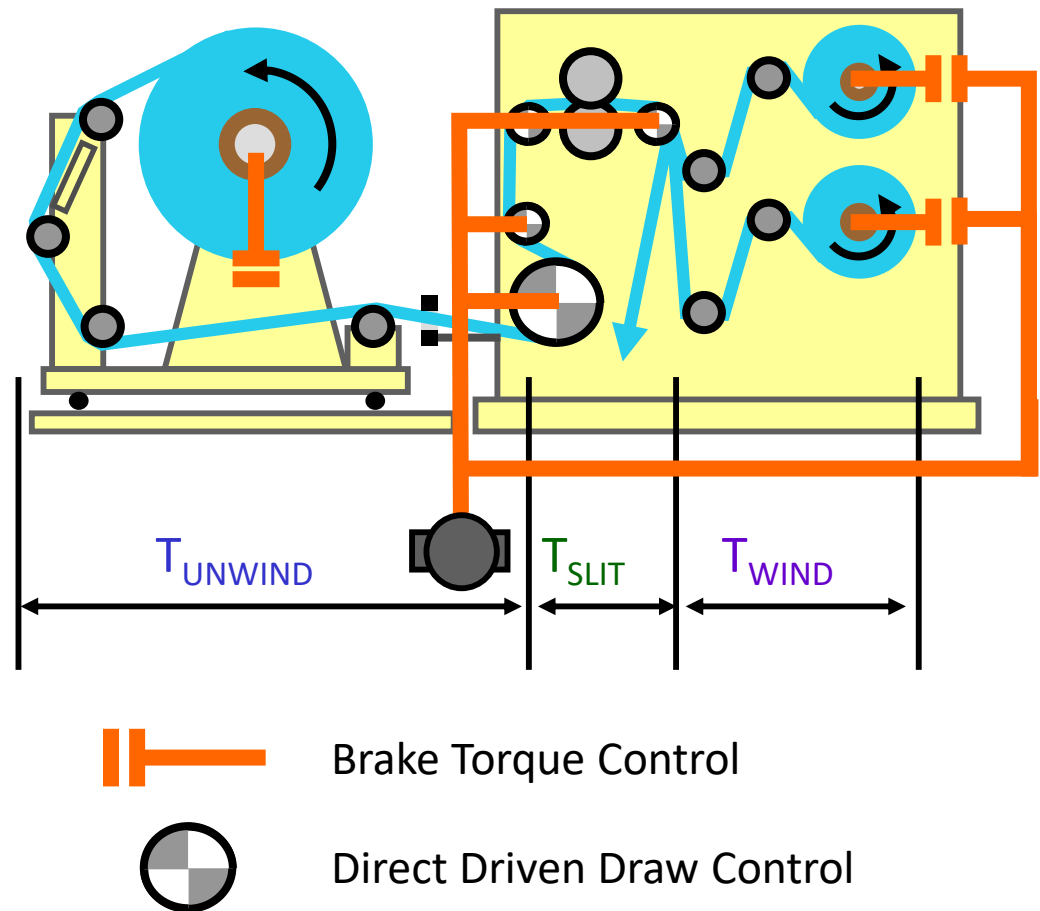
Typical Slitter Rewinder

Many slitters are considered to have four tension zones.

Unwind tension controlled by brake or motor.

Slitting tension control by draw.

Two rewind tension controlled by clutch or differential shaft.



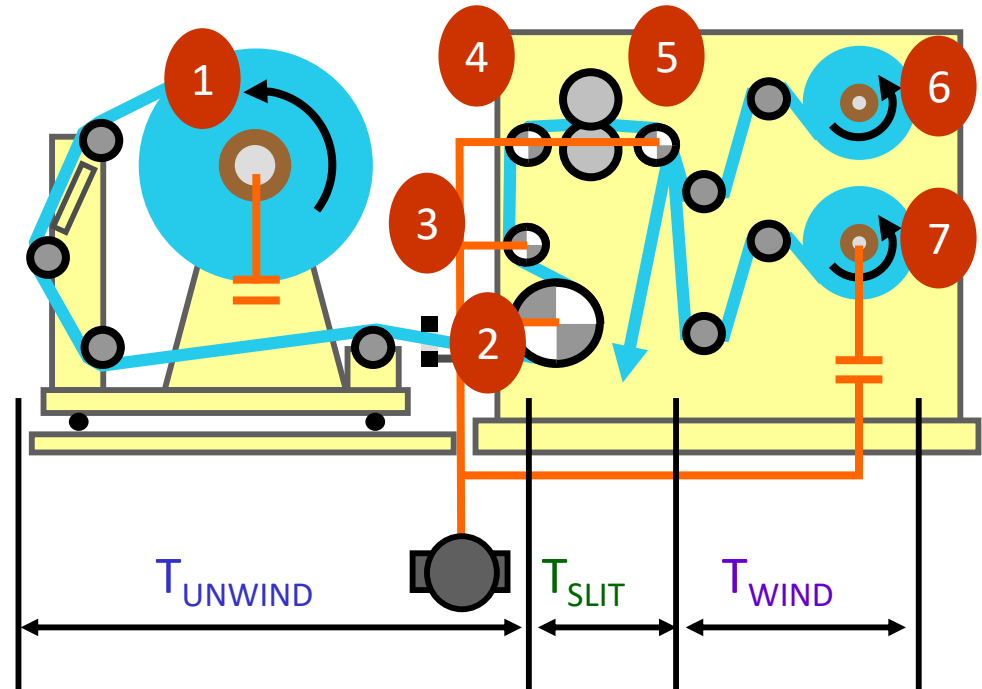
Typical Slitter Rewinder

This slitter actually has six tension zones.

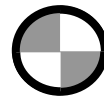
There are 7 drive points, so there are 6 tension zones.

1-2, 2-3, 3-4, 4-5, 5-6, and 5-7.

2-3, 3-4, 4-5 set draw ratio zones set by the designed.



Brake Torque Control



Direct Driven Draw Control



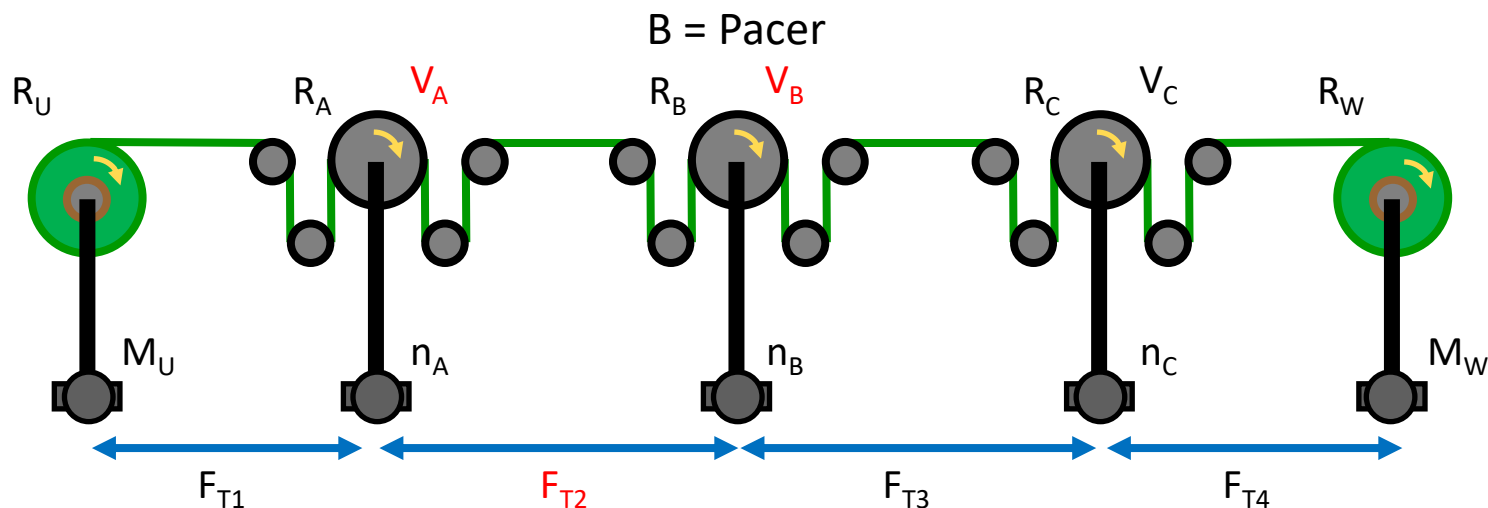
Clutch Torque Control

Tension Isolation?

Tension isolation is a term used to describe a drive roller with traction forces greater than the tension differential before and after the drive.

However, downstream tensions will always be a function of upstream tension and no amount of friction will prevent strain transport.

Tension from Speed Ratios (Draw)



What determines tension in open-loop draw controlled systems?

M_X = Torque at X

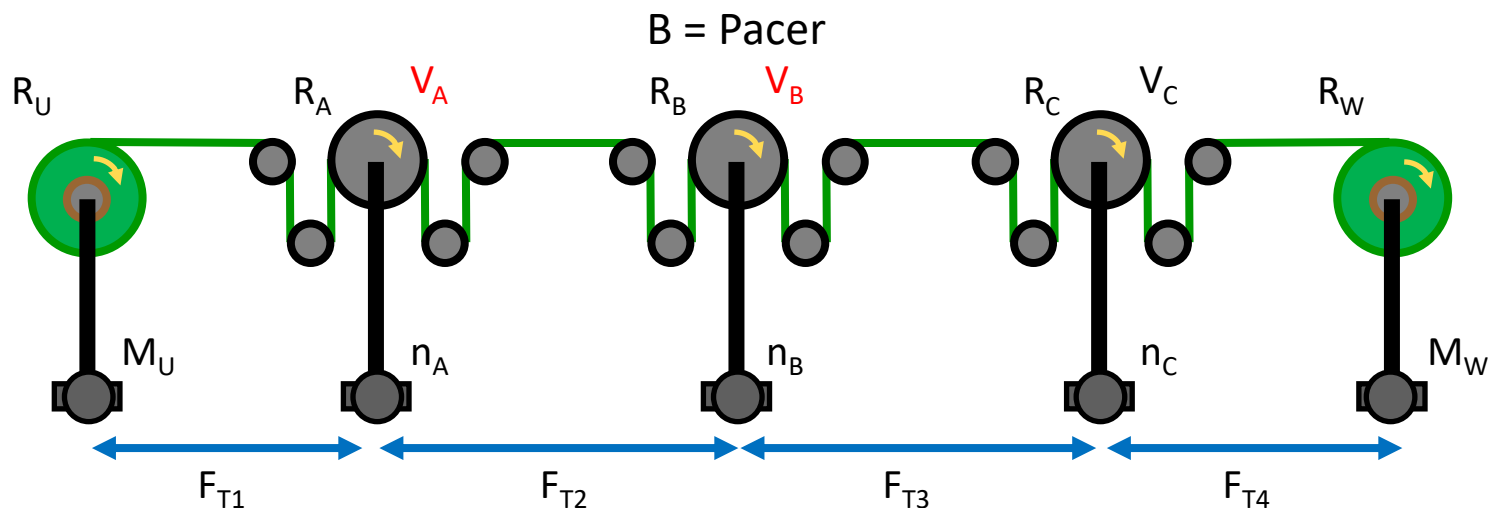
n_X = Rotational speed at X

V_X = Velocity at X

F_{TX} = Force of tension in zone X

R_X = Radius of roll or roller X

Tension from Speed Ratios (Draw)



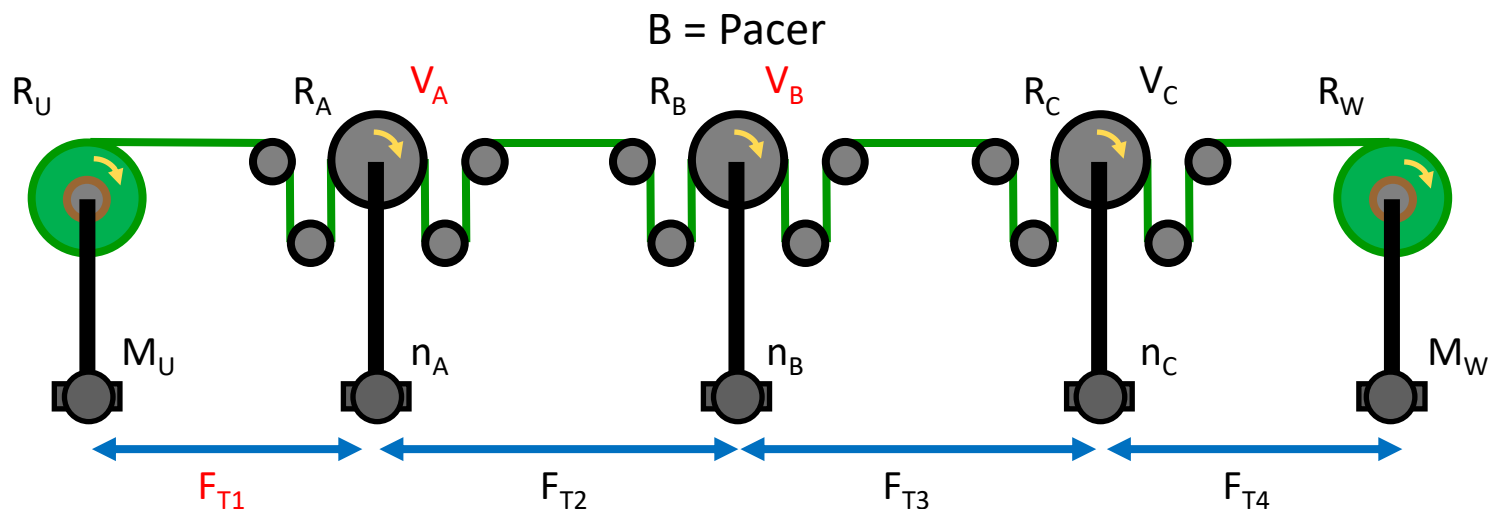
What determines tension in open-loop draw controlled systems?

- M_X = Torque at X
- n_X = Rotational speed at X
- V_X = Velocity at X
- F_{TX} = Force of tension in zone X
- R_X = Radius of roll or roller X

Follow the strain and convert strain to tension via thickness, width, and modulus.

$$V_X = 2\pi R_X n_X$$

Tension from Speed Ratios (Draw)



What determines tension in open-loop draw controlled systems?

$$\frac{V_1}{(1 + \varepsilon_1)} = \frac{V_2}{(1 + \varepsilon_2)}$$

$$\varepsilon_2 = \frac{V_2(1 + \varepsilon_1)}{V_1} - 1$$

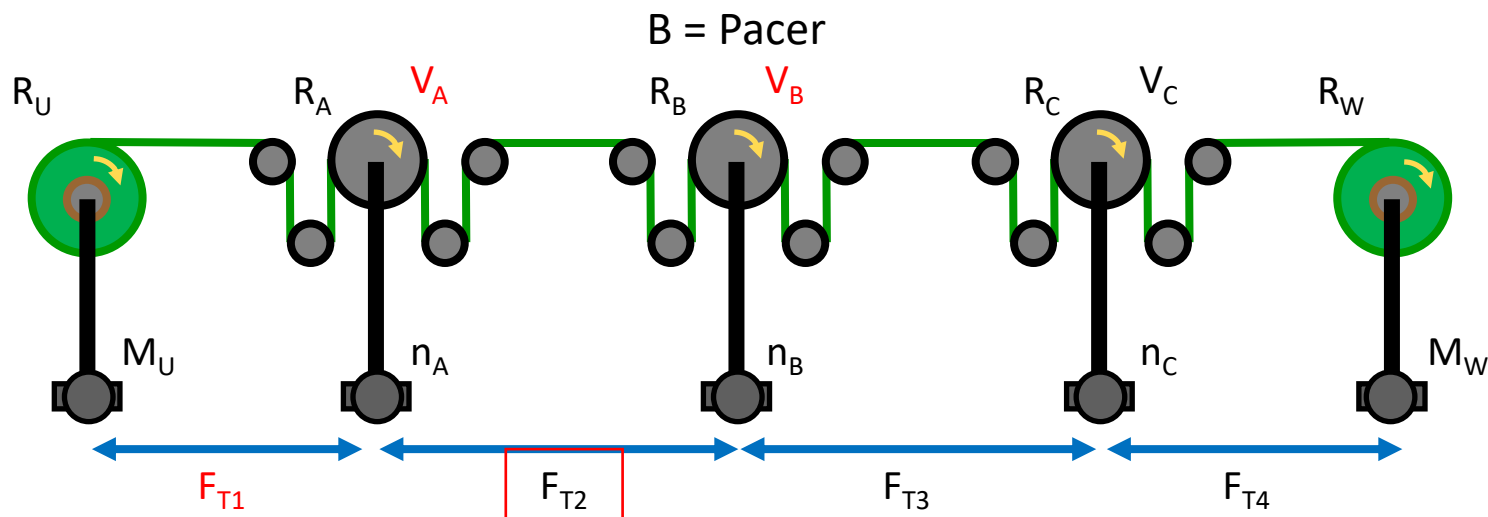
$$\varepsilon_2 = \frac{V_2 - V_1}{V_1} + \frac{V_2 \varepsilon_1}{V_1}$$

$$Draw = \frac{(V_2 - V_1)}{V_1}$$

$$\varepsilon_2 \cong Draw + \varepsilon_1$$

Since $V_1/V_2 \approx 1$

Tension from Speed Ratios (Draw)



What determines tension in open-loop draw controlled systems?

ϵ_X = Strain in zone X

t_X = Web thickness in zone X

w_X = Web width in zone X

E_X = Web elastic modulus in zone X

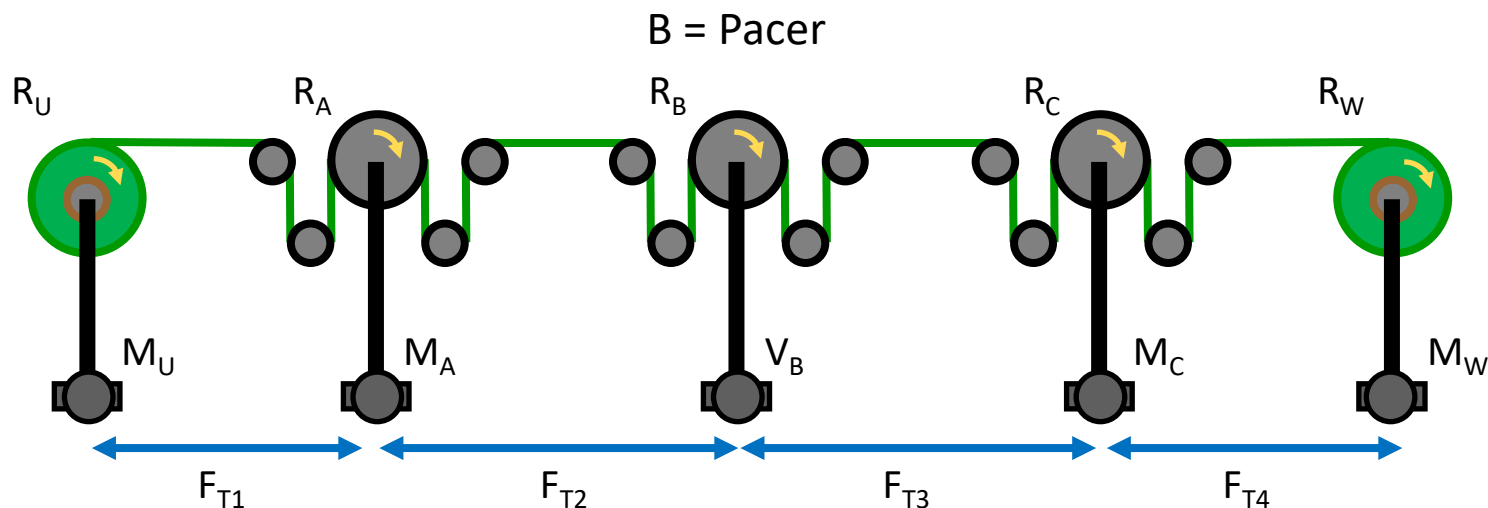
$$\epsilon_1 = \frac{F_{T1}}{t_1 w_1 E_1}$$

$$\epsilon_2 \cong \frac{(V_2 - V_1)}{V_1} + \frac{F_{T1}}{t_1 w_1 E_1}$$

$$F_{T2} \cong \left[\frac{(V_2 - V_1)}{V_1} + \frac{F_{T1}}{t_1 w_1 E_1} \right] (t_2 w_2 E_2)$$

This example assumes elasticity.

Tension from Torque



What determines tension in open-loop torque controlled systems?

M_X = Torque at X

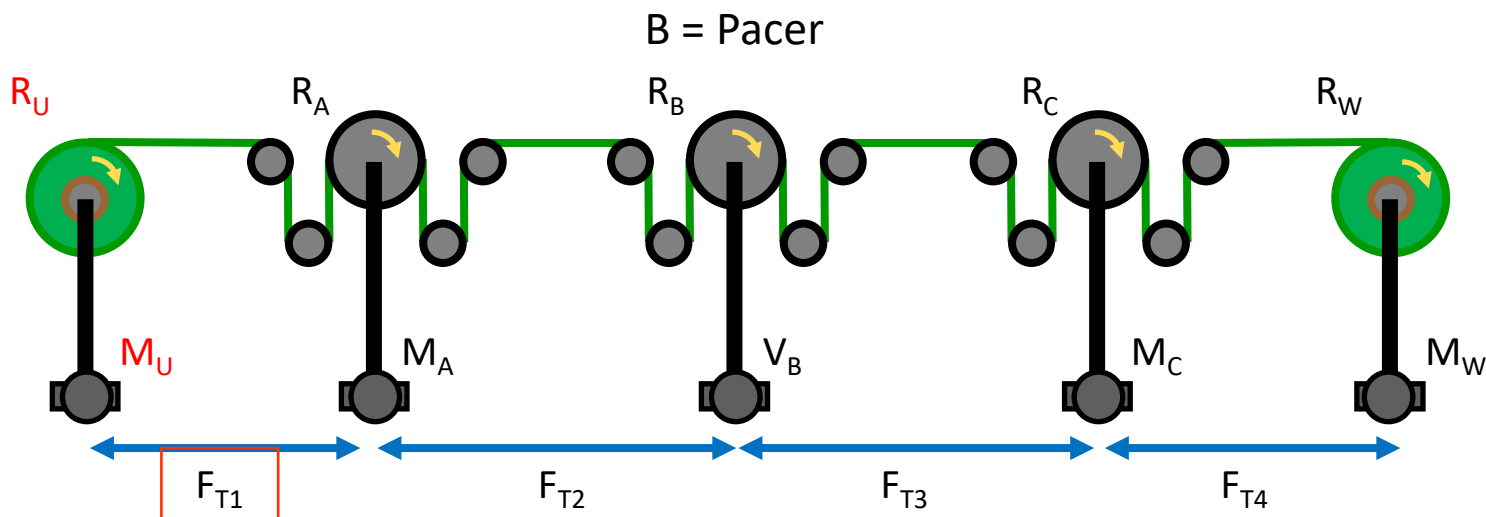
F_{TX} = Force of tension in zone X

R_X = Radius of roll or roller X

Torque, M , will be negative for backwards or braking torque and positive for forward or driving torque.

Tension, F_T , will be negative pulling upstream and position pulling downstream.

Tension from Torque



What determines unwind tension force, F_{T1} ?

$$\Sigma M = 0$$

$$M_U + F_{T1}R_U = 0$$

$$F_{T1} = -\frac{M_U}{R_U}$$

M_X = Torque at X

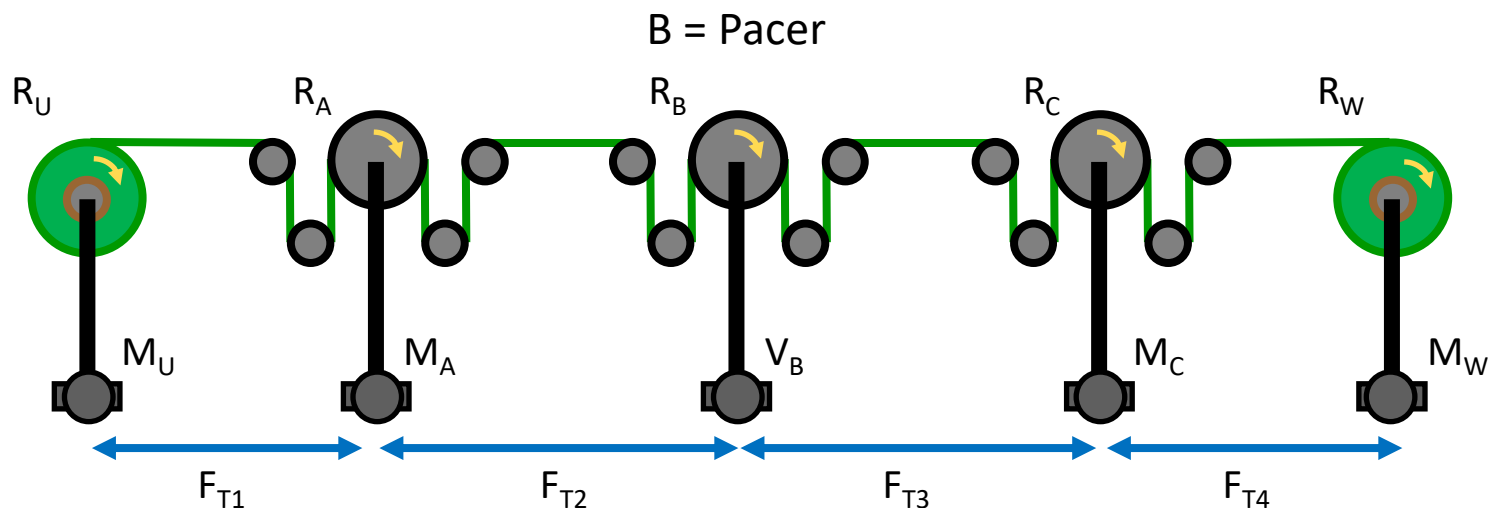
F_{TX} = Force of tension in zone X

R_X = Radius of roll or roller X

Unwind tension will increase as unwinding roll radius decreases.

Unwind tension should be constant if applied torque is set proportional to radius.

Tension from Torque



What determines unwind tension force, F_{T1} , including inertial torques?

$$\Sigma M = 0$$

$$M_U + F_{T1}R_U + I_U\alpha_U = 0$$

$$F_{T1} = -\frac{(M_U + I_U\alpha_U)}{R_U}$$

Rotation acceleration will be positive during acceleration and negative during deceleration.

M_X = Torque at X

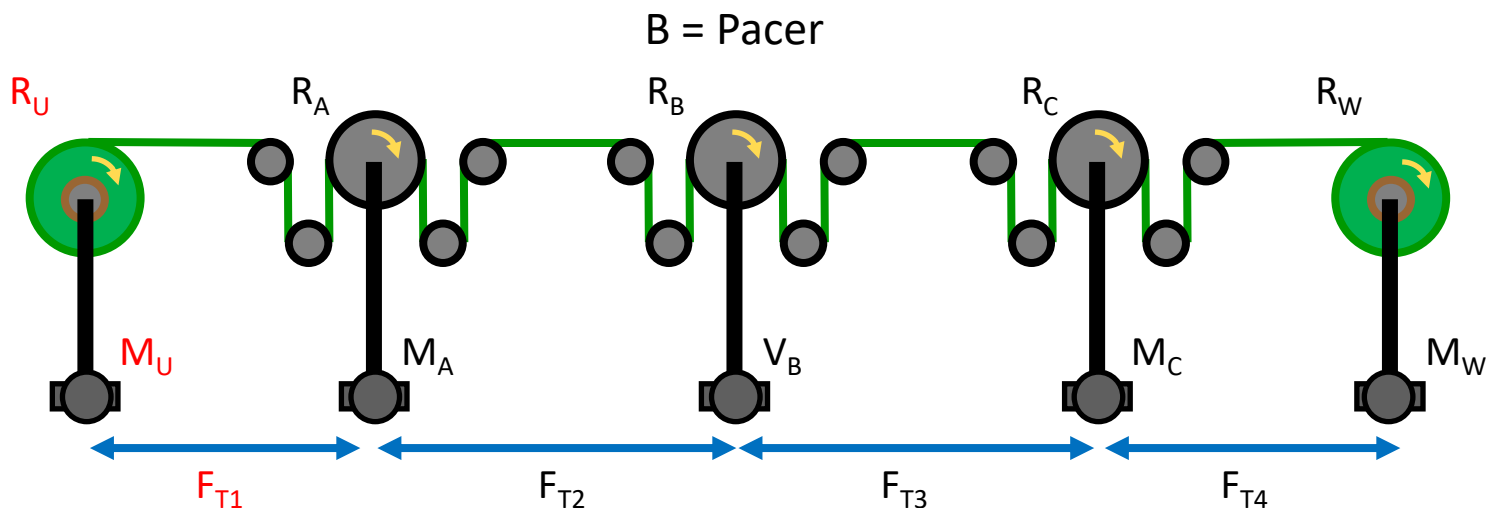
F_{TX} = Force of tension in zone X

R_X = Radius of roll or roller X

I_X = Inertia of roll or roller X

α_X = Rotational acceleration of X

Tension from Torque



What determines unwind tension force, F_{T1} , including inertial torques?

$$\Sigma M = 0$$

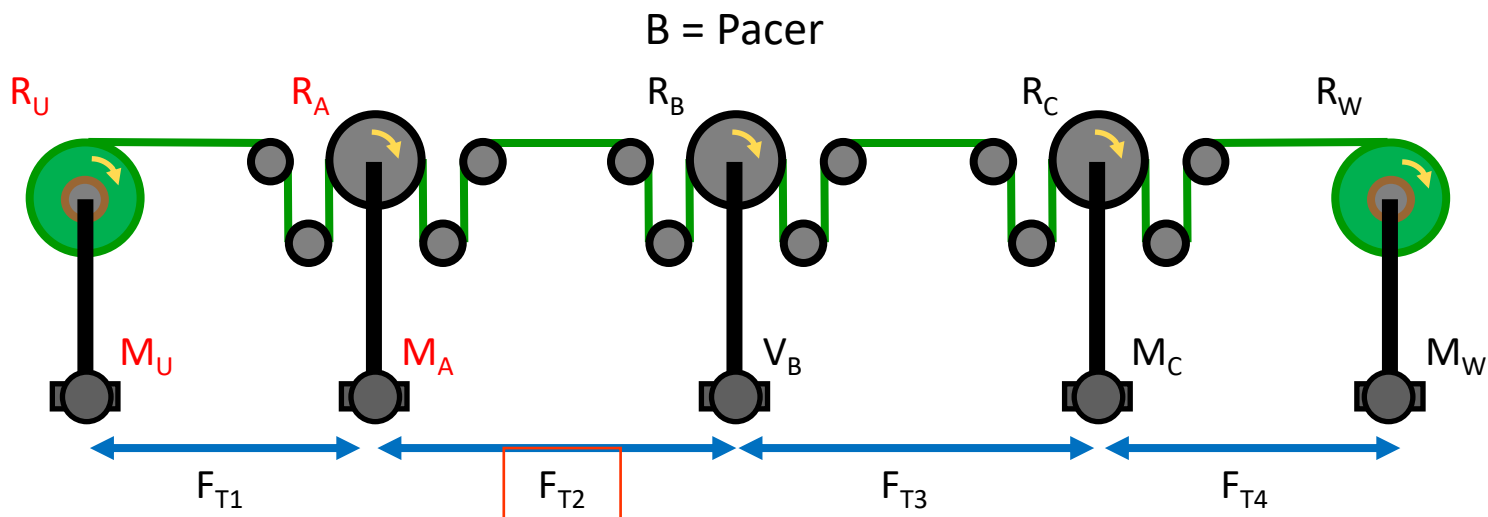
$$M_U + F_{T1}R_U + \overbrace{I_U\alpha_U} = 0$$

$$F_{T1} = -\frac{(M_U + I_U\alpha_U)}{R_U}$$

Inertial torques create tension variations even with constant applied torque.

Roll inertia and rotational accelerations change with roll radius changes.

Tension from Torque



What determines tension force in zone 2, F_{T2}?

$$\Sigma M = 0$$

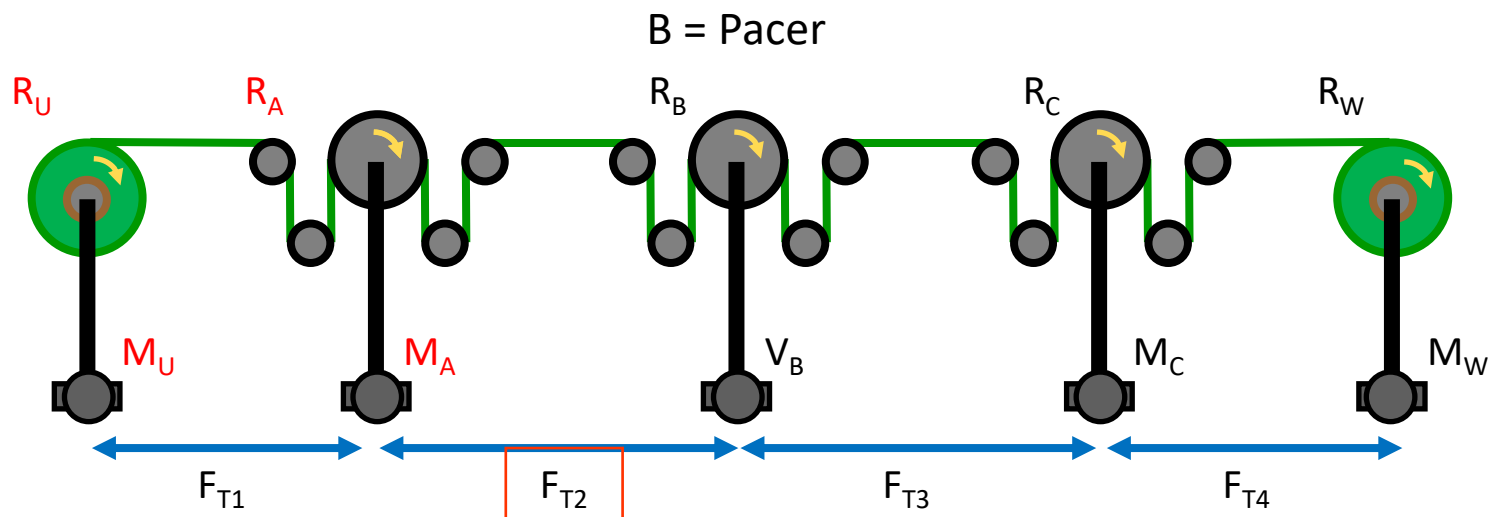
$$M_A + F_{T1}R_A - F_{T2}R_A = 0$$

$$F_{T2} = \frac{M_A}{R_A} + F_{T1}$$

$$\Delta F_{T12} = \frac{M_A}{R_A}$$

$$F_{T2} = \frac{M_A}{R_A} + \frac{M_U}{R_U}$$

Tension from Torque



What determines tension force in zone 2, F_{T2} , including **inertial torques**?

$$\Sigma M = 0$$

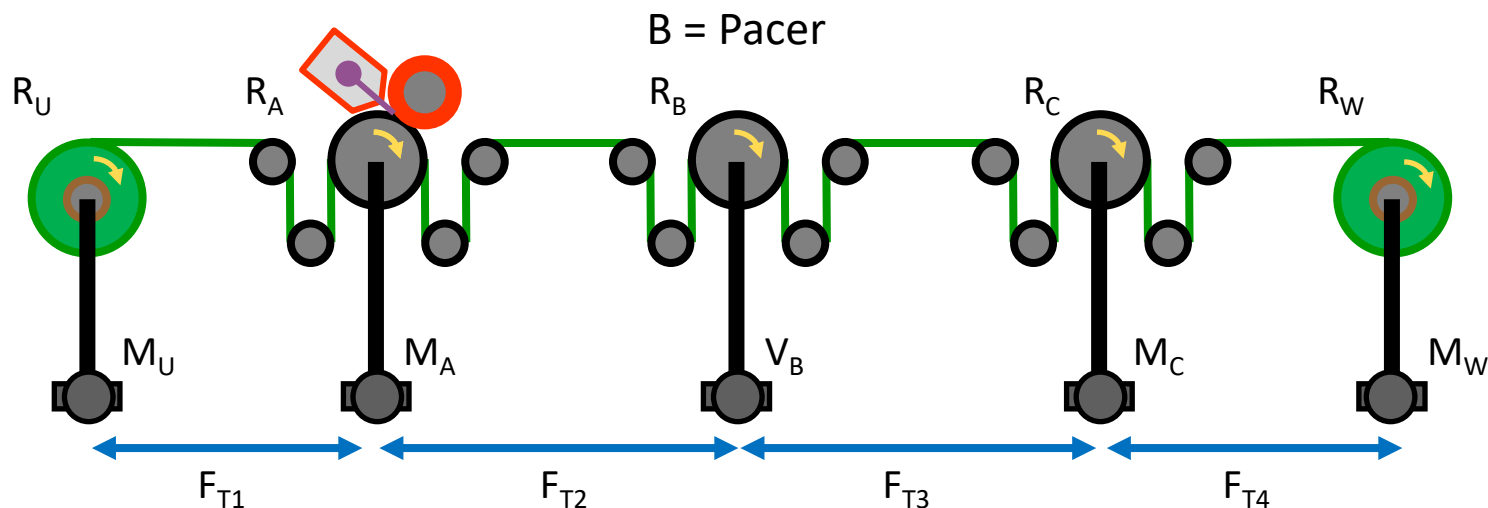
$$M_A + F_{T1}R_A - F_{T2}R_A + I_A\alpha_A = 0$$

$$F_{T2} = \frac{(M_A + I_A\alpha_A)}{R_A} + F_{T1}$$

$$\Delta F_{T12} = \frac{(M_A + I_A\alpha_A)}{R_A}$$

$$F_{T2} = \frac{(M_A + I_A\alpha_A)}{R_A} + \frac{(M_U + I_U\alpha_U)}{R_U}$$

Tension from Torque



What determines tension force in zone 2, F_{T2} , including inertial and **process torques**?

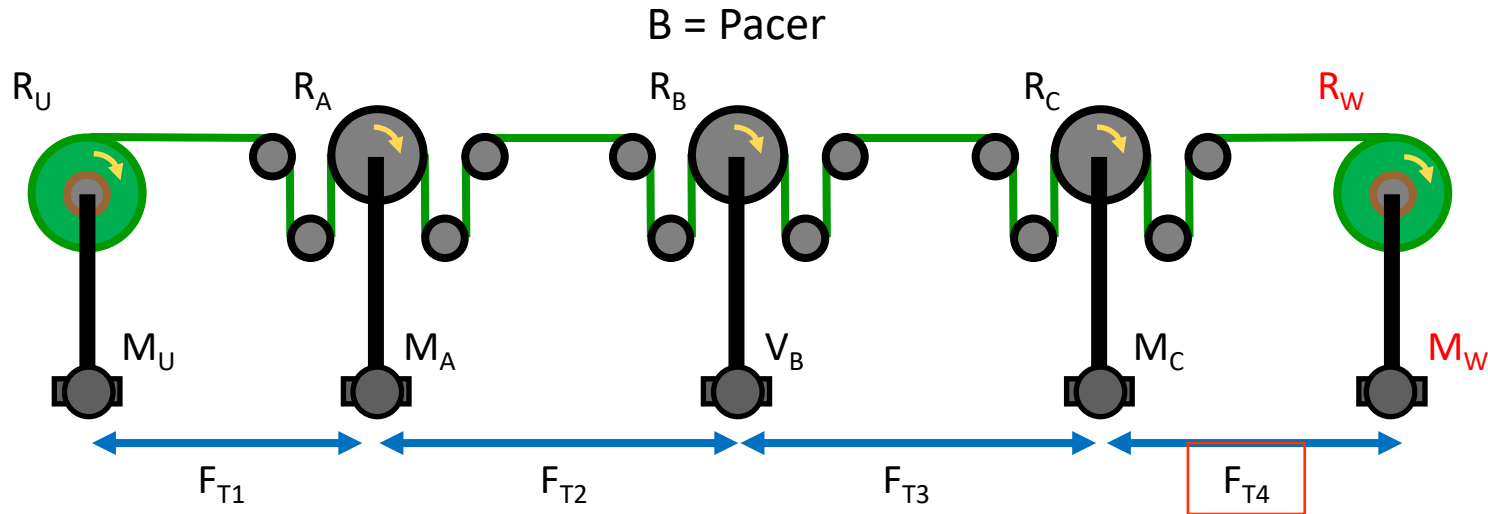
$$\Sigma M = 0$$

$$M_A + F_{T1}R_A - F_{T2}R_A + I_A\alpha_A + \overbrace{F_{PA}R_A} = 0$$

$$F_{T2} = \frac{(M_A + I_A\alpha_A)}{R_A} + F_{PA} + F_{T1}$$

Process forces include:
coating, calendaring, laminating,
embossing, and slitting.

Tension from Torque



What determines winding tension force, F_{T4} ?

$$\Sigma M = 0$$

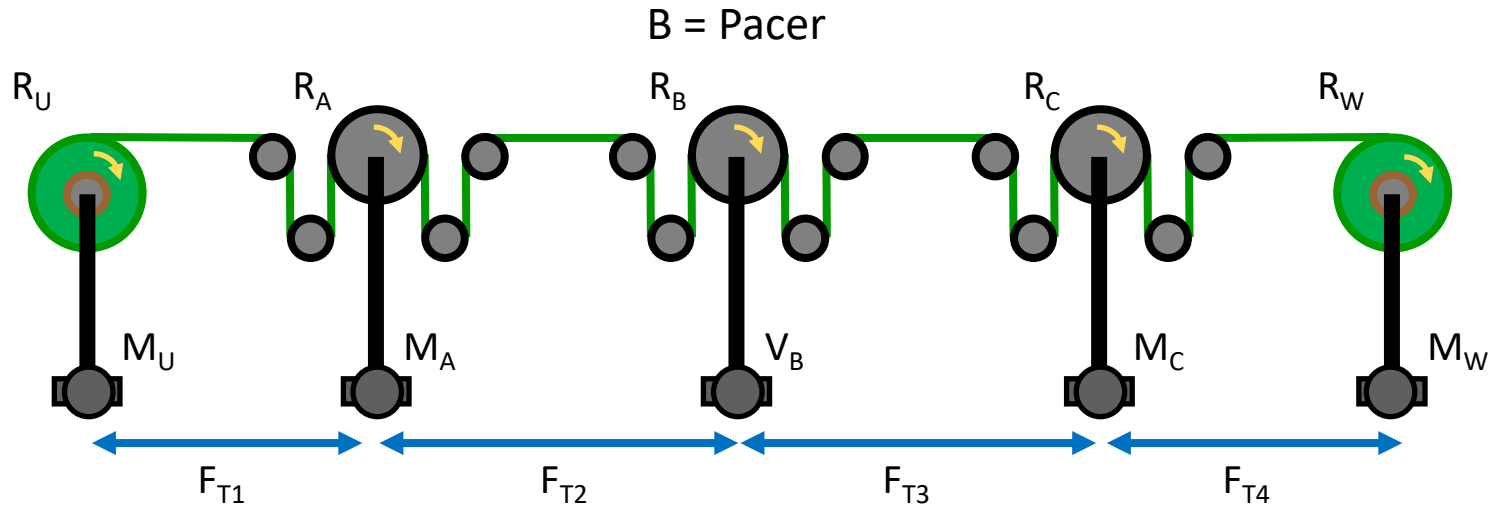
$$M_W + F_{T4} R_W = 0$$

$$F_{T4} = -\frac{(M_W)}{R_W}$$

Winding tension will decrease as winding roll radius decreases.

Winding tension should be constant if applied torque is set proportional to radius.

Tension from Torque



What determines winding tension force, F_{T4} , including inertial torques?

Inertial torques create tension variations even with constant applied torque. Roll inertia and rotational accelerations change with roll radius changes.

$$\Sigma M = 0$$

$$M_W + F_{T4}R_W + \overbrace{I_W\alpha_W} = 0$$

$$F_{T4} = -\frac{(M_W + I_W\alpha_W)}{R_W}$$



2.1 Speed, Length, Diameter

Important dimensions in web handling

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2.1 Learning Objectives

- ▶ How is web speed and length measured?
- ▶ What causes errors in length measurement?
- ▶ How is diameter of winding roll determined?

Speed vs Velocity

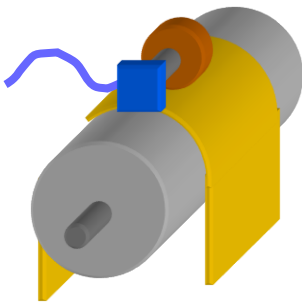
Most people would use these term interchangeably.

- ▶ Speed is a scalar (value only)
 - ▶ The car is going 60 mph.
 - ▶ Speed = distance / time

- ▶ Velocity is a vector (value and direction)
 - ▶ The care is going 60 mph towards the north.
 - ▶ $V = \text{change in position} / \text{change in time} = \Delta x / \Delta t$

Measuring Speed or Length

Surface Speed Monitor



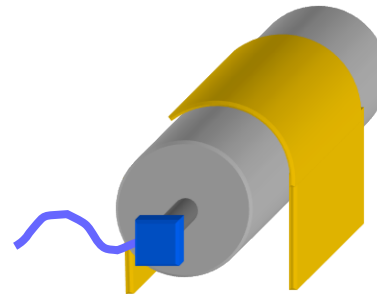
Gravity or spring loaded wheel rides on the web.

Encoder counts pulses of wheel revolutions

$$V = (\text{rpm}_{\text{wheel}})(\text{Pi})(D_{\text{wheel}})$$

$$L = (V)(\text{time})$$

Roller Speed Monitor



Tensioned web wraps metal roller, either driven or idling, nipped or unnipped.

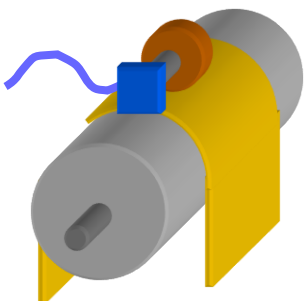
Encoder counts pulses of roller revolutions

$$V = (\text{rpm}_{\text{roller}})(\text{Pi})(D_{\text{roller}})$$

$$L = (V)(\text{time})$$

Measuring Speed or Length

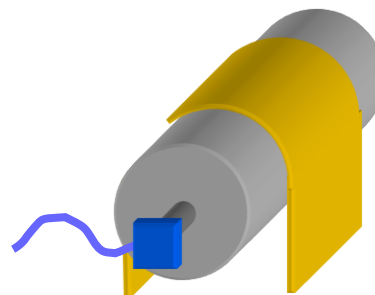
Surface Speed Monitor



Pro: Nip reduced air lubrication. May include on-device visual tally of length. Easy to add to existing eqmt.

Con: May slip. Rubber wheel may deform, bounce, or wear. Nip may wrinkle or deform sensitive webs.

Roller Speed Monitor



Pro: Large metal roller has accurate diameter, won't deform, and with proper roughness, unlikely to slip.

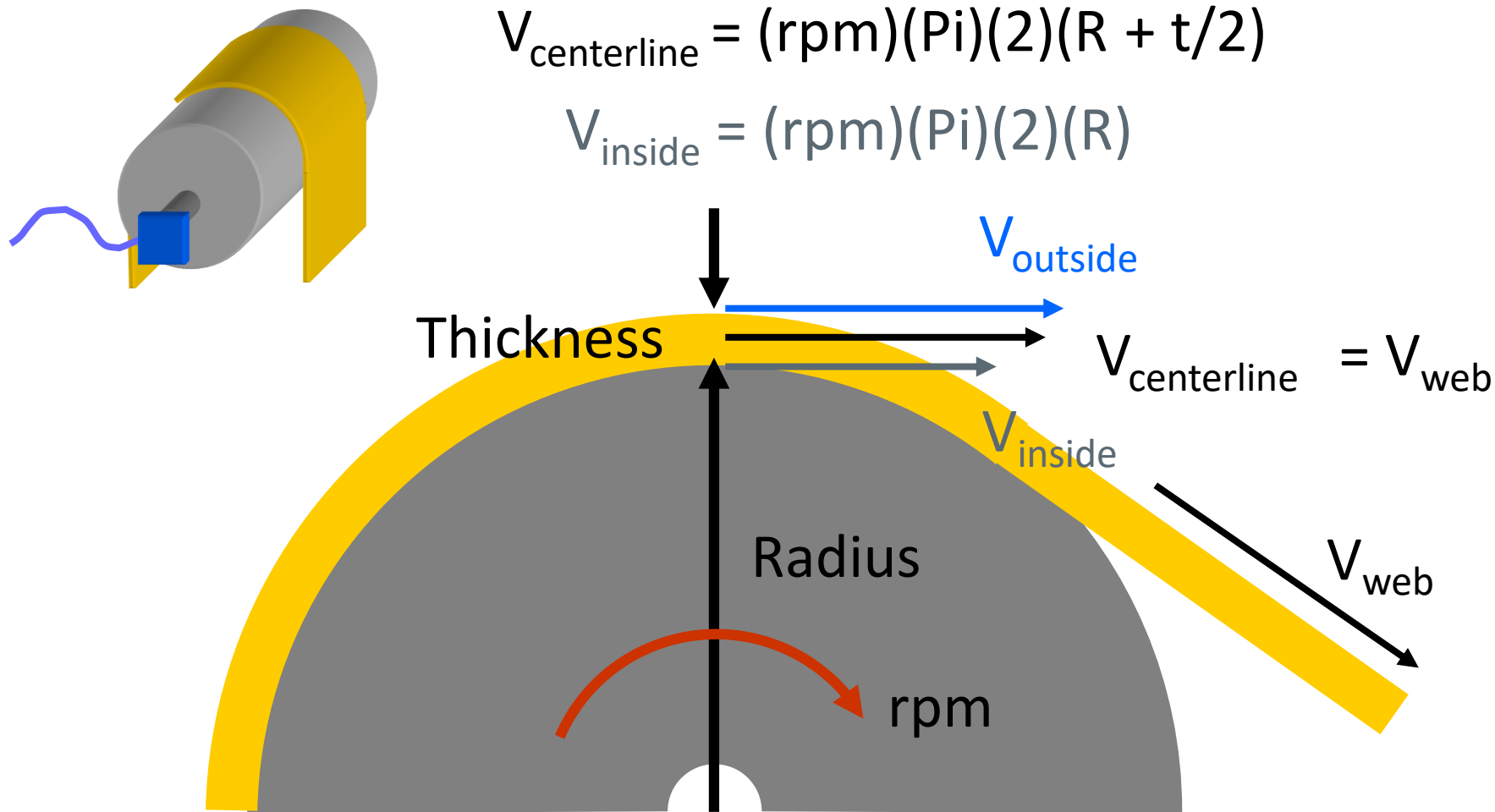
Con: More difficult to add to existing equipment. May need rough or textured roller surface.

Effect of Thickness and Radius

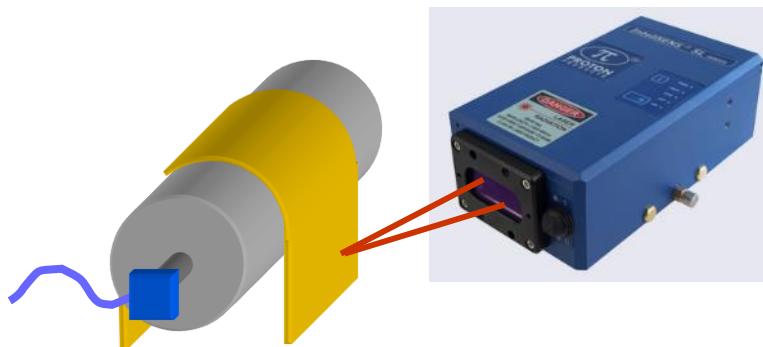
$$V_{\text{outside}} = (\text{rpm})(\text{Pi})(2)(R + t)$$

$$V_{\text{centerline}} = (\text{rpm})(\text{Pi})(2)(R + t/2)$$

$$V_{\text{inside}} = (\text{rpm})(\text{Pi})(2)(R)$$



Laser Velocimeter



Specification	Maximum	Units
Accuracy*	0.05	%
Repeatability	0.02	%
Acceleration	500	m/s ²
Internal measurement rate	100	kHz
Measurement output interval		μs
Laser spot size	3	mm

Pro: Measure actual surface speed.

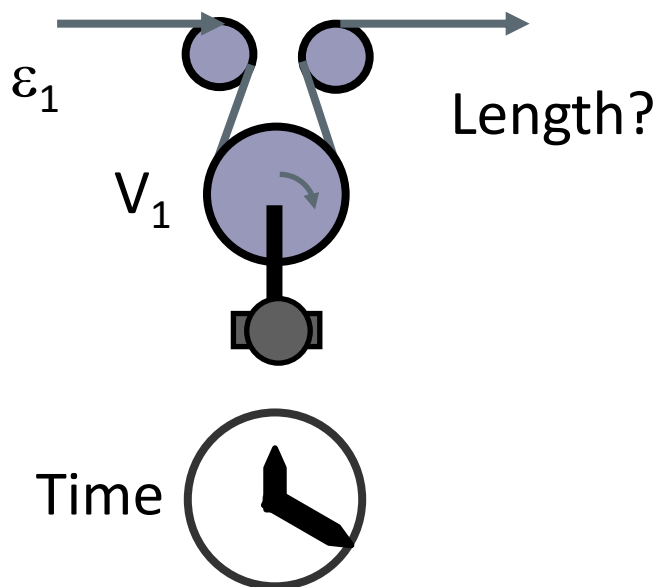
Con: Expensive.

Model	SL1525	Units
Stand-off distance	150	mm
	5.91	inch
Depth-of-field	25	mm
	0.98	inch
Minimum speed	0.1	m/min
	0.3	ft/min
Maximum speed	2500	m/min
	8200	ft/min



Courtesy of Proton Products

Does Length = Velocity x Time?



$$L_1 = V_1 t$$

$$L_0 = \frac{L_1}{(1 + \epsilon_1)}$$

The first estimate of length feed rate is velocity x time.

$$(100 \text{ m/minute})(100 \text{ minutes}) = 10,000 \text{ m}$$

This calculate ignores that the web is strained by tension and will have a shorter length when tension is removed.

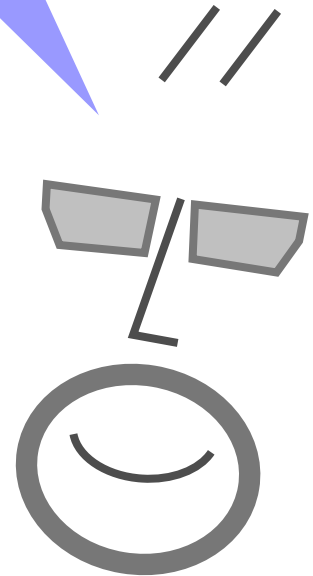
If strain 1 = 0.01 = 1%, the 10,000 m when measured without tension will measure $(10000)/(1+0.01) = 9901\text{m}$

It will seem as if someone lost 99m!

Torque Control Qs

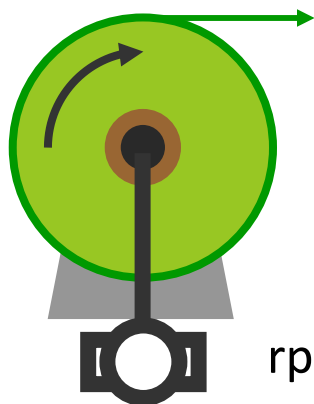
How is diameter measured? Why is diameter important?

- ▶ Translating speed (in rpms) into velocity (surface speed).
- ▶ Translating center torque into surface tension or tension change.
- ▶ Calculating tension set point from taper tension functions.
- ▶ Calculating torque of rotational inertia during speed changes.
- ▶ Positioning gap rollers at winding (though photo eye methods are preferred).



Need for Diameter at Winding and Unwinding

Unwinding



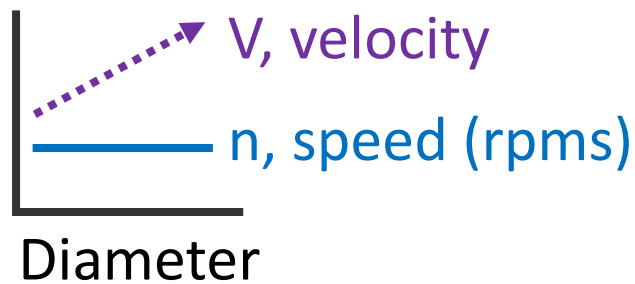
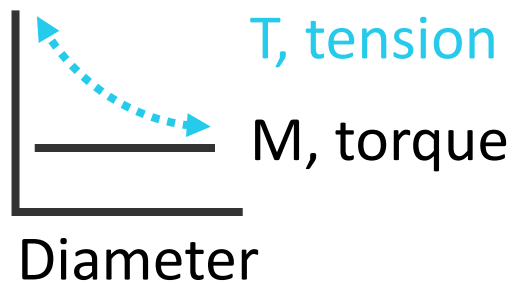
Winding



Motor
rpm, torque

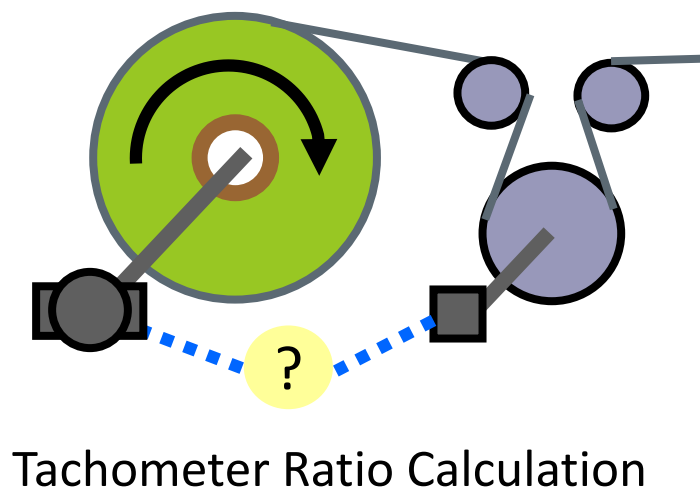
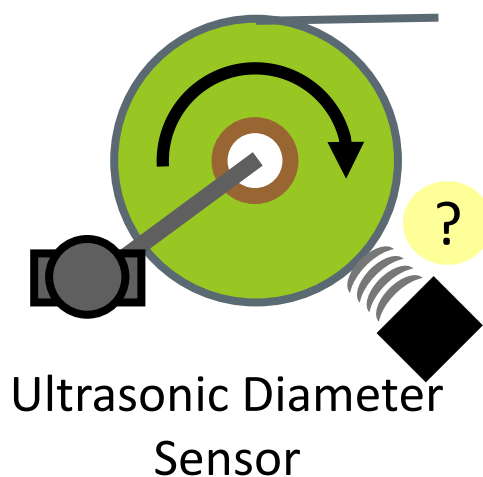
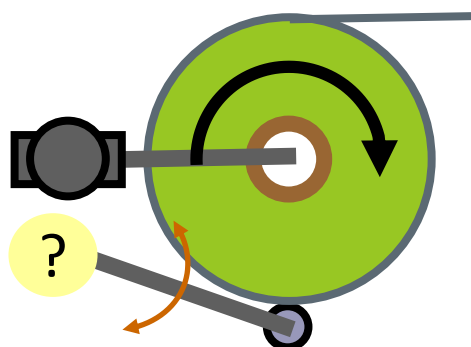
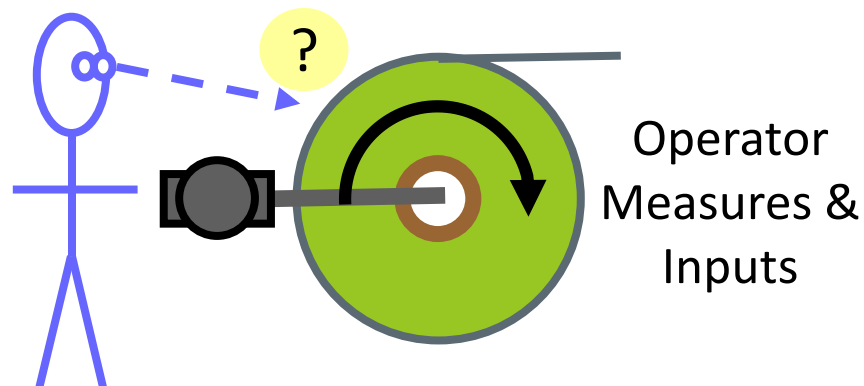
Without diameter compensation at winding and unwinding:

- Constant speed (in rpms) will change surface velocity directly with diameter
- Constant torque will change tension inversely with roll diameter.



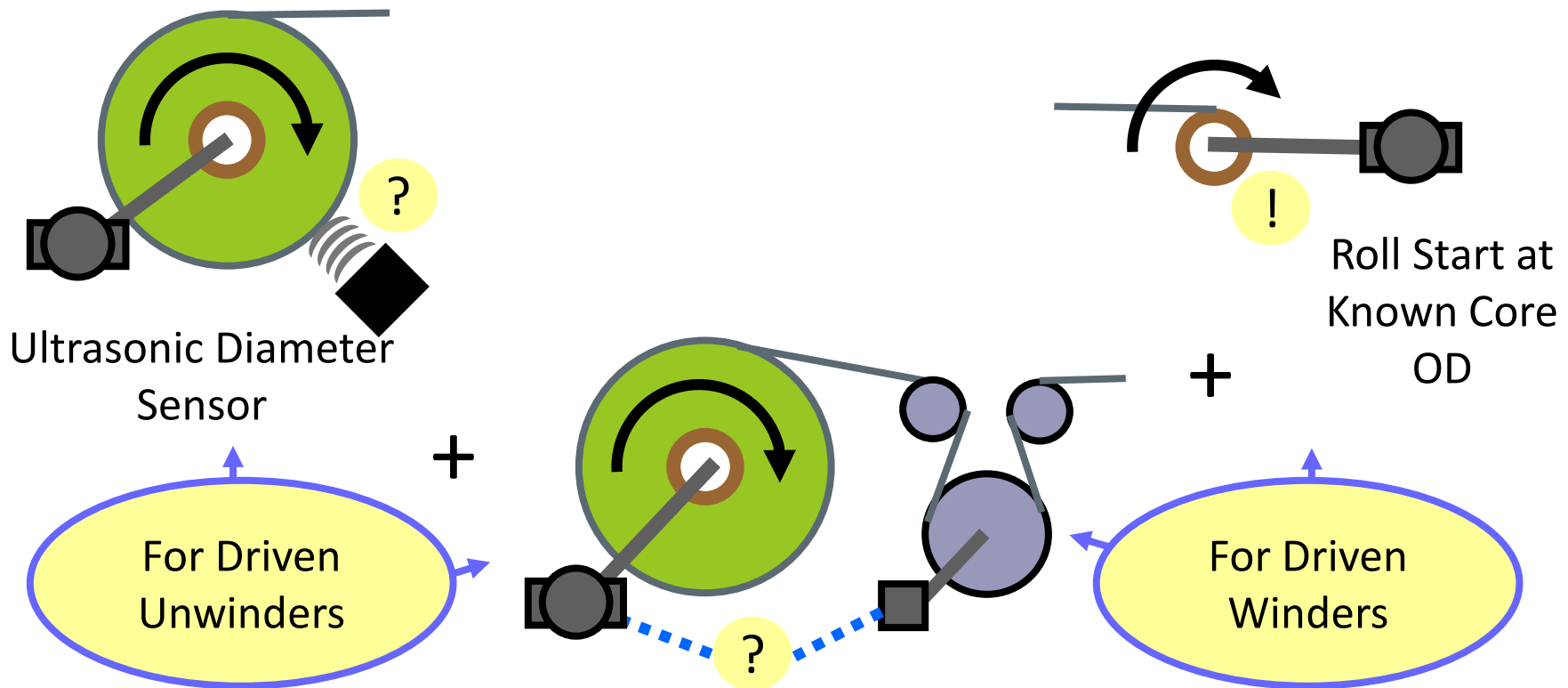
Diameter Compensation

The most common radius detection methods are:



Diameter Feedback

Two diameter measurement methods are commonly combined on advance unwinder and winders:



Tachometer Ratio Calculation

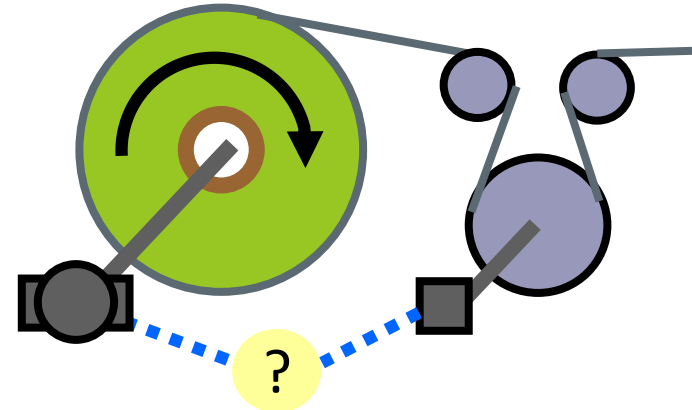
Tachometer Ratio Diameter Calculation

Inputs:

1. Unwinding or winding roll rpms.
2. Rpm's of non-slipping roller (often steel driven roller)
3. Diameter of driven roller

Calculate:

Diameter of unwinding or winding roll.



$$D_{roll} = D_{roller} \frac{(rpm_{roller})}{(rpm_{roll})}$$

Drawback of this method:
Can't calculate when speed is zero.

Thickness Buildup Diameter Calculation

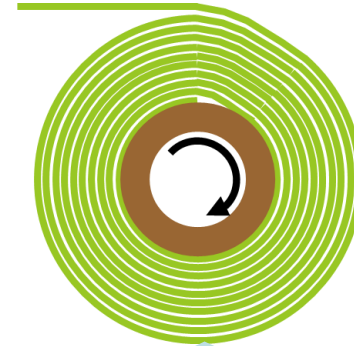
Inputs:

1. Core diameter, D_{core}
2. Web thickness, t
3. Roll revolutions, n
4. Compressibility factor, C

Calculate:

Diameter of unwinding or winding roll.

$$D_{roll} = D_{core} + 2ntC$$



Compressibility factor, C , is an unknown function of material properties and winding conditions. C can be < 1 for soft or porous materials or > 1 for rough materials or due to entrained air or thickness recovery.

Dancer Position Diameter Calculation

For winders with:

- 1) Motor position feedback and
- 2) Dancer position feedback

Diameter can be calculated from a small controlled motor rotation during stalled tensioning. A controlled angular roll rotation will create a measureable dancer arm rotation.

$$r_{roll}\theta_{roll} = 2r_{dancer}\theta_{dancer}$$

$$r_{roll} = 2r_{dancer}\left(\frac{\theta_{dancer}}{\theta_{roll}}\right)$$

